

# HIGH PERFORMANCE COMPUTING (HPC) MARKET

GLOBAL FORECAST TO 2027

BY COMPONENT (SOLUTIONS (SERVERS, STORAGE, NETWORKING DEVICES, AND SOFTWARE), AND SERVICES), COMPUTATION TYPE, DEPLOYMENT, ORGANIZATION SIZE, SERVER PRICES BAND, VERTICAL, AND REGION

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## LIST OF ABBREVIATIONS

| ABBREVIATION | FULL FORM                                                  |
|--------------|------------------------------------------------------------|
| AI           | Artificial Intelligence                                    |
| AICTO        | Arab Information and Communication Technology Organization |
| ANL          | Energy's Argonne National Laboratory                       |
| ANSI         | American National Standards Institute                      |
| APAC         | Asia Pacific                                               |
| APU          | Accelerated Processing Unit                                |
| ASP          | Average Selling Price                                      |
| ASSP         | American Society of Safety Professionals                   |
| AWS          | Amazon Web Services                                        |
| BFSI         | Banking, Financial Services, and Insurance                 |
| CAE          | Computer-aided Engineering                                 |
| CAGR         | Compound Annual Growth Rate                                |
| CENELEC      | European Electrotechnical Standardization Organization     |
| CEO          | Chief Executive Officer                                    |
| CGG          | Center of Good Governance                                  |
| CITC         | Communications and Information Technology Commission       |
| CNRS         | National Center for Scientific Research                    |
| CPM          | Computational Memory                                       |
| CPU          | Central Processing Unit                                    |
| CR           | Computational Resources                                    |
| CSL          | China's Cyber Security Law                                 |
| CSP          | Cloud Service Provider                                     |
| CT           | Computed Tomography                                        |
| CXO          | Chief Experience Officer                                   |
| DAS          | Directly Attached Storage                                  |
| DBaaS        | Database-as-a-Service                                      |
| DCG          | Data Center Group                                          |
| DDN          | Data Direct Networks                                       |
| DNA          | Deoxyribonucleic Acid                                      |

|        |                                                                        |
|--------|------------------------------------------------------------------------|
| DOE    | US Department of Energy                                                |
| DOE SC | US Department of Energy - the Office of Science                        |
| ECLAC  | Economic Commission for Latin America                                  |
| ESCAP  | United Nations Economic and Social Commission for Asia and the Pacific |
| EU     | European Union                                                         |
| EUDCA  | European Cloud Association                                             |
| FITARA | Federal Information Technology Acquisition Reform Act                  |
| GDPR   | General Data Protection Regulation                                     |
| GPU    | Graphics Processing Unit                                               |
| HD     | High Definition                                                        |
| HDTV   | High-Definition Television                                             |
| HIPAA  | Health Insurance Portability and Accountability Act                    |
| HPC    | High-Performance Computing                                             |
| HPE    | Hewlett Packard Enterprises                                            |
| I/O    | Input/Output                                                           |
| IC     | Integrated Circuit                                                     |
| IDG    | Intelligent Devices Group                                              |
| IEC    | International Electrotechnical Commission                              |
| IEEE   | Institute of Electrical and Electronics Engineers                      |
| IIoT   | Industrial Internet of Things                                          |
| IoT    | Internet of Things                                                     |
| IP     | Intellectual Property                                                  |
| ISO    | International Organization for Standardization                         |
| ISVs   | Independent Software Vendors                                           |
| IT     | Information Technology                                                 |
| ITA    | International Trade Administration                                     |
| ITC    | International Trade Centre                                             |
| ITSC   | Information Technology Standards Committee                             |
| JDSC   | Japan Data Center Council                                              |
| LAN    | Local Area Network                                                     |
| LHC    | Large Hadron Collider                                                  |
| LLNL   | Lawrence Livermore National Laboratory                                 |
| MAS    | Monetary Authority of Singapore                                        |

|      |                                               |
|------|-----------------------------------------------|
| MEA  | Middle East & Africa                          |
| ML   | Machine Learning                              |
| MLPS | Multi-Level Protection Scheme                 |
| MPP  | Massively Parallel Processing                 |
| MRI  | Magnetic Resonance Imaging                    |
| MTBF | Mean Time Between Failure                     |
| MTCS | Multi-Tier Cloud Security                     |
| NAS  | Network-Attached Storage                      |
| NGS  | Next-Generation Sequencing                    |
| NNSA | National Nuclear Security Administration      |
| ODM  | Original Design Manufacturer                  |
| OEM  | Original Equipment Manufacturer               |
| ORNL | Oak Ridge National Laboratory                 |
| OSHA | Occupational Safety and Health Administration |
| PCB  | Printed Circuit Board                         |
| PoC  | Proof of Concept                              |
| POPI | Protection of Personal Information Act        |
| PUE  | Power Usage Effectiveness                     |
| R&D  | Research and Development                      |
| RISC | Reduced Instruction Set Computer              |
| ROI  | Return on Investment                          |
| RoW  | Rest of World                                 |
| SAN  | Storage Area Networks                         |
| SI   | System Integrator                             |
| SME  | Small- and Medium-Sized Enterprises           |
| SOC  | System-on-Chip                                |
| UK   | United Kingdom                                |
| US   | United States of America                      |
| USD  | United States Dollar                          |
| UVic | University of Victoria                        |
| VFX  | Visual Effect                                 |

# 1 INTRODUCTION

---

## 1.1 STUDY OBJECTIVES

- To define and forecast the high-performance computing (HPC) market based on component, computation type, deployment, organization size, service price band, vertical, and region in terms of value
- To describe and forecast the size of the HPC market in four key regions, namely, North America, Europe, Asia Pacific (APAC), and Rest of the World (RoW), along with their respective countries
- To provide detailed information regarding drivers, restraints, opportunities, and challenges influencing the growth of the market
- To strategically analyze micromarkets<sup>1</sup> with respect to individual growth trends, prospects, and contributions to the overall market size
- To study the complete value chain of the HPC market
- To provide information regarding the impact of the COVID-19 on the HPC market, its segments, and the market players
- To analyze opportunities in the market for stakeholders by identifying high-growth segments of the HPC ecosystem
- To strategically profile the key players and comprehensively analyze their market positions in terms of their rankings and core competencies<sup>2</sup>, and provide a detailed competitive landscape
- To analyze competitive developments, such as product launches, acquisitions, collaborations, agreements, and partnerships, in the HPC market

## 1.2 DEFINITION

High-performance computing (HPC) refers to the practice of aggregating computing power to solve large problems in science, engineering, or business. HPC is the use of “supercomputers” to solve computational problems that are too large and require a long time to resolve for standard computers. A desktop computer has a single processing chip called the central processing unit (CPU). An HPC system, on the other hand, is a network of nodes, each of which comprises one or more processing chips and its own memory. The HPC technology exhibits excellent computing power and high performance to run advanced applications. It uses parallel supercomputers and clusters of computers where aggregation of computing power is needed for delivering higher performance.

---

1. Micromarkets are defined as the segments and subsegments of the high-performance computing market.

2. Core competencies of companies are captured in terms of their key developments and strategies adopted to sustain their positions in the high-performance computing market.

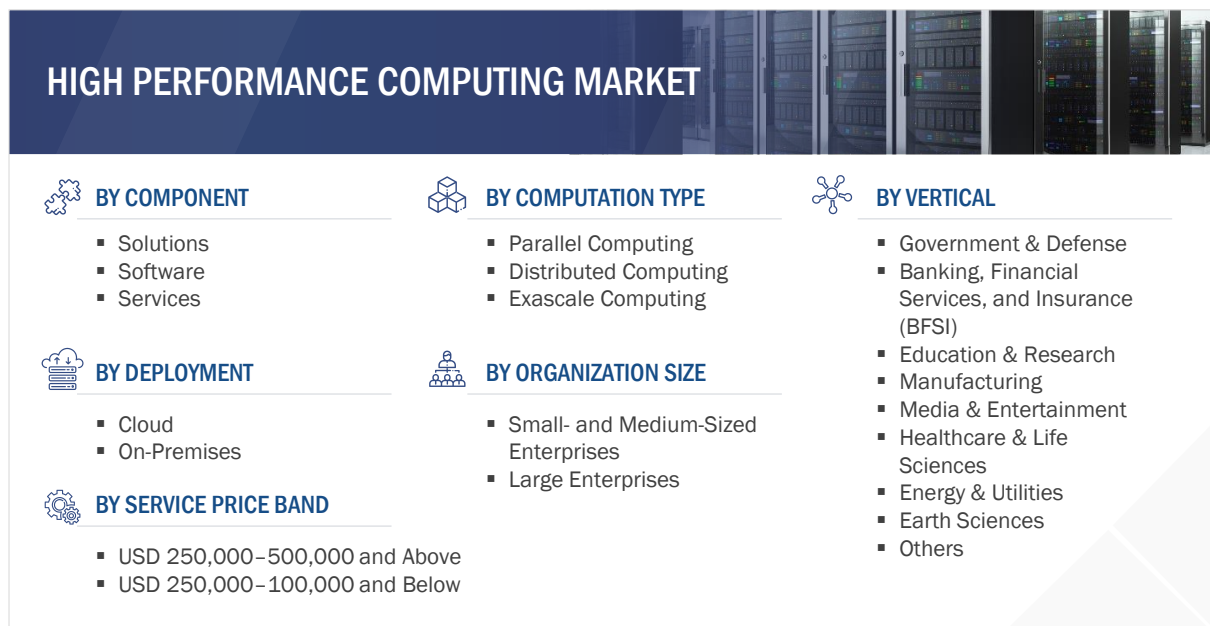
### 1.2.1 INCLUSIONS AND EXCLUSIONS

| PARTICULARS        | INCLUSION                                                                                                                                                                                                       | EXCLUSION                                                                                              |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Component          | Solutions, software, and services                                                                                                                                                                               | Supporting HPC infrastructure solutions such as power and cooling                                      |
| Computation Type   | Parallel computing, distributed computing, and Exascale computing (only qualitative writeup)                                                                                                                    | NA                                                                                                     |
| Deployment         | Cloud and on-premises                                                                                                                                                                                           | NA                                                                                                     |
| Organization Size  | Small- and medium-sized enterprises and large enterprises                                                                                                                                                       | Third-party providers such as resellers and distributors and companies offering free and open software |
| Service Price Band | USD 250,000–500,000 and above and USD 250,000–100,000 and below                                                                                                                                                 | Pricing of HPC accessories                                                                             |
| Vertical           | Government & defense, banking, financial services, and insurance (BFSI), education & research, manufacturing, media & entertainment, healthcare & life sciences, energy & utilities, earth sciences, and others | Non-profit organizations                                                                               |
| Geography          | North America (US, Canada, and Mexico), Europe (Germany, UK, France, and Rest of Europe), APAC (China, Japan, South Korea, and Rest of APAC), and RoW (Middle East & Africa and South America)                  | NA                                                                                                     |

## 1.3 STUDY SCOPE

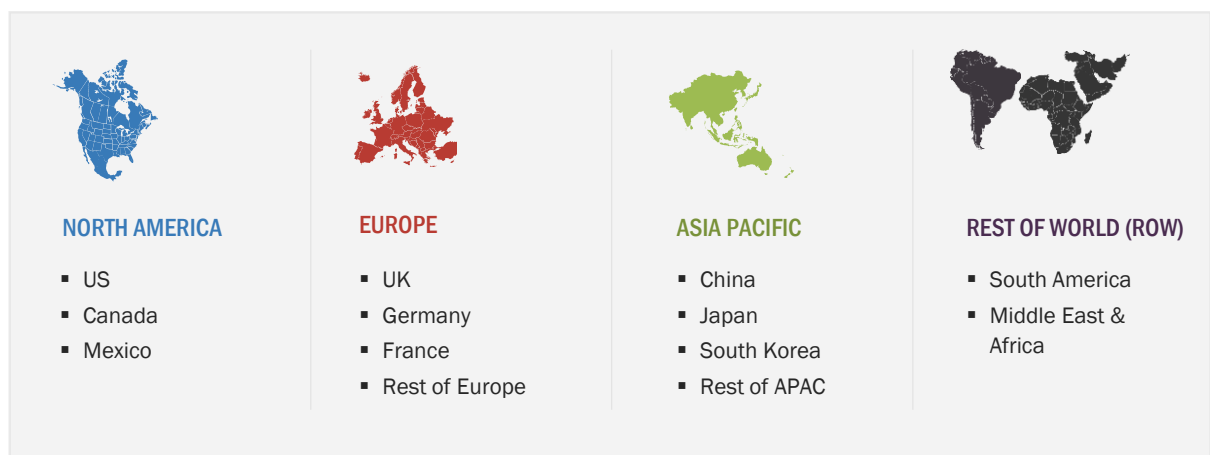
### 1.3.1 MARKETS COVERED

The report covers both demand and supply side segmentation of the high-performance computing market. The supply side segmentation includes component, computation type, deployment, organization size, and service price band, whereas the demand side segmentation includes vertical and region. The following figure provides an overview of the micromarkets covered in the report.

**FIGURE 1** HIGH-PERFORMANCE COMPUTING MARKET: SEGMENTATION

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

### 1.3.2 GEOGRAPHIC SCOPE



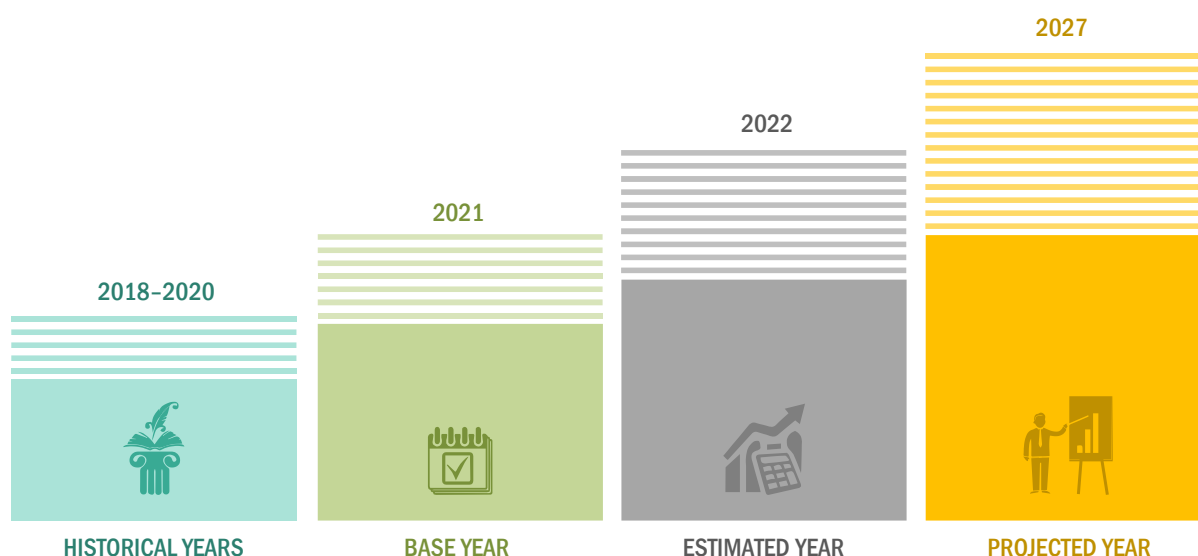
Note: Rest of Europe mainly includes Spain, Italy, the Netherlands, and other European countries.

Rest of APAC includes India, Singapore, and other Asian countries.

Rest of the World includes Qatar, Kuwait, Brazil, Argentina, Chile, and other countries in the Middle East and Latin America.

Source: MarketsandMarkets Analysis

### 1.3.3 YEARS CONSIDERED



Note: The forecast period is from 2022 to 2027. The base year considered for the company profiles is 2021. Wherever the information for the base year was unavailable, information for 2020 has been considered.

## 1.4 CURRENCY

The currency used in the report is the US dollar (USD), with the market size indicated in USD million/billion.

- ISO 4217 codes have been used to depict currency names.
- The market has mainly been studied in terms of USD million; the denomination of the market size has been based on values, enabling uniform accommodation of these values in tables, figures, or any form of the study of a particular segment.
- Revenues of companies have been acquired from their latest annual reports. For companies that provide revenues in USD, they have been sourced as they are, while for those reporting their revenues in other currencies, the average annual currency exchange rates have been used for the particular year (from the ofx.com website) to convert the values to USD.

## 1.5 LIMITATIONS

- Industry players have kept quantitative information for some of the market segments confidential. For such segments, the qualitative insights gathered during the study have been used to arrive at the market size in terms of value.
- Several micromarkets are considered to arrive at the total size of the high-performance computing market. Some of the related markets are not considered to avoid overlapping
- Only recent and important developments from January 2019 to April 2022 have been considered to analyze strategies adopted by companies to compete in the high-performance computing market.
- A limited number of industry experts are available in emerging markets across APAC, the Middle East & Africa, and South America. In such cases, the regional market size has been derived based on the weightage assigned to these markets depending on the qualitative insights gathered from the global industry experts and typical technology adoption trends observed in these regional markets.

- The size of the high-performance computing market in RoW has been estimated for the Middle East & Africa and South America. However, the country-wise market sizes for these regions have not been included in the study due to data granularity.

## 1.6 STAKEHOLDERS

- HPC software developers
- HPC hardware providers
- HPC service providers
- HPC cloud solution providers
- Application design and development service providers
- System integrators/migration service providers
- Consultancy firms/advisory firms
- Training and education service providers
- Data integration service providers
- Managed service providers
- Independent software vendors (ISVs)
- Original equipment manufacturers (OEMs)
- Technology service providers
- Government organizations
- Consultants/consultancies/advisory firms
- Support and maintenance service providers
- Information Technology (IT) infrastructure providers
- System integrators (SIs)
- Regional associations
- Value-added resellers and distributors
- End users of high-performance computing

## 1.7 SUMMARY OF CHANGES

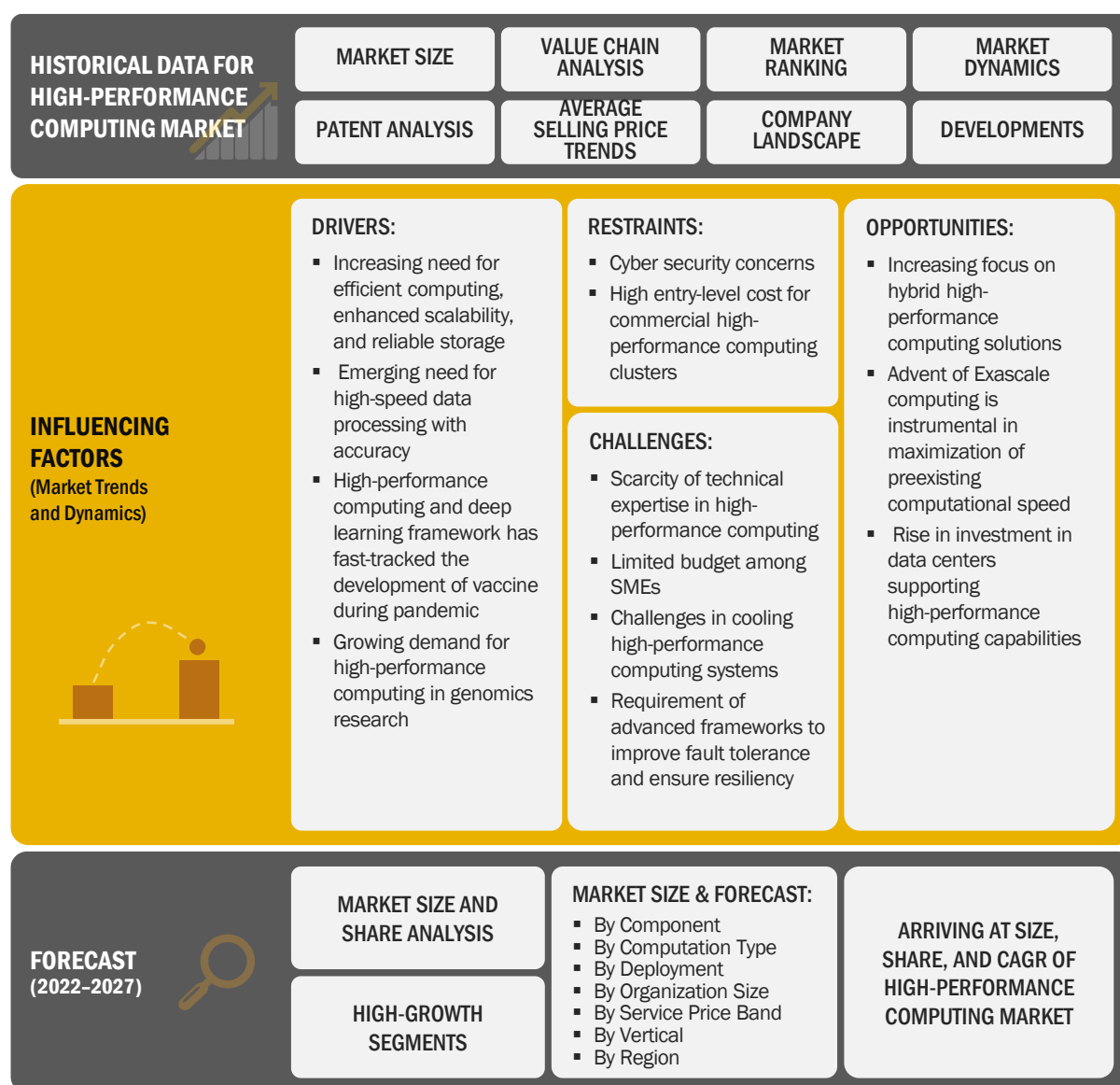
- In this latest edition of the report, the global HPC market has been segmented into component, computation type, deployment, organization size, server price bands, vertical, and region.
- The historical data tables from 2018 to 2021 are provided in the study.
- The forecast data for each segment/region from 2022 to 2027 is provided in each market table.
- The market overview section of the report is refined, and new pointers such as average selling prices (ASPs) of key market players, key events and conferences, and key stakeholders and buying criteria are added to the report
- The impact of COVID-19 is showcased wherever applicable, along with infographics depicting pre-COVID-19 and post-COVID-19 scenarios.
- The key player strategies/right to win and SME matrix are added to the competitive landscape section.
- Updated financial information/product portfolio of players is provided for each listed company in a graphical representation.
- Market developments of profiled players between 2019 and 2022 are included in this edition.

## 2 RESEARCH METHODOLOGY

### 2.1 RESEARCH DATA

We have studied secondary sources, directories, and databases, such as Hoovers, Bloomberg BusinessWeek, Factiva, and OneSource, to identify and collect information useful for this extensive technical, market-oriented, and commercial study of the high-performance computing market. Primary sources such as experts from related companies have been interviewed to obtain and verify critical information, as well as assess the prospects of the market. The key players in the high-performance computing market have been identified through secondary research, and their market rankings have been determined through primary and secondary research. This research includes the study of annual reports of the top companies and interviews with key opinion leaders, such as chief executive officers (CEO), directors, and marketing personnel.

**FIGURE 2** HIGH-PERFORMANCE COMPUTING MARKET: RESEARCH DESIGN



In the secondary research process, various secondary sources have been referred to for identifying and collecting information relevant to this study. The secondary sources include annual reports, press releases, and investor presentations of companies; white papers, certified publications, and articles by recognized authors; directories; and databases. Secondary research has been mainly carried out to obtain key information regarding the supply and value chains of the high-performance computing ecosystem, total pool of key players, market classification and segmentation according to the industry trends to the bottom-most level, geographic markets, and key developments from both market- and technology-oriented perspectives.

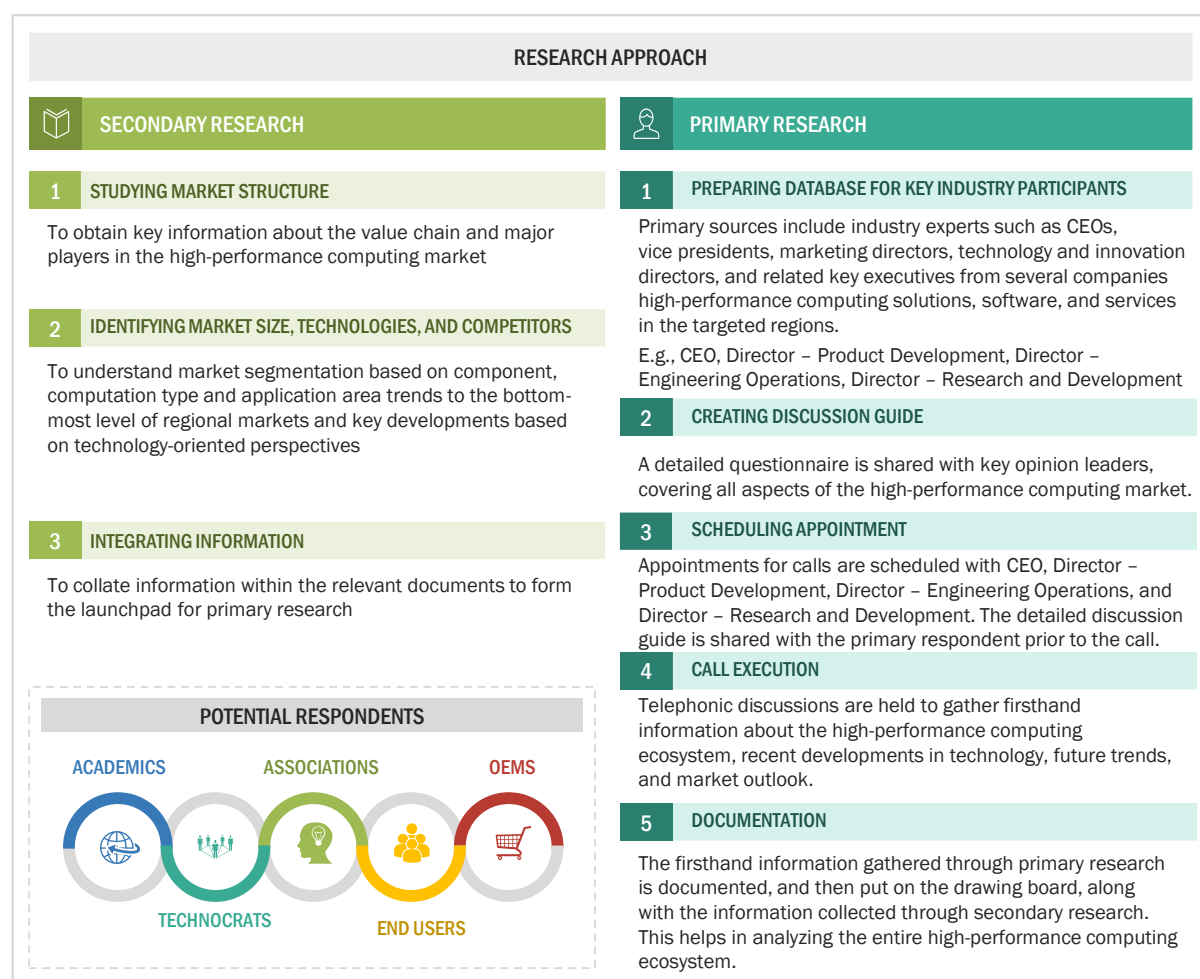
In the primary research process, various primary sources from both supply and demand sides have been interviewed to obtain qualitative and quantitative information relevant to this report. The primary sources from the supply side include industry experts such as chief executive officers, vice presidents, marketing directors, technology and innovation directors, software developers, and related executives from various key companies and organizations operating in the high-performance computing market.

After the complete market engineering (including calculations for the market statistics, the market breakdown, the data triangulation, the market size estimations, and the market forecasting), extensive primary research has been conducted to verify and validate the critical numbers obtained.

Primary research has also been conducted to identify segmentation types, industry trends, and key players in the market, as well as to analyze the competitive landscape; key market dynamics such as drivers, restraints, challenges, and opportunities; and key player strategies. In the complete market engineering process, both top-down and bottom-up approaches have been extensively used, along with several data triangulation methods, to estimate and forecast the sizes of the market, as well as its segments and subsegments listed in this report. Extensive qualitative and quantitative analyses have been carried out on the complete market engineering process to list the key information/insights.

### 2.1.1 SECONDARY AND PRIMARY RESEARCH

The figure below illustrates the way primary and secondary research has been carried out for gathering and analyzing the data related to the high-performance computing market.



Source: MarketsandMarkets Analysis

### 2.1.2 SECONDARY DATA

In the secondary research process, various secondary sources have been referred to for identifying and collecting information pertinent to the study. These secondary sources include annual reports; press releases; investor presentations of companies; white papers; certified publications; and articles from recognized authors, directories, and databases. Secondary research has been mainly used to obtain key information about the industry's supply chain, market's monetary chain, total pool of key players, and market segmentation according to industry trends, geographic markets, and key developments from both market- and technology-oriented perspectives. The secondary data has been collected and analyzed to arrive at the overall market size and further validated through primary research.

### 2.1.2.1 List of secondary sources

| SOURCE                                                                         | WEB LINK                                                        |
|--------------------------------------------------------------------------------|-----------------------------------------------------------------|
| International Trade Administration (ITA)                                       | <a href="https://www.trade.gov/">https://www.trade.gov/</a>     |
| European Cloud Association (EUDCA)                                             | <a href="https://eosc.eu/">https://eosc.eu/</a>                 |
| Telecommunications Industry Association                                        | <a href="https://tiaonline.org/">https://tiaonline.org/</a>     |
| United Nations Economic and Social Commission for Asia and the Pacific (ESCAP) | <a href="https://www.unescap.org/">https://www.unescap.org/</a> |
| Economic Commission for Latin America (ECLAC)                                  | <a href="https://www.cepal.org/en">https://www.cepal.org/en</a> |
| Arab Information and Communication Technology Organization (AICTO)             | <a href="http://www.aicto.org/">http://www.aicto.org/</a>       |

Source: MarketsandMarkets Analysis

### 2.1.2.2 Key secondary sources

| PARAMETER                                                                                                                                          | SOURCE                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>MARKET SIZE</b><br>(In Terms of Value)                      | <ul style="list-style-type: none"> <li>▪ Company Financials</li> <li>▪ Magazines</li> <li>▪ Journals</li> <li>▪ Press Releases</li> <li>▪ Paid Databases</li> <li>▪ MarketsandMarkets Data Repository</li> </ul> |
|  <b>REVENUE OF COMPANIES</b>                                    | <ul style="list-style-type: none"> <li>▪ Annual Reports</li> <li>▪ Company Website</li> <li>▪ Public Databases</li> <li>▪ MarketsandMarkets Data Repository</li> </ul>                                           |
|  <b>QUALITATIVE INFORMATION</b><br>(Market Dynamics and Trends) | <ul style="list-style-type: none"> <li>▪ Company Website</li> <li>▪ Annual Reports</li> <li>▪ Press Releases</li> <li>▪ MarketsandMarkets Data Repository</li> </ul>                                             |

### 2.1.3 PRIMARY DATA

In the primary research process, various primary sources from both supply and demand sides have been interviewed to obtain qualitative and quantitative information important for this report. Primary sources from the supply side include industry experts such as chief executive officers (CEOs), vice presidents, marketing directors, technology and innovation directors, and related executives from various key companies and organizations operating in the high-performance computing market. After the complete market engineering (which includes calculations for the market statistics, market breakdown, data triangulation, market estimation, and market forecasting), extensive primary research has been conducted to gather information and to verify and validate the critical numbers that were obtained.

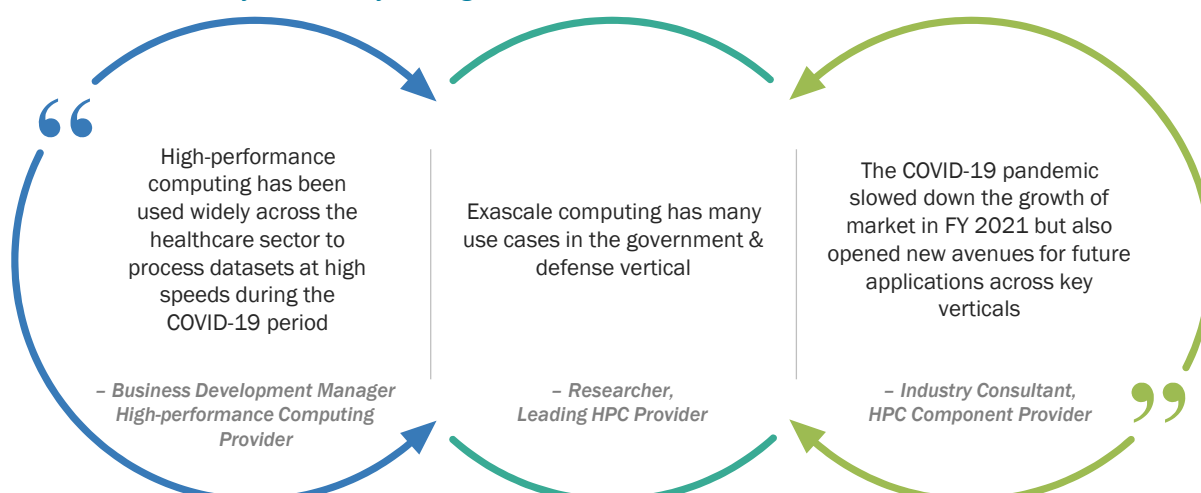
Primary research has also been conducted to identify the segmentation types, industry trends, and key players and analyze the competitive landscape and key market dynamics such as drivers, restraints, opportunities, challenges, industry trends, Porter's five forces analysis, as well as growth strategies adopted by various key market players. Approximately 30% and 70% of primary interviews have been conducted for the demand and supply sides, respectively.

### 2.1.3.1 Primary interviews with experts

| COMPANY                      | ROLE IN ECOSYSTEM      | REGIONAL EXPERTISE | DESIGNATION                                              |
|------------------------------|------------------------|--------------------|----------------------------------------------------------|
| NEC Corporation              | HPC Solutions Provider | Asia Pacific       | Senior Solutions Specialist                              |
| NEC Corporation              | HPC Solutions Provider | Asia Pacific       | Senior Technical Lead                                    |
| NEC Corporation              | HPC Solutions Provider | Asia Pacific       | HPC Application Tuning Specialist                        |
| Aspen Systems                | HPC Solutions Provider | North America      | Sales Director                                           |
| Inspur                       | HPC Solutions Provider | Asia Pacific       | Business Development Manager                             |
| Inspur                       | HPC Solutions Provider | Europe             | HPC Sales Engineer                                       |
| Penguin Computing            | HPC Solutions Provider | North America      | HPC Software Engineer                                    |
| Advanced HPC                 | HPC Solutions Provider | North America      | Senior Sales Engineer                                    |
| Advanced Micro Devices (AMD) | HPC Solutions Provider | Asia Pacific       | Senior Software Engineer                                 |
| Arm Group                    | HPC Solutions Provider | Middle East        | Staff Engineer (HPC Performance Analysis & Benchmarking) |

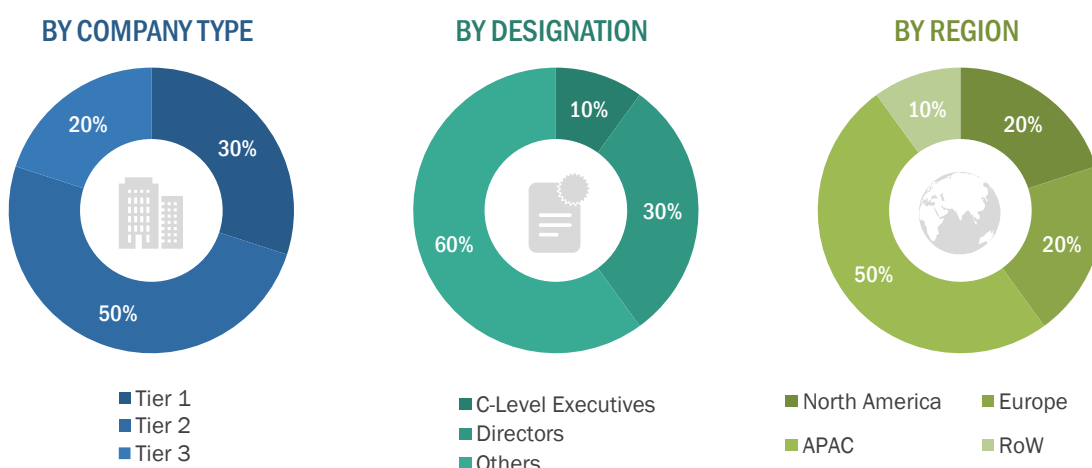
Source: Interviews with Experts

### 2.1.3.2 Key industry insights



Source: Interviews with Experts

### 2.1.3.3 Breakdown of primaries



Note: RoW mainly includes the Middle East & Africa and South America. Other designations include product managers, sales managers, and marketing managers. Three tiers of companies have been defined based on their total revenue as of 2021; tier 3: revenue less than USD 100 million; tier 2: revenue between USD 100 million and USD 1 billion; and tier 1: revenue more than USD 1 billion.

## 2.2 MARKET SIZE ESTIMATION

Top-down and bottom-up approaches have been used to estimate and validate the sizes of the high-performance computing market and other dependent submarkets. The leading players in the market have been identified through secondary research, and their market shares in the key regions have been determined through primary and secondary research. This entire procedure includes studying the annual and financial reports of the top players and extensive interviews with the industry experts such as chief executive officers, vice presidents, directors, and marketing executives for the key insights.

All percentage shares, splits, and breakdowns have been determined using secondary sources and verified through primary sources. All the possible parameters that can affect the markets covered in this study have been accounted for, viewed in detail, verified through primary research, and analyzed to get the final quantitative and qualitative data. This data has been consolidated and supplemented with detailed inputs and analysis from MarketsandMarkets and presented in this report. Market size estimation methodology figures represent the overall market size estimation process employed for this study. MARKET SIZE

### 2.2.1 BOTTOM-UP APPROACH

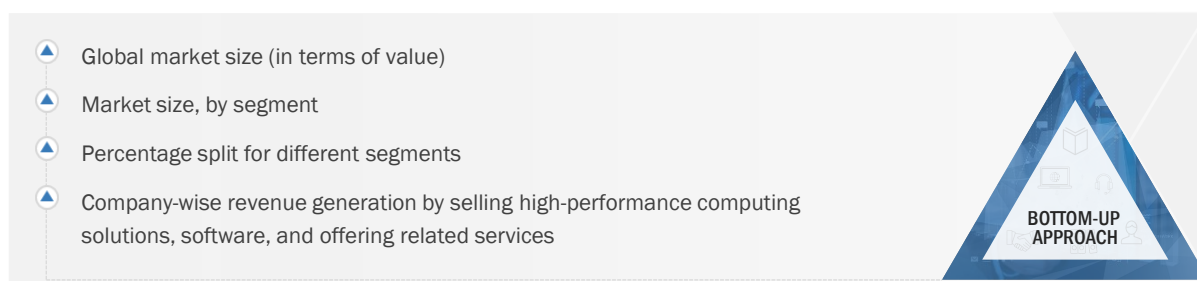
The bottom-up approach has been used to arrive at the overall size of the high-performance computing market from the revenues of key players and their market shares. Calculations based on revenues of the key companies identified in the market led to estimating their overall market size.

#### 2.2.1.1 Approach for arriving at market size using bottom-up analysis

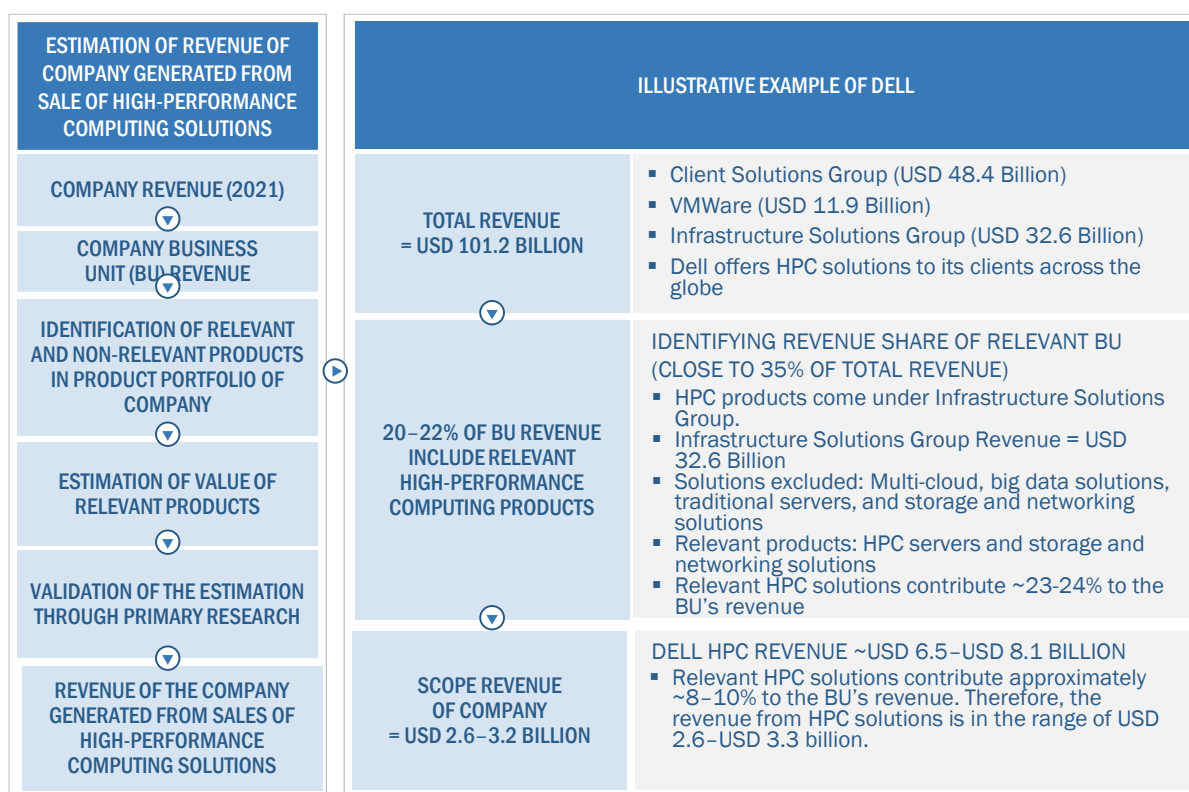
- Identifying end users that are either using or are expected to use high-performance computing.
- Tracking leading companies and other key manufacturers operating across various sectors.
- Deriving the size of the high-performance computing market through the data sanity method; analyzing revenues of more than 25 key providers through their annual reports and press releases and summing them up to estimate the overall market size.

- Understanding and analyzing Y-o-Y projections to derive the projected market values of each component, deployment, and vertical; initially, the Y-o-Y projections showed a moderate growth till 2020. However, with the unfolding of the impact of the COVID-19, the high-performance computing market witnessed a slight reduction in growth in FY 2020 and Q2 of FY 2021 but made a significant recovery from Q3 of FY 2021 and FY 2022. The market is expected to witness a steady increase in growth after that, considering the surging demand for high-performance computing from different verticals.
- Carrying out the market trend analysis to obtain the CAGR of the high-performance computing market by understanding the industry penetration rate and analyzing the demand and supply of high-performance computing in different industries.
- Assigning a percentage to the overall revenue or, in a few cases, to segmental revenues of each company to derive their revenues from the sales of high-performance computing. This percentage for each company has been assigned based on their product portfolios and the range of their high-performance computing solutions, software, and services.
- Verifying and crosschecking estimates at every level through discussions with the key opinion leaders, including CXOs, directors, and operation managers, and with domain experts at MarketsandMarkets.
- Studying various paid and unpaid information sources such as annual reports, press releases, white papers, and databases.
- Tracking the ongoing projects and identifying the upcoming high-performance computing implementation projects by companies and forecasting the market size based on these developments and other critical parameters.

**FIGURE 3** MARKET SIZE ESTIMATION METHODOLOGY: BOTTOM-UP APPROACH



**FIGURE 4** MARKET SIZE ESTIMATION METHODOLOGY: APPROACH 1 BOTTOM-UP (SUPPLY SIDE)—ILLUSTRATION OF REVENUE ESTIMATION OF COMPANIES FROM SALES OF HIGH-PERFORMANCE COMPUTING



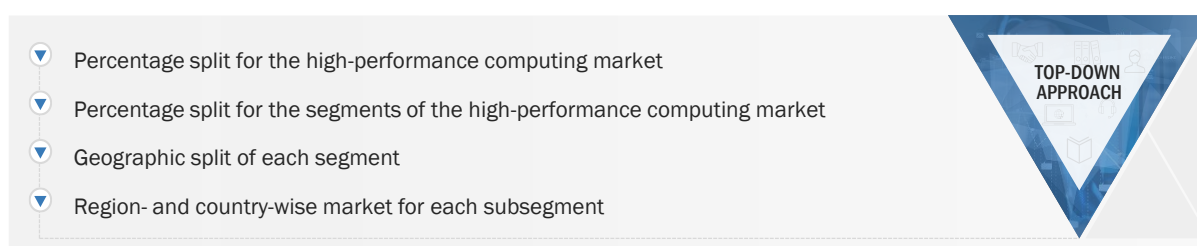
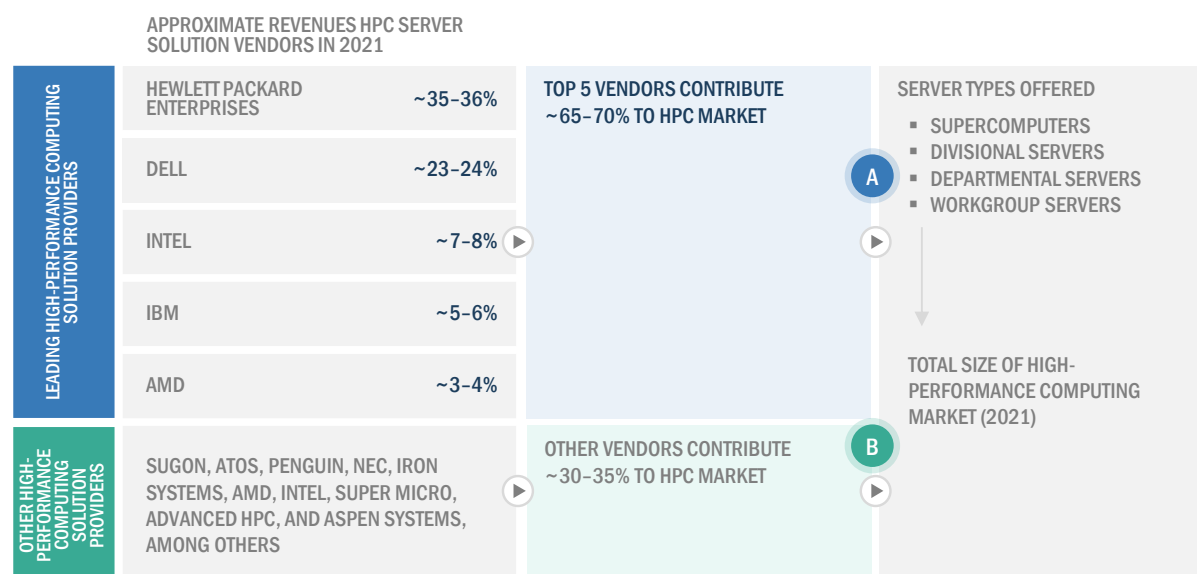
Source: Annual Reports, SEC Filings, Investor Presentations, Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

## 2.2.2 TOP-DOWN APPROACH

In the top-down approach, the overall market size has been used to estimate the size of the individual markets (mentioned in the market segmentation) through percentage splits from secondary and primary research.

### 2.2.2.1 Approach for capturing market size using top-down analysis

- Focusing on top-line investments and expenditures being made in the ecosystem
- Listing key developments, analyzing updated technologies in the marketplace, and evaluating the market by further splitting it into various segments
- Analyzing the information related to the market revenue generated by the key high-performance computing solutions manufacturing companies
- Conducting multiple on-field discussions with key opinion leaders involved in the development of various high-performance computing solutions and software and offering related services
- Estimating geographic splits by using secondary sources based on various factors, such as the number of players in a specific country and region, components used in a high-performance computing offering, and level of product penetration

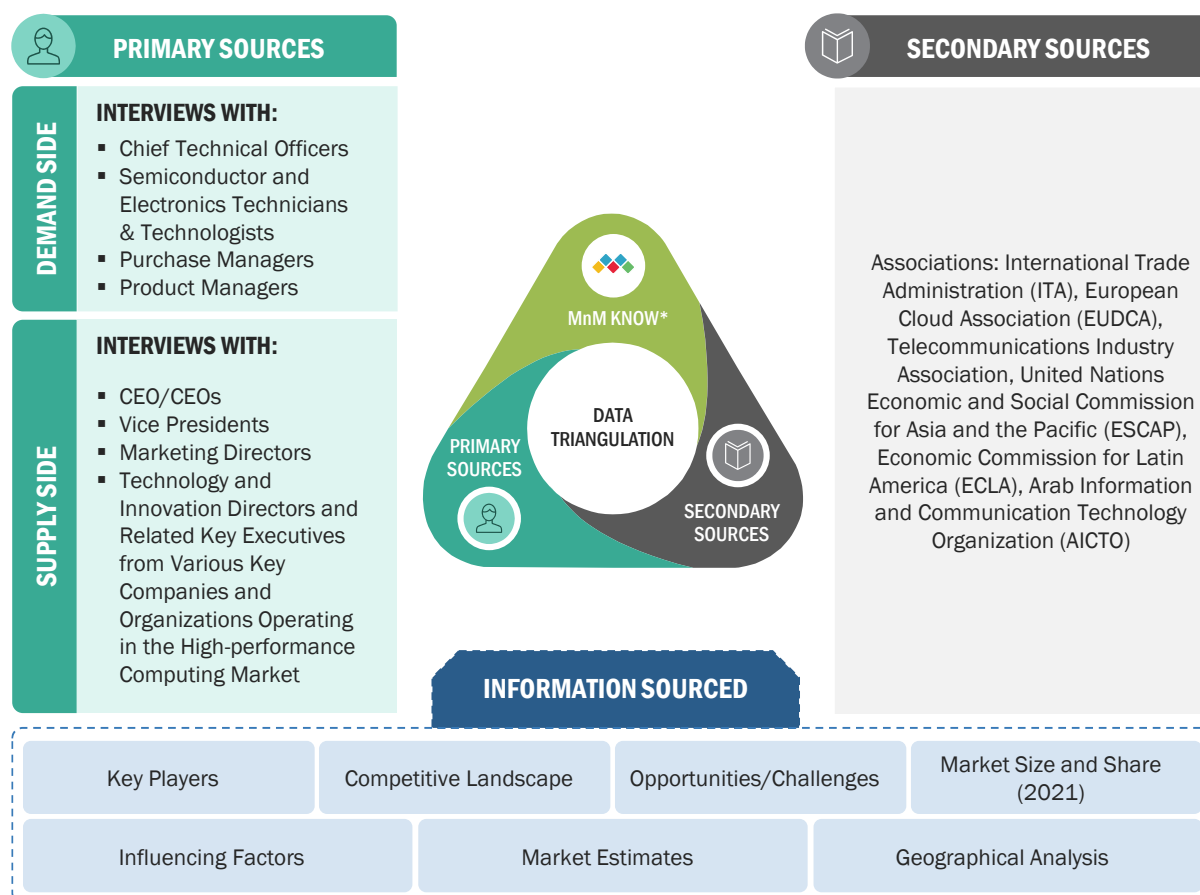
**FIGURE 5** MARKET SIZE ESTIMATION METHODOLOGY: TOP-DOWN APPROACH**FIGURE 6** MARKET SIZE ESTIMATION METHODOLOGY: APPROACH 2 (SUPPLY SIDE)—REVENUE GENERATED FROM SALE OF HIGH-PERFORMANCE COMPUTING SERVER SOLUTIONS

Source: Annual Reports, SEC Filings, Investor Presentations, Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

## 2.3 MARKET BREAKDOWN AND DATA TRIANGULATION

After arriving at the overall market size from the market size estimation process explained above, the total market has been split into several segments and subsegments. The market breakdown and data triangulation procedures have been employed wherever applicable to complete the overall market engineering process and arrive at the exact statistics for all segments and subsegments. The data has been triangulated by studying various factors and trends from the demand and supply sides. Along with this, the market size has been validated using both top-down and bottom-up approaches.

**FIGURE 7 DATA TRIANGULATION**



Note: MnM KNOW stands for MarketsandMarkets' "Knowledge Asset Management" framework. In this context, it stands for existing market research knowledge repository of over 5,000 granular markets, our flagship competitive intelligence and market research platform "Knowledge Store," subject matter experts, and independent consultants. MnM KNOW acts as an independent source that helps us validate information gathered from primary and secondary sources.

## 2.4 RESEARCH ASSUMPTIONS

The following assumptions have been considered to complete the engineering of the high-performance computing market.

| PARAMETER                                     | ASSUMPTION                                                                                                                                                                                                                                                                                |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GLOBAL ECONOMIC OUTLOOK</b>                | The global economy has a direct impact on any market. Regarding the global economic landscape, macroeconomic collapses and recessions have been considered while forecasting the market size in this research study owing to the COVID-19 pandemic and the ongoing Russia-Ukraine crisis. |
| <b>EXCHANGE RATES AND CURRENCY CONVERSION</b> | For the conversion of various currencies to the US dollars, average historical exchange rates have been used according to the year specified. For all historical and current exchange rates required for calculations and currency conversions, the OANDA website has been used.          |
| <b>GOVERNMENT SPENDING PATTERNS</b>           | A stable and moderate government spending pattern has been assumed during the forecast period.                                                                                                                                                                                            |
| <b>MARKET SATURATION</b>                      | New avenues are expected to arise in emerging economies such as China, South Korea, and India.                                                                                                                                                                                            |
| <b>DATA AUTHENTICITY</b>                      | Company revenues and segment-specific information have been derived from the annual reports of the respective companies. The information provided in the annual reports has been assumed to be authentic.                                                                                 |

Source: Annual Reports, SEC Filings, Investor Presentations, Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

## 2.5 RISK ASSESSMENTS

**TABLE 1** LIMITATIONS AND ASSOCIATED RISKS

| HIGH-PERFORMANCE COMPUTING MARKET | COVERED                                                                                                                                                                                                                                                                                                         | RISK                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Component                         | The high-performance computing market is further bifurcated based on various components such as solutions, software, and services                                                                                                                                                                               | The costs of various high-performance computing products and services can vary based on different providers and region availability. Thus, the associated risk may have a 5–6% impact on the global market.                                                                                                                                                                         |
| Country-wise Market Size          | The country-level quantitative information for the Middle East & Africa and South America has not been provided in the report owing to the limited availability of information from primary and secondary sources.                                                                                              | NA                                                                                                                                                                                                                                                                                                                                                                                  |
| Vertical                          | Government & defense, banking, financial services, and insurance (BFSI), education & research, manufacturing, media & entertainment, healthcare & life sciences, energy & utilities, earth sciences, and others have been considered while evaluating the global size of the high-performance computing market. | High-performance computing is an emerging market, and hence it is changing dynamically over the period of time due to the introduction of new applications. The COVID-19 pandemic has been responsible for the opening of new avenues for a few sectors and has been detrimental to a few sectors as well. Thus, the associated risk may have an 8–10% impact on the global market. |
| Deployment                        | The high-performance computing market has been split based on deployment into cloud and on-premises.                                                                                                                                                                                                            | The pricing of a high-performance computing solution, software, and service may vary according to the vendors in different regions. Thus, the associated risk may have a 3–4% impact on the total global market.                                                                                                                                                                    |

Source: Annual Reports, SEC Filings, Investor Presentations, Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

## 2.6 LIMITATIONS

### METHODOLOGY-RELATED LIMITATIONS



- Spendings from verticals (demand side) have been considered to cross-verify market size with supply side
- All forecast values are based on historic data. The varying costs of high-performance computing solutions, software, and services may change the market size
- The impact of COVID-19 has been considered. It may have short-term impact on the high-performance computing market

### SCOPE-RELATED LIMITATIONS



- The market size for Exascale computing is not considered due to lack of use cases across various key verticals
- Certain applications of high-performance computing are not considered since their usage is still under the view of research

Source: MarketsandMarkets Analysis

### 3 EXECUTIVE SUMMARY

#### 3.1 GROWTH RATE ASSUMPTIONS/GROWTH FORECAST

In this section, we have defined the growth forecast model based on the assessment of qualitative and quantitative factors impacting the market growth.

| DURATION             | IMPACT ANALYSIS      | ANALYST VIEW                                                                                                                                                                                                                                                                 | KEY COMMENTS                                                                                                                                                 |
|----------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022                 | Slow Growth Rate     | According to recent company financials, few top companies in the HPC market are witnessing a negative growth trend; however, with growing demand for HPC in the healthcare & life sciences and government & defense verticals, they are expected to witness positive growth. | The HPC market is expected to exhibit slow y-o-y growth of 1.20% owing to the COVID-19 crisis.                                                               |
| 2022–2023            | Moderate Growth Rate | Partial demand recovery will be observed, which will be lower than sustainable growth                                                                                                                                                                                        | COVID-19 impact is expected to decline, and business activities are likely to recover at partial to full levels.                                             |
| 2023–2024            | Steady Growth Rate   | A steady demand may be observed, which will be more or less equal to sustainable growth.                                                                                                                                                                                     | Business activities will resume completely, and steady growth is expected owing to the rise in demand due to the spillover effect from the previous 2 years. |
| 2024–2027 and beyond | Steady Growth Rate   | Demand could be slightly higher or equivalent compared to past projections.                                                                                                                                                                                                  | Steady growth of 5–6% could be observed.                                                                                                                     |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

High-performance computing (HPC) refers to the practice of aggregating computing power to solve large problems in science, engineering, or business. The adoption of HPC is gaining traction as enterprises across key verticals are focusing on scaling up their infrastructure for cloud computing and edge computing by integrating them with digital technologies such as AI, IoT, and big data analytics. HPC enables the community of researchers, scientists, and engineers working in various fields to solve complex, compute-intensive problems in a short period. Benefits associated with HPC are high network performance, fast storage, and the ability to process large volumes of data. HPC also enables organizations to develop new, innovative products and services to maintain a competitive edge.

HPC has gradually evolved over the decades and undergone various technological innovations and advancements. In the early 1970s, the vector computing system concept was used with distributed memory systems and had the processing capability of Megaflops/second (Mflops/s). In the early 1990s, the Massively Parallel Processing (MPP) theory was used in large enterprises and SMEs, where systems were equipped with “off-the-shelf components” and had a processing speed of up to multiple Gigaflops/second (Gflops/s). In 2000, this speed increased to teraflops, and by 2008 the computational speed reached petaflops. Since then, supercomputers have become billion times faster, and the HPC market has gained tremendous traction. With the increasing technological advancements, HPC clusters are moving toward Exascale computing, which is expected to fuel the growth of the HPC market in the next 5 years.

COVID-19 has negatively impacted the growth of the HPC market due to disruptions in supply chains and a shortage of workforce. However, HPC vendors are striving to support employees, customers, and channel partners to stabilize their business operations with the help of cloud HPC solutions. Hence, the COVID-19 crisis has also opened new avenues for market growth.

This study has been carried out to understand the dynamics of the HPC market while estimating and forecasting the market size. It also helps identify the key technologies and verticals related to HPC, along with geographic areas that are expected to provide various growth opportunities for the market. The key verticals using HPC include government & defense, banking, financial services, and insurance (BFSI), education & research, manufacturing, media & entertainment, healthcare & life sciences, energy & utilities, and earth sciences.

Product launch was the key strategy adopted by players to increase their shares in the HPC market between March 2019 and March 2022. Hewlett Packard Enterprises (US), Dell (US), Intel (US), IBM (US), and Advanced Micro Devices (AMD) (US) are the leading market players that have adopted this strategy to boost their business. Partnerships, contracts, agreements, and collaborations were the other major strategies adopted by key market players.

**TABLE 2 HPC MARKET, BY SERVER SHIPMENTS 2018–2021 (MILLION UNITS)**

| Year   | 2018 | 2019 | 2020 | 2021 |
|--------|------|------|------|------|
| Server | 13.8 | 14.2 | 14.5 | 15.1 |

Source: International Trade Administration (ITA), European Cloud Association (EUDCA), Telecommunications Industry Association, United Nations Economic and Social Commission for Asia and the Pacific (ESCAP), Economic Commission for Latin America (ECLA), Arab Information and Communication Technology Organization (AICTO), Interviews with Experts, Secondary Research, Annual Reports, Press Releases, Investor Presentations, White Papers, Journals and Magazines, and MarketsandMarkets Analysis

**TABLE 3 HPC MARKET, BY SERVER SHIPMENTS 2022–2027 (MILLION UNITS)**

| Year   | 2022 | 2023 | 2024 | 2025 | 2026 | 2027 | CAGR<br>(2022–2027) |
|--------|------|------|------|------|------|------|---------------------|
| Server | 15.7 | 16.6 | 17.6 | 18.6 | 19.6 | 20.8 | 5.9%                |

Source: International Trade Administration (ITA), European Cloud Association (EUDCA), Telecommunications Industry Association, United Nations Economic and Social Commission for Asia and the Pacific (ESCAP), Economic Commission for Latin America (ECLA), Arab Information and Communication Technology Organization (AICTO), Interviews with Experts, Secondary Research, Annual Reports, Press Releases, Investor Presentations, White Papers, Journals and Magazines, and MarketsandMarkets Analysis

**TABLE 4 HPC MARKET, 2018–2021 (USD BILLION)**

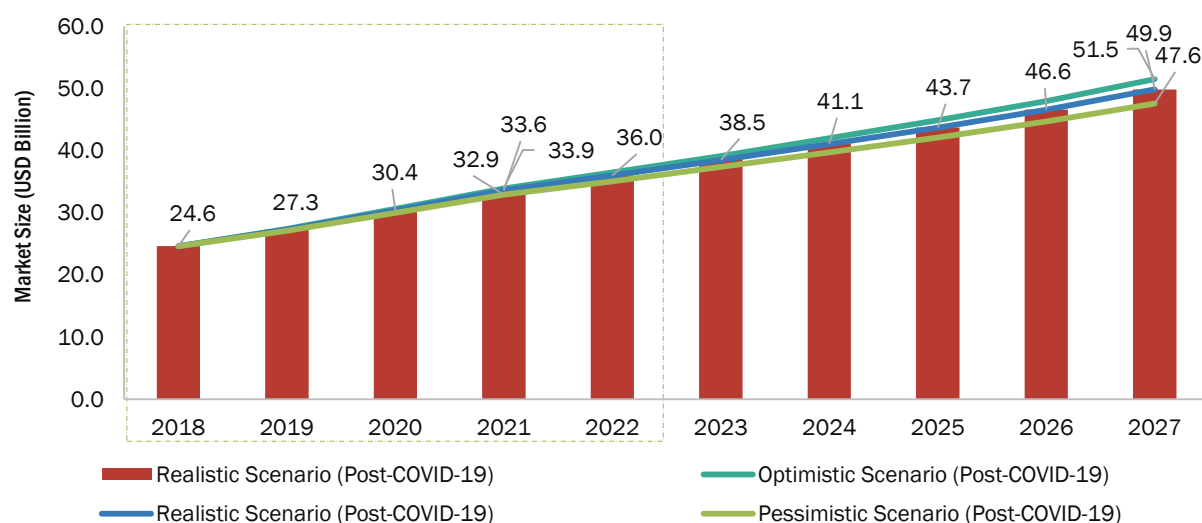
| Year        | 2018 | 2019 | 2020 | 2021 |
|-------------|------|------|------|------|
| Market Size | 24.6 | 27.3 | 30.4 | 33.6 |

Source: International Trade Administration (ITA), European Cloud Association (EUDCA), Telecommunications Industry Association, United Nations Economic and Social Commission for Asia and the Pacific (ESCAP), Economic Commission for Latin America (ECLA), Arab Information and Communication Technology Organization (AICTO), Interviews with Experts, Secondary Research, Annual Reports, Press Releases, Investor Presentations, White Papers, Journals and Magazines, and MarketsandMarkets Analysis

**TABLE 5 HPC MARKET, 2022–2027 (USD BILLION)**

| Year        | 2022 | 2023 | 2024 | 2025 | 2026 | 2027 | CAGR (2022–2027) |
|-------------|------|------|------|------|------|------|------------------|
| Market Size | 36.0 | 38.5 | 41.1 | 43.7 | 46.6 | 49.9 | 6.7%             |

Source: International Trade Administration (ITA), European Cloud Association (EUDCA), Telecommunications Industry Association, United Nations Economic and Social Commission for Asia and the Pacific (ESCAP), Economic Commission for Latin America (ECLA), Arab Information and Communication Technology Organization (AICTO), Interviews with Experts, Secondary Research, Annual Reports, Press Releases, Investor Presentations, White Papers, Journals and Magazines, and MarketsandMarkets Analysis

**FIGURE 8** IMPACT OF COVID-19 ON HPC MARKET

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

### 3.2 POST-COVID-19 SCENARIO

**TABLE 6** POST-COVID-19 SCENARIO: HPC MARKET, 2022–2027 (USD BILLION)

| Year        | 2022 | 2023 | 2024 | 2025 | 2026 | 2027 | CAGR<br>(2022–2027) |
|-------------|------|------|------|------|------|------|---------------------|
| Market Size | 36.0 | 38.5 | 41.1 | 43.7 | 46.6 | 49.9 | 6.7%                |

Note: Values are represented in 1 decimal to exhibit minute differences.

Source: Annual Reports, Press Releases, Investor Presentations, Interviews with Experts, Institute of Electrical and Electronics Engineers (IEEE), Texas A&M University, University of Victoria (UVic), Company Websites, and MarketsandMarkets Analysis

### 3.3 OPTIMISTIC SCENARIO (POST-COVID-19)

**TABLE 7** OPTIMISTIC SCENARIO (POST-COVID-19): HPC MARKET, 2022–2027 (USD BILLION)

| Year        | 2022 | 2023 | 2024 | 2025 | 2026 | 2027 | CAGR<br>(2022–2027) |
|-------------|------|------|------|------|------|------|---------------------|
| Market Size | 36.5 | 39.1 | 42.0 | 44.9 | 48.0 | 51.5 | 7.1%                |

Source: International Trade Administration (ITA), European Cloud Association (EUDCA), Telecommunications Industry Association, United Nations Economic and Social Commission for Asia and the Pacific (ESCAP), Economic Commission for Latin America (ECLA), Arab Information and Communication Technology Organization (AICTO), Interviews with Experts, Secondary Research, Annual Reports, Press Releases, Investor Presentations, White Papers, Journals and Magazines, and MarketsandMarkets Analysis

### 3.4 PESSIMISTIC SCENARIO (POST-COVID-19)

**TABLE 8** PESSIMISTIC SCENARIO (POST-COVID-19): HPC MARKET, 2022–2027 (USD BILLION)

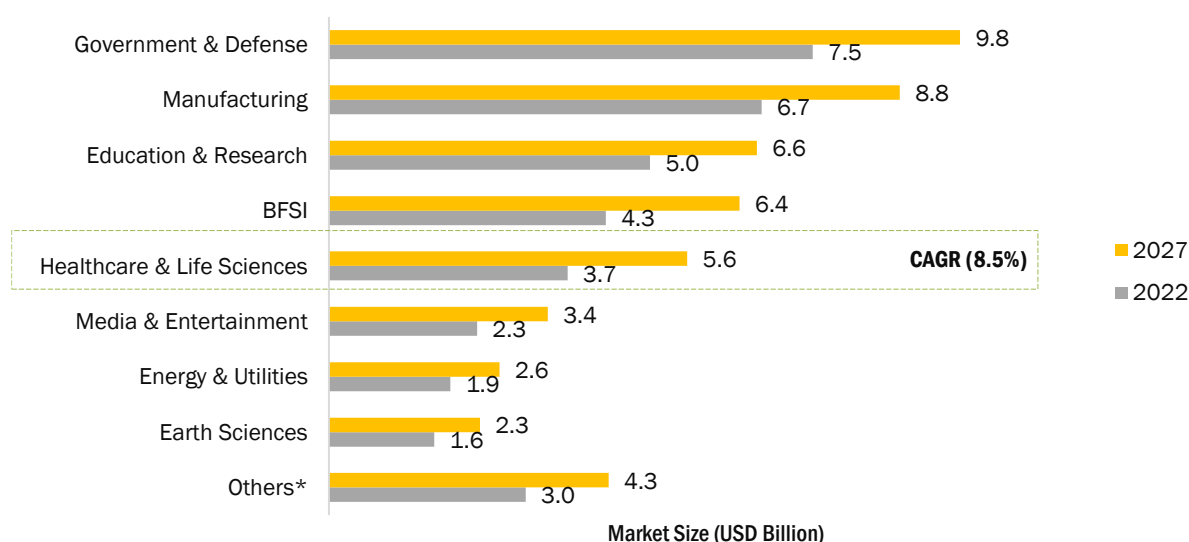
| Year        | 2022 | 2023 | 2024 | 2025 | 2026 | 2027 | CAGR<br>(2022–2027) |
|-------------|------|------|------|------|------|------|---------------------|
| Market Size | 35.1 | 37.4 | 39.7 | 42.1 | 44.7 | 47.6 | 6.3%                |

Source: International Trade Administration (ITA), European Cloud Association (EUDCA), Telecommunications Industry Association, United Nations Economic and Social Commission for Asia and the Pacific (ESCAP), Economic Commission for Latin America (ECLA), Arab Information and Communication Technology Organization (AICTO), Interviews with Experts, Secondary Research, Annual Reports, Press Releases, Investor Presentations, White Papers, Journals and Magazines, and MarketsandMarkets Analysis

The HPC market has historically been showcasing sustainable growth owing to its rapid adoption across key verticals such as government & defense, healthcare & life sciences, manufacturing, and earth sciences. However, with the outbreak and widespread of the COVID-19 pandemic, the market growth has declined, and the trend is expected to continue till the disrupted supply chains across the globe normalize. In the post-COVID-19 period, the market is expected to witness substantial growth due to the rise in demand for HPC in the processing of patient datasets and the requirement for high computational power across data centers, weather forecasting simulation, climate change simulation modeling, and seismic simulation modeling.

As the COVID-19 pandemic continues to persist in certain parts of North America and a few countries in Asia Pacific, the manufacturing sector is facing enormous challenges. The US and Chinese manufacturing units, which are some of the largest in the world, have been experiencing intermittent stagnation since the Q3 of 2020. Various manufacturing companies have reduced their production activities across the region. Besides, the extreme scarcity of hardware accessories required to manufacture HPC systems has deterred the market's growth for a few years.

**FIGURE 9** HEALTHCARE & LIFE SCIENCES SEGMENT TO EXHIBIT HIGHEST CAGR IN HPC MARKET FROM 2022 TO 2027

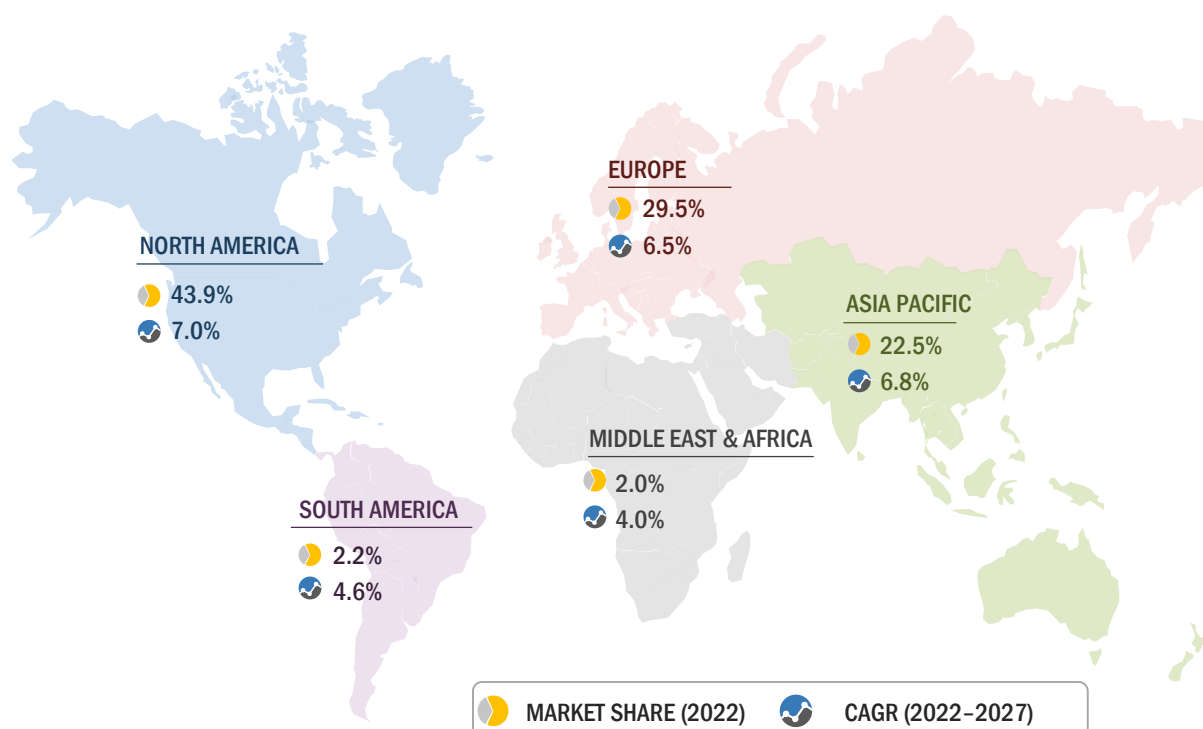


Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

Note: Others include retail, transportation, travel & hospitality, and real estate

Healthcare & Life Sciences segment is expected to grow from USD 3.7 billion in 2022 to USD 5.6 billion in 2027. Advancements in healthcare systems, healthcare research, and communication technologies have led to the development of better standard practices within the healthcare & life sciences vertical. With rising technological innovations, healthcare institutes face the need to deliver various healthcare services through active collaboration between hospital administrators and doctors while improving the overall efficiency and quality of patient care. HPC is being implemented widely in various applications within the healthcare & life sciences vertical for running multiple simulations, data management and modeling processes, and next-generation sequencing. HPC solutions provide the necessary computational power for these applications and help tackle challenges pertaining to the high computational power requirements of the healthcare & life sciences vertical. Hence, the healthcare & life sciences vertical is expected to grow at the highest CAGR during the forecast period.

**FIGURE 10** NORTH AMERICA TO REGISTER HIGHEST GROWTH RATE IN HPC MARKET IN 2022



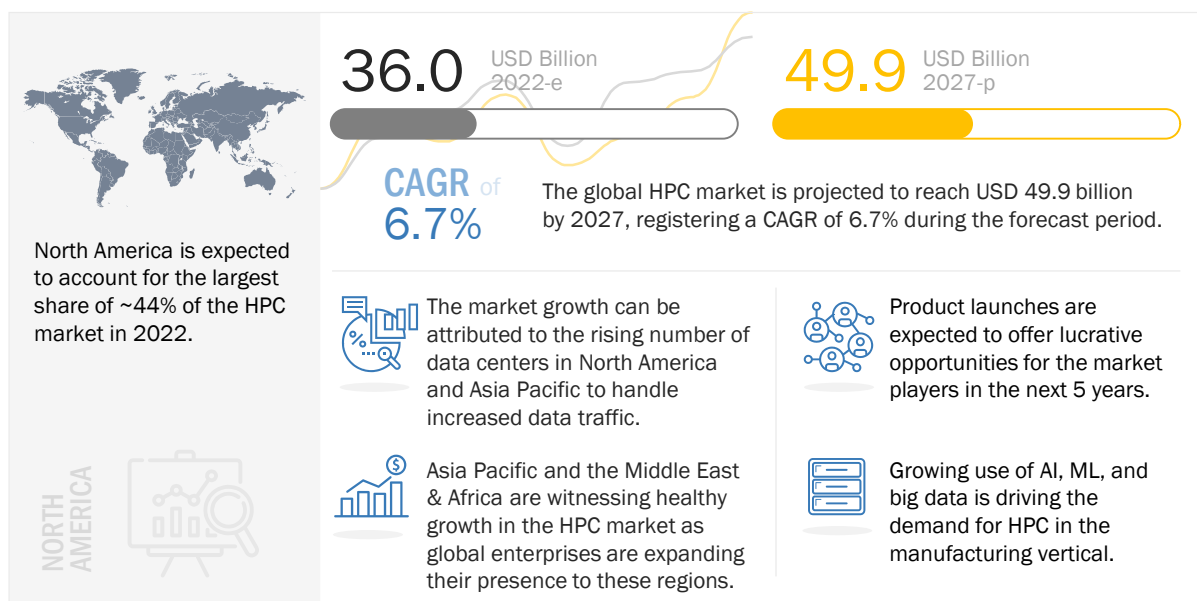
Source: International Trade Administration (ITA), European Cloud Association (EUDCA), Telecommunications Industry Association, United Nations Economic and Social Commission for Asia and the Pacific (ESCAP), Economic Commission for Latin America (ECLA), Arab Information and Communication Technology Organization (AICTO), Interviews with Experts, Secondary Research, Annual Reports, Press Releases, Investor Presentations, White Papers, Journals and Magazines, and MarketsandMarkets Analysis

North America is expected to grow at the highest CAGR of 7.0% in HPC market during the forecast period. North America is expected to grow from USD 15.8 billion in 2022 to USD 22.2 billion in 2027. Growth in IT infrastructure and adoption of digital technologies such as AI, IoT, and big data is expected to drive the growth of HPC in this region. Over the past decade, a number of new data centers have been established in US, and Canada. These data centers require high computational power to handle connectivity traffic across the globe. HPC solutions help in satisfying this need for this computational power thereby driving the growth of market in this region.

## 4 PREMIUM INSIGHTS

### 4.1 ATTRACTIVE GROWTH OPPORTUNITIES IN HIGH-PERFORMANCE COMPUTING MARKET

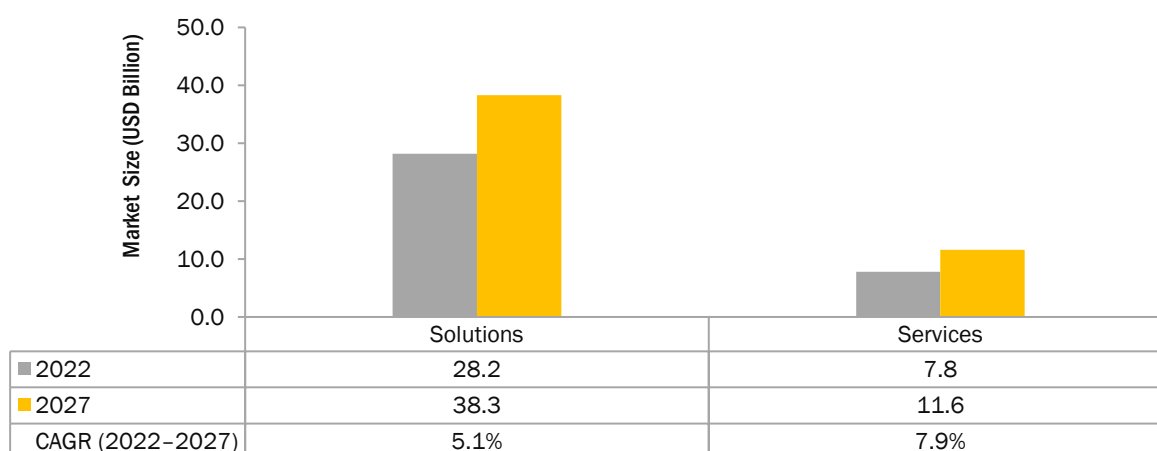
**FIGURE 11** INCREASING DEMAND FOR HIGH-PERFORMANCE COMPUTING IN BANKING, FINANCIAL SERVICES & INSURANCE AND HEALTHCARE & LIFE SCIENCES VERTICALS TO DRIVE MARKET GROWTH FROM 2022 TO 2027



Source: Annual Reports, Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

### 4.2 HIGH-PERFORMANCE COMPUTING MARKET, BY COMPONENT

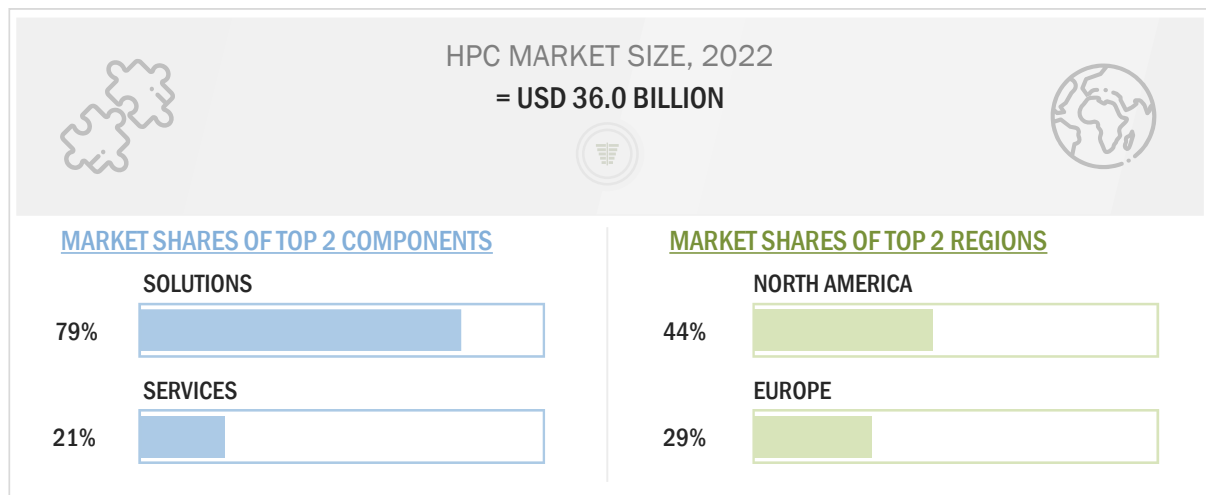
**FIGURE 12** SOLUTIONS SEGMENT TO ACCOUNT FOR LARGER SHARE OF HIGH-PERFORMANCE COMPUTING MARKET FROM 2022 TO 2027



Source: Annual Reports, Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

### 4.3 HIGH-PERFORMANCE COMPUTING MARKET, BY COMPONENT AND REGION

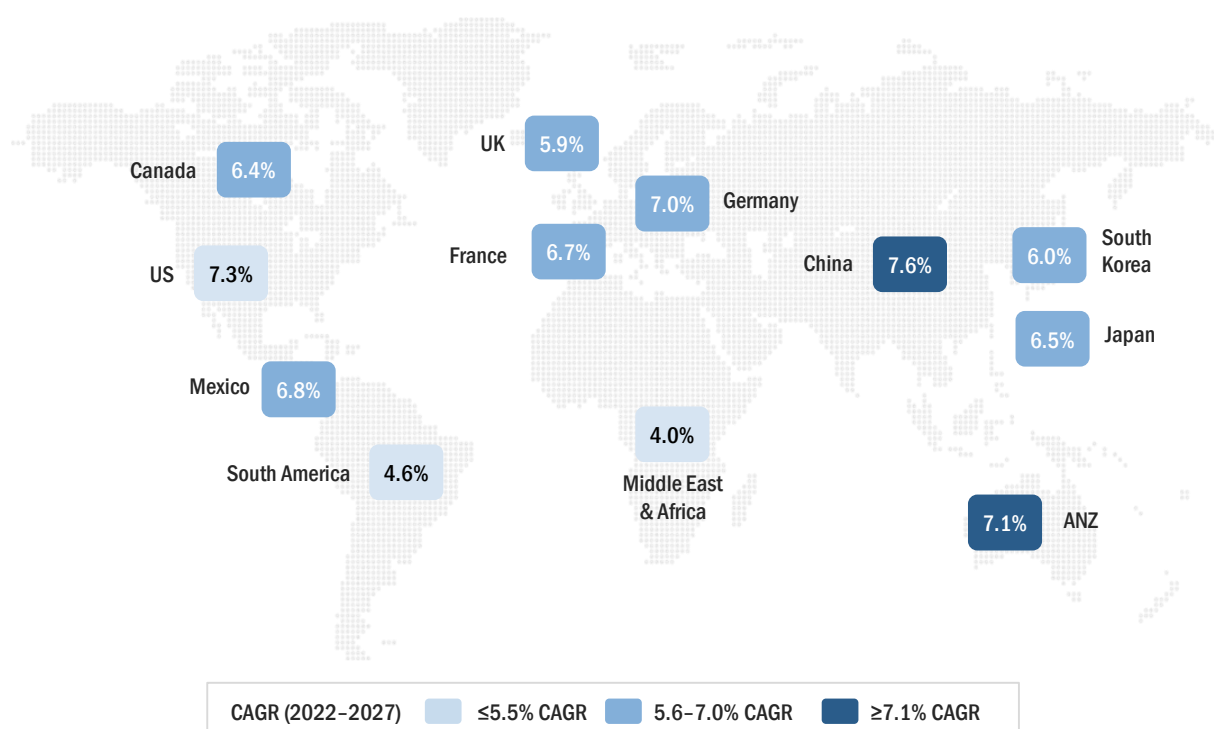
**FIGURE 13 SOLUTIONS SEGMENT AND NORTH AMERICA TO HOLD LARGEST SHARES OF HIGH-PERFORMANCE COMPUTING MARKET IN 2022**



Source: Annual Reports, Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

### 4.4 HIGH-PERFORMANCE COMPUTING MARKET, BY COUNTRY

**FIGURE 14 HIGH-PERFORMANCE COMPUTING MARKET IN CHINA TO GROW AT HIGHEST CAGR FROM 2022 TO 2027**



Source: Annual Reports, Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

## 5 MARKET OVERVIEW

### 5.1 INTRODUCTION

High-performance computing (HPC) is a combination of algorithms, programs, systems, software, and architecture. It can quickly and efficiently solve complex problems related to scientific and engineering research in the fields of physics, aeronautics, and pharmaceutical. HPC systems are mainly used for process research and high computing purposes. The banking, financial services, and insurance (BFSI) sector exhibits high growth potential for the HPC market. HPC allows financial institutions to access information faster, execute applications more efficiently, analyze data at faster rates, and expedite procedures by going beyond the capabilities of a standard PC. Some banks are conducting Proof-of-Concept (PoC) experiments using HPC, which is likely to propel its demand in the BFSI sector.

### 5.2 MARKET DYNAMICS

In this chapter, we have provided an overview of market dynamics, such as drivers, restraints, opportunities, and challenges, influencing the growth of the HPC market. These factors can help market players design strategies and devise action plans to grow and improve their performance in the HPC market.

**FIGURE 15** HIGH-PERFORMANCE COMPUTING MARKET: DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES

|                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br><b>DRIVERS</b>       | <ul style="list-style-type: none"> <li>▪ Increasing need for efficient computing, enhanced scalability, and reliable storage</li> <li>▪ Emerging need for high-speed data processing with accuracy</li> <li>▪ HPC and deep learning framework has Fastrack the development of vaccine during pandemic</li> <li>▪ Growing demand for HPC in genomics research</li> </ul> |
| <br><b>RESTRAINTS</b>    | <ul style="list-style-type: none"> <li>▪ Cyber security concerns</li> <li>▪ High entry-level cost for commercial HPC clusters</li> </ul>                                                                                                                                                                                                                                |
| <br><b>OPPORTUNITIES</b> | <ul style="list-style-type: none"> <li>▪ Increasing focus on hybrid HPC Solutions</li> <li>▪ Advent of Exascale computing is adding in maximization of preexisting computational speed</li> <li>▪ Rise in investment in Data centers supporting HPC capability</li> </ul>                                                                                               |
| <br><b>CHALLENGES</b>    | <ul style="list-style-type: none"> <li>▪ Scarcity of technical expertise in HPC</li> <li>▪ Limited budget among SMEs</li> <li>▪ Challenges in cooling HPC systems</li> <li>▪ Requirement of advanced frameworks to improve fault tolerance and ensure resiliency</li> </ul>                                                                                             |

Source: Interviews with Experts and MarketsandMarkets Analysis

## 5.2.1 DRIVERS

### 5.2.1.1 Growing need for efficient computing, high scalability, and reliable storage

The need to develop energy-efficient HPC systems and software is rising as the use of data and data processing is increasing significantly in all fields globally. Companies from various verticals, such as manufacturing, energy & utilities, chemical, government & defense, as well as academic institutions, need to resolve complex problems and calculations related to any natural phenomenon, such as weather forecasting, climate change, and space explorations. IT professionals in these verticals deal with web data, databases, financial data, and applications that require higher processing power and speed.

The emergence of big data has increased the demand for systems that can handle data-intensive workloads. HPC clusters are systems that can handle large volumes of data with ease and support high-performance data analytics (HPDA). Advanced data collection technologies and applications require high processing power and speed for data analysis and accurate results. Better scans, high-definition television (HDTV), multilayer scans, and enhanced information-gathering capabilities all add to the requirements for more Central Processing Units (CPUs) and quicker storage solutions that are delivered through HPC.

### 5.2.1.2 Increasing demand for high-speed data processing with accuracy

The growth of the HPC market is mainly driven by the ability of HPC systems to process large volumes of data at high speed and accuracy. Companies from different verticals, such as finance, medical, research, seismic exploration, and government & defense, are witnessing slow and inefficient data processing issues. The highly dynamic nature of the financial market makes vitality a crucial parameter so that the derivative valuations are processed faster and accurately. Likewise, in the medical field, applications such as computed tomography (CT) scanning and magnetic resonance imaging (MRI) are required to deliver quick and accurate results by processing complex algorithms. HPC systems in this field help reduce the compute time and process the CT scan and MRI data quickly and at high accuracy. Thus, the increasing demand for high computational speed for data processing in the abovementioned verticals is driving the growth of the HPC market.

### 5.2.1.3 Increased use of high-performance computing and deep learning frameworks in development of COVID-19 vaccines

HPC has played an important role in the development of COVID-19 vaccines. Scientists worldwide have highly relied on the high computing power of HPC systems and supercomputers to run complicated mathematical models, which transform massive volumes of changing COVID-19 data into simulations of biological and chemical processes. This also helped them predict new strains of the virus and develop new treatments by computing complex molecular human body changes. For instance, in an article published by Hewlett Packard Enterprises (HPE) in April 2020, the company has supported initiatives of using HPC and AI systems and expertise in research institutions such as the US Department of Energy's Argonne National Laboratory (ANL), Lawrence Livermore National Laboratory (LLNL), and French National Center for Scientific Research (CNRS), and French National Infrastructure for HPC Resources and Facilities.

### 5.2.1.4 Growing demand for high-performance computing (HPC) systems in genomics research

Genomics research enables researchers to analyze and identify genetic variants associated with various diseases and their responses to treatments. Genomic sequencing helps detect all types of genetic variations across the entire genome. HPC systems have increased the speed, accuracy, and reliability of genomic research by efficiently analyzing a large amount of sequencing data. The advent of Next-Generation Sequencing (NGS) technologies has allowed the investigation of multiple areas of human health through different omics techniques.

**FIGURE 16 HPC MARKET: IMPACT ANALYSIS OF DRIVERS**

| DRIVERS                                                                                      | 2 YEARS | 5 YEARS | DESCRIPTION                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------|---------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Increasing need for efficient computing, enhanced scalability, and reliable storage research | ●       | ●       | The ability to develop energy efficient software and HPC system is essential, as the use of data and data processing keeps increasing in all areas of the world. The need for power efficient computing is not only due to the environmental impact. Rather, we need energy efficient computing in order to even deliver on the trends predicted. |
| Emerging need for high-speed data processing with accuracy                                   | ●       | ●       | The main factor driving the growth of the HPC market is the ability of HPC solutions to process large volumes of data with speed and accuracy.                                                                                                                                                                                                    |
| HPC and deep learning framework has Fastrack the development of vaccine during pandemic      | ●       | ●       | High- performance computing (HPC) has played important role in fighting COVID-19 by supporting computational need in vaccine development. Scientists around the globe were dependent on the enormous computing power of HPC and supercomputers to run complicated mathematical models                                                             |
| Growing demand for HPC in genomics                                                           | ●       | ●       | The goal of genomics research is to analyze and identify genetic variants associated with disease and response to treatment. Genomic sequencing enables to detect all types of genetic variations across the entire genome.                                                                                                                       |
| IMPACT LEVEL:      ●      HIGH      ●      MEDIUM      ●      LOW      ●      VERY LOW       |         |         |                                                                                                                                                                                                                                                                                                                                                   |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

## 5.2.2 RESTRAINTS

### 5.2.2.1 Cyber security concerns

With the growing use of HPC systems, concerns regarding cyber security are also rising as several entities access HPC systems, making data security a major concern. The nature of HPC exposes it to multiple potential security threats. For instance, in an airport, if HPC is an integral part of the security system, it also becomes the prime target for cyberattacks. Furthermore, HPC environments are clustered, making them vulnerable to many security threats. Since clustered HPC systems are heterogeneous, they require multiple management systems to operate. This slows down the implementation of security policies and security management processes; as a result, vulnerabilities are not remediated. Thus, security concerns are expected to restrain the growth of the HPC market.

### 5.2.2.2 High deployment costs associated with commercial high-performance computing (HPC) clusters

The cost of HPC systems depends on the number of nodes in their clusters. High computing power requires a large number of nodes, similar to supercomputing. However, achieving even some form of HPC requires at least 8 to 12 cores. Developing a functional cluster with 8–12 cores, networking, storage, and operating capabilities, requires a large investment. For instance, an entry-level commercial cluster may require a budget of anywhere between USD 10,000 and 25,000. The deployment of advanced systems with a large number of nodes is more expensive.

**FIGURE 17 HPC MARKET: IMPACT ANALYSIS OF RESTRAINTS**

| RESTRAINTS                                        | 2 YEARS                                                                                | 5 YEARS                                                                                  | DESCRIPTION                                                                                                                                                                         |
|---------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cyber security concerns                           |       |         | With the growing use of HPC, cyber security concerns are also rising. The nature of HPC exposes it to a variety of potential security threats.                                      |
| High entry-level cost for commercial HPC clusters |       |         | The cost of HPC systems depends on the number of nodes in clusters. Massive computing power requires a large number of nodes as available in supercomputing.                        |
| IMPACT LEVEL:                                     |  HIGH |  MEDIUM |  LOW  VERY LOW |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

## 5.2.3 OPPORTUNITIES

### 5.2.3.1 Increasing focus on adoption of hybrid high-performance computing (HPC) systems

With the growing demand for HPC systems and increasing technological advancements, organizations nowadays are focusing on adopting hybrid HPC solutions. IT teams are constantly working on balancing cloud and on-premises HPC solutions. Enterprises are scaling out on-premises HPC resources as needed. Simultaneously, cloud service providers (CSPs) have also started offering turnkey HPC systems according to the specific workload needs of enterprises. A hybrid HPC system exhibits high efficiency. Besides, it resolves security and privacy concerns and lowers maintenance costs. Therefore, the rise in the adoption of hybrid HPC solutions is expected to create lucrative opportunities for the market players.

### 5.2.3.2 Introduction of Exascale computing

The introduction of Exascale computing is expected to increase the benefits of HPC. The Exascale Computing Project is a collaborative effort of two organizations of the US Department of Energy (DOE)—the Office of Science (DOE SC) and National Nuclear Security Administration (NNSA). Exascale refers to a computing system 50 times faster than the current supercomputers. Exascale computing is expected to provide advanced modeling and simulation solutions to address the key challenges in scientific discovery, energy assurance, economic competitiveness, and national security. With the rising volume of data, HPC systems are demanding high memory capacity and bandwidth to operate efficiently. Exascale computing can cater to these demands.

### 5.2.3.3 Rising investments in data centers supporting HPC capability

Enterprises across the globe are increasing investments in improving the data centers supporting HPC capability for business processes. They are adopting data center capabilities to cater to customers and maintain ecosystems, such as partners, suppliers, and distributors. IT technologies such as AI, data analytics, simulation Industrial Internet of Things (IIoT), networking, and storage enable simple workflows in manufacturing firms and business processes of enterprises. These technologies enable the simplicity and automation of workflows and business processes. Thus, with the increasing investments in servers, networking devices, and data centers, the adoption of HPC is expected to increase.

**FIGURE 18 HPC MARKET: IMPACT ANALYSIS OF OPPORTUNITIES**

| OPPORTUNITIES                                                                                                                                                                                                                                                                                                                                                                      | 2 YEARS                                                                           | 5 YEARS                                                                           | DESCRIPTION                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Increasing focus on hybrid HPC Solutions                                                                                                                                                                                                                                                                                                                                           |  |  | A hybrid HPC solution results in better efficiencies. Apart from this, it also resolves security and privacy concerns as well as lowers the maintenance cost. Therefore, the increasing focus on hybrid HPC solutions is expected to provide tremendous opportunities for the growth of the HPC market. |
| Advent of Exascale computing is adding in maximization of preexisting computational speed                                                                                                                                                                                                                                                                                          |  |  | The arrival of exascale computing is seen as one of the biggest opportunities that will maximize the benefits of HPC. Exascale refers to computing systems that are at least 50 times faster than the most powerful supercomputers in use today.                                                        |
| Rise in investment in Data centers supporting HPC capability                                                                                                                                                                                                                                                                                                                       |  |  | Enterprises across the globe are increasing investments in improving the Data centers supporting HPC capability for business processes for simplicity. They are adopting data center capabilities for catering to customers and maintaining ecosystems, such as partners, suppliers, and distributors.  |
| IMPACT LEVEL :  HIGH  MEDIUM  LOW  VERY LOW |                                                                                   |                                                                                   |                                                                                                                                                                                                                                                                                                         |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

## 5.2.4 CHALLENGES

### 5.2.4.1 Less technical expertise related to high-performance computing (HPC)

HPC was a mature and conventional area of information technology. It was distinguished by a narrow category of applications used by a fundamentally fixed set of industries, primarily using on-premise systems. But over the last few years, all of that has started to change. New technologies, the cloud, and a developing set of commercial use cases in the field of machine learning and data analytics have set in motion of change for HPC. This is no longer a tool for only scientists and researchers. HPC is quickly becoming a strategic necessity for all industries that want to gain a competitive advantage in their markets or keep pace with their industry peers to survive. However, there is a scarcity of skills and available training in HPC to prepare and expand the workforce for the HPC and Big Data research and development.

### 5.2.4.2 Limited budgets of SMEs

The potential users in many SMEs lack awareness about the benefits of HPC and do not have the budget to set up such systems. SMEs in many developing countries are still skeptical about adopting HPC due to the high investment costs involved. Many of them are unaware of HPC's various advantages, such as high performance and customizable delivery. However, cloud computing can boost the use of HPC among SMEs, as it will considerably cut down the costs for them.

### 5.2.4.3 Challenges in cooling HPC systems

HPC systems have massive power and storage demands. As the power and storage demands of these HPC deployments continue to grow, the air-cooling technology does not offer a viable solution. High-density servers are required for HPC, which generate heat faster, resulting in high-temperature levels. Liquid cooling is more suitable for high-density facilities as it is both cost- and energy-efficient. Liquid cooling solutions are said to reduce cooling energy consumption by 40–50% and improve Power Usage Effectiveness (PUE). This reduction in energy consumption leads to a reduction in carbon footprint, which is both environmental and economical. These benefits provided by liquid cooling are expected to attract widespread demand from HPC facilities.

#### 5.2.4.4 Requirement of advanced frameworks to improve fault tolerance and ensure resiliency

HPC systems are susceptible to soft errors in both logic circuits and memory subsystems and fail-stop errors in the entire system. Scientific simulations that run on HPC devices use iterative logic to solve complex problems, which require massive processing power. Such massive computing power requires parallelism with at least 100,000 nodes to achieve sustained Petascale performance. However, systems having 100,000 nodes and a large Mean Time Between Failure (MTBF) may encounter a failure every nine hours on average. Even larger systems may encounter failures every hour. Designing a fault-tolerant system is therefore challenging and requires multiple approaches at different levels of the software stack. Advanced algorithms are needed to ensure proper process checkpointing at the lowest level (operating system) as well as at the system level. The advanced distributed checkpointing framework may be needed to design rollback recovery procedures at the parallel application level.

**FIGURE 19 HPC MARKET: IMPACT ANALYSIS OF CHALLENGES**

| CHALLENGES                                                                                                                             | 2 YEARS | 5 YEARS | DESCRIPTION                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------|---------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scarcity of technical expertise in HPC                                                                                                 | ●       | ●       | Over the last few years, New technologies, cloud, and a developing set of commercial use cases in the field of machine learning and data analytics have set in motion of change for HPC.                                                                                                                            |
| Limited budget among SMEs                                                                                                              | ●       | ●       | The potential users in many SMEs lack awareness about the benefits of HPC and do not the budget to set up such systems. SMEs in many developing countries are still skeptical about adopting HPC due to the high investment costs involved.                                                                         |
| Challenges in cooling HPC systems                                                                                                      | ●       | ●       | HPC systems have massive power and storage demands. As the power and storage demands of these HPC deployments continue to grow, the air-cooling technology does not offer a viable solution. High-density servers are required for HPC, which generate heat at a faster rate, resulting in high temperature levels. |
| Requirement of advanced frameworks to improve fault tolerance and ensure resiliency                                                    | ●       | ●       | Massive computing power requires parallelism with at least 100,000 nodes to achieve sustained Peta scale performance. However, systems having 100,000 nodes and large Mean Time Between Failure (MTBF) may encounter a failure every nine hours on average.                                                         |
| <b>IMPACT LEVEL :</b> <span>●</span> <b>HIGH</b> <span>●</span> <b>MEDIUM</b> <span>●</span> <b>LOW</b> <span>●</span> <b>VERY LOW</b> |         |         |                                                                                                                                                                                                                                                                                                                     |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

### 5.3 VALUE CHAIN ANALYSIS

Value chain analysis is a systematic approach to examining all activities occurring in an industry to analyze the sources of competitive advantage. The model works by segregating the industry into its strategically relevant activities to understand the behavior of costs and identify sources of differentiation. This analysis is used for the strategic planning of the entire supply chain and distribution network. The value chain of the HPC ecosystem gives an insight into the industry and showcases the interconnections between various stakeholders in the ecosystem.

In this section, we have described the overall structure of the value chain that comprises research and development, raw material suppliers, original equipment manufacturers (OEMs), system integrators, distribution channels, and end users.

**FIGURE 20** RAW MATERIAL SUPPLIERS AND ORIGINAL EQUIPMENT MANUFACTURERS COLLECTIVELY ADD MAJOR VALUE TO FINAL PRODUCT

| VALUE ADDITION                                                                                                                                         |                                                                                                                                         |                                                                                                                                                                       |                                                                                                                                    |                                                                                                                                                                         |                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0–10%                                                                                                                                                  | 10–20%                                                                                                                                  | 20–75%                                                                                                                                                                | 75–80%                                                                                                                             | 80–95%                                                                                                                                                                  | 95–100%                                                                                                                                                                                                                                |
| PARTICIPANTS                                                                                                                                           |                                                                                                                                         |                                                                                                                                                                       |                                                                                                                                    |                                                                                                                                                                         |                                                                                                                                                                                                                                        |
| <b>RESEARCH &amp; DEVELOPMENT</b><br>Includes various research institutes and funded study groups that work on exploring potential applications of HPC | <b>RAW MATERIAL SUPPLIERS</b><br>Includes various companies that provide industry grade raw materials for manufacturing of HPC hardware | <b>ORIGINAL EQUIPMENT MANUFACTURERS</b><br>Includes manufacturers of HPC products, and solutions. Dell, Lenovo, Hewlett Packard Enterprises are key HPC manufacturers | <b>INTEGRATORS</b><br>Integrators integrate HPC systems into other devices and provide a complete end-to-end solution to consumers | <b>DISTRIBUTORS</b><br>Distributors include brick and mortar shops as well as online 3rd party portals selling HPC products and solutions as well as providing services | <b>END USERS</b><br>End users of HPC products and solutions include key industry verticals such as BFSI, healthcare & life sciences, education and & research, government & defense, manufacturing, energy & utilities, earth sciences |
|                                                                                                                                                        | MOST VALUE ADDITION                                                                                                                     |                                                                                                                                                                       |                                                                                                                                    |                                                                                                                                                                         |                                                                                                                                                                                                                                        |

Source: Interviews with Experts, Secondary Research, Annual Reports, Press Releases, Investor Presentations, White Papers, Journals and Magazines, and MarketsandMarkets Analysis

**Research and Development:** The manufacturers of HPC systems conduct in-house R&D or in collaboration with research institutes to develop prototypes of the systems.

**Raw Material and Component Suppliers:** The raw material and component suppliers provide parts and software, such as servers, storage devices, and networking devices, which are essential for developing HPC systems. Component manufacturers supply components to OEMs, who assemble the components to make the final product HPC.

**Original Equipment Manufacturers (OEMs):** The original equipment manufacturing phase plays a significant role in the entire value chain. Most prominent HPC providers have component manufacturing facilities, which help them achieve cost-effectiveness, especially while manufacturing HPC systems and related products. Advanced Micro Devices, Inc. (US), Intel (US), HPE (US), IBM (US), and Lenovo (China) are the major OEMs operating in the HPC market.

**System Integrators:** System integrators integrate HPC systems into devices and provide a complete end-to-end solution to customers.

**Distributors:** Distributors are the local suppliers that stock and sell different products and/or offer services of HPC component manufacturers, from local to global manufacturers. Distribution channels play an essential role in expanding the geographic reach of manufacturers. These channels can be divided into two types: offline channels (direct suppliers) and online channels. Direct suppliers are the premium outlets or exclusive suppliers for large-scale regional and global HPC system manufacturing companies, such as Advanced Micro Devices, Inc. (US), Intel (US), HPE (US), IBM (US), and Lenovo (China). The majority of the sales of HPC systems are achieved through distributors who are engaged in transactional sales. Direct suppliers primarily target contractual sales and often act as distributors within a particular region. The value addition of this stage varies depending on the functioning of channel partners, from stocking to the resale of HPC systems.

End Users: The last phase of this value chain is end users from different industries and sectors, including BFSI, healthcare, life sciences, education & research, government & defense, manufacturing, energy & utilities, earth sciences, and others.

## 5.4 TECHNOLOGY ANALYSIS

### 5.4.1 EXASCALE COMPUTING IN HPC SYSTEMS

Supercomputers are used to model and simulate complex, dynamic systems, which are costly, unrealistic, or impossible to demonstrate physically. Supercomputers are transforming how scientists discover the evolution of biological systems, the universe, weather forecasting, and renewable energy.

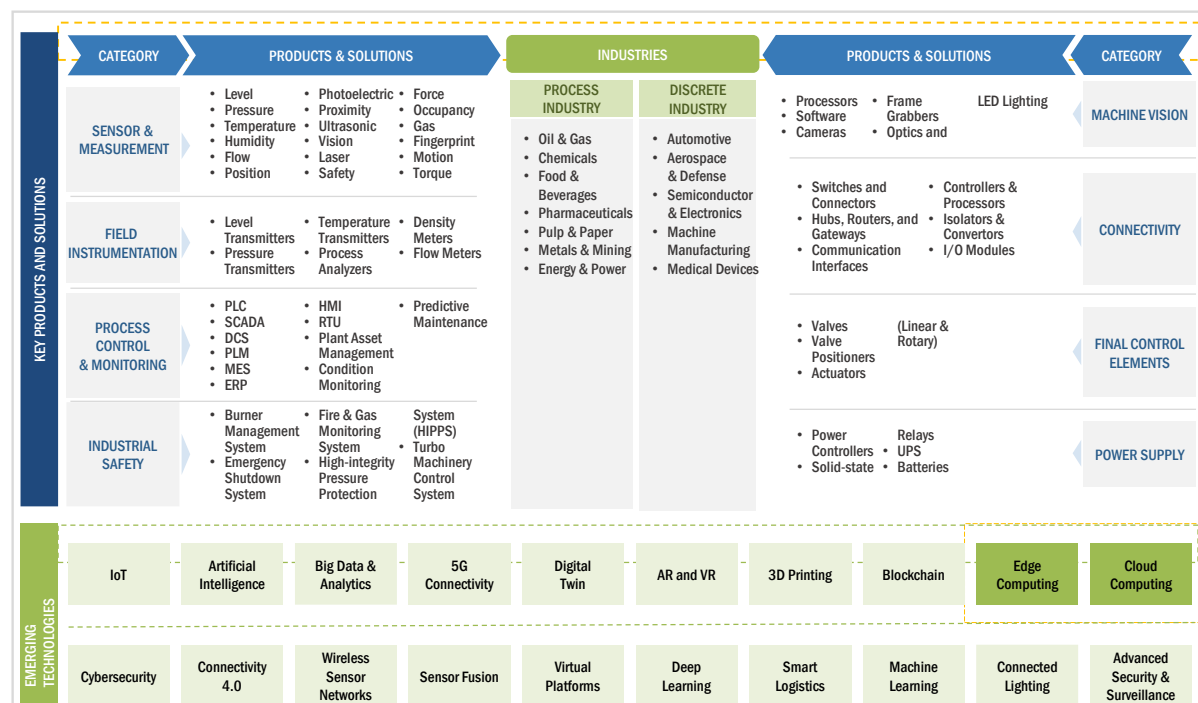
Exascale computing refers to computing systems capable of performing at least one exaflop calculation per second ( $10^{18}$ ). Exascale supercomputers are the next generation of supercomputers, allowing scientists to better simulate the complicated processes involved in energy, medicine, biotechnology, advanced manufacturing, material design, and the universe's physics quickly with high definition.

### 5.4.2 CLOUD COMPUTING IN HPC SYSTEMS

Cloud computing is a methodology that provides access to engineers and IT departments to drive their daily organizational operations without the fear of slowing down. This, in turn, shortens the time to develop and launch products. Cloud HPC solutions are developed on the latest technologies. They are easy to use and cost-effective and form the basis for implementing HPC products and solutions.

## 5.5 ECOSYSTEM ANALYSIS

**FIGURE 21 HPC MARKET: ECOSYSTEM**



Source: Interviews with Experts, Secondary Research, Annual Reports, Press Releases, Investor Presentations, White Papers, Journals and Magazines, and MarketsandMarkets Analysis

Advanced Micro Devices (US), Intel (US), HPE (US), IBM (US), Dell (US), Lenovo (China), Fujitsu (Japan), Atos (France), CISCO (US), Nvidia (Japan), and NEC Corporation (Japan) are the major companies that offer HPC systems. The ecosystem comprises material suppliers, OEMs, distributors, and end users.

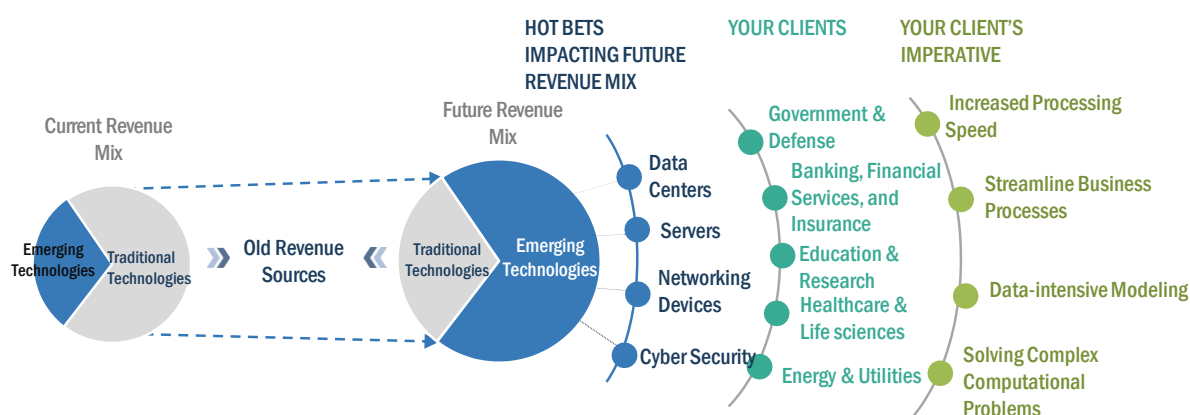
**TABLE 9**      **HPC MARKET: ECOSYSTEM**

| COMPANY                          | ROLE IN ECOSYSTEM     |
|----------------------------------|-----------------------|
| Advanced Micro Devices           | HPC System Provider   |
| Intel                            | HPC System Provider   |
| Hewlett Packard Enterprise       | HPC System Provider   |
| IBM                              | HPC System Provider   |
| Dell                             | HPC System Provider   |
| Lenovo                           | HPC System Provider   |
| Fujitsu                          | HPC System Integrator |
| Atos                             | HPC System Provider   |
| CISCO                            | HPC System Provider   |
| NVIDIA                           | HPC System Provider   |
| NEC Corporation                  | HPC System Integrator |
| Amazon Web Services              | HPC System Provider   |
| Oracle                           | HPC System Provider   |
| Microsoft                        | HPC System Provider   |
| Inspur                           | HPC System Integrator |
| NetApp                           | HPC System Integrator |
| IRON Global                      | HPC System Provider   |
| Aspen System                     | HPC System Provider   |
| Advanced Clustering Technologies | HPC System Integrator |

Source: Interviews with Experts, Secondary Research, Annual Reports, Press Releases, Investor Presentations, White Papers, Journals and Magazines, and MarketsandMarkets Analysis

## 5.6 TRENDS/DISRUPTIONS IMPACTING CUSTOMERS

The MarketsandMarkets engagement model can help the providers of HPC systems prioritize high-growth niche opportunities to achieve the targeted revenue.



Source: Interviews with Experts, Secondary Research, Annual Reports, Press Releases, Investor Presentations, White Papers, Journals and Magazines, and MarketsandMarkets Analysis

The increasing shift toward digitization and miniaturization of devices is driving the growth of the oil & gas, chemical, and food & beverage industries, which is consequently fueling the growth of the HPC market. Cloud deployment of HPC is expected to bring new revenue sources for HPC system providers. In data centers, the use of HPC enables high-speed processing of data, thereby saving operational time. However, the use of HPC requires extremely reliable data storage systems. Therefore, data accuracy and sensitivity are critical for the optimal use of HPC. Several companies are still lagging in tapping the opportunities in the market for Exascale HPC.

## 5.7 CASE STUDIES

### 5.7.1 CASE STUDY 1: EDUCATION

|                   |                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Objective | To study 100 simulation projects for COVID-19 proteins                                                                                                                                                                                                                                                                                                                         |
| Description       | Dr. Greg Bowman, an associate professor of biochemistry and molecular biophysics at Washington University School of Medicine, was supervising Folding@Home, the largest crowd-sourced computational biology project. The project relied on the power of the tens of thousands of home computers to process complex calculations to simulate the dynamics of COVID-19 proteins. |
| Solution Offered  | AMD deployed the high-performance AMD EPYC processor to accelerate research on COVID-19, cancer, and neurodegenerative diseases.                                                                                                                                                                                                                                               |
| Benefits          | Assisted in boosting study throughput                                                                                                                                                                                                                                                                                                                                          |

Source: Company Website and Press Releases

### 5.7.2 CASE STUDY 2: ENERGY AND UTILITIES

|                   |                                                                                                                                                                                                                                                                                                                 |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Objective | To collect huge throughput of data from detectors on the Large Hadron Collider (LHC)                                                                                                                                                                                                                            |
| Description       | CERN, a European organization for nuclear research, is one of the world's largest and most respected centers for scientific research. The company is preparing for LHC restart in 2021. The company was looking for a hardware platform capable of processing huge data throughput from every server to server. |
| Solution Offered  | AMD deployed AMD EPYC 7742 processors to offer processing performance, memory speed, and bandwidth required for the project.                                                                                                                                                                                    |
| Benefits          | Processors saved more than a terabit data flowing from servers<br>Provided cost-effective, compact systems                                                                                                                                                                                                      |

Source: Company Website and Press Releases

### 5.7.3 CASE STUDY 3: EDUCATION

|                   |                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Objective | To find a supercomputing solution that supports complex calculations                                                                                                                                                                                                                                                                                            |
| Description       | DiRAC, a multiple service provider to particular science workflows, and the University of Leicester came together to process and study projects on theoretical astrophysics, particle physics, cosmology, and nuclear physics. These projects help in understanding how the universe works. They demand super supercomputing solutions to perform calculations. |
| Solution Offered  | HPE designed and developed systems needed by DiRAC and the University of Leicester.                                                                                                                                                                                                                                                                             |
| Benefits          | Speed of calculations increased by 20 times                                                                                                                                                                                                                                                                                                                     |

Source: Company Website and Press Releases

## 5.8 PORTER'S FIVE FORCES ANALYSIS

Porter's Five Forces framework analyzes the competition in a particular market. It provides insights into the industry to help determine the competitive intensity and the attractiveness of an industry in terms of profitability.

### 5.8.1 BARGAINING POWER OF SUPPLIERS

The bargaining power of suppliers assesses the ease with which suppliers can drive up the prices of HPC products and solutions by leveraging the unique proposition of their products or add pressure on buyers by raising prices, lowering product quality, or reducing the availability of their products. There is high competition among HPC product and solution providers owing to their significant presence in the market. Besides, the differentiation between HPC systems offered by different manufacturers is moderate, as products with the same or unique specifications are available. As a result, the bargaining power of suppliers is expected to be low in this market.

### 5.8.2 BARGAINING POWER OF BUYERS

The bargaining power of buyers indicates the ease at which buyers drive down the prices of products in a market. The impact of this force on the market was moderate in 2021. Advanced Micro Devices, Inc is the largest market player offering its products across the globe. However, local players that offer HPC systems at lower prices and shorter lead times increase the competition for the major players. But, their inability to

provide products in larger volumes is leading end users to opt for major players. Thus, the bargaining power of buyers is moderate in this market.

### 5.8.3 THREAT OF NEW ENTRANTS

The threat of new entrants refers to the force of new, potential competitors in any market who can capture the market share of existing companies. The threat of new entrants in the HPC market is low as many global and domestic players have an established presence in the market. Advanced Micro Devices, Inc. (US), Intel (US), HPE (US), IBM (US), Dell (US), Lenovo (China), Fujitsu (Japan), Atos (France), CISCO (US), Nvidia (Japan), and NEC Corporation (Japan) are examples of some of the established players in this market. These players collectively account for a significant share of the global market. Besides, there are many local players in developed and developing countries. The existing players with an established brand name are making significant R&D investments to remain competitive by offering customized products, which also poses a challenge for new entrants. The existing players are focusing on expanding their product portfolio, developing safety systems for industries with high growth opportunities, and providing their products across different regions to remain competitive. Hence, the threat of new entrants is estimated to be low during the forecast period.

### 5.8.4 THREAT OF SUBSTITUTES

The threat of substitutes arises when customers or buyers in the market can achieve the same results using products/solutions/services other than those in the market under consideration. The key factors that determine the threat of substitutes in the HPC market include price, quality, and differentiation. There are no external substitutes for HPC. The established market players have high technical expertise related to HPC. Thus, the overall impact of the threat of substitutes is expected to be low during the review period.

### 5.8.5 INTENSITY OF COMPETITIVE RIVALRY

Currently, the intensity of competitive rivalry in the HPC market is high, and the trend is expected to continue during the forecast period. Market players are focusing on expanding their product portfolio; most of them are adopting different approaches to develop technologically advanced products. Also, they are trying to gain a competitive edge by implementing different features and variations in their products. The intense competition in the market has influenced companies to adopt powerful competitive strategies to guarantee sustainability.

**TABLE 10** IMPACT OF PORTER'S FIVE FORCES ON HPC MARKET, 2021

| FORCE                            | IMPACT (2021) |
|----------------------------------|---------------|
| Intensity of Competitive Rivalry | High          |
| Bargaining Power of Suppliers    | Low           |
| Bargaining Power of Buyers       | Moderate      |
| Threat of Substitutes            | Low           |
| Threat of New Entrants           | Low           |
| Average Impact of Each Force     | Moderate      |

Source: Interviews with Experts, Secondary Research, Annual Reports, Press Releases, Investor Presentations, White Papers, Journals and Magazines, and MarketsandMarkets Analysis

## 5.9 PRICING ANALYSIS

There are various offerings under HPC. These offerings include servers, software, networking devices, and storage. The table below describes the pricing analysis of HPC vendors as per their solution offerings in the market:

**TABLE 11 HIGH-PERFORMANCE COMPUTING MARKET: PRICING ANALYSIS (2021–2022)**

| SR. NO | COMPANY NAME | PRODUCT                            | PRODUCT FEATURE                                                                                                                                                                                              | PRICING RANGE                                                                                                                                                                                                                                                 |
|--------|--------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | Dell         | PowerEdge Rack Servers             | <ul style="list-style-type: none"> <li>Balanced performance</li> <li>Storage capacity for future growth</li> </ul>                                                                                           | <ul style="list-style-type: none"> <li>1-Socket (supports 1 processor): Starting at USD 559.00</li> <li>2-Socket (supports up to 2 processors): Starting at USD 1,059.00</li> <li>4-Socket (supports up to 4 processors): Starting at USD 7,628.99</li> </ul> |
|        |              | Modular Infrastructure             | <ul style="list-style-type: none"> <li>Meets specific business demands</li> <li>Controls costs</li> <li>Streamlines management with flexible processing, storage, and networking modules</li> </ul>          | <ul style="list-style-type: none"> <li>PowerEdge M Series: Starting at USD 1,649.00</li> <li>PowerEdge MX: Starting at USD 3,249.00</li> <li>PowerEdge FX Series: Starting at USD 2,609.00</li> </ul>                                                         |
|        |              | PowerEdge Tower Servers            | <ul style="list-style-type: none"> <li>From mini-towers to rack-capable platforms</li> <li>More cache and speed</li> </ul>                                                                                   | <ul style="list-style-type: none"> <li>1-Socket (supports 1 processor): Starting at USD 499.00</li> <li>2-Socket (supports up to 2 processors): Starting at USD 1,159.00</li> </ul>                                                                           |
| 2      | NVIDIA       | NVIDIA's GPU-accelerated solutions | <ul style="list-style-type: none"> <li>Streamlined with access to IT management experts</li> <li>Access massive computing power on demand and with ease</li> <li>Supports all top cloud platforms</li> </ul> | <ul style="list-style-type: none"> <li>Free trials available</li> <li>Wide range of prices and leasing options</li> </ul>                                                                                                                                     |
| 3      | Intel        | X86 2x626V processor               | <ul style="list-style-type: none"> <li>48 Cores</li> <li>2 TB memory capacity</li> <li>2400 MHz memory frequency</li> </ul>                                                                                  | USD 5,800                                                                                                                                                                                                                                                     |
| 4      | AMD          | AMD EPYC 1x7702P processor         | <ul style="list-style-type: none"> <li>64 Cores</li> <li>4 TB memory capacity</li> <li>3200 MHz memory frequency</li> </ul>                                                                                  | USD 4,425                                                                                                                                                                                                                                                     |

Note: The pricing showcased is indicative as a few players do not provide pricing details on their websites

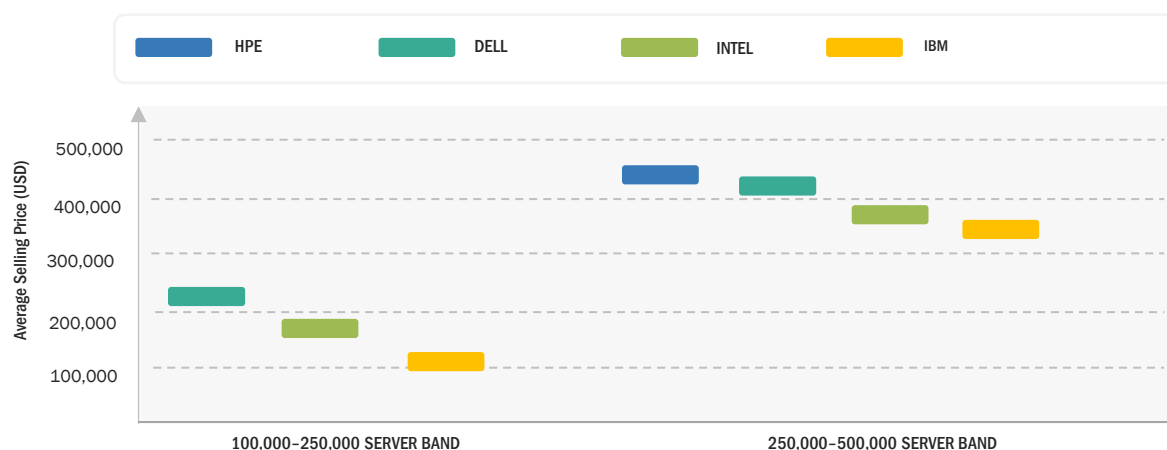
Source: Company Website

Above mentioned, historic and forecast prices were compiled from various sources such as Tier-1 supply portals, press releases, company presentations, industry experts, white papers, and the MnM data repository. Upon analysis, the projected price was derived based on the OEM announcements, Tier-1 new product developments, future demand, historical price trends, and penetration of HPC in the overall market space.

### 5.9.1 AVERAGE SELLING PRICES OF 4 KEY PLAYER OFFERINGS, BY HPC SERVERS

This section includes indicative price ranges of products of 4 players for HPC servers across key verticals.

**FIGURE 22** AVERAGE SELLING PRICES OF 4 KEY PLAYER OFFERINGS, BY HPC SERVERS

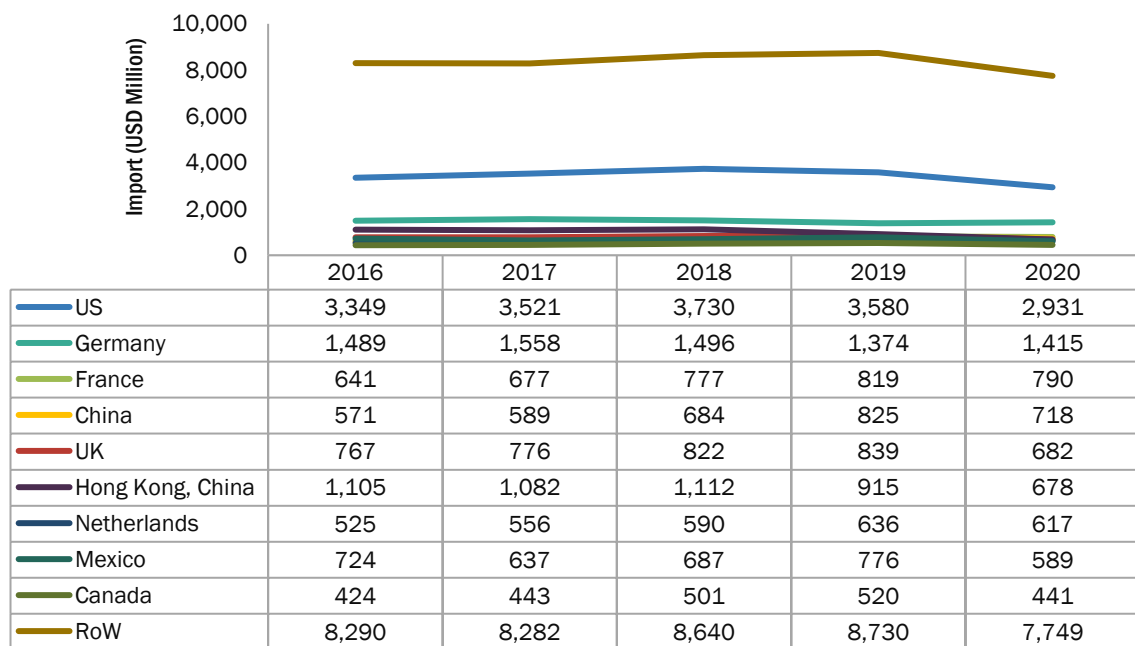


Source: Secondary Research

## 5.10 TRADE ANALYSIS

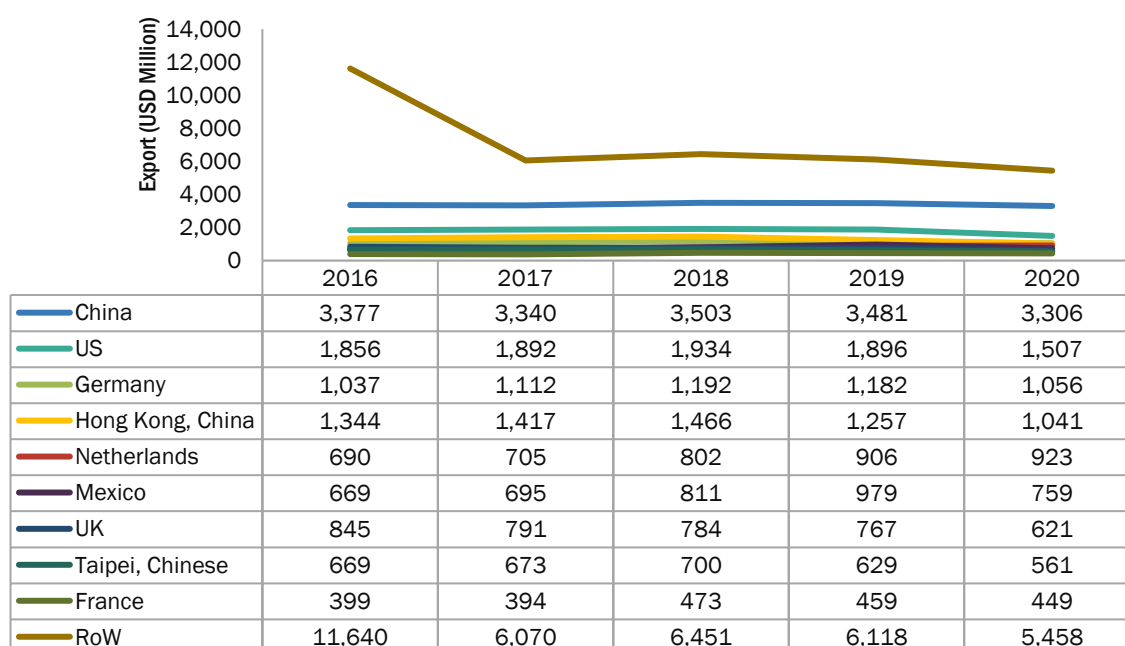
In this section, we have provided the trade data of components used in HPC. The global import and export information for the HS Code 8471, which includes automatic data-processing machines and units thereof; magnetic or optical readers, machines for transcribing data onto data media in coded form, and machines for processing data, is provided below.

**FIGURE 23** IMPORT DATA, BY COUNTRY, 2016–2020 (USD MILLION)



Source: International Trade Centre (ITC) Trade Map

In 2020, the US was the largest importer of automatic data-processing machines and units thereof; magnetic or optical readers, machines for transcribing data onto data media in coded form, and machines for processing data, with imports valued at USD 104,802 million. It held a 25.6% share of the total imports, while most countries held a single-digit share of imports in 2020. Germany registered the second-highest imports share in 2020, worth USD 33,925 million, followed by France with USD 27,733 million and China with USD 208,23 million.

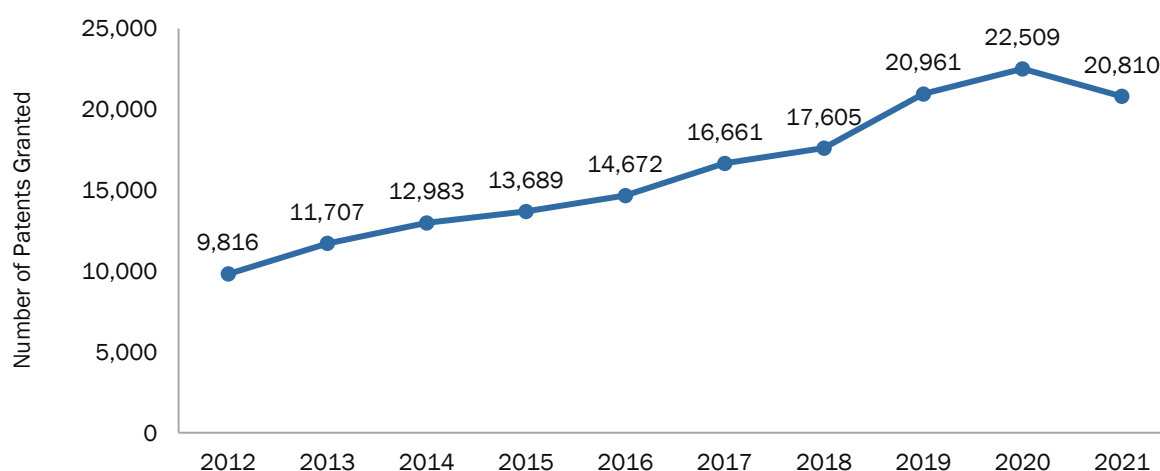
**FIGURE 24** EXPORT DATA, BY COUNTRY, 2016–2020 (USD MILLION)

Source: International Trade Centre (ITC) Trade Map

China was the largest exporter in 2020, with exports valued at USD 170,176 million – accounting for a 44.5% share of the exports in 2020. The US was the second-largest exporter, with exports valued at USD 31,997 million in 2020. The top three countries exporting these components after China and the US are Germany, Hong Kong (China), and the Netherlands, with exports valued at USD 24,788 million, USD 24,230 million, and USD 15,745 million, respectively, in 2020.

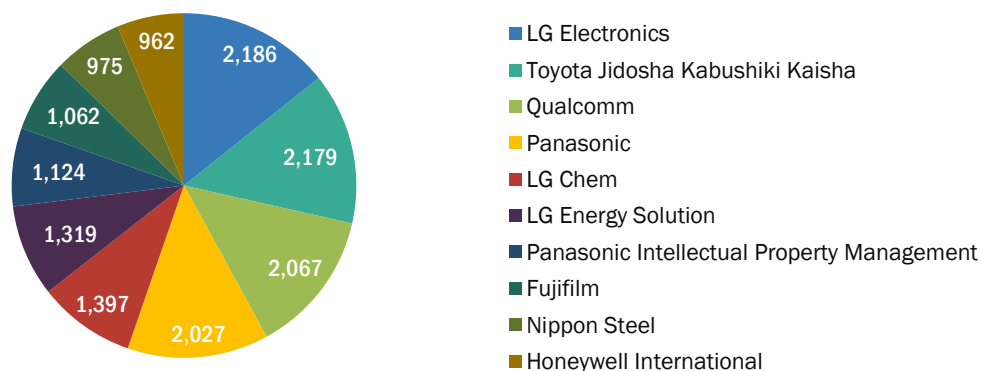
### 5.10.1 PATENTS ANALYSIS

In this section, we have provided an indicative analysis of some of the global patents pertaining to the HPC market.

**FIGURE 25** NUMBER OF PATENTS GRANTED PER YEAR FROM 2012 TO 2021

Source: Lens.org

**FIGURE 26** TOP 10 COMPANIES WITH HIGHEST NUMBER OF PATENT APPLICATIONS IN LAST 10 YEARS



Source: Lens.org

**TABLE 12** TOP 20 PATENT OWNERS IN LAST 10 YEARS

| PATENT OWNER                           | NO. OF PATENTS |
|----------------------------------------|----------------|
| Raytheon Technologies                  | 3,105          |
| IBM                                    | 1,412          |
| Intel                                  | 1,147          |
| Micron Technology                      | 830            |
| United Technologies                    | 642            |
| General Electric Company               | 515            |
| Bristol-Myers Squibb                   | 511            |
| Nvidia                                 | 441            |
| Novartis                               | 407            |
| Oracle International                   | 389            |
| Intermolecular                         | 341            |
| Fujitsu                                | 338            |
| Fujifilm                               | 267            |
| Samsung Electronics                    | 251            |
| Schlumberger Technology                | 250            |
| Hewlett Packard Enterprise Development | 244            |
| Microsoft Technology Licensing         | 236            |
| Microsoft                              | 182            |
| Amazon Technologies                    | 173            |
| Novo Nordisk                           | 153            |

Source: Lens.org

**TABLE 13** LIST OF FEW PATENTS IN HPC MARKET, 2021–2022

| MONTH & YEAR   | TITLE                                                     | PUBLICATION NUMBER | INVENTOR             | DESCRIPTION                                                                                                                                                                                                                                                                                                                                   |
|----------------|-----------------------------------------------------------|--------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| February 2022  | High Performance Computing System                         | EP 3945425 A1      | Sipearl              | This patent discloses an HPC system (HPC) having multiple computational cores (C1, C2, C3, C4, and CD), each computational core is associated with a computational memory (CPM), arranged to form a computational resource (CR) being utilized for performing computation.                                                                    |
| October 2021   | High-Performance Computing Memory                         | CN 214336329 U     | Linkzol Beijing Tech | This patent reveals an HPC memory, which comprises a protection bin with an opening on one side, a memory body, a positioning component, and a sliding component.                                                                                                                                                                             |
| September 2021 | Obfuscation for High-Performance Computing Systems        | US 11122079 B1     | Architecture Tech    | This patent states about the obfuscation computing system and communication with nodes in a distributed computing platform. The nodes include compute nodes that provide resources in the distributed computing platform and a controller node that performs resource management of the resources.                                            |
| May 2021       | Cloud Computing Equipment with High Computing Performance | CN 214751723 U     | Wang Feng            | The utility model relates to the technical field of cloud computing, in particular, to cloud computing equipment with high computing performance, which comprises a heat dissipation box, a heat dissipation cavity is arranged in the heat dissipation box, an exhaust hood is fixedly connected to the top end of the heat dissipation box. |
| March 2021     | HPC on a Public Grid                                      | US 10956628 B2     | Autodesk Inc         | This patent reveals the method and system that provide performance computing on a public grid. A computational mesh to solve a received engineering simulation is computed.                                                                                                                                                                   |

Source: Lens.org

## 5.11 KEY CONFERENCES AND EVENTS BETWEEN 2022 AND 2023

**TABLE 14** HPC MARKET: DETAILED LIST OF CONFERENCES AND EVENTS

| YEAR/ QUARTER | NORTH AMERICA                                                                                                                                                                    | EUROPE                                                                                                     | ASIA-PACIFIC                                                                                                          | REST OF THE WORLD                                                                                |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Q1'22         | [March] <a href="#">Skip to content</a><br><a href="#">2023 Energy HPC Conference</a>                                                                                            | [March] <a href="#">ISC High Performance 2022</a>                                                          | [March] <a href="#">Supercomputing Frontiers Asia 2022</a>                                                            |                                                                                                  |
| Q2'22         | [June] <a href="#">International Conference on High-Performance Reconfiguration Computing Systems and Applications</a>                                                           | [May] <a href="#">International Conference on HPC and Statistics ICHPCS</a>                                | [May] <a href="#">International Conference on HPC and Networking ICHPCN</a>                                           | [May] <a href="#">International Conference on HPC for Power Systems</a>                          |
| Q3'22         | [August] <a href="#">International Conference on HPC and Cluster Technology</a>                                                                                                  |                                                                                                            | [August] <a href="#">International Conference on Recent Advances in High-Performance Smart Computing Applications</a> | [August] <a href="#">International Conference on HPC</a>                                         |
| Q4'22         | [October] <a href="#">International Conference on Experimental Mathematics and HPC</a>                                                                                           | [November] <a href="#">International Conference on High-Performance Computer, Computing and Statistics</a> | [December] <a href="#">29th IEEE International Conference on HPC, Data, and Analytics</a>                             |                                                                                                  |
| Q1'23         | [January] <a href="#">International Conference on HPC and Decision Procedures</a><br>[January] <a href="#">International Conference on High-Performance Scientific Computing</a> | [February] <a href="#">International Conference on HPC and Algorithms</a>                                  |                                                                                                                       | [February] <a href="#">International Conference on High-Throughput and Performance Computing</a> |
| Q2'23         | [April] <a href="#">International Conference on Applied Physics and HPC</a>                                                                                                      |                                                                                                            |                                                                                                                       |                                                                                                  |
| Q3'23         |                                                                                                                                                                                  | [July] <a href="#">International Conference on High-Performance Reconfigurable Computing Architectures</a> |                                                                                                                       |                                                                                                  |
| Q4'23         |                                                                                                                                                                                  | [December] <a href="#">HPC Conferences in Rome</a>                                                         | [December] <a href="#">International Conference on Compute Engineering and Applications</a>                           |                                                                                                  |

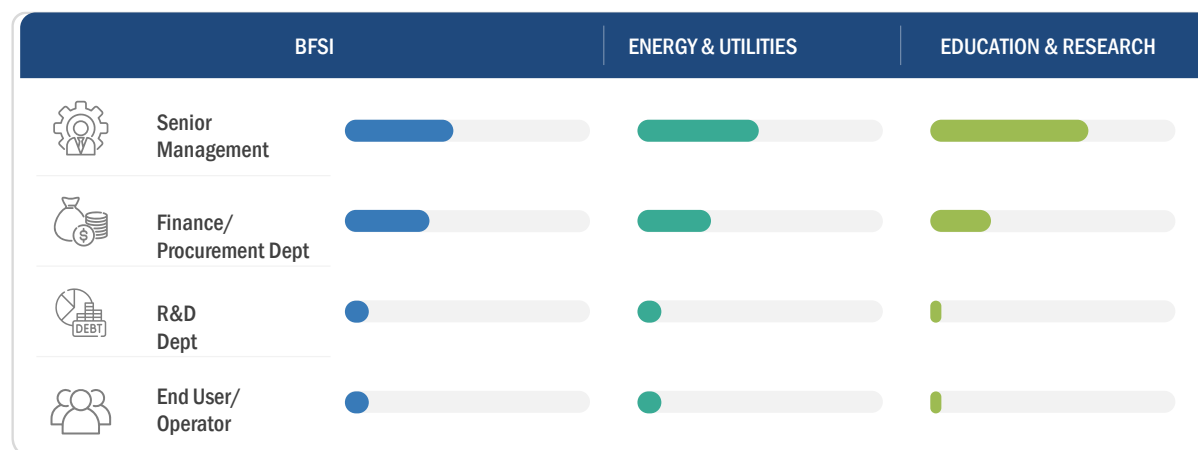
Source: Secondary Research

## 5.12 KEY STAKEHOLDERS AND BUYING CRITERIA

This section identifies the key stakeholders and the extent of their influence in deciding what to buy and who to buy from for each of the top 3 verticals in the HPC market.

### 5.12.1 KEY STAKEHOLDERS IN BUYING PROCESS

**FIGURE 27** INFLUENCE OF STAKEHOLDERS IN BUYING PROCESS FOR TOP 3 VERTICALS



Source: Secondary Research, Company Websites, and MarketsandMarkets Analysis

**TABLE 15** INFLUENCE OF STAKEHOLDERS IN BUYING PROCESS FOR TOP 3 VERTICALS (%)

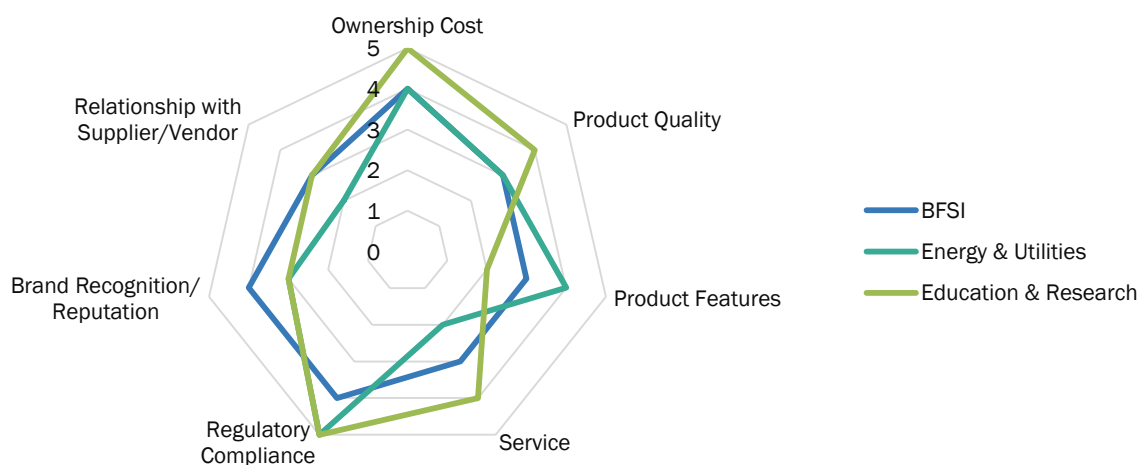
| KEY STAKEHOLDERS         | BFSI       | ENERGY & UTILITIES | EDUCATION & RESEARCH |
|--------------------------|------------|--------------------|----------------------|
| Senior Management        | 45         | 50                 | 65                   |
| Finance/Procurement Dept | 35         | 30                 | 25                   |
| R&D Dept                 | 10         | 10                 | 05                   |
| End User/Operator        | 10         | 10                 | 05                   |
| <b>TOTAL</b>             | <b>100</b> | <b>100</b>         | <b>100</b>           |

Source: Secondary Research, Company Websites, and MarketsandMarkets Analysis

### 5.12.2 BUYING CRITERIA

This section identifies the key criteria for selecting a supplier or vendor and their weights for each of the top 3 end-uses of the bus seat market.

**FIGURE 28** KEY BUYING CRITERIA FOR TOP 3 INDUSTRY VERTICALS



Source: Secondary Research, Company Websites, and MarketsandMarkets Analysis

**TABLE 16** KEY BUYING CRITERIA FOR TOP 3 VERTICALS

| BUYING CRITERIA                   | BFSI | ENERGY & UTILITIES | EDUCATION & RESEARCH |
|-----------------------------------|------|--------------------|----------------------|
| Ownership Cost                    | 5    | 5                  | 5                    |
| Product Quality                   | 3    | 3                  | 5                    |
| Product Features                  | 2    | 3                  | 4                    |
| Brand Recognition/Reputation      | 2    | 3                  | 4                    |
| Service                           | 2    | 3                  | 4                    |
| Relationship with Supplier/Vendor | 3    | 3                  | 4                    |
| Regulatory Compliance             | 3    | 3                  | 5                    |

Note: Respondents were rated for each criterion on a scale of 1–5, where 1 indicates “Least Important” and 5 indicates “Most Important” or “Critical”.

Source: Secondary Research, Company Websites, and MarketsandMarkets Analysis

## 5.13 REGULATIONS AND STANDARDS

Various governments and agencies globally develop standards, regulations, and certifications for every industry. Manufacturers must comply with all the required safety standards across different countries.

In the global HPC market, the key government organizations that implement standards and provide certifications to manufacturers include the American National Standards Institute (ANSI), International Organization for Standardization (ISO), European Agency for Safety and Health at Work, International Electrotechnical Commission (IEC), American Society of Safety Professionals (ASSP), European Electrotechnical Standardization Organization (CENELEC), and Occupational Safety and Health Administration (OSHA). The key regulations and standards that the manufacturers follow in the HPC ecosystem are mentioned below.

**TABLE 17** REGULATIONS AND STANDARDS FOR HPC MARKET

| REGULATORY ORGANIZATION | REGULATION/STANDARDS | DESCRIPTION                                                                                                                                                                                                                                                                  |
|-------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO                     | ISO 45001:2018       | ISO 45001:2018 specifies requirements for an occupational health and safety (OH&S) management system,                                                                                                                                                                        |
| ANSI                    | ASSP A10.3-2020      | This standard specifies the Safety Requirements for Powder-Actuated Fastening Systems.                                                                                                                                                                                       |
| ANSI/ASSP               | ANSI/ASSP Z359.2     | The ANSI/ASSP Z359 standard focuses on fall protection equipment and systems for climbing, work positioning, fall arrest, rescue, evacuation, and other fall hazards.                                                                                                        |
| IEC                     | IEC 61508            | IEC 61508 is the international standard for electrical, electronic, and programmable electronic safety-related systems                                                                                                                                                       |
| IEC                     | IEC 62061            | This standard describes the implementation of safety-related control systems on machinery and examines the whole lifecycle from the concept phase through to decommissioning.                                                                                                |
| IEC                     | IEC 61511            | IEC 61511 provides requirements for the specification, design, installation, operation, and maintenance of a safety instrumented system.                                                                                                                                     |
| IEC                     | IEC 62443            | IEC 62443 is a set of security standards for the secure development of industrial automation and control systems.                                                                                                                                                            |
| OSHA                    | OSHA Standards       | There are four groups of OSHA standards: General Industry, Construction, Maritime, and Agriculture. (General Industry is the set that applies to the largest number of workers and worksites). These standards are designed to protect workers from a wide range of hazards. |

Source: International Organization for Standardization (ISO), European Agency for Safety and Health at Work, International Electrotechnical Commission (IEC), American Society of Safety Professionals (ASSP), ASTM International, and iso.org

## 5.14 REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

This section should include a region-wise list of key regulatory bodies, government agencies, industry associations, and non-profit organizations relevant to the market.

**TABLE 18** NORTH AMERICA: LIST OF REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

| COUNTRY | ORGANIZATION NAME                                                          | ORGANIZATION TYPE     | SHORT DESCRIPTION                                                                                                              |
|---------|----------------------------------------------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------|
| U.S.    | <a href="#">Occupational Safety and Health Administration (OSHA)</a>       | Government Regulatory | OSHA regulates safe and healthful working conditions for workers.                                                              |
| Canada  | <a href="#">Canadian Centre for Occupational Health and Safety (CCOHS)</a> | Government Regulatory | Canadian Centre for Occupational Health and Safety (CCOHS) works towards implementing workplace health and safety regulations. |

Source: International Organization for Standardization (ISO), European Agency for Safety and Health at Work, International Electrotechnical Commission (ICE), American Society of Safety Professionals (ASSP), ASTM International, and iso.org

**TABLE 19** EUROPE: LIST OF REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

| COUNTRY        | ORGANIZATION NAME                                                                                              | ORGANIZATION TYPE    | SHORT DESCRIPTION                                                                                                                                                                                                                                                                                                                      |
|----------------|----------------------------------------------------------------------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| European Union | <a href="https://www.itf-oecd.org/">European Agency for Safety and Health at workhttps://www.itf-oecd.org/</a> | Government Agency    | The EU-OSHA is the European Union information agency for occupational safety and health. EU-OSHA contributes to the European Commission's Strategic Framework on Health and Safety at Work from 2021 to 2027 and other relevant EU strategies.                                                                                         |
| Germany        | <a href="#">Federal Institute for Occupational Safety and Health</a>                                           | Government Regulator | <a href="#">Federal Institute for Occupational Safety and Health</a> is responsible for regulating scientific divisions such as changing the world of work, products, and work systems and handling hazardous substances.                                                                                                              |
| UK             | <a href="#">Office for Product Safety &amp; Standards</a>                                                      | Government Agency    | It is the UK government's <a href="#">Office for Product Safety &amp; Standards</a> and is part of the Department for Business, Energy & Industrial Strategy. The Office for Product Safety and Standards (OPSS) was created in January 2018 to deliver consumer protection and support business confidence, productivity, and growth. |

Source: International Organization for Standardization (ISO), European Agency for Safety and Health at Work, International Electrotechnical Commission (ICE), American Society of Safety Professionals (ASSP), ASTM International, and iso.org

**TABLE 20** REST OF WORLD: LIST OF REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

| COUNTRY | ORGANIZATION NAME                                                  | ORGANIZATION TYPE    | SHORT DESCRIPTION                                                                          |
|---------|--------------------------------------------------------------------|----------------------|--------------------------------------------------------------------------------------------|
| UAE     | <a href="#">United Arab Emirates Health and Safety Legislation</a> | Government Regulator | Safety is governed by both the federal law of the UAE and the law of the emirate of Dubai. |

Source: International Organization for Standardization (ISO), European Agency for Safety and Health at Work, International Electrotechnical Commission (IEC), American Society of Safety Professionals (ASSP), ASTM International, and [iso.org](https://www.iso.org)

## 6 HIGH-PERFORMANCE COMPUTING MARKET, BY COMPONENT

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### KEY FINDINGS

- The HPC market for the services segment is expected to grow at a CAGR of 8.2% during the forecast period.
- The market for the solutions segment is projected to reach USD 49.9 Billion by 2027, registering a CAGR of 6.7% during the forecast period.
- The demand for HPC solutions is expected to grow at a significant rate during the review period owing to the rising number of data centers across the globe.

## 6.1 INTRODUCTION

The HPC market based on component has been segmented into solutions, software, and services. The scope of the solutions segment includes servers, storage devices, networking devices, and software, whereas services considered are design and consulting, integration and deployment, and support, maintenance, and management. HPC solutions are expected to gain traction among various industries and sectors owing to their excellent performance and scalability features. Moreover, HPC services are widely used for designing and supporting the integration and deployment of HPC solutions.

**FIGURE 29** HIGH-PERFORMANCE COMPUTING MARKET, BY COMPONENT



Source: International Trade Administration (ITA), European Cloud Association (EUDCA), Telecommunications Industry Association, United Nations Economic and Social Commission for Asia and the Pacific (ESCAP), Economic Commission for Latin America (ECLA), Arab Information and Communication Technology Organization (AICTO), Interviews with Experts, Secondary Research, Annual Report, Press Releases, Investor Presentations, White Papers, Journals and Magazines, and MarketsandMarkets Analysis

### 6.1.1 DRIVERS: HIGH-PERFORMANCE COMPUTING MARKET FOR COMPONENT

- The growing need for enhancing the performance of applications to maximize Return on Investment (ROI) on HPC infrastructure is driving the demand for software solutions.
- The growing adoption of self-guided technical support and self-driving vehicles is expected to fuel the demand for HPC solutions.
- The rising use of cloud applications in enterprises worldwide is expected to drive the growth of the HPC market.

### 6.1.2 COVID-19 IMPACT

- In the wake of the COVID-19 crisis, enterprises are carrying out business operations by utilizing digital technologies such as AI, data analytics, and simulation. The increased use of digital technologies has fueled the demand for HPC solutions and related services in the COVID-19 period.
- The adoption of HPC solutions is expected to rise exponentially in the coming years as several organizations have planned to shift enterprise workloads to the cloud.

**TABLE 21** HIGH-PERFORMANCE COMPUTING MARKET, BY COMPONENT, 2018–2021 (USD BILLION)

| Component        | 2018        | 2019        | 2020        | 2021        |
|------------------|-------------|-------------|-------------|-------------|
| <b>Solutions</b> | 19.8        | 21.8        | 24.2        | 26.5        |
| <b>Services</b>  | 4.9         | 5.4         | 6.3         | 7.1         |
| <b>Total</b>     | <b>24.6</b> | <b>27.3</b> | <b>30.4</b> | <b>33.6</b> |

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors.

Source: Annual Reports, Investor Presentations, Third-party Databases, Vendor Interviews, Channel Partner Interviews, Independent Subject Matter Experts, Cloud Associations, Company Datasheets, Press Releases, Company Websites, Government Websites, Vendor Surveys, and MarketsandMarkets' Adjacent Market Repositories

**TABLE 22** HIGH-PERFORMANCE COMPUTING MARKET, BY COMPONENT, 2022–2027 (USD BILLION)

| Component        | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | CAGR<br>(2022–2027) |
|------------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| <b>Solutions</b> | 28.2        | 30.1        | 32.0        | 33.9        | 36.0        | 38.3        | 6.3%                |
| <b>Services</b>  | 7.8         | 8.5         | 9.1         | 9.8         | 10.6        | 11.6        | 8.2%                |
| <b>Total</b>     | <b>36.0</b> | <b>38.5</b> | <b>41.1</b> | <b>43.7</b> | <b>46.6</b> | <b>49.9</b> | <b>6.7%</b>         |

Note: .

Numbers are rounded off to the nearest digit, resulting in rounding-off errors.

Source: Annual Reports, Investor Presentations, Third-party Databases, Vendor Interviews, Channel Partner Interviews, Independent Subject Matter Experts, Cloud Associations, Company Datasheets, Press Releases, Company Websites, Government Websites, Vendor Surveys, and MarketsandMarkets' Adjacent Market Repositories

The solutions segment is expected to hold the largest market share during the forecast period. With an increasing number of datasets generated across different verticals, such as BFSI, healthcare & life sciences, and manufacturing, due to the growing adoption of digital technologies such as AI, big data, and IoT, the demand for high-speed computation solutions is rising significantly. The use of HPC solutions can fulfil this demand by processing various datasets in a short time and with extreme precision.

## 6.2 SOLUTIONS

HPC uses the concept of parallel processing, where a cluster of computers, known as supercomputers, are used to solve complex computing problems. HPC provides improved analytical capabilities and fast product development cycles and facilitates the modeling, simulation, and analyses of data, which requires a computing infrastructure. HPC solutions help in overcoming the supercomputing barriers and in finding results for complex engineering and data analysis problems. The server solution contributes almost 50% of the total HPC market revenue, and the trend is expected to continue for the next 5 years. The major vendors offering HPC solutions are IBM, HPE, Fujitsu, and Dell.

**TABLE 23 SOLUTIONS: HIGH-PERFORMANCE COMPUTING MARKET, BY TYPE, 2018–2021 (USD BILLION)**

| Solutions       | 2018        | 2019        | 2020        | 2021        |
|-----------------|-------------|-------------|-------------|-------------|
| Server          | 13.8        | 14.2        | 14.5        | 15.1        |
| Storage         | 3.1         | 4.8         | 6.0         | 7.2         |
| Network Devices | 2.9         | 2.8         | 3.7         | 4.3         |
| <b>Total</b>    | <b>19.8</b> | <b>21.8</b> | <b>24.2</b> | <b>26.5</b> |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 24 SOLUTIONS: HIGH-PERFORMANCE COMPUTING MARKET, BY TYPE, 2022–2027 (USD BILLION)**

| Type               | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | CAGR<br>(2022–2027) |
|--------------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| Server             | 15.7        | 16.6        | 17.6        | 18.6        | 19.6        | 20.8        | 5.9%                |
| Storage            | 8.0         | 8.7         | 9.4         | 10.2        | 11.0        | 12.0        | 8.4%                |
| Networking Devices | 4.6         | 4.7         | 4.9         | 5.1         | 5.3         | 5.5         | 3.6%                |
| <b>Total</b>       | <b>28.2</b> | <b>30.1</b> | <b>32.0</b> | <b>33.9</b> | <b>36.0</b> | <b>38.3</b> | <b>6.3%</b>         |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

## 6.2.1 SERVER

Delivering high-density, cost-effective, and robust servers for data center infrastructures has become essential for companies from various verticals due to the increasing connectivity traffic worldwide. Organizations require hardware that support dynamic workloads in applications such as Web 2.0 and cloud hosting. HPC servers, including supercomputer and divisional systems and departmental and workgroup systems, deliver core count, high frequency, rapid calculations, scalability, and power efficiency benefits required in these applications. They comprise various operating systems, such as Windows, Linux, and UNIX, as well as processors, memory, cables, communication boards, and system software. IBM, HPE, Dell, and Fujitsu are the major players offering HPC servers.

### 6.2.1.1 Supercomputer and divisional systems

Supercomputers have 512 nodes or above, whereas divisional systems are servers with nodes between 128 and 512. These systems fall into the price band ranging from USD 250,000 to USD 500,000. HPC systems are rapidly being deployed in the supercomputing and divisional systems environment. These systems incur huge expenses and require high technical expertise for proper functioning. Moreover, they are only designed to perform specialized and dedicated tasks. HPC systems are cost-effective and can deploy many servers to work together on a single task. They aid in accelerating tasks and provide rapid execution with the help of supercomputing and divisional systems.

### 6.2.1.2 Departmental and workgroup systems

Departmental systems are servers with a node count between 16 and 128, whereas workgroup systems are servers with 16 or fewer nodes. The costs of these systems range from USD 100,000 to USD 499,999. Departmental and workgroup systems help individuals by providing them with the computing power needed to increase productivity, computation speed, and parallel computation; assist in reducing design and development cycles; improve efficiency; and reduce time-to-solution. In addition, these systems can be

deployed in small workspaces. Workgroup systems do not require special power or cooling infrastructure. They are highly preferred for applications in the computer-aided engineering, BFSI, and healthcare & life sciences sectors.

### 6.2.2 STORAGE

Over the past few decades, there has been an exponential increase in the volumes of structured and unstructured data originating from mobile devices and cloud-native applications. Furthermore, as organizations have started adopting digital technologies such as AI and IoT, the requirement for storage solutions has surged. HPC is also used for computational analysis, information-escalated research, rich media, 3D modeling, seismic handling, data mining, and large-scale simulation, which require on-premise or cloud storage solutions. Moreover, the HPC applications are required to handle huge volumes of data within a short time and, therefore, require high capacity storage with high speed and expandability. The HPC storage delivers a faster processing speed and larger storage capacity. Considering this, the vendors build and provide storage solutions dedicated to HPC applications. The major vendors offering storage solutions include Dell, GIGA-BYTE, and HPE.

### 6.2.3 NETWORKING DEVICES

HPC infrastructure requirements differ from organization to organization. Thus, a standard networking infrastructure cannot be used. Depending upon the organization's demand, the HPC infrastructure has to be integrated with application-specific networking infrastructure to leverage the HPC capabilities. Furthermore, traditional networks are highly static and associated with high operational overheads and manual intervention. They act as a bottleneck in supporting varied HPC applications and are usually configured to handle a specific application. Thus, organizations prefer highly scalable and easily manageable network infrastructure. HPC networking devices cater to these requirements, enabling organizations to use HPC solutions optimally. HPC networking devices include ethernet switches, ethernet network interface cards, Local Area Network (LAN) interconnections, InfiniBand, adapters, and routers. Among these, Ethernet and InfiniBand are gaining significant traction in the HPC market. Major vendors offering networking devices include DDN, Dell, and Cisco.

### 6.2.4 SOFTWARE

The most productive HPC systems feature a finely tuned combination of hardware and software. HPC software consist of platform and library tools that integrate with HPC hardware to leverage the optimum capabilities of HPC systems for various applications that require high computational speeds. As HPC systems are becoming complex featuring different heterogeneous hardware components to accommodate different workloads, programmers need software development tools to ease application development and aid with targeted performance tuning and optimization. A typical HPC software suite comprises a comprehensive collection of highly tuned linear algebra subroutines designed to help extract maximum performance from the HPC system with the least amount of effort. Companies such as Hewlett Packard Enterprises, Dell, Lenovo, and Intel are market leaders developing HPC software.

## 6.3 SERVICES

HPC services are required to be highly customized to be free from product design overheads. HPC services help employees within an organization to meet their ultimate objective of improved operability and energy and resource utilization. The growing complexities in the installation and deployment of HPC systems are expected to boost the demand for HPC services. HPC services are only available for deployment via the cloud. HPC systems can be hosted over managed cloud services such as AWS (Amazon Web Services) and Microsoft Azure. By making use of HPC over the managed cloud, organizations can tap into the power of these HPC systems without requiring setting up HPC systems manually. The use of HPC hosted over the cloud follows a pay-as-you-go pricing model.

### 6.3.1 DESIGN AND CONSULTING

Design and consulting services address the business requirements and challenges related to servers, storage, networking devices, and software solutions. With the help of consulting services, organizations can get help in designing HPC systems that fit their business and technical requirements. They aid in designing business-specific solutions for HPC deployments. They also help end users in integrating these solutions into their IT environment. The consulting services aid organizations in switching from their legacy infrastructure to modern HPC systems, eliminating the risk of choosing an accurate HPC solution that does not fit their business needs. This helps end users adopt new HPC technologies quickly, with an end-to-end consultative approach for infrastructure modernization.

Moreover, these services help address challenges end users face regarding technical expertise, system management, and cost optimization. Consulting services include partial or complete system designs or blueprints, planning, evaluation analysis, building prototypes, technical evaluation, and other test platforms. These services are provided by industry experts, qualified domain experts, and professionals. Moreover, design and consulting services are crucial for all the organizations, especially for non-IT companies, as they have limited know-how regarding HPC solutions.

### 6.3.2 INTEGRATION AND DEPLOYMENT

The integration and deployment services include onsite installation of HPC solutions, such as servers, storage, networking devices, and software to meet client-specific requirements. These services majorly include functions, such as unboxing, staging, and racking of HPC solutions, as well as the support infrastructure that includes power, cooling, and network connections. Small and large organizations use integration and deployment services during the deployment of HPC solutions. Integration and deployment service providers act as a one-stop shop for the hassle-free deployment of HPC systems across client organizations, helping organizations minimize downtimes. Integration and deployment services provide advantages such as increased operational efficiency, reduced IT costs, enhanced customer experience, and risk mitigation, support, maintenance, and management.

The support, maintenance, and management services include various onsite and offsite services. Onsite services include support activities, such as device repair and replacement, and various de-installation services, such as disk wiping, and resale and recycling of devices. Remote support or offsite services include assessing the speed problems and remote troubleshooting, as well as the maintenance activities such as air conditioning, electrical wiring, fire suppression system, and physical access. The support, maintenance, and management services help end users understand the changing business conditions, client insights, market trends, and service inconveniences to help smoothen the support process within an organization.

## 7 HIGH-PERFORMANCE COMPUTING MARKET, BY COMPUTATION TYPE

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### KEY FINDINGS

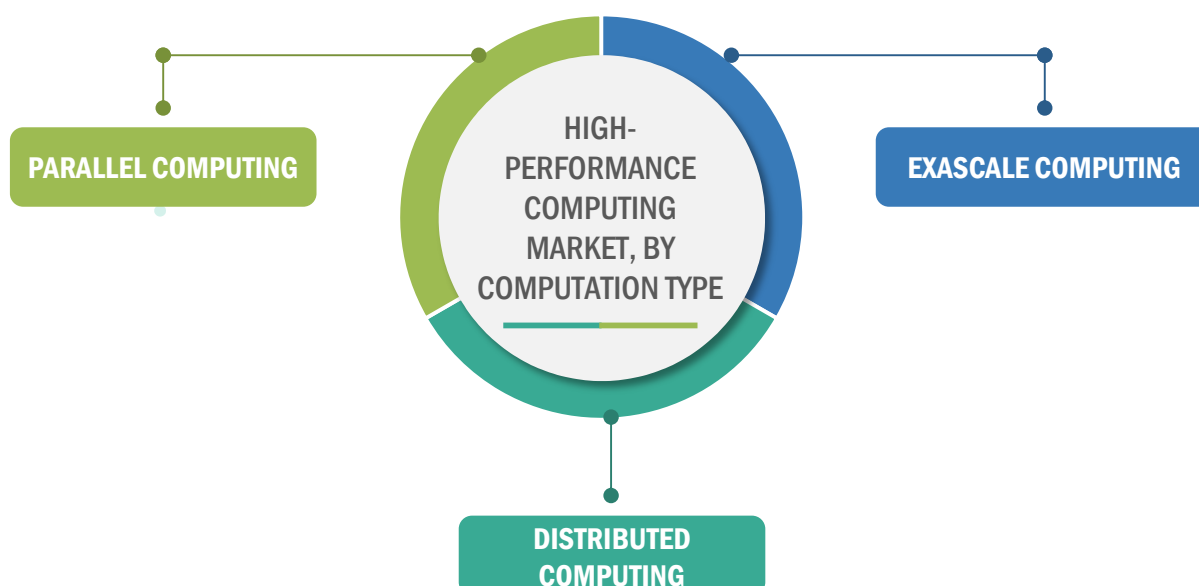
- The HPC market for the parallel computing segment is projected to reach USD 29 Billion by 2027, registering a CAGR of 7.3% during the forecast period.
- The market for the distributed computing segment is projected to reach USD 19.5 Billion by 2027, registering a CAGR of 5.5% during the forecast period.
- The market for the Exascale computing segment is projected to grow at a CAGR of 14.6% during the forecast period.

## 7.1 INTRODUCTION

HPC is one of the applications of supercomputers to solve computing problems that are either too large or time-consuming for conventional computers. HPC refers to the ability of supercomputers to do complex calculations and process large volumes of data quickly. HPC systems are used in various industries, including retail, education & research, telecommunications, and healthcare, and can be installed on-premises, at the edge, or on the cloud.

The HPC market has experienced significant growth due to various industrial and research applications that require the processing of large volumes of organizational data. HPC has shown tremendous growth potential across key industries over the last 5 years due to the rapid penetration of artificial intelligence (AI) and automation technologies, which fueled the demand for cloud services. With the increased demand for cloud services, the need for HPC has grown significantly. In this chapter, the high-speed computing market based on computation type has been segmented into parallel computing, distributed computing, and Exascale computing.

**FIGURE 30** HIGH-PERFORMANCE COMPUTING MARKET, BY COMPUTATION TYPE



Source: International Trade Administration (ITA), European Cloud Association (EUDCA), Telecommunications Industry Association, United Nations Economic and Social Commission for Asia and the Pacific (ESCAP), Economic Commission for Latin America (ECLA), Arab Information and Communication Technology Organization (AICTO), Interviews with Experts, Secondary Research, Annual Reports, Press Releases, Investor Presentations, White Papers, Journals and Magazines, and MarketsandMarkets Analysis

**TABLE 25** HIGH-PERFORMANCE COMPUTING MARKET, BY COMPUTATION TYPE, 2018–2021 (USD BILLION)

| Computation Type | 2018        | 2019        | 2020        | 2021        |
|------------------|-------------|-------------|-------------|-------------|
| Parallel         | 13.6        | 15.2        | 17.0        | 18.9        |
| Distributed      | 11.0        | 11.7        | 12.9        | 14.1        |
| Exascale         | -           | 0.4         | 0.5         | 0.6         |
| <b>Total</b>     | <b>24.6</b> | <b>27.3</b> | <b>30.4</b> | <b>33.6</b> |

Source: Vendor Interviews, Channel Partner Interviews, Independent Subject Matter Experts, Cloud Associations, Company Datasheets, Annual Reports, Investor Presentations, Third-party Databases, Press Releases, Company Websites, Government Websites, Vendor Surveys, and MarketsandMarkets' Adjacent Market Repositories

**TABLE 26** HIGH-PERFORMANCE COMPUTING MARKET, BY COMPUTATION TYPE, 2022–2027 (USD BILLION)

| Computation Type | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | CAGR<br>(2022–2027) |
|------------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| Parallel         | 20.4        | 21.9        | 23.5        | 25.3        | 27.2        | 29.0        | 7.3%                |
| Distributed      | 14.9        | 15.8        | 16.7        | 17.5        | 18.3        | 19.5        | 5.5%                |
| Exascale         | 0.7         | 0.8         | 0.9         | 1.0         | 1.1         | 1.4         | 14.6%               |
| <b>Total</b>     | <b>36.0</b> | <b>38.5</b> | <b>41.1</b> | <b>43.7</b> | <b>46.6</b> | <b>49.9</b> | <b>6.7%</b>         |

Note: e: estimated; p: projected

Source: Vendor Interviews, Channel Partner Interviews, Independent Subject Matter Experts, Cloud Associations, Company Datasheets, Annual Reports, Investor Presentations, Third-party Databases, Press Releases, Company Websites, Government Websites, Vendor Surveys, and MarketsandMarkets' Adjacent Market Repositories

## 7.2 PARALLEL COMPUTING

### 7.2.1 PARALLEL COMPUTING HELPS IN SOLVING COMPLEX MATHEMATICAL PROBLEMS

In parallel computing, multiple processing elements run simultaneously to solve complex problems. These problems are divided into smaller problems, and a dedicated computing resource is assigned to solve them. Parallel computing has gained popularity across various applications in the industrial sector. The use of other computing techniques has physical constraints that prevent frequency scaling, thereby reducing the system's capability of operating at optimal computational speeds. Parallel computing has become the dominant standard in computer architecture, as it is being increasingly used in modern multi-core processors due to its less power consumption and heat generation features in data processing.

In parallel computing, a single computer is equipped with multiple processors to perform numerous operations. The system may have a shared or distributed memory architecture that improves the system performance in applications related to computer vision, such as fault detection. Parallel computing is also being used in video and image processing, deep learning and machine learning, web services, natural language processing, autonomous driving, medicine, drug discovery, and biotechnology.

### 7.2.2 CASE STUDY: AWS AND ASTRAZENECA

In 2021, AstraZeneca (UK), one of the world's leading pharmaceutical companies, invested heavily in research of genomic studies. AstraZeneca has accelerated the use of genomics in the development of precision medicine and advanced the translation of genomics to transform drug discovery. Accordingly, the company has collaborated with Amazon Web Services to achieve its intended goal. The company uses petabytes of genomic sequencing data to conduct drug research and development. To rapidly process data at scale, AstraZeneca uses Amazon Web Services' (AWS) cloud solution to build a fast, efficient computing solution for extracting impactful genomics insights. This is done with the help of a high-performance computing cloud solution with a parallel computing infrastructure.

### 7.2.3 BIT-LEVEL PARALLELISM

A bit-level parallelism is a form of parallel computing that increases the processor's word size. A word is defined as an entity that comprises computing bits. Increasing the word size reduces the number of instructions the processor is required to execute to perform a process on variables whose sizes are more than the length of the word. For example, a processor with a 32-bit word size can perform 4 independent 1-byte additions simultaneously. If the processor size is 1 byte, it will perform 4 operations.

### 7.2.4 INSTRUCTION-LEVEL PARALLELISM

Instruction-level parallelism is an architecture in which multiple operations can be performed parallelly in a particular process with a set of resources such as identifiers, address space, registers, state, and program counters. Instruction-level parallelism refers to the compiler design techniques and processors designed to execute operations such as memory load and store, integer addition, and float multiplication. These techniques enhance the performance of processors. Examples of architectures that use instruction-level parallelism are very long instruction words and superscalar computing architectures. The execution hardware in a typical instruction-level parallelism processor is similar to that of a reduced instruction set computer (RISC) machine. The difference between a machine with instruction-level parallelism and one without is that there may be more hardware components present, such as a number of integer adders rather than just one hardware component, and that the control will allow, and possibly arrange, simultaneous access to whatever execution hardware is present.

### 7.2.5 TASK PARALLELISM

Task parallelism, also known as functional or control parallelism, uses algorithmic decomposition to break down a massive task into several subtasks, which helps in multiplying computing resources. The key advantage associated with this technique is multiple portions of the visualization task to be executed are performed in parallel mode. The main disadvantage of this technique is that the number of independent tasks that can be processed simultaneously, as well as the number of CPUs available to accomplish it, limits the maximum amount of parallelism. In addition, it can be difficult to load-balance the tasks and, therefore, can be challenging to take full advantage of the available resources. Task parallelism is used effectively in the entertainment industry, where several frames in an animated production are rendered in parallel. Specific hardware choices for improving the performance of task parallelism are dependent upon the details of the required tasks.

## 7.3 DISTRIBUTED COMPUTING

### 7.3.1 DISTRIBUTED COMPUTING HELPS IN LEVERAGING HPC RESOURCES USING LOW-VALUE COMMODITY HARDWARE

Distributed computing is also called distributed processing. A distributed computer system consists of multiple software components that are spread across various computers within the network but are operated as a single system. The computers in a distributed system can be connected by a local network or a wide area network. A distributed system can consist of many possible configurations, such as mainframes, personal computers, workstations, and minicomputers. The goal of distributed computing is to make a mesh of computer networks work as a single computer.

High-performance computing makes use of distributed computing facilities for solving problems that need high computing power. Distributed computing in HPC connects multiple computer servers over a network into a cluster to share data and coordinate data processing. Such a cluster is referred to as a “distributed system.” Distributed computing offers advantages such as scalability, performance resilience, and cost-effectiveness using low-cost, commodity hardware.

### 7.3.2 USE CASE: DELL AND WALT DISNEY ANIMATION STUDIO

Media files used in the entertainment industry are usually large and require high computing power to process. High-performance computing makes it possible to render special effects, create animations, and transcode large, complex media files rapidly. In 2020, Walt Disney Animation Studio (US) collaborated with Dell (US) to renovate its existing computing system infrastructure. Accordingly, Dell equipped Disney's studios with state-of-the-art HPC solutions. This helped the studios to animate, render special effects, and handle large media files at high speeds, significantly reducing the production process time required for making animated movies.

### 7.3.3 GRID COMPUTING

Grid computing works by running specialized software on every computer participating in the data grid. The software acts as the manager of the entire system and coordinates various tasks across the grid. Specifically, the software assigns subtasks to each computer so they can work simultaneously on their respective subtasks. After completing subtasks, the outputs are collected to complete a large-scale task. The software allows each computer to communicate over the network with the other computers to share information on what portion of the subtasks each computer is running and how to consolidate and deliver outputs.

### 7.3.4 CLUSTER COMPUTING

Cluster computing refers to various computer systems connected across a network that perform like a single entity. Each computer that is connected to the network is called a node. Cluster computing offers solutions to solve complicated problems by providing fast computational speed and enhanced data integrity. The connected computers work in tandem to execute operations, thus creating the impression of a single system (virtual machine). This process is termed transparency of the system. Based on the principle of distributed systems, this networking technology performs its operations. In cluster computing, the local area network (LAN) is the connection unit.

### 7.3.5 CLOUD COMPUTING

Cloud Computing refers to a computing infrastructure and software model that enables access to shared pools of configurable resources such as computer networks, servers, storage, applications, and services. Cloud computing allows individual users and companies to store and process data in either a privately owned cloud or on a third-party server located in a remote data center to make the data accessing

mechanism more efficient and reliable. Cloud computing relies on utility-type sharing of resources to achieve economies of scale.

Cloud HPC is a technology used to onboard HPC resources in the cloud. HPC hardware can be adjusted to the specific needs of each engineering department and can be scaled up or down according to changing needs. Workload throughput can be increased by running more jobs with a mix of batch and autoscaling clusters. Key companies offering cloud HPC are Hewlett Packard Enterprise (HPE) (US), Google (US), IBM (US), Dell (US), and Fujitsu (Japan).

#### **7.3.5.1 USE CASE: FRAUD DETECTION IN BFSI SECTOR**

Some of the major frauds in the financial industry include Ponzi schemes, misleading investment schemes, identity fraud, phishing, card fraud, counterfeiting, skimming, fake prizes, and inheritance scams. A centralized system for fraud detection can gather information from all available sources. Such systems can help in the detection and prevention of cyber-attacks, as well as increase data privacy, data security, and support digital forensic investigations. Given the massive amounts of data, the system in place must be able to detect anomalies and intrusion attempts. Despite their enormous capacities, legacy data centers are incapable of handling such large volumes of data. In this scenario, high-performance computing is useful in modern fraud detection and risk modeling, allowing financial institutions to detect attacks on their systems and prevent financial fraud. Recently, PayPal (US) implemented an HPC environment, and within a year of its adoption, the company was able to save USD 710 million from cyber-attacks.

### **7.4 EXASCALE COMPUTING**

#### **7.4.1 EXASCALE COMPUTING HELPS IN MAKING NEW SCIENTIFIC DISCOVERIES BY PROCESSING LARGE AMOUNTS OF DATA IN SHORT TIME**

The advent of Exascale computing is one of the major opportunities that will maximize the benefits of HPC. Exascale computing refers to computing systems that are at least 50 times faster than the most powerful supercomputers in use today. Exascale computing can provide advanced modeling and simulation solutions that address the key challenges in scientific discovery, energy assurance, economic competitiveness, and national security. The Exascale Computing Project is a collaborative effort of two US Department of Energy (DOE) organizations—the Office of Science (DOE SC) and National Nuclear Security Administration (NNSA). With the increasing volumes of data, HPC systems are demanding both greater memory capacity and high-performance bandwidth to operate efficiently. Exascale computing addresses both these challenges.

#### **7.4.2 USE CASE: VACCINE DEVELOPMENT**

During the vaccine development phase of the COVID-19 pandemic, the availability of a vast amount of COVID-19 material posed a considerable challenge for researchers. There were millions of known proteins to model against the critical COVID spike, over 30 terabytes of accumulated medical data to process, millions of publications that could be key to discovery, and more than 150 billion facts of medical knowledge available for analysis. Researchers could not analyze this vast amount of data within a short period and come up with a viable plan for COVID-19 vaccine development. Exascale computing gave these researchers breakthrough capabilities by helping them do their work faster. By allowing researchers to process data in two minutes, which would usually take 2 hours even with the help of a supercomputer, HPC helped immensely in fast-tracking the development of COVID-19 vaccines.

## 8 HIGH-PERFORMANCE COMPUTING MARKET, BY DEPLOYMENT

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### KEY FINDINGS

- The deployment of HPC solutions is rising in large organizations from the manufacturing, BFSI, and energy & utility sectors as they require high computing power in on-premise infrastructures.
- The HPC market for the cloud segment is expected to grow at a CAGR of 8.1 % during the forecast period.
- The market for the on-premises segment is expected to reach USD 41.1 billion by 2027, registering a CAGR of 6.4% during the forecast period.

## 8.1 INTRODUCTION

The HPC market based on deployment type has been segmented into on-premises and cloud. HPC systems are being increasingly deployed in various enterprises and data centers worldwide as they have started adopting cloud technology on a large scale. Cloud deployment enables organizations to leverage the benefits of the pay-as-you-go pricing model. As cloud operations require low latency and bandwidth for computing, there is a tremendous scope for the use of HPC systems in enterprises and data centers that have already started using cloud solutions.

**FIGURE 31** HIGH-PERFORMANCE COMPUTING MARKET, BY DEPLOYMENT



Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

### 8.1.1 DRIVERS: HIGH-PERFORMANCE COMPUTING MARKET FOR DEPLOYMENT TYPE

- Access to specialized hardware such as GPUs, InfiniBand, Large Memory, and new processor architectures is expected to drive the demand for cloud HPC.
- The growing need to avoid capacity utilization risks while making optimal use of existing cloud capacity is expected to drive the demand for cloud HPC solutions.
- The adoption of on-premises HPC is driven by users who want to control their HPC solutions completely and do not send any data into the data lake.

### 8.1.2 COVID-19 IMPACT

COVID-19 forced many organizations across the globe to implement Work from Home (WFH) policies to run businesses. This increased the adoption of cloud infrastructure and cloud deployment of HPC systems.

**TABLE 27** HIGH-PERFORMANCE COMPUTING MARKET, BY DEPLOYMENT, 2018–2021 (USD BILLION)

| Deployment   | 2018        | 2019        | 2020        | 2021        |
|--------------|-------------|-------------|-------------|-------------|
| On-premises  | 20.8        | 23.0        | 25.5        | 28.1        |
| Cloud        | 3.8         | 4.3         | 4.9         | 5.5         |
| <b>Total</b> | <b>24.6</b> | <b>27.3</b> | <b>30.4</b> | <b>33.6</b> |

Source: Vendor Interviews, Channel Partner Interviews, Independent Subject Matter Experts, Cloud Associations, Company Datasheets, Annual Reports, Investor Presentations, Third-party Databases, Press Releases, Company Websites, Government Websites, Vendor Surveys, and MarketsandMarkets' Adjacent Market Repositories

**TABLE 28** HIGH-PERFORMANCE COMPUTING MARKET, BY DEPLOYMENT, 2022–2027 (USD BILLION)

| Deployment Type | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | CAGR<br>(2022–2027) |
|-----------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| On-premises     | 30.1        | 32.1        | 34.1        | 36.2        | 38.5        | 41.1        | 6.4%                |
| Cloud           | 6.0         | 6.5         | 7.0         | 7.5         | 8.1         | 8.8         | 8.1%                |
| <b>Total</b>    | <b>36.0</b> | <b>38.5</b> | <b>41.1</b> | <b>43.7</b> | <b>46.6</b> | <b>49.9</b> | <b>6.7%</b>         |

Note: e: estimated; p: projected

Source: Vendor Interviews, Channel Partner Interviews, Independent Subject Matter Experts, Cloud Associations, Company Datasheets, Annual Reports, Investor Presentations, Third-party Databases, Press Releases, Company Websites, Government Websites, Vendor Surveys, and MarketsandMarkets' Adjacent Market Repositories

The cloud deployment segment is expected to grow at a higher growth rate during the forecast period, owing to the benefits associated with cloud deployment of HPC systems, such as low maintenance costs, high performance, and power optimization.

## 8.2 CLOUD

HPC is no longer limited to supercomputer users. The on-demand solutions are also significantly gaining traction due to their cost-effectiveness. Cloud deployment minimizes the organization's operational costs by eliminating the need to buy additional computing resources. Moreover, it helps small enterprises lower their operational costs as they do not have to develop HPC infrastructure. Major players that offer cloud-based HPC solutions include Amazon, IBM, and Microsoft.

**TABLE 29** CLOUD: HIGH-PERFORMANCE COMPUTING MARKET, BY VERTICAL, 2018–2021 (USD BILLION)

| Vertical                   | 2018       | 2019       | 2020       | 2021       |
|----------------------------|------------|------------|------------|------------|
| BFSI                       | 0.76       | 0.85       | 0.95       | 1.05       |
| Earth Science              | 0.45       | 0.51       | 0.58       | 0.66       |
| Education & Research       | 0.48       | 0.53       | 0.60       | 0.67       |
| Energy & Utilities         | 0.71       | 0.79       | 0.89       | 1.00       |
| Government & Defense       | 0.26       | 0.29       | 0.33       | 0.38       |
| Healthcare & Life sciences | 0.45       | 0.52       | 0.60       | 0.68       |
| Manufacturing              | 0.19       | 0.21       | 0.24       | 0.26       |
| Media & Entertainment      | 0.18       | 0.20       | 0.23       | 0.25       |
| Other                      | 0.37       | 0.42       | 0.47       | 0.53       |
| <b>Total</b>               | <b>3.8</b> | <b>4.3</b> | <b>4.9</b> | <b>5.5</b> |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 30** CLOUD: HIGH-PERFORMANCE COMPUTING MARKET, BY VERTICAL, 2022–2027 (USD BILLION)

| Vertical                   | 2022       | 2023       | 2024       | 2025       | 2026       | 2027       | CAGR<br>(2022–2027) |
|----------------------------|------------|------------|------------|------------|------------|------------|---------------------|
| BFSI                       | 1.12       | 1.20       | 1.28       | 1.37       | 1.46       | 1.56       | 6.7%                |
| Earth Science              | 0.73       | 0.80       | 0.87       | 0.95       | 1.04       | 1.14       | 9.3%                |
| Education & Research       | 0.72       | 0.77       | 0.83       | 0.89       | 0.95       | 1.02       | 7.3%                |
| Energy & Utilities         | 1.08       | 1.16       | 1.25       | 1.34       | 1.44       | 1.56       | 7.6%                |
| Government & Defense       | 0.41       | 0.45       | 0.49       | 0.53       | 0.58       | 0.64       | 9.1%                |
| Healthcare & Life sciences | 0.75       | 0.83       | 0.91       | 1.00       | 1.10       | 1.21       | 9.9%                |
| Manufacturing              | 0.28       | 0.31       | 0.33       | 0.35       | 0.38       | 0.41       | 7.6%                |
| Media & Entertainment      | 0.28       | 0.30       | 0.33       | 0.35       | 0.38       | 0.42       | 8.5%                |
| Other                      | 0.58       | 0.62       | 0.68       | 0.73       | 0.79       | 0.85       | 8.1%                |
| <b>Total</b>               | <b>6.0</b> | <b>6.5</b> | <b>7.0</b> | <b>7.5</b> | <b>8.1</b> | <b>8.8</b> | <b>8.1%</b>         |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

### 8.3 ON-PREMISES

The on-premises deployment consists of various dedicated computing nodes that are attached to their parent nodes and further connected to workstations. This deployment type supports specialized hardware, such as InfiniBand and Graphics Processing Unit (GPU). However, the emergence of HPC in the cloud has affected the growth of the on-premises deployment type segment. Yet, organizations are deploying HPC solutions using the on-premises deployment type because of the security and confidentiality factors. Many vendors are making continued investments in the on-premises deployment type, as well as developing new cloud business models for HPC. Major vendors offering on-premise HPC solutions include HPE, IBM, and Dell.

**TABLE 31** ON-PREMISES: HIGH-PERFORMANCE COMPUTING MARKET, BY VERTICAL, 2018–2021 (USD BILLION)

| Vertical                   | 2018        | 2019        | 2020        | 2021        |
|----------------------------|-------------|-------------|-------------|-------------|
| BFSI                       | 4.6         | 5.0         | 5.5         | 6.0         |
| Earth Science              | 2.3         | 2.6         | 2.9         | 3.3         |
| Education & Research       | 3.0         | 3.3         | 3.7         | 4.0         |
| Energy & Utilities         | 4.0         | 4.4         | 4.9         | 5.3         |
| Government & Defense       | 1.2         | 1.4         | 1.6         | 1.7         |
| Healthcare & Life sciences | 1.9         | 2.1         | 2.4         | 2.7         |
| Manufacturing              | 1.1         | 1.2         | 1.3         | 1.5         |
| Media & Entertainment      | 0.9         | 1.0         | 1.1         | 1.3         |
| Other                      | 1.7         | 1.8         | 2.1         | 2.3         |
| <b>Total</b>               | <b>20.8</b> | <b>23.0</b> | <b>25.5</b> | <b>28.1</b> |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 32** ON-PREMISES: HIGH-PERFORMANCE COMPUTING MARKET, BY REGION, 2022–2027 (USD BILLION)

| Vertical                   | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | CAGR<br>(2022–2027) |
|----------------------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| <b>BFSI</b>                | 6.4         | 6.7         | 7.1         | 7.4         | 7.8         | 8.2         | 5.2%                |
| Earth Science              | 3.6         | 3.9         | 4.2         | 4.5         | 4.8         | 5.2         | 8.0%                |
| Education & Research       | 4.3         | 4.5         | 4.8         | 5.0         | 5.3         | 5.6         | 5.7%                |
| Energy & Utilities         | 5.6         | 5.9         | 6.2         | 6.6         | 6.9         | 7.3         | 5.3%                |
| Government & Defense       | 1.9         | 2.0         | 2.2         | 2.4         | 2.5         | 2.8         | 7.9%                |
| Healthcare & Life sciences | 2.9         | 3.2         | 3.4         | 3.7         | 4.0         | 4.3         | 8.1%                |
| Manufacturing              | 1.6         | 1.7         | 1.8         | 2.0         | 2.1         | 2.2         | 6.9%                |
| Media & Entertainment      | 1.4         | 1.5         | 1.6         | 1.7         | 1.8         | 1.9         | 7.2%                |
| Other                      | 2.5         | 2.6         | 2.8         | 3.0         | 3.2         | 3.5         | 7.1%                |
| <b>Total</b>               | <b>30.1</b> | <b>32.1</b> | <b>34.1</b> | <b>36.2</b> | <b>38.5</b> | <b>41.1</b> | <b>6.4%</b>         |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

## 9 HIGH-PERFORMANCE COMPUTING MARKET, BY ORGANIZATION SIZE

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### KEY FINDINGS

- The HPC market for the large enterprises segment is expected to hold a higher share (7.4%) of the overall market in 2027 due to the growing adoption of cloud technology.
- The market for the SMEs segment is projected to grow at a higher CAGR of 6.5% during the
- The growth of the SMEs segment can be attributed to the ease of using HPC on the cloud, which helps reduce the operational expenses for SMEs.

## 9.1 INTRODUCTION

The HPC market based on organization size has been segmented into SMEs and large enterprises. Currently, large enterprises are adopting HPC solutions at a significant rate, and the trend is expected to continue during the forecast period. SMEs have started adopting cloud-based HPC solutions due to their cost-effectiveness, high performance, and scalability.

**FIGURE 32** HIGH-PERFORMANCE COMPUTING MARKET, BY ORGANIZATION SIZE



Source: Industry Experts and MarketsandMarkets Analysis

### 9.1.1 DRIVERS: HIGH-PERFORMANCE COMPUTING MARKET FOR ORGANIZATION SIZE

- The increasing focus of governments and global vendors on digital transformation and business expansions are expected to be the major driving factors for the growth of the HPC market.
- The growing need to reduce the expenses related to deploying HPC systems among SMEs is expected to increase the adoption of the cloud during the forecast period.

### 9.1.2 COVID-19 IMPACT

- Countries across the globe announced several financial packages to restore their economy from the adverse effects of the COVID-19 pandemic. For instance, the US passed a Senate vote to roll out a USD 2 trillion rescue package, whereas Italy, one of the highly impacted countries by the pandemic crisis, came up with an emergency plan of USD 28 billion that helped it move slowly toward normalcy.
- Businesses across the globe faced numerous challenges to stay up and running to sustain in the market. SMEs were adversely affected by this pandemic situation.

**TABLE 33** HIGH-PERFORMANCE COMPUTING MARKET, BY ORGANIZATION SIZE, 2018–2021 (USD BILLION)

| Organization Size            | 2018        | 2019        | 2020        | 2021        |
|------------------------------|-------------|-------------|-------------|-------------|
| Small and Medium Enterprises | 18.6        | 20.5        | 22.8        | 25.2        |
| Large Enterprises            | 6.1         | 6.8         | 7.6         | 8.4         |
| <b>Total</b>                 | <b>24.6</b> | <b>27.3</b> | <b>30.4</b> | <b>33.6</b> |

Note: Numbers are rounded off to the nearest integer, resulting in rounding off errors

Source: Annual Reports, Investor Presentations, Third-party Databases, Vendor Interviews, Channel Partner Interviews, Independent Subject Matter Experts, SMEs Associations, Company Datasheets, Press Releases, Company Websites, Government Websites, Vendor Surveys, and MarketsandMarkets' Adjacent Market Repositories

**TABLE 34** HIGH-PERFORMANCE COMPUTING MARKET, BY ORGANIZATION SIZE, 2022–2027 (USD BILLION)

| Organization Size | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | CAGR<br>(2022–2027) |
|-------------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| SMEs              | 26.91       | 28.71       | 30.56       | 32.45       | 34.52       | 36.85       | 6.5%                |
| Large Enterprises | 9.11        | 9.80        | 10.52       | 11.26       | 12.08       | 13.01       | 7.4%                |
| <b>Total</b>      | <b>36.0</b> | <b>38.5</b> | <b>41.1</b> | <b>43.7</b> | <b>46.6</b> | <b>49.9</b> | <b>6.7%</b>         |

Numbers are rounded off to the nearest integer, resulting in rounding off errors.

Source: Vendor Interviews, Channel Partner Interviews, Independent Subject Matter Experts, SMEs Associations, Company Datasheets, Annual Reports, Investor Presentations, Third-party Databases, Press Releases, Company Websites, Government Websites, Vendor Surveys, and MarketsandMarkets' Adjacent Market Repositories

Large enterprises have high financial resources, which they can invest in setting up modern HPC infrastructure; however, SMEs have limited financial capabilities. Hence, the large enterprises segment is expected to hold a higher share of the overall HPC market.

## 9.2 SMALL- AND MEDIUM-SIZED ENTERPRISES

Organizations having an employee base of fewer than 1,000 people can be termed SMEs. Desktop computers are inefficient in fulfilling the computational and storage demand for high-end applications in SMEs. Thus, SMEs are shifting toward solutions that can cater to their high-computational demands. SMEs face three critical challenges: capital, skills, and scalability. To tackle these challenges, SMEs are adopting cloud-based HPC solutions, which offer the flexibility to scale their IT infrastructure as per their requirement. SMEs face intense competition from large enterprises; thus, to gain a competitive edge, they are increasingly deploying HPC solutions that aid them in increasing productivity, and the trend is expected to continue during the forecast period.

## 9.3 LARGE ENTERPRISES

Organizations with an employee base of more than 1,000 people are considered large enterprises. As the adoption of new digital technologies is increasing, many large enterprises have replaced their traditional data center infrastructure technologies and other management processes with HPC systems and solutions. Large enterprises are usually characterized by their requirement for high-server densities and high computational power. These organizations require highly reliable infrastructure. HPC solutions can facilitate data center providers with fast processing abilities and deliver quick results, increasing their profitability. Moreover, HPC solutions aid large enterprises in processing numerous applications simultaneously within a short time, enabling a significant reduction in downtime.

## 10 HIGH-PERFORMANCE COMPUTING MARKET, BY SERVER PRICE BAND

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### KEY FINDINGS

- The HPC market for the USD 250,000–500,000 and above segment is projected to reach USD 35.6 Billion by 2027, registering a CAGR of 6.9% during the forecast period.
- The market for the USD 250,000–100,000 and below segment is projected to reach USD 14.3 Billion by 2027, registering a CAGR of 6.3% during the forecast period.
- HPC systems available in the USD 250,000–100,000 and below server price band are generally used by small enterprises with limited budgets, whereas large enterprises prefer HPC systems available in the USD 250,000–500,000 and above server price band.

## 10.1 INTRODUCTION

The HPC market based on server price band has been segmented into USD 250,000–500,000 and above and USD 250,000–100,000 and below. These server price bands, depending upon the user's requirement, include servers of different groups, such as supercomputers, divisional, departmental, and workgroup, which are used to process both large and moderate volumes of data.

**FIGURE 33** HIGH-PERFORMANCE COMPUTING MARKET, BY SERVER PRICE BAND



Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 35** HIGH-PERFORMANCE COMPUTING MARKET, BY SERVER PRICE BAND, 2018–2021 (USD BILLION)

| Server Price Band               | 2018        | 2019        | 2020        | 2021        |
|---------------------------------|-------------|-------------|-------------|-------------|
| USD 250,000 – 500,000 and above | 17.3        | 19.2        | 21.5        | 23.8        |
| USD 250,000 – 100,000 and below | 7.3         | 8.1         | 8.9         | 9.8         |
| <b>Total</b>                    | <b>24.6</b> | <b>27.3</b> | <b>30.4</b> | <b>33.6</b> |

Source: Vendor Interviews, Channel Partner Interviews, Independent Subject Matter Experts, Cloud Associations, Company Datasheets, Annual Reports, Investor Presentations, Third-party Databases, Press Releases, Company Websites, Government Websites, Vendor Surveys, and MarketsandMarkets' Adjacent Market Repositories

**TABLE 36** HIGH-PERFORMANCE COMPUTING MARKET, BY SERVER PRICE BAND, 2022–2027 (USD BILLION)

| Server Price Band             | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | CAGR<br>(2022–2027) |
|-------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| USD 250,000–500,000 and Above | 25.5        | 27.3        | 29.2        | 31.1        | 33.2        | 35.6        | 6.9%                |
| USD 250,000–100,000 and Below | 10.5        | 11.2        | 11.9        | 12.6        | 13.4        | 14.3        | 6.3%                |
| <b>Total</b>                  | <b>36.0</b> | <b>38.5</b> | <b>41.1</b> | <b>43.7</b> | <b>46.6</b> | <b>49.9</b> | <b>6.7%</b>         |

Source: Annual Reports, Investor Presentations, Third-party Databases, Vendor Interviews, Channel Partner Interviews, Independent Subject Matter Experts, Cloud Associations, Company Datasheets, Press Releases, Company Websites, Government Websites, Vendor Surveys, and MarketsandMarkets' Adjacent Market Repositories

The USD 250,000–100,000 and below segment is expected to grow at a higher CAGR of 3.7% during the forecast period, owing to the increasing demand for cost-effective solutions exhibiting high performance from SMEs.

## **10.2 USD 250,000–500,000 AND ABOVE**

### **10.2.1 USD 250,000–500,000 AND ABOVE SERVER PRICE BAND COMPRISES SYSTEMS THAT HELP IN SOLVING HIGHLY COMPLEX MATHEMATICAL PROBLEMS**

The USD 250,000–500,000 and above server price band comprises supercomputer and divisional systems groups. These systems are developed based on state-of-the-art HPC infrastructures to solve highly complex problems in the shortest time. They comprise many connected computing nodes with distributed memory, which are controlled by a single master unit. These servers operate through 2 types of operating systems, i.e., Linux and Windows.

## **10.3 USD 250,000–100,000 AND BELOW**

### **10.3.1 USD 250,000–100,000 AND BELOW SERVER PRICE BAND COMPRISE SYSTEMS THAT CAN SOLVE MODERATELY COMPLEX MATHEMATICAL PROBLEMS**

The USD 250,000–100,000 and below price band includes departmental and workgroup systems. Departmental systems are servers that cost around USD 100,000–249,999, whereas workgroup systems cost under USD 100,000. These servers are primarily used by SMEs to solve moderate or small problems at a low cost. The servers are compatible with traditional architecture and technologies; however, they can be upgraded by adding new accelerators and processors and enhancing the processing speed to Petaflops/second (Pf/s).

## 11 HIGH-PERFORMANCE COMPUTING MARKET, BY VERTICAL

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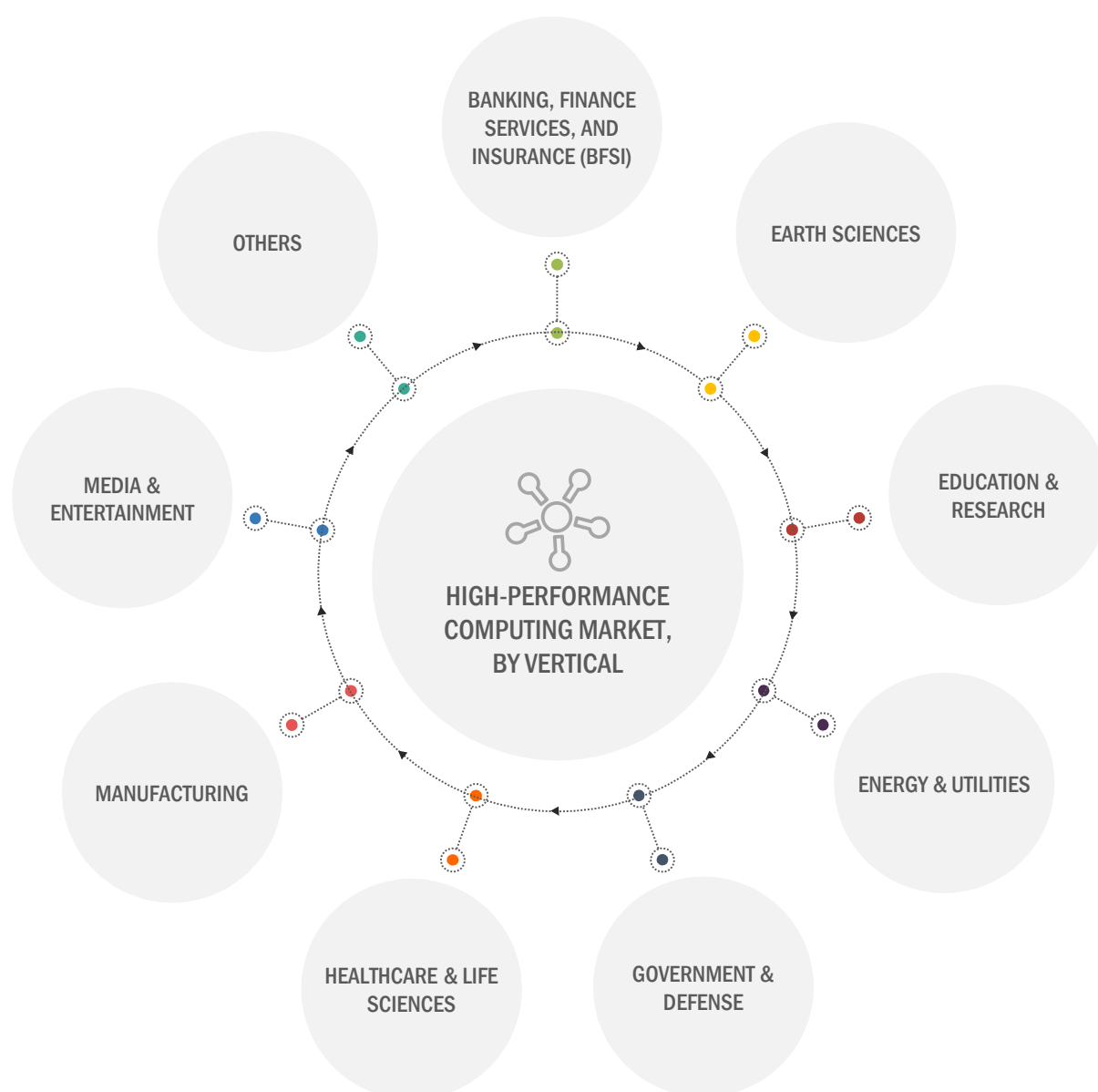
### KEY FINDINGS

- The HPC market for the banking, financial services, and insurance vertical is expected to hold the largest share of the overall market in 2022 due to the rising demand for HPC solutions for banking, financial services, and insurance analytics applications.
- The HPC market for the healthcare & life sciences vertical is projected to reach USD 5.6 Billion by 2027, at the highest CAGR of 8.5% during the forecast period.
- The market for the BFSI and media & entertainment verticals are projected to grow at the CAGRs of 5.5% and 7.4%, respectively, as the demand is shifting toward advanced infrastructure for storing and processing data.
- The growing infusion of modern technologies such as AI, IoT, and big data analytics is increasing across manufacturing facilities. Hence, there is a growing demand for HPC solutions in the manufacturing sector.
- The rapid penetration of AI and machine learning frameworks in the healthcare industry is fueling the demand for HPC solutions.

## 11.1 INTRODUCTION

The HPC market based on vertical has been segmented into government & defense, BFSI, education & research, manufacturing, media & entertainment, healthcare & life sciences, energy & utilities, earth sciences, and others. HPC systems and software are required to solve complex problems owing to the enormous quantity of data produced by modern analysis methods, including high-output sequencing. These systems and software include supercomputing clusters and machine learning (ML) frameworks that use GPU acceleration to achieve high computational speeds. The use of HPC requires access to high data storage since the processing and analysis of data need to be completed within specific time frames. For instance, in the context of clinical research and industrial process monitoring, the use of GPU acceleration, distributed computing, and cutting-edge ML tools helps in speeding up analysis execution times, enabling quick processing of large data sets.

HPC makes use of supercomputers. It uses moderate computing power, higher than a desktop's computing power. Different architectural system frameworks are specially designed to support the operations of the HPC system. These frameworks are algorithmically designed to take advantage of high-performance architecture in verticals such as BFSI, owing to the increasing demand for high-frequency transactions and rising government regulations in areas such as core banking, insurance, and other financial services. Moreover, HPC solutions are being widely adopted for education and research purposes to facilitate research requirements in structural analysis, chemistry, astrophysics, and nuclear physics. HPC solutions have been increasingly adopted in the healthcare & life sciences vertical to cater to the growing requirement for computing power generated due to the high influx of patient data processing during the pandemic period.

**FIGURE 34** HPC MARKET, BY VERTICAL

Note: Other verticals include retail, transportation, travel & hospitality, and real estate.

Source: Interviews with Experts, Secondary Research, Annual Reports, Press Releases, Investor Presentations, White Papers, Journals and Magazines, and MarketsandMarkets Analysis

### 11.1.1 DRIVERS: HIGH-PERFORMANCE COMPUTING MARKET FOR VERTICAL

- In the media & entertainment vertical, large media files of different audio and video formats require fast processing capabilities to provide a quick and seamless experience for end users. This can be achieved with the help of HPC solutions.
- The adoption of AI and HPC is rising in the BFSI vertical to analyze daily financial activities in areas such as risk management and fraud detection.
- In scientific research, HPC enables the development of complex simulations to analyze vast volumes of data in a short time.

- The growing digitalization in government and public agencies, rising demand for reliable data storage, and increasing need to meet regulatory, security, and compliance requirements are expected to fuel the adoption of HPC solutions in the government & defense vertical.
- The need to process large volumes of customer banking data in a short time is fueling the adoption of HPC solutions in the BFSI sector.
- The growing use of IoT in manufacturing plants is leading to the generation of high data volumes that require low latency, which is fueling the demand for HPC systems in the manufacturing vertical.

### 11.1.2 COVID-19 IMPACT

- Different verticals have witnessed different impacts of the COVID-19 crisis. Verticals, where cloud and multi-cloud technologies are used with HPC systems, were marginally affected. These areas require on-site deployments and have depicted relatively slow adoption of HPC solutions. The rise in demand for high bandwidth by customers and employees working remotely across verticals such as government & defense, BFSI, and manufacturing has surged the need for HPC systems.
- Vendors worldwide are enhancing their product offerings to tackle the challenges imposed due to the COVID-19 crisis.

**TABLE 37 HIGH-PERFORMANCE COMPUTING MARKET, BY VERTICAL, 2018-2021 (USD BILLION)**

| Vertical                   | 2018        | 2019        | 2020        | 2021        |
|----------------------------|-------------|-------------|-------------|-------------|
| BFSI                       | 5.4         | 5.9         | 6.5         | 7.1         |
| Earth Science              | 2.8         | 3.1         | 3.5         | 3.9         |
| Education & Research       | 3.5         | 3.9         | 4.3         | 4.7         |
| Energy & Utilities         | 4.8         | 5.2         | 5.8         | 6.3         |
| Government & Defense       | 1.5         | 1.7         | 1.9         | 2.1         |
| Healthcare & Life sciences | 2.4         | 2.7         | 3.0         | 3.4         |
| Manufacturing              | 1.3         | 1.4         | 1.6         | 1.8         |
| Media & Entertainment      | 1.1         | 1.2         | 1.4         | 1.5         |
| Others                     | 2.0         | 2.3         | 2.5         | 2.8         |
| <b>Total</b>               | <b>24.6</b> | <b>27.3</b> | <b>30.4</b> | <b>33.6</b> |

Note: . Other verticals include retail, transportation, travel & hospitality, and real estate.

Source: Interviews with Experts, Secondary Research, Annual Reports, Press Releases, Investor Presentations, White Papers, Journals and Magazines, and MarketsandMarkets Analysis

**TABLE 38** HIGH-PERFORMANCE COMPUTING MARKET, BY VERTICAL, 2022–2027 (USD BILLION)

| Vertical                   | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | CAGR<br>(2022–2027) |
|----------------------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| <b>BFSI</b>                | 7.5         | 7.9         | 8.3         | 8.8         | 9.2         | 9.8         | 5.5%                |
| Earth Science              | 4.3         | 4.7         | 5.0         | 5.4         | 5.9         | 6.4         | 8.2%                |
| Education & Research       | 5.0         | 5.3         | 5.6         | 5.9         | 6.2         | 6.6         | 5.9%                |
| Energy & Utilities         | 6.7         | 7.1         | 7.5         | 7.9         | 8.3         | 8.8         | 5.7%                |
| Government & Defense       | 2.3         | 2.5         | 2.7         | 2.9         | 3.1         | 3.4         | 8.1%                |
| Healthcare & Life sciences | 3.7         | 4.0         | 4.4         | 4.7         | 5.1         | 5.6         | 8.5%                |
| Manufacturing              | 1.9         | 2.0         | 2.2         | 2.3         | 2.5         | 2.6         | 7.0%                |
| Media & Entertainment      | 1.6         | 1.8         | 1.9         | 2.0         | 2.2         | 2.3         | 7.4%                |
| Others                     | 3.0         | 3.3         | 3.5         | 3.8         | 4.0         | 4.3         | 7.3%                |
| <b>Total</b>               | <b>36.0</b> | <b>38.5</b> | <b>41.1</b> | <b>43.7</b> | <b>46.6</b> | <b>49.9</b> | <b>6.7%</b>         |

Note: Other verticals include retail, transportation, travel & hospitality, and real estate.

Source: Interviews with Experts, Secondary Research, Annual Reports, Press Releases, Investor Presentations, White Papers, Journals and Magazines, and MarketsandMarkets Analysis

The BFSI and healthcare and life sciences verticals are experiencing increased demand for HPC solutions for analyzing data and finding insights to overcome business problems in a short period. HPC helps to find treatments for diseases by processing large volumes of patient datasets within a short time. The companies in the government & defense vertical invest a high amount in deploying HPC solutions for applications such as weather forecasting, investigations, and storage.

## 11.2 GOVERNMENT & DEFENSE

The adoption of cutting-edge IT infrastructure solutions is growing significantly in the government & defense vertical to enhance computing efficiency. Citizens have requirements pertaining to seamless delivery and faster response times while availing the services related to government initiatives. Thus, the capabilities of HPC solutions can be leveraged to provide citizens with a seamless experience in availing of various government initiatives in a short period owing to their high-speed processing capabilities. The defense sector is also increasingly deploying HPC solutions. For instance, in September 2021, Penguin Computing was awarded two contracts totaling USD 68 million from the US Department of Defense. The US Department of Defense launched the High-performance Computing Modernization Program to use Penguin Computing's True HPC supercomputing platform in the defense sector. HPC solutions can help address defense-related requirements, including real-time data acquisition and analysis, emergency operations preparation, disaster modeling and analysis simulations, surveillance, and encryption/decryption of confidential information. HPC systems are used to run large computations for structural analysis, fluid dynamics, and computer-assisted engineering and design applications used extensively in the civil and military aviation sectors.

**TABLE 39** GOVERNMENT & DEFENSE: HIGH-PERFORMANCE COMPUTING MARKET, BY REGION, 2018–2021 (USD BILLION)

| Region        | 2018        | 2019        | 2020        | 2021        |
|---------------|-------------|-------------|-------------|-------------|
| North America | 0.661       | 0.746       | 0.847       | 0.954       |
| Europe        | 0.392       | 0.436       | 0.488       | 0.541       |
| APAC          | 0.380       | 0.429       | 0.487       | 0.548       |
| RoW           | 0.050       | 0.056       | 0.062       | 0.069       |
| <b>Total</b>  | <b>1.48</b> | <b>1.67</b> | <b>1.88</b> | <b>2.11</b> |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 40** GOVERNMENT & DEFENSE: HIGH-PERFORMANCE COMPUTING MARKET, BY REGION, 2022–2027 (USD BILLION)

| Region        | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | CAGR<br>(2022–2027) |
|---------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| North America | 1.042       | 1.134       | 1.232       | 1.334       | 1.447       | 1.576       | 8.6%                |
| Europe        | 0.582       | 0.623       | 0.667       | 0.711       | 0.759       | 0.813       | 6.9%                |
| APAC          | 0.598       | 0.650       | 0.705       | 0.763       | 0.827       | 0.900       | 8.5%                |
| RoW           | 0.074       | 0.079       | 0.084       | 0.089       | 0.095       | 0.101       | 6.5%                |
| <b>Total</b>  | <b>2.29</b> | <b>2.49</b> | <b>2.69</b> | <b>2.90</b> | <b>3.13</b> | <b>3.39</b> | <b>8.1%</b>         |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 41** GOVERNMENT & DEFENSE: HIGH-PERFORMANCE COMPUTING MARKET, BY DEPLOYMENT, 2018–2021 (USD BILLION)

| Deployment   | 2018       | 2019       | 2020       | 2021       |
|--------------|------------|------------|------------|------------|
| On-premises  | 1.2        | 1.4        | 1.6        | 1.7        |
| Cloud        | 0.3        | 0.3        | 0.3        | 0.4        |
| <b>Total</b> | <b>1.5</b> | <b>1.7</b> | <b>1.9</b> | <b>2.1</b> |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 42** GOVERNMENT & DEFENSE: HIGH-PERFORMANCE COMPUTING MARKET, BY DEPLOYMENT, 2022–2027 (USD BILLION)

| Deployment   | 2022       | 2023       | 2024       | 2025       | 2026       | 2027       | CAGR<br>(2022–2027) |
|--------------|------------|------------|------------|------------|------------|------------|---------------------|
| On-premises  | 1.9        | 2.0        | 2.2        | 2.4        | 2.5        | 2.8        | 7.9%                |
| Cloud        | 0.4        | 0.4        | 0.5        | 0.5        | 0.6        | 0.6        | 9.1%                |
| <b>Total</b> | <b>2.3</b> | <b>2.5</b> | <b>2.7</b> | <b>2.9</b> | <b>3.1</b> | <b>3.4</b> | <b>8.1%</b>         |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

### 11.3 BANKING, FINANCIAL SERVICES, AND INSURANCE (BFSI)

The BFSI sector comprises organizations that offer banking services such as core banking, corporate, retail, investment banking, hedge funds, as well as financial services (payment gateways, stockbroking, and mutual funds) and insurance services covering both life and general risks. Financial institutes such as insurance companies and banks require innovative IT solutions to manage operational risks, achieve sustainable growth, and deliver optimal performance. Moreover, the increase in high-frequency trading has resulted in innovations in connectivity, data access, and computational power systems used in these organizations. For instance, in February 2022, SambaNova launched a new HPC-based offering called SambaNova GPT Banking for performing sentiment analysis, language translation, and entity recognition in banking services. Regulatory standards related to the financial industry have further driven the need for installing robust HPC solutions for meeting various compliance needs. Thus, organizations are increasingly installing robust, reliable, and secured HPC systems capable of delivering high-end computing outputs.

**TABLE 43** BANKING, FINANCIAL SERVICES, AND INSURANCE: HIGH-PERFORMANCE COMPUTING MARKET, BY REGION, 2018–2021 (USD BILLION)

| Region        | 2018        | 2019        | 2020        | 2021        |
|---------------|-------------|-------------|-------------|-------------|
| North America | 2.32        | 2.55        | 2.82        | 3.09        |
| Europe        | 1.70        | 1.84        | 2.01        | 2.17        |
| APAC          | 1.14        | 1.26        | 1.41        | 1.55        |
| RoW           | 0.20        | 0.22        | 0.24        | 0.26        |
| <b>Total</b>  | <b>5.36</b> | <b>5.87</b> | <b>6.47</b> | <b>7.07</b> |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 44** BANKING, FINANCIAL SERVICES, AND INSURANCE: HIGH-PERFORMANCE COMPUTING MARKET, BY REGION, 2022–2027 (USD BILLION)

| Region        | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | CAGR<br>(2022–2027) |
|---------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| North America | 3.29        | 3.49        | 3.69        | 3.90        | 4.12        | 4.37        | 5.9%                |
| Europe        | 2.27        | 2.37        | 2.47        | 2.57        | 2.67        | 2.79        | 4.2%                |
| APAC          | 1.66        | 1.77        | 1.88        | 1.99        | 2.11        | 2.25        | 6.4%                |
| RoW           | 0.28        | 0.29        | 0.31        | 0.32        | 0.34        | 0.36        | 5.3%                |
| <b>Total</b>  | <b>7.49</b> | <b>7.92</b> | <b>8.35</b> | <b>8.78</b> | <b>9.25</b> | <b>9.77</b> | <b>5.5%</b>         |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 45** BANKING, FINANCIAL SERVICES, AND INSURANCE: HIGH-PERFORMANCE COMPUTING MARKET, BY DEPLOYMENT, 2018–2021 (USD BILLION)

| Deployment   | 2018       | 2019       | 2020       | 2021       |
|--------------|------------|------------|------------|------------|
| On-premises  | 4.6        | 5.0        | 5.5        | 6.0        |
| Cloud        | 0.8        | 0.8        | 0.9        | 1.0        |
| <b>Total</b> | <b>5.4</b> | <b>5.9</b> | <b>6.5</b> | <b>7.1</b> |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 46** BANKING, FINANCIAL SERVICES, AND INSURANCE: HIGH-PERFORMANCE COMPUTING MARKET, BY DEPLOYMENT, 2022–2027 (USD BILLION)

| Deployment   | 2022       | 2023       | 2024       | 2025       | 2026       | 2027       | CAGR<br>(2022–2027) |
|--------------|------------|------------|------------|------------|------------|------------|---------------------|
| On-premises  | 6.370      | 6.716      | 7.066      | 7.413      | 7.791      | 8.217      | 5.2%                |
| Cloud        | 1.124      | 1.202      | 1.284      | 1.366      | 1.456      | 1.557      | 6.7%                |
| <b>Total</b> | <b>7.5</b> | <b>7.9</b> | <b>8.3</b> | <b>8.8</b> | <b>9.2</b> | <b>9.8</b> | <b>5.5%</b>         |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

## 11.4 EDUCATION & RESEARCH

Increasing computational power has paved the way to execute high-end research projects, which was previously impossible in various core research areas that require high computational power. HPC solutions can process large volumes of research data for applications such as structural analysis, computational biology, chemistry, genetics and medicine, electromagnetism, astrophysics and nuclear physics, climate change and meteorology, geophysics statistics, and mathematical modeling. Moreover, with the capabilities of HPC solutions, researchers can effectively carry out research activities related to DNA modeling, human genome mapping, and the creation of deep neural networks. HPC vendors are highly focused on delivering customized HPC solutions for various research applications. These purpose-built solutions are easily deployable and equipped with common configuration and management tools that can adapt to the increasing workloads quickly.

### 11.4.1 RECENT DEVELOPMENTS

In April 2022, the US Department of Energy (DOE) announced an investment of USD 20 million in basic research to explore potentially high-impact approaches in extreme-scale science and scientific computing. Several academic and research institutes are adopting HPC solutions to deliver computational efficiency across various research stages.

**TABLE 47** EDUCATION & RESEARCH: HIGH-PERFORMANCE COMPUTING MARKET, BY REGION, 2018–2021 (USD BILLION)

| Region        | 2018        | 2019        | 2020        | 2021        |
|---------------|-------------|-------------|-------------|-------------|
| North America | 1.42        | 1.57        | 1.75        | 1.93        |
| Europe        | 1.03        | 1.13        | 1.26        | 1.38        |
| APAC          | 0.89        | 0.97        | 1.07        | 1.16        |
| RoW           | 0.17        | 0.18        | 0.20        | 0.21        |
| <b>Total</b>  | <b>3.51</b> | <b>3.86</b> | <b>4.27</b> | <b>4.68</b> |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 48** EDUCATION & RESEARCH: HIGH-PERFORMANCE COMPUTING MARKET, BY REGION, 2022–2027 (USD BILLION)

| Region        | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | CAGR<br>(2022–2027) |
|---------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| North America | 2.06        | 2.20        | 2.34        | 2.48        | 2.64        | 2.82        | 6.4%                |
| Europe        | 1.48        | 1.58        | 1.67        | 1.78        | 1.88        | 2.01        | 6.3%                |
| APAC          | 1.23        | 1.29        | 1.36        | 1.42        | 1.49        | 1.57        | 5.1%                |
| RoW           | 0.21        | 0.22        | 0.22        | 0.22        | 0.23        | 0.23        | 1.7%                |
| <b>Total</b>  | <b>4.98</b> | <b>5.28</b> | <b>5.59</b> | <b>5.90</b> | <b>6.24</b> | <b>6.63</b> | <b>5.9%</b>         |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 49** EDUCATION & RESEARCH: HIGH-PERFORMANCE COMPUTING MARKET, BY DEPLOYMENT, 2018–2021 (USD BILLION)

| Deployment   | 2018       | 2019       | 2020       | 2021       |
|--------------|------------|------------|------------|------------|
| On-premises  | 3.0        | 3.3        | 3.7        | 4.0        |
| Cloud        | 0.5        | 0.5        | 0.6        | 0.7        |
| <b>Total</b> | <b>3.5</b> | <b>3.9</b> | <b>4.3</b> | <b>4.7</b> |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 50** EDUCATION & RESEARCH: HIGH-PERFORMANCE COMPUTING MARKET, BY DEPLOYMENT, 2022–2027 (USD BILLION)

| Deployment   | 2022       | 2023       | 2024       | 2025       | 2026       | 2027       | CAGR<br>(2022–2027) |
|--------------|------------|------------|------------|------------|------------|------------|---------------------|
| On-premises  | 4.3        | 4.5        | 4.8        | 5.0        | 5.3        | 5.6        | 5.7%                |
| Cloud        | 0.7        | 0.8        | 0.8        | 0.9        | 1.0        | 1.0        | 7.3%                |
| <b>Total</b> | <b>5.0</b> | <b>5.3</b> | <b>5.6</b> | <b>5.9</b> | <b>6.2</b> | <b>6.6</b> | <b>5.9%</b>         |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

## 11.5 MANUFACTURING

The manufacturing industry faces challenges in implementing innovative technologies, resolving operational issues, and managing supply chains. HPC helps manufacturers at each phase of product development and supply chain, from running innovative design simulations to automating processes and predicting maintenance issues. Moreover, manufacturing companies need to reduce their development cost and the time to market their products to compete effectively. Therefore, manufacturing industries are using computer-aided engineering (CAE) and simulation to develop enhanced products at reduced costs. These design and simulation systems require high computational power to run multiple product simulations and derive outputs.

Moreover, as these simulations are computation-intensive, they need high execution time. Thus, the industry players are benefitting from the use of HPC for applications such as computational structural mechanics, computational fluid dynamics, and computational electromagnetics. Various companies in the manufacturing industry are actively looking for HPC solutions for improving computing speed, efficiency, and overall performance by enabling faster access to data. HPC vendors also provide energy-efficient, modular HPC solutions that are flexible in terms of configuration and capable of addressing a wide range of problems.

**TABLE 51** MANUFACTURING: HIGH-PERFORMANCE COMPUTING MARKET, BY REGION, 2018–2021 (USD BILLION)

| Region        | 2018        | 2019        | 2020        | 2021        |
|---------------|-------------|-------------|-------------|-------------|
| North America | 0.603       | 0.671       | 0.752       | 0.835       |
| Europe        | 0.333       | 0.367       | 0.406       | 0.446       |
| APAC          | 0.292       | 0.327       | 0.368       | 0.410       |
| RoW           | 0.047       | 0.052       | 0.057       | 0.062       |
| <b>Total</b>  | <b>1.28</b> | <b>1.42</b> | <b>1.58</b> | <b>1.75</b> |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 52** MANUFACTURING: HIGH-PERFORMANCE COMPUTING MARKET, BY REGION, 2022–2027 (USD BILLION)

| Region        | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | CAGR<br>(2022–2027) |
|---------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| North America | 0.900       | 0.968       | 1.038       | 1.111       | 1.190       | 1.281       | 7.3%                |
| Europe        | 0.474       | 0.504       | 0.534       | 0.564       | 0.597       | 0.634       | 6.0%                |
| APAC          | 0.444       | 0.479       | 0.516       | 0.554       | 0.596       | 0.644       | 7.7%                |
| RoW           | 0.065       | 0.069       | 0.072       | 0.076       | 0.079       | 0.083       | 5.0%                |
| <b>Total</b>  | <b>1.88</b> | <b>2.02</b> | <b>2.16</b> | <b>2.30</b> | <b>2.46</b> | <b>2.64</b> | <b>7.0%</b>         |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 53** MANUFACTURING: HIGH-PERFORMANCE COMPUTING MARKET, BY DEPLOYMENT, 2018–2021 (USD BILLION)

| Deployment   | 2018       | 2019       | 2020       | 2021       |
|--------------|------------|------------|------------|------------|
| On-premises  | 1.1        | 1.2        | 1.3        | 1.5        |
| Cloud        | 0.2        | 0.2        | 0.2        | 0.3        |
| <b>Total</b> | <b>1.3</b> | <b>1.4</b> | <b>1.6</b> | <b>1.8</b> |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 54** MANUFACTURING: HIGH-PERFORMANCE COMPUTING MARKET, BY DEPLOYMENT, 2022–2027 (USD BILLION)

| Deployment   | 2022       | 2023       | 2024       | 2025       | 2026       | 2027       | CAGR<br>(2022–2027) |
|--------------|------------|------------|------------|------------|------------|------------|---------------------|
| On-premises  | 1.600      | 1.713      | 1.830      | 1.950      | 2.082      | 2.231      | 6.9%                |
| Cloud        | 0.284      | 0.306      | 0.330      | 0.354      | 0.380      | 0.410      | 7.6%                |
| <b>Total</b> | <b>1.9</b> | <b>2.0</b> | <b>2.2</b> | <b>2.3</b> | <b>2.5</b> | <b>2.6</b> | <b>7.0%</b>         |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

## 11.6 MEDIA & ENTERTAINMENT

Companies from the media & entertainment vertical have started adopting HPC to develop and deliver their products in new and innovative ways and expand their presence. HPC solutions help them collect, store, and use a large volume of data to get better insights into the psyche of their customers. Software used by media and entertainment companies for 3D modeling and animation, web and graphic design, VFX and gaming, and filmmaking require devices with powerful processors with compatible RAM and ROM. Applications such as video rendering and high-resolution image editing require very high computing capabilities. HPC solutions have helped improve communication, content creation, distribution, and consumption across all available media and entertainment platforms.

Moreover, the media and entertainment companies require advanced IT capabilities for applications such as live streaming, virtual reality, and drone-based video capture. HPC has proved to be the best solution to manage the increasing IT workloads and constantly changing landscape of the media & entertainment industry. HPC solutions can help media and entertainment companies by providing the right set of solutions and services to optimize resources, manage productivity, track consumer behavior, and reduce production costs.

**TABLE 55** MEDIA & ENTERTAINMENT: HIGH-PERFORMANCE COMPUTING MARKET, BY REGION, 2018–2021 (USD BILLION)

| Region        | 2018        | 2019        | 2020        | 2021        |
|---------------|-------------|-------------|-------------|-------------|
| North America | 0.494       | 0.553       | 0.623       | 0.696       |
| Europe        | 0.321       | 0.359       | 0.404       | 0.451       |
| APAC          | 0.233       | 0.257       | 0.284       | 0.312       |
| RoW           | 0.042       | 0.046       | 0.051       | 0.056       |
| <b>Total</b>  | <b>1.09</b> | <b>1.22</b> | <b>1.36</b> | <b>1.52</b> |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 56** MEDIA AND ENTERTAINMENT: HIGH-PERFORMANCE COMPUTING MARKET, BY REGION, 2022–2027 (USD BILLION)

| Region        | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | CAGR(2022–2027) |
|---------------|-------------|-------------|-------------|-------------|-------------|-------------|-----------------|
| North America | 0.754       | 0.816       | 0.880       | 0.947       | 1.021       | 1.104       | 7.9%            |
| Europe        | 0.488       | 0.528       | 0.568       | 0.611       | 0.658       | 0.711       | 7.8%            |
| APAC          | 0.333       | 0.354       | 0.375       | 0.396       | 0.420       | 0.446       | 6.0%            |
| RoW           | 0.059       | 0.063       | 0.066       | 0.070       | 0.074       | 0.078       | 5.6%            |
| <b>Total</b>  | <b>1.63</b> | <b>1.76</b> | <b>1.89</b> | <b>2.02</b> | <b>2.17</b> | <b>2.34</b> | <b>7.4%</b>     |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 57** MEDIA & ENTERTAINMENT: HIGH-PERFORMANCE COMPUTING MARKET, BY DEPLOYMENT, 2018–2021 (USD BILLION)

| Deployment   | 2018        | 2019        | 2020        | 2021        |
|--------------|-------------|-------------|-------------|-------------|
| On-premises  | 0.91        | 1.02        | 1.14        | 1.26        |
| Cloud        | 0.18        | 0.20        | 0.23        | 0.25        |
| <b>Total</b> | <b>1.09</b> | <b>1.22</b> | <b>1.36</b> | <b>1.52</b> |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 58** MEDIA AND ENTERTAINMENT: HIGH-PERFORMANCE COMPUTING MARKET, BY DEPLOYMENT, 2022–2027 (USD BILLION)

| Deployment   | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | CAGR (2022–2027) |
|--------------|-------------|-------------|-------------|-------------|-------------|-------------|------------------|
| On-premises  | 1.36        | 1.46        | 1.56        | 1.67        | 1.79        | 1.92        | 7.2%             |
| Cloud        | 0.28        | 0.30        | 0.33        | 0.35        | 0.38        | 0.42        | 8.5%             |
| <b>Total</b> | <b>1.63</b> | <b>1.76</b> | <b>1.89</b> | <b>2.02</b> | <b>2.17</b> | <b>2.34</b> | <b>7.4%</b>      |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

## 11.7 HEALTHCARE & LIFE SCIENCES

Advancements in healthcare systems, healthcare research, and communication technologies have led to the development of better standard practices within the healthcare & life sciences vertical. With rising technological innovations, healthcare institutes face the need to deliver various healthcare services through active collaboration between hospital administrators and doctors while improving the overall efficiency and quality of patient care. Life sciences deal with the use of informatics to understand and unveil previously unknown facts about living beings. HPC is being implemented widely in various applications within the healthcare & life sciences vertical for running multiple simulations, data management and modeling processes, and next-generation sequencing. HPC solutions provide the necessary computational power for these applications and help tackle challenges pertaining to the high computational power requirements of the healthcare & life sciences vertical. Therefore, HPC vendors provide state-of-the-art solutions to perform accelerated simulations and data analytics. Healthcare organizations, pharmaceutical companies, universities, and biotech companies are widely deploying HPC solutions to improve the effectiveness of clinical treatments and stay competitive within the market. For instance, In May 2020, teams from the University of Tennessee and Oak Ridge National Laboratory (ORNL) published a paper on Repurposing Therapeutics for COVID-19 using HPC for processing research data. Furthermore, the University of Texas at Austin is also leveraging the capabilities of HPC for cancer treatment. Working on this project, researchers from the University of Texas scanned petabytes of data to determine the relationship between a cancer patient's genome and the composition of their tumors.

**TABLE 59** HEALTHCARE & LIFE SCIENCES: HIGH-PERFORMANCE COMPUTING MARKET, BY REGION, 2018–2021 (USD BILLION)

| Region        | 2018        | 2019        | 2020        | 2021        |
|---------------|-------------|-------------|-------------|-------------|
| North America | 1.095       | 1.241       | 1.416       | 1.600       |
| Europe        | 0.703       | 0.796       | 0.906       | 1.022       |
| APAC          | 0.493       | 0.549       | 0.613       | 0.680       |
| RoW           | 0.067       | 0.074       | 0.081       | 0.089       |
| <b>Total</b>  | <b>2.36</b> | <b>2.66</b> | <b>3.02</b> | <b>3.39</b> |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 60** HEALTHCARE & LIFE SCIENCES: HIGH-PERFORMANCE COMPUTING MARKET, BY REGION, 2022–2027 (USD BILLION)

| Region        | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | CAGR<br>(2022–2027) |
|---------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| North America | 1.754       | 1.917       | 2.090       | 2.272       | 2.473       | 2.702       | 9.0%                |
| Europe        | 1.118       | 1.220       | 1.328       | 1.441       | 1.566       | 1.708       | 8.8%                |
| APAC          | 0.730       | 0.782       | 0.836       | 0.890       | 0.949       | 1.015       | 6.8%                |
| RoW           | 0.095       | 0.101       | 0.107       | 0.112       | 0.118       | 0.125       | 5.6%                |
| <b>Total</b>  | <b>3.70</b> | <b>4.02</b> | <b>4.36</b> | <b>4.71</b> | <b>5.11</b> | <b>5.55</b> | <b>8.5%</b>         |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 61** HEALTHCARE & LIFE SCIENCES: HIGH-PERFORMANCE COMPUTING MARKET, BY DEPLOYMENT, 2018–2021 (USD BILLION)

| Deployment   | 2018       | 2019       | 2020       | 2021       |
|--------------|------------|------------|------------|------------|
| On-premises  | 1.9        | 2.1        | 2.4        | 2.7        |
| Cloud        | 0.5        | 0.5        | 0.6        | 0.7        |
| <b>Total</b> | <b>2.4</b> | <b>2.7</b> | <b>3.0</b> | <b>3.4</b> |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 62** HEALTHCARE & LIFE SCIENCES: HIGH-PERFORMANCE COMPUTING MARKET, BY DEPLOYMENT, 2022–2027 (USD BILLION)

| Deployment   | 2022       | 2023       | 2024       | 2025       | 2026       | 2027       | CAGR<br>(2022–2027) |
|--------------|------------|------------|------------|------------|------------|------------|---------------------|
| On-premises  | 2.946      | 3.191      | 3.449      | 3.717      | 4.011      | 4.345      | 8.1%                |
| Cloud        | 0.752      | 0.829      | 0.911      | 0.998      | 1.095      | 1.206      | 9.9%                |
| <b>Total</b> | <b>3.7</b> | <b>4.0</b> | <b>4.4</b> | <b>4.7</b> | <b>5.1</b> | <b>5.6</b> | <b>8.5%</b>         |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

## 11.8 ENERGY & UTILITIES

The energy & utilities vertical is witnessing steady growth in the demand for HPC solutions. In the energy & utilities sector, the oil & gas industry has witnessed high adoption of HPC solutions due to the growing need for accurate reservoir identification to reduce the cost of operation and mitigate drilling risks. HPC solutions facilitate efficient acquisition and processing of seismic data, basin modeling, and reservoir modeling, thereby providing the required computational power, reliability, storage efficiency, and power optimization. Oil and gas companies are adopting HPC solutions for advanced seismic imaging to explore and develop new energy sources. For instance, in 2019, Eni S.p.A (Italy) installed HPC5, one of the world's green and most powerful HPC infrastructures, utilizing over 1800 Dell EMC PowerEdge servers that have helped the company increase oil and gas exploration and production.

HPC vendors such as Dell, Intel, Lenovo, and Inspur provide customized solutions with maximum operational performance and scalability. Vendors also offer one-stop supercomputers that can be deployed in compact areas such as boats and operate effectively in harsh environments.

**TABLE 63** ENERGY & UTILITIES: HIGH-PERFORMANCE COMPUTING MARKET, BY REGION, 2018–2021 (USD BILLION)

| Region        | 2018        | 2019        | 2020        | 2021        |
|---------------|-------------|-------------|-------------|-------------|
| North America | 2.01        | 2.21        | 2.46        | 2.70        |
| Europe        | 1.34        | 1.46        | 1.60        | 1.74        |
| APAC          | 1.07        | 1.18        | 1.32        | 1.45        |
| RoW           | 0.34        | 0.36        | 0.39        | 0.42        |
| <b>Total</b>  | <b>4.75</b> | <b>5.22</b> | <b>5.76</b> | <b>6.31</b> |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 64** ENERGY & UTILITIES: HIGH-PERFORMANCE COMPUTING MARKET, BY REGION, 2022–2027 (USD BILLION)

| Region        | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | CAGR<br>(2022–2027) |
|---------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| North America | 2.88        | 3.07        | 3.26        | 3.45        | 3.66        | 3.89        | 6.2%                |
| Europe        | 1.83        | 1.92        | 2.01        | 2.10        | 2.20        | 2.31        | 4.8%                |
| APAC          | 1.56        | 1.66        | 1.77        | 1.88        | 2.01        | 2.14        | 6.6%                |
| RoW           | 0.43        | 0.45        | 0.46        | 0.47        | 0.48        | 0.50        | 2.9%                |
| <b>Total</b>  | <b>6.70</b> | <b>7.10</b> | <b>7.50</b> | <b>7.90</b> | <b>8.34</b> | <b>8.84</b> | <b>5.7%</b>         |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 65** ENERGY & UTILITIES: HIGH-PERFORMANCE COMPUTING MARKET, BY DEPLOYMENT, 2018–2021 (USD BILLION)

| Deployment   | 2018       | 2019       | 2020       | 2021       |
|--------------|------------|------------|------------|------------|
| On-premises  | 4.0        | 4.4        | 4.9        | 5.3        |
| Cloud        | 0.7        | 0.8        | 0.9        | 1.0        |
| <b>Total</b> | <b>4.8</b> | <b>5.2</b> | <b>5.8</b> | <b>6.3</b> |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 66** ENERGY & UTILITIES: HIGH-PERFORMANCE COMPUTING MARKET, BY DEPLOYMENT, 2022–2027 (USD BILLION)

| Deployment   | 2022       | 2023       | 2024       | 2025       | 2026       | 2027       | CAGR<br>(2022–2027) |
|--------------|------------|------------|------------|------------|------------|------------|---------------------|
| On-premises  | 5.6        | 5.9        | 6.2        | 6.6        | 6.9        | 7.3        | 5.3%                |
| Cloud        | 1.1        | 1.2        | 1.3        | 1.3        | 1.4        | 1.6        | 7.6%                |
| <b>Total</b> | <b>6.7</b> | <b>7.1</b> | <b>7.5</b> | <b>7.9</b> | <b>8.3</b> | <b>8.8</b> | <b>5.7%</b>         |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

## 11.9 EARTH SCIENCES

The earth sciences vertical faces challenges in building complex climate models, predicting seismic events, accurate weather forecasting, simulating climatic changes, and disaster forecasting. HPC solutions can help deliver faster results while improving the spatial accuracy of climatic models. HPC solutions provide the necessary power and speed, improving the accuracy and timeliness of outputs. HPC vendors provide solutions that offer high computational power and sustain the energy requirements and deliver accurate results. Vendors also provide ruggedized HPC systems capable of providing high computational power, radar operation support, video monitoring, and complex data management capabilities for various outdoor applications in the earth sciences vertical.

**TABLE 67** EARTH SCIENCES: HIGH-PERFORMANCE COMPUTING MARKET, BY REGION, 2018–2021 (USD BILLION)

| Region        | 2018        | 2019        | 2020        | 2021        |
|---------------|-------------|-------------|-------------|-------------|
| North America | 1.17        | 1.30        | 1.46        | 1.63        |
| Europe        | 0.95        | 1.08        | 1.23        | 1.39        |
| APAC          | 0.50        | 0.57        | 0.64        | 0.72        |
| RoW           | 0.14        | 0.16        | 0.18        | 0.20        |
| <b>Total</b>  | <b>2.77</b> | <b>3.11</b> | <b>3.52</b> | <b>3.94</b> |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 68** EARTH SCIENCES: HIGH-PERFORMANCE COMPUTING MARKET, BY REGION, 2022–2027 (USD BILLION)

| Region        | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | CAGR<br>(2022–2027) |
|---------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| North America | 1.76        | 1.89        | 2.03        | 2.18        | 2.34        | 2.51        | 7.4%                |
| Europe        | 1.53        | 1.67        | 1.83        | 1.99        | 2.17        | 2.38        | 9.2%                |
| APAC          | 0.79        | 0.86        | 0.93        | 1.00        | 1.09        | 1.18        | 8.4%                |
| RoW           | 0.21        | 0.22        | 0.24        | 0.25        | 0.27        | 0.29        | 6.4%                |
| <b>Total</b>  | <b>4.29</b> | <b>4.65</b> | <b>5.03</b> | <b>5.43</b> | <b>5.87</b> | <b>6.36</b> | <b>8.2%</b>         |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 69** EARTH SCIENCES: HIGH-PERFORMANCE COMPUTING MARKET, BY DEPLOYMENT, 2018–2021 (USD BILLION)

| Deployment   | 2018       | 2019       | 2020       | 2021       |
|--------------|------------|------------|------------|------------|
| On-premises  | 2.3        | 2.6        | 2.9        | 3.3        |
| Cloud        | 0.4        | 0.5        | 0.6        | 0.7        |
| <b>Total</b> | <b>2.8</b> | <b>3.1</b> | <b>3.5</b> | <b>3.9</b> |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 70** EARTH SCIENCES: HIGH-PERFORMANCE COMPUTING MARKET, BY DEPLOYMENT, 2022–2027 (USD BILLION)

| Deployment   | 2022       | 2023       | 2024       | 2025       | 2026       | 2027       | CAGR<br>(2022–2027) |
|--------------|------------|------------|------------|------------|------------|------------|---------------------|
| On-premises  | 3.6        | 3.9        | 4.2        | 4.5        | 4.8        | 5.2        | 8.0%                |
| Cloud        | 0.7        | 0.8        | 0.9        | 1.0        | 1.0        | 1.1        | 9.3%                |
| <b>Total</b> | <b>4.3</b> | <b>4.7</b> | <b>5.0</b> | <b>5.4</b> | <b>5.9</b> | <b>6.4</b> | <b>8.2%</b>         |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

## 11.10 OTHERS

The scope of the others segment includes retail, transportation, travel & hospitality, and real estate. The usage of HPC in the retail sector facilitates grocery stores to efficiently track inventory, optimize warehouse operations, manage orders, and track deals. The travel & hospitality sector is steadily adopting HPC solutions to increase operational efficiency, enhance guest experiences, and reduce costs. Travel and hospitality companies are also deploying HPC solutions to optimize internal IT operations and synchronize them with global networks. The players in the real estate sector are installing HPC solutions to facilitate smooth delivery, drive revenues, and reduce operational costs of several real estate development projects. Moreover, the real estate companies in the construction business are installing HPC solutions at high-end workstations and implementing standard server systems to meet different project requirements

**TABLE 71** OTHERS: HIGH-PERFORMANCE COMPUTING MARKET, BY REGION, 2018–2021 (USD BILLION)

| Region        | 2018        | 2019        | 2020        | 2021        |
|---------------|-------------|-------------|-------------|-------------|
| North America | 0.896       | 1.002       | 1.126       | 1.256       |
| Europe        | 0.577       | 0.640       | 0.714       | 0.790       |
| APAC          | 0.494       | 0.554       | 0.624       | 0.698       |
| RoW           | 0.063       | 0.069       | 0.076       | 0.082       |
| <b>Total</b>  | <b>2.03</b> | <b>2.26</b> | <b>2.54</b> | <b>2.83</b> |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 72** OTHERS: HIGH-PERFORMANCE COMPUTING MARKET, BY REGION, 2022–2027 (USD BILLION)

| Region        | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | CAGR<br>(2022–2027) |
|---------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| North America | 1.358       | 1.460       | 1.571       | 1.685       | 1.810       | 1.952       | 7.5%                |
| Europe        | 0.847       | 0.904       | 0.965       | 1.027       | 1.094       | 1.170       | 6.7%                |
| APAC          | 0.756       | 0.815       | 0.880       | 0.946       | 1.018       | 1.101       | 7.8%                |
| RoW           | 0.087       | 0.091       | 0.095       | 0.100       | 0.104       | 0.109       | 4.7%                |
| <b>Total</b>  | <b>3.05</b> | <b>3.27</b> | <b>3.51</b> | <b>3.76</b> | <b>4.03</b> | <b>4.33</b> | <b>7.3%</b>         |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 73** OTHERS: HIGH-PERFORMANCE COMPUTING MARKET, BY DEPLOYMENT, 2018–2021 (USD BILLION)

| Deployment   | 2018       | 2019       | 2020       | 2021       |
|--------------|------------|------------|------------|------------|
| On-premises  | 1.7        | 1.8        | 2.1        | 2.3        |
| Cloud        | 0.4        | 0.4        | 0.5        | 0.5        |
| <b>Total</b> | <b>2.0</b> | <b>2.3</b> | <b>2.5</b> | <b>2.8</b> |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 74** OTHERS: HIGH-PERFORMANCE COMPUTING MARKET, BY DEPLOYMENT, 2022–2027 (USD BILLION)

| Deployment   | 2022       | 2023       | 2024       | 2025       | 2026       | 2027       | CAGR<br>(2022–2027) |
|--------------|------------|------------|------------|------------|------------|------------|---------------------|
| On-premises  | 2.5        | 2.6        | 2.8        | 3.0        | 3.2        | 3.5        | 7.1%                |
| Cloud        | 0.6        | 0.6        | 0.7        | 0.7        | 0.8        | 0.9        | 8.1%                |
| <b>Total</b> | <b>3.0</b> | <b>3.3</b> | <b>3.5</b> | <b>3.8</b> | <b>4.0</b> | <b>4.3</b> | <b>7.3%</b>         |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

## 12 HIGH-PERFORMANCE COMPUTING MARKET, BY REGION

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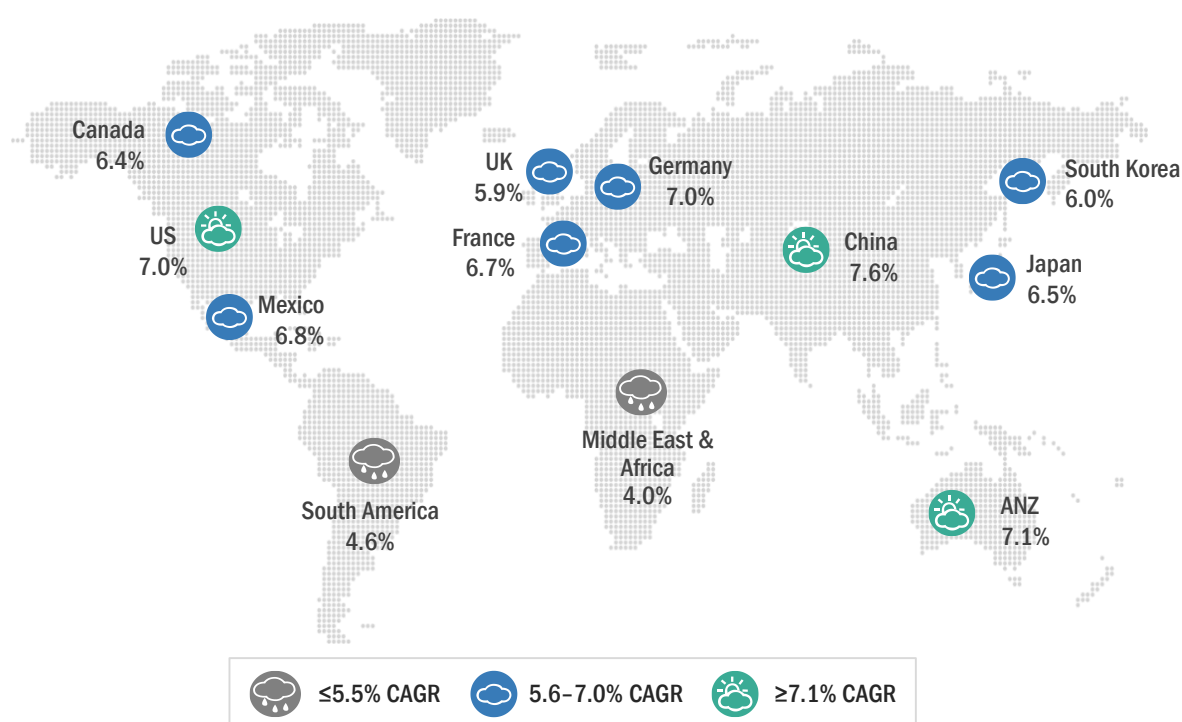
### KEY FINDINGS

- North America held the largest share (43.8%) of the global market in 2022. It is projected to reach USD 22.2 billion by 2027, registering a CAGR of 7.0% during the forecast period.
- The primary factors for the growth of the North American market are growing awareness regarding the use of technology and the presence of a large number of providers of HPC solutions.
- The HPC market in Europe is expected to grow steadily during the forecast period due to the increasing demand for HPC systems in the earth sciences sector for applications such as climate change simulation modeling and weather forecasting analysis.
- The increasing number of data centers and IT enterprises in high-growth markets such as China and India is expected to fuel the adoption of HPC solutions.
- The HPC market in South America will be driven by tourism industry as high computational speeds are required to handle guest data in large scale hotels.

## 12.1 INTRODUCTION

The HPC market based on region has been segmented into North America, Europe, APAC, and Rest of World (RoW). North America is expected to hold the largest market share (~43%) during the forecast period due to the early adoption of digital technologies such as AI, IoT, and big data across various key verticals, such as BFSI, manufacturing, and government & defense. The US government is funding various defense research projects to enhance its border defense infrastructure due to increasing illegal border crossing cases across the southern states of Texas and New Mexico. HPC solutions can process the data related to all illegal infiltrations in a short time, optimizing the immigration process. European countries are focusing on reducing carbon footprint and funding several projects to develop simulations of climate change patterns that may occur in the future. These simulations can be carried out with the help of HPC solutions, boosting their demand. Over the last decade, most of the countries in Asia Pacific have ramped up the development of their IT infrastructure and become a hub of data centers in the world. Data centers handle a large volume of connectivity traffic and require high computational power to ensure smooth operations, where HPC solutions play a vital role.

**FIGURE 35** HIGH-PERFORMANCE COMPUTING MARKET IN CHINA TO REGISTER HIGHEST CAGR DURING FORECAST PERIOD



Source: Vendor Interviews, Channel Partner Interviews, Independent Subject Matter Experts, Cloud Associations, Company Datasheets, Annual Reports, Investor Presentations, Third-party Databases, Press Releases, Company Websites, Government Websites, Vendor Surveys, and MarketsandMarkets' Adjacent Market Repositories

**TABLE 75** HIGH-PERFORMANCE COMPUTING MARKET, BY REGION, 2018–2021 (USD BILLION)

| Region        | 2018        | 2019        | 2020        | 2021        |
|---------------|-------------|-------------|-------------|-------------|
| North America | 10.7        | 11.9        | 13.3        | 14.7        |
| Europe        | 7.3         | 8.1         | 9.0         | 9.9         |
| APAC          | 5.5         | 6.1         | 6.8         | 7.5         |
| RoW           | 1.1         | 1.2         | 1.3         | 1.4         |
| <b>Total</b>  | <b>24.6</b> | <b>27.3</b> | <b>30.4</b> | <b>33.6</b> |

Source: Vendor Interviews, Channel Partner Interviews, Independent Subject Matter Experts, Cloud Associations, Company Datasheets, Annual Reports, Investor Presentations, Third-party Databases, Press Releases, Company Websites, Government Websites, Vendor Surveys, and MarketsandMarkets' Adjacent Market Repositories

**TABLE 76** HIGH-PERFORMANCE COMPUTING MARKET, BY REGION, 2022–2027 (USD BILLION)

| Region        | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | CAGR<br>(2022–2027) |
|---------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| North America | 15.8        | 16.9        | 18.1        | 19.4        | 20.7        | 22.2        | 7.0%                |
| Europe        | 10.6        | 11.3        | 12.0        | 12.8        | 13.6        | 14.5        | 6.5%                |
| APAC          | 8.1         | 8.7         | 9.2         | 9.9         | 10.5        | 11.3        | 6.8%                |
| RoW           | 1.5         | 1.6         | 1.6         | 1.7         | 1.8         | 1.9         | 4.3%                |
| <b>Total</b>  | <b>36.0</b> | <b>38.5</b> | <b>41.1</b> | <b>43.7</b> | <b>46.6</b> | <b>49.9</b> | <b>6.7%</b>         |

Source: Vendor Interviews, Channel Partner Interviews, Independent Subject Matter Experts, Cloud Associations, Company Datasheets, Annual Reports, Investor Presentations, Third-party Databases, Press Releases, Company Websites, Government Websites, Vendor Surveys, and MarketsandMarkets' Adjacent Market Repositories

## 12.2 NORTH AMERICA

North America is expected to hold the largest market share (~44%) during the forecast period as it is a technologically advanced region and has a significant presence of major market players. Moreover, the region has the largest presence of cloud service providers and is witnessing an increase in the investments in the technological development of existing infrastructures, which is fueling the deployment rate of HPC systems. The region is also experiencing a high demand for compute and on-demand access to process and analyze data. Thus, the market for HPC server solutions is expected to hold a major share of the regional market during the forecast period. The top countries contributing to the growth of the HPC market in North America include the US and Canada. Many major IT companies are based in the US and have invested heavily in setting up high-capacity storage and high-speed data analysis facilities that handle and analyze complex global data.

Moreover, the US and Canadian governments have recognized the importance of computational science in important areas of biotechnology and nanomaterials and are thus increasingly investing in the installation of HPC solutions. For instance, the National Science Foundation's Advanced Computing Infrastructure Program has set up a study committee, which examines anticipated priorities and associated tradeoffs for advanced computing in support of the National Science Foundation-sponsored science and engineering research projects. Such initiatives are expected to boost the demand for HPC in the future. Most market players, including HPE, Dell, Lenovo, IBM, Emerson, NVIDIA, and Cisco, have their headquarters in the US. These players are focusing on introducing the next-generation data center technologies to attract customers, which is expected to further fuel the regional market growth.

### 12.2.1 DRIVERS: NORTH AMERICAN HIGH-PERFORMANCE COMPUTING MARKET

- The rising deployment of IT infrastructures, such as servers, storage, and network devices, to enhance data computing and increase data accuracy by enterprises is expected to fuel the market growth in North America.
- Cloud services offered by the Canadian government are expected to enhance the operations of local businesses and promote the adoption of emerging and modern technologies.

### 12.2.2 COVID-19 IMPACT

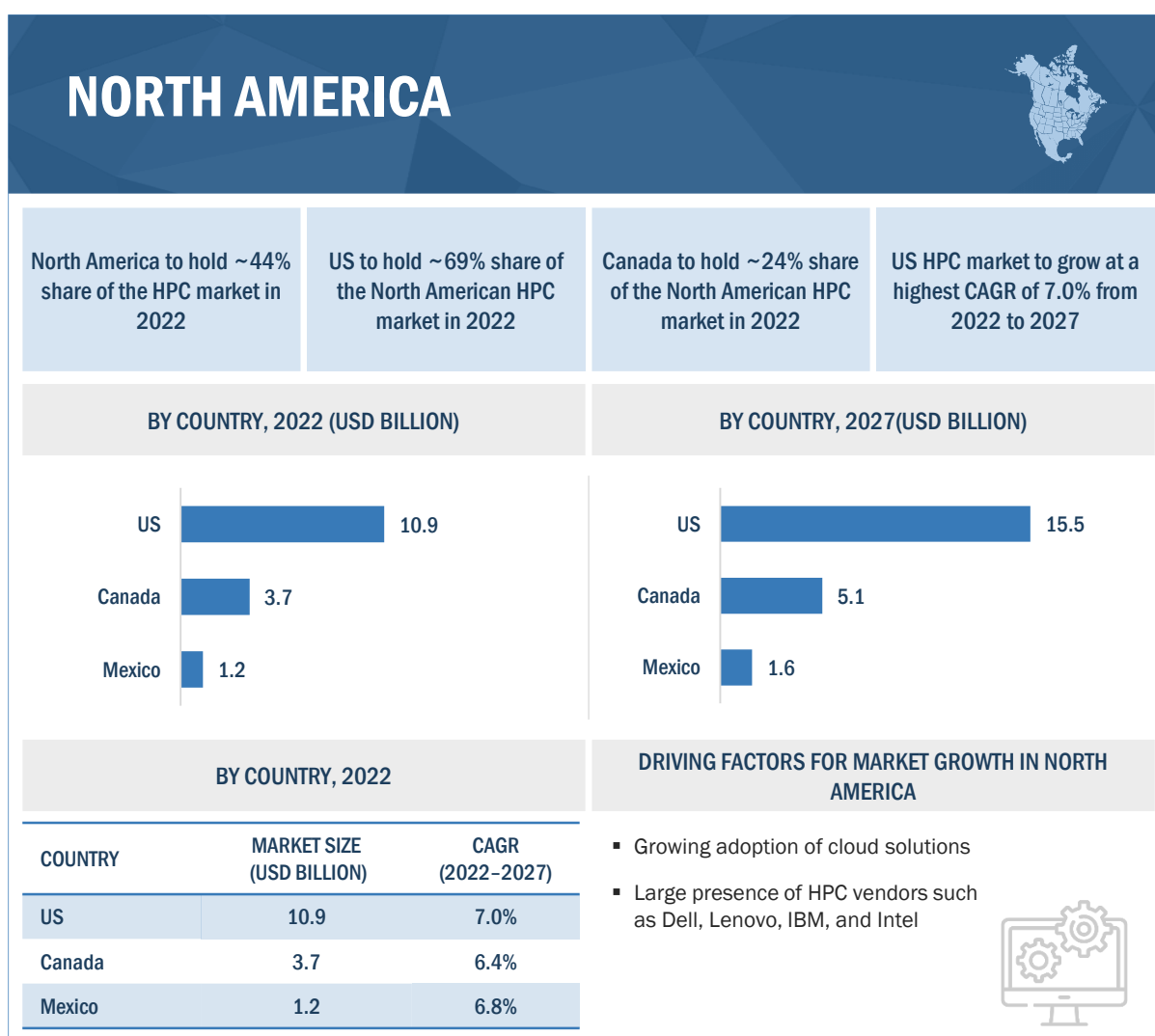
- North America has been one of the hardest-hit regions by the COVID-19 crisis.
- The North American HPC market experienced a negative impact of the pandemic crisis due to disruptions in supply chains and a decline in transportation activities.
- Industry leaders such as IBM, Intel, and HPE have undertaken various initiatives to work with governments to fight COVID-19 by offering their cloud services to enhance the operations of healthcare systems.

### 12.2.3 NORTH AMERICA: REGULATIONS

| REGULATION/LAW | DESCRIPTION                                                           |
|----------------|-----------------------------------------------------------------------|
| US             | Federal Information Technology Acquisition Reform Act (FITARA)        |
| US             | Health Insurance Portability and Accountability Act (HIPAA)           |
| US             | Federal Coordinating Council for Science, Engineering, and Technology |
| US             | Consumer Data Privacy and Security Act                                |
| US             | Consumer Data Privacy Legislation                                     |

*Note: The list is non-exhaustive and derived on the best effort basis.*

*Source: Secondary Research and MarketsandMarkets Analysis*

**FIGURE 36** NORTH AMERICA: HIGH-PERFORMANCE COMPUTING MARKET

Source: Interviews with Experts, Secondary Research, White Papers, Journals and Magazines, and MarketsandMarkets Analysis

**TABLE 77** NORTH AMERICA: HIGH-PERFORMANCE COMPUTING MARKET, BY COUNTRY, 2018-2021 (USD BILLION)

| Country      | 2018        | 2019        | 2020        | 2021        |
|--------------|-------------|-------------|-------------|-------------|
| US           | 7.3         | 8.1         | 9.1         | 10.1        |
| Canada       | 2.6         | 2.8         | 3.2         | 3.5         |
| Mexico       | 0.8         | 0.9         | 1.0         | 1.1         |
| <b>Total</b> | <b>10.7</b> | <b>11.9</b> | <b>13.3</b> | <b>14.7</b> |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 78** NORTH AMERICA: HIGH-PERFORMANCE COMPUTING MARKET, BY COUNTRY, 2022–2027 (USD BILLION)

| Country      | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | CAGR<br>(2022–2027) |
|--------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| US           | 10.904      | 11.720      | 12.573      | 13.450      | 14.414      | 15.505      | 7.3%                |
| Canada       | 3.730       | 3.975       | 4.227       | 4.483       | 4.763       | 5.079       | 6.4%                |
| Mexico       | 1.169       | 1.250       | 1.335       | 1.422       | 1.517       | 1.625       | 6.8%                |
| <b>Total</b> | <b>15.8</b> | <b>16.9</b> | <b>18.1</b> | <b>19.4</b> | <b>20.7</b> | <b>22.2</b> | <b>7.0%</b>         |

Numbers are rounded off to the nearest integer, resulting in rounding-off errors.

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

## 12.2.4 US

### 12.2.4.1 Presence of advanced IT infrastructure is fueling demand for high-performance computing (HPC) solutions in US

The US is a technologically advanced country and has a large presence of leading market players such as IBM, Intel, and Dell. The country is expected to hold a larger share of the North American market during the forecast period. It has witnessed growth in HPC adoption due to the presence of standard regulations, advanced IT infrastructure, a large number of enterprises, and the availability of highly skilled technical experts. According to the Institute of Electrical and Electronics Engineers (IEEE) Computer Society, various branches of the US government have been investing in installing new HPC technologies and solutions for years. The US military heavily invests in deploying classified data processing, surveillance, and data protection solutions. Law enforcement is also a major sector that uses HPC systems in operations such as DNA analysis and public video analysis. All these factors are driving the growth of the US HPC market.

## 12.2.5 CANADA

### 12.2.5.1 Government-led support to increase adoption of digital technologies to fuel demand for high-performance computing (HPC) solutions

Canada is also a major market for high-performance computing in North America. The country is exhibiting high potential for adopting HPC solutions due to the expanding IT infrastructure and various subsidies provided by the government to fuel the adoption of HPC systems in various verticals. The Government of Canada has proactively supported the adoption of digital technologies. Government organizations are the major end users of big data technology in Canada to explore the use of big data analytics to enhance disease research and air quality monitoring. Digitization is fueling the adoption of cloud-based security solutions and services in the public sector. Under the High-Speed Internet Access Programme, the Canadian government is committed to connecting 98% of its population to high-speed internet by 2026 and all by 2030. Such government initiatives are boosting the adoption of digital solutions in various businesses and are, in turn, expected to fuel the demand for HPC solutions and services.

## 12.2.6 MEXICO

### 12.2.6.1 Expanding industrial sector to fuel demand for high-performance computing (HPC)

Mexico is likely to witness moderate growth in the North American high-performance computing market during the review period due to the lack of modern IT infrastructure and limited scope of potential HPC use cases. However, the country is expected to exhibit high growth potential for the high-performance computing market as it is an evolving technological hub in North America owing to its favorable business environment, cheap labor rates, and proximity to the US market. Such conditions might attract the providers of HPC systems and solutions to the country. The developing industrial sector in Mexico is expected to fuel the adoption of IT services, creating demand for new infrastructure and, thus, for HPC systems.

## 12.3 EUROPE

Europe is expected to hold the second-largest share (~29%) of the global HPC market during the forecast period. Many enterprises in the region are deploying HPC solutions to enhance the computational power required for key applications in various industries, increasing the penetration of HPC solutions. The European market is exhibiting high growth potential for the high-performance computing market as HPC manufacturers are focused on offering better quality services to their end users in a highly competitive market landscape. The demand for HPC solutions is on the rise to process a large number of patient datasets in the medical sector. The region comprises major economies, such as the UK, Germany, the Netherlands, France, and Spain, which are increasingly investing in installing HPC solutions. Ministers from the seven European countries, including the Netherlands, France, Germany, Italy, Luxembourg, Portugal, and Spain, have signed a declaration to support the next generation of computing and data infrastructures. This is expected to further fuel the demand for HPC in Europe.

### 12.3.1 DRIVERS: EUROPEAN HIGH-PERFORMANCE COMPUTING MARKET

- The implementation of the General Data Protection Regulation (GDPR), the EU law on data protection and privacy policy, has increased investments in data centers. Data centers constantly need storage solutions and services for monitoring and business operations. This leads to the creation of large datasets of daily business operations, which are processed with the use of HPC solutions for creating future business operation plans.
- Increased investments in cloud-based services by European automobile organizations would foster smart manufacturing. Smart manufacturing involves simulation of the production process before the products are manufactured. HPC helps in running these simulations with pinpoint precision.
- The rising need to meet regulatory compliance and data protection laws due to stringent data security policies imposed by the EU is expected to boost the market growth.

### 12.3.2 COVID-19 IMPACT

- The spread of COVID-19 affected major economies across Europe, with the UK, Germany, France, Italy, and Ireland being the most affected. As a result, several countries in Europe had decreased their on-field workforce and had urged people to work remotely. This fueled the demand for HPC solutions in the region.

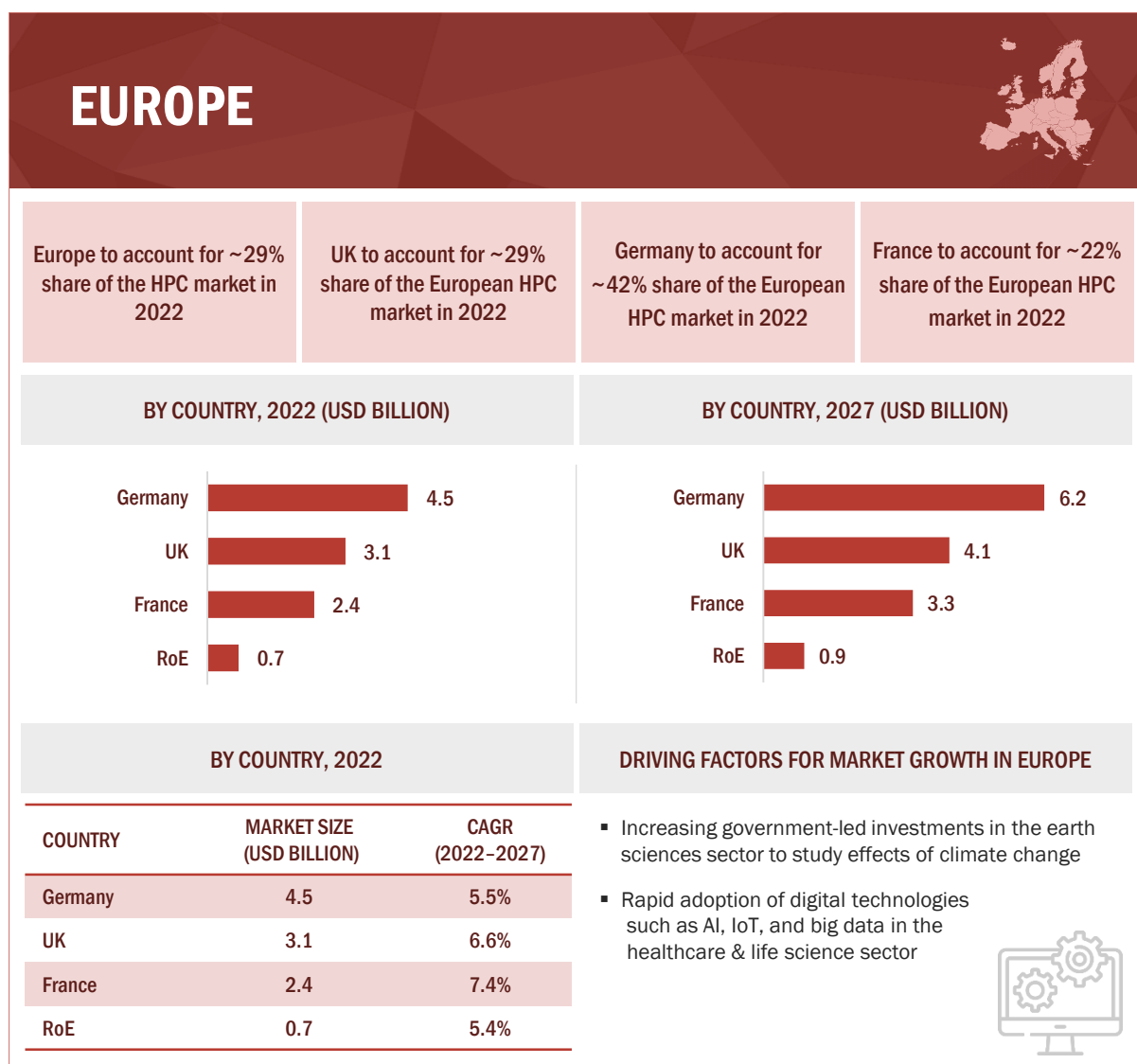
### 12.3.3 EUROPE: REGULATIONS

| REGULATION/LAW                                | DESCRIPTION                               |
|-----------------------------------------------|-------------------------------------------|
| General Data Protection Regulation (GDPR), EU | General Data Protection Regulation (GDPR) |
| Europe                                        | The Council Regulation (EU) 2018/1488     |
| Europe                                        | The EU Cybersecurity Act                  |

Note: The list is non-exhaustive and derived on the best effort basis.

Source: Secondary Research and MarketsandMarkets Analysis

**FIGURE 37** SNAPSHOT OF HIGH-PERFORMANCE COMPUTING MARKET IN EUROPE



Note: Rest of Europe mainly includes Spain, Italy, the Netherlands, and other European countries.

Source: Interviews with Experts, Secondary Research, White Papers, Journals and Magazines, and MarketsandMarkets Analysis

**TABLE 79** EUROPE: HIGH-PERFORMANCE COMPUTING MARKET, BY COUNTRY, 2018–2021 (USD BILLION)

| Country      | 2018       | 2019       | 2020       | 2021       |
|--------------|------------|------------|------------|------------|
| UK           | 2.2        | 2.4        | 2.6        | 2.9        |
| Germany      | 3.0        | 3.4        | 3.7        | 4.1        |
| France       | 1.6        | 1.8        | 2.0        | 2.2        |
| RoE          | 0.5        | 0.6        | 0.6        | 0.7        |
| <b>Total</b> | <b>7.3</b> | <b>8.1</b> | <b>9.0</b> | <b>9.9</b> |

Note: Rest of Europe mainly includes the Netherlands, Spain, Italy, and other European countries.

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 80** EUROPE: HIGH-PERFORMANCE COMPUTING MARKET, BY COUNTRY, 2022–2027 (USD BILLION)

| Country      | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | CAGR<br>(2022–2027) |
|--------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| UK           | 3.1         | 3.3         | 3.5         | 3.7         | 3.9         | 4.1         | 5.9%                |
| Germany      | 4.5         | 4.8         | 5.1         | 5.4         | 5.8         | 6.2         | 7.0%                |
| France       | 2.4         | 2.5         | 2.7         | 2.9         | 3.0         | 3.3         | 6.7%                |
| RoE          | 0.7         | 0.8         | 0.8         | 0.8         | 0.9         | 0.9         | 4.9%                |
| <b>Total</b> | <b>10.6</b> | <b>11.3</b> | <b>12.0</b> | <b>12.8</b> | <b>13.6</b> | <b>14.5</b> | <b>6.5%</b>         |

Rest of Europe mainly includes the Netherlands, Spain, Italy, and other European countries.

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

### 12.3.4 UK

#### 12.3.4.1 Increased adoption of high-performance computing (HPC) systems in medical sector to drive market growth

The UK is one of the economically and technologically well-developed countries in Europe. The country has been increasingly adopting HPC solutions and services over the last 5 years. Verticals such as government & defense, media & entertainment, and education & research adopt HPC solutions and services on a large scale. For instance, in April 2022, the Center of Good Governance (CGG) signed a lease to build a new European HPC hub in Southeast England and increase its cloud HPC capacity by up to 100 petaflops to develop a computing system for researchers to speed up the scientific and industrial computation processes. The UK government is committed to the adoption of AI and big data analytics across the country, resulting in the generation of a large volume of datasets and, in turn, accelerating the adoption of HPC solutions. Despite difficult economic conditions experienced in the country during the COVID-19 pandemic, the high-performance computing market has remained remarkably resilient.

### 12.3.5 GERMANY

#### 12.3.5.1 Rising deployment of high-performance computing (HPC) systems due to high internet penetration to drive market growth

Germany is one of the most promising markets for high-performance computing in Europe. With the emergence of big data analytics, IoT, AI, and Industry 4.0, HPC is gaining traction in the country. According to the World Bank, Germany had 95% internet penetration across the country in 2021. During the COVID-19 pandemic, most of the IT workers in Germany were working from home. This created the need for high computational power to compensate for work over workstations in offices. Hence, various IT enterprises across the country adopted HPC solutions to address this challenge.

Germany is one of the major automobile manufacturing hubs in Europe. The country is well known worldwide for manufacturing automobiles. Various software, such as AutoCAD, BricsCAD, and LibreCAD, are used to build designs of automobiles and run simulations to test multiple parameters, such as passenger safety, operational efficiency, and fuel economy. This requires high computational power to run all these simulations smoothly; hence, HPC solutions are being highly adopted in the country's automotive industry.

### 12.3.6 FRANCE

#### 12.3.6.1 Government initiatives to increase adoption of AI and ML technologies in public sector to fuel market growth

With an advanced economy and flourishing IoT, AI, and machine learning (ML) markets, France has emerged as a lucrative market for HPC. The French government is increasing its spending on fueling the adoption of AI and ML in various applications in the public sector. With the help of these technologies, HPC solutions solve large problems in science and engineering fields and businesses. France is expected to grow at the highest growth rate in the HPC market in Europe. The country is adopting HPC at a rapid pace, contributing 22.2% to the European HPC market in 2022.

### 12.3.7 REST OF EUROPE

The scope of the Rest of Europe market includes the Netherlands, Spain, Italy, and other European countries. Most enterprises in these countries have adopted advanced technologies, such as AI, big data analytics, and IoT, to process data and perform complex calculations at high speeds. The adoption of HPC is expected to be driven by the growing customer base in the public and private sectors in various countries in this region.

## 12.4 ASIA PACIFIC

Asia Pacific is estimated to be the fastest-growing regional market for HPC; it is expected to grow at the highest CAGR of 6.8% during the forecast period. The growing demand for low energy consuming devices in key verticals to solve various complex mathematical problems in seconds with the support of Exascale computing has fueled the demand for HPC solutions in Asia Pacific. Many major companies have started deploying cloud-based HPC services to cater to end users' demand for HPC solutions and services. Countries such as China and Japan highly contribute to the regional market's growth. Moreover, India, Australia, Singapore, and Hong Kong are also expected to witness considerable growth in terms of the deployment of HPC solutions across key verticals in the coming years due to the developing IT infrastructure. The rapid growth of the social media and gaming industries in Asia Pacific has further increased the demand for scalable architectures capable of handling complex operations. This demand can be met by the effective deployment of HPC solutions, further contributing to the growth of this market.

#### 12.4.1 DRIVERS: ASIA PACIFIC HIGH-PERFORMANCE COMPUTING MARKET

- The increasing adoption of IT technologies and the rising number of data centers in the region are projected to drive the growth of the HPC market.
- The rising adoption of cloud storage, AI, data analytics, IoT, and related services and economic growth are attracting high investments in the region, eventually fueling the demand for HPC solutions and services.
- China and India are emerging economies globally. They have witnessed increased investments from IT giants, which have helped them scale their IT infrastructure, thereby propelling the demand for HPC products, solutions, and services.
- The growing adoption of online gaming, IoT, AI, and machine learning in developing countries is expected to create lucrative opportunities for the HPC market.

#### 12.4.2 COVID-19 IMPACT

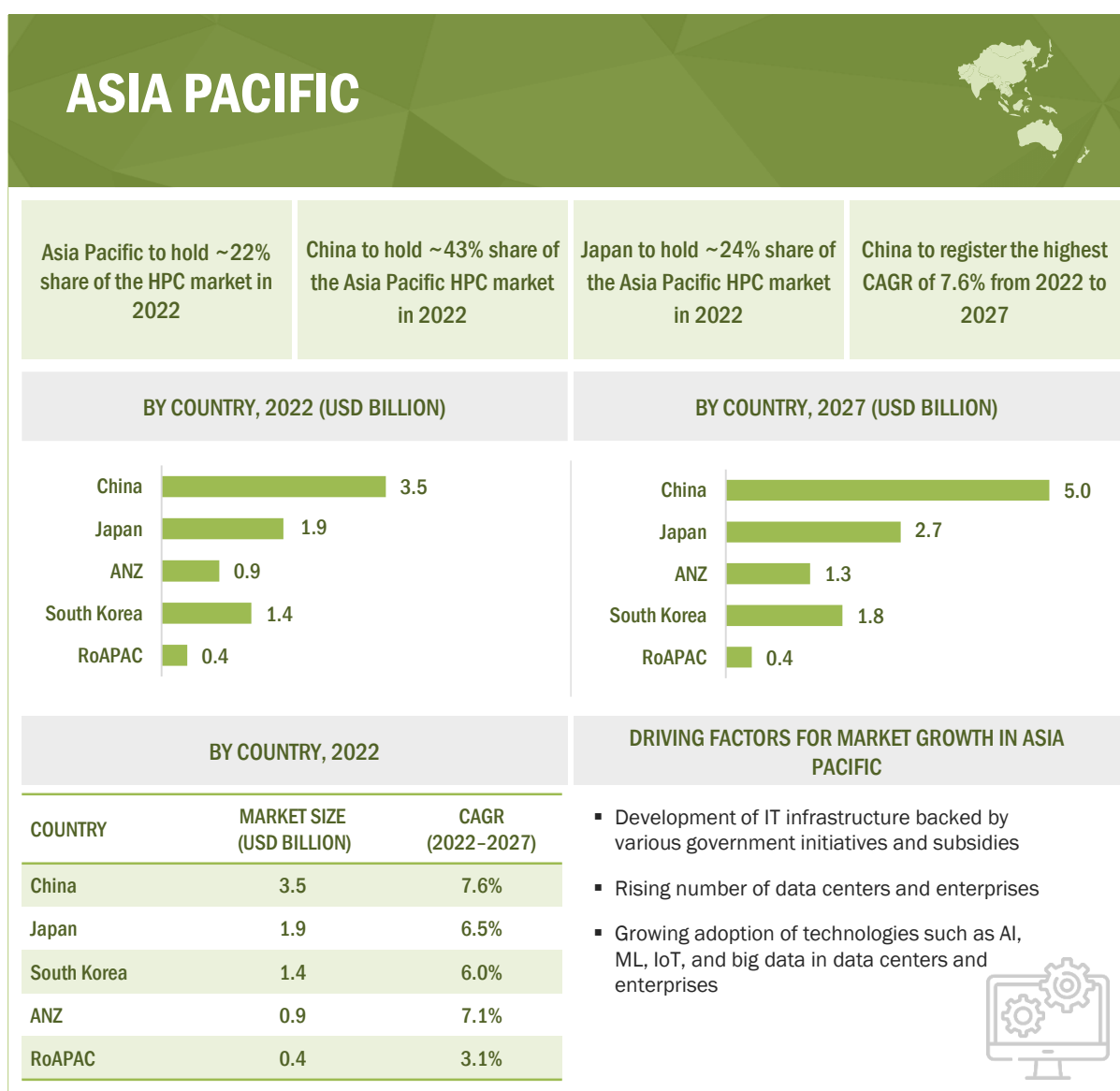
- While technology spending in Asia Pacific has increased, the setback due to the COVID-19 pandemic has been imminent. This resulted in increased adoption of cloud technology across key verticals, which helped in stabilizing business operations that would have been otherwise adversely impacted by COVID-19.
- According to statements given by leading cloud solutions providers (CSPs) in the region, the spending on public cloud data center hardware and software grew by 1%, while the spending on traditional data centers and private cloud declined by 1% in 2021.
- The growing adoption of cutting-edge technologies such as AI, data analytics, and IoT in China is expected to fuel the demand for HPC solutions, especially in the country's manufacturing sector.

### 12.4.3 ASIA PACIFIC: REGULATIONS

| REGULATION/LAW                                     | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Multi-Tier Cloud Security (MTCS), Singapore        | <ul style="list-style-type: none"> <li>The MTCS standard was prepared under the direction of the Information Technology Standards Committee (ITSC) of the Infocomm Development Authority of Singapore (IDA).</li> <li>The purpose of MTCS is to provide:</li> <li>A common standard that CSPs can apply to address customer concerns about the security and confidentiality of data in the cloud and how the usage of cloud solutions will impact the customers' business.</li> </ul> |
| Monetary Authority of Singapore (MAS)              | MAS is the central bank and financial regulatory authority of Singapore. MAS has established Guidelines on Outsourcing (MAS Guidelines) to govern the outsourcing practices of Financial Institutions (FIs). MAS recognizes cloud services as a form of outsourcing.                                                                                                                                                                                                                  |
| Cloud Computing Service Security Assessment, China | To improve the security and controllability of cloud computing services purchased by government agencies and key information infrastructure operators, the National Internet Information Office, National Development and Reform Commission, Ministry of Industry and Information Technology, and Ministry of Finance have formulated the Cloud Computing Service Security Assessment.                                                                                                |
| Multi-Level Protection Scheme (MLPS) 2.0, China    | MPLS is a regulation that outlines security requirements for different types of data. It requires all network operators wishing to provide cloud services in China to store data in infrastructure within the country's borders.                                                                                                                                                                                                                                                      |
| The Singapore Standard (SS) 564                    | SS 564 provides a recognized framework and logical and consistent methodology to achieve continuous improvement for data center facilities. This standard is modeled after the ISO 50001 standard on energy management; however, it is specifically tailored to meet the needs of data centers in Singapore.                                                                                                                                                                          |
| Japan Data Center Council (JDCC)                   | JDCC has formulated "Data Center Facility Standard" to construct a proprietary facility standard that conforms to the actual situation in Japan. This modifies those portions of the Tier standard that seem excessive for Japan and adds Japan-specific elements. JDCC uses Power Usage Effectiveness (PUE) and Data Center Infrastructure Efficiency (DCIE) recommended by The Green Grid from the US as indicators to measure the energy efficiency of a data center.              |
| Internet Data Center (IDC), China                  | Any business providing value-added data services through a data center in China, including cloud services, has to apply for an IDC license. Any foreign company operating under such a license in China is limited to 50% co-ownership with a Chinese firm in a joint venture.                                                                                                                                                                                                        |
| China's Cyber Security Law (CSL)                   | CSL is a broad piece of legislation that dictates how companies should approach security and privacy within the country. It also has mandatory requirements around breach notification, the appointment of a head of cybersecurity, and incident response plans. It governs almost every aspect of online and network activities and gives law enforcement agencies the authority to inspect and monitor businesses, including in-person inspections on company premises.             |

*Note: The list is non-exhaustive and derived on a best-effort basis.*

*Source: Secondary Research and MarketsandMarkets Analysis*

**FIGURE 38** SNAPSHOT HIGH-PERFORMANCE COMPUTING MARKET IN ASIA PACIFIC

Note: Rest of APAC mainly includes India, Singapore, and other Asian countries.

Source: Interviews with Experts, Secondary Research, White Papers, Journals and Magazines, and MarketsandMarkets Analysis

**TABLE 81** ASIA PACIFIC: HIGH-PERFORMANCE COMPUTING MARKET, BY COUNTRY, 2018-2021 (USD BILLION)

| Country      | 2018       | 2019       | 2020       | 2021       |
|--------------|------------|------------|------------|------------|
| China        | 2.32       | 2.59       | 2.91       | 3.24       |
| Japan        | 1.34       | 1.48       | 1.65       | 1.82       |
| South Korea  | 0.95       | 1.05       | 1.16       | 1.27       |
| ANZ          | 0.62       | 0.69       | 0.78       | 0.86       |
| RoAPAC       | 0.27       | 0.29       | 0.32       | 0.34       |
| <b>Total</b> | <b>5.5</b> | <b>6.1</b> | <b>6.8</b> | <b>7.5</b> |

Rest of APAC mainly includes India, Singapore, and other Asian countries.

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 82** ASIA PACIFIC: HIGH-PERFORMANCE COMPUTING MARKET, BY COUNTRY, 2022–2027 (USD BILLION)

| Country      | 2022       | 2023       | 2024       | 2025       | 2026        | 2027        | CAGR<br>(2022–2027) |
|--------------|------------|------------|------------|------------|-------------|-------------|---------------------|
| China        | 3.50       | 3.78       | 4.06       | 4.36       | 4.68        | 5.05        | 7.6%                |
| Japan        | 1.95       | 2.08       | 2.21       | 2.35       | 2.50        | 2.67        | 6.5%                |
| South Korea  | 1.36       | 1.44       | 1.53       | 1.61       | 1.71        | 1.82        | 6.0%                |
| ANZ          | 0.93       | 1.00       | 1.07       | 1.14       | 1.22        | 1.31        | 7.1%                |
| RoAPAC       | 0.35       | 0.37       | 0.38       | 0.39       | 0.40        | 0.41        | 3.1%                |
| <b>Total</b> | <b>8.1</b> | <b>8.7</b> | <b>9.2</b> | <b>9.9</b> | <b>10.5</b> | <b>11.3</b> | <b>6.8%</b>         |

Rest of APAC mainly includes India, Singapore, and other Asian countries.

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

## 12.4.4 AUSTRALIA AND NEW ZEALAND (ANZ)

### 12.4.4.1 Rising use of high-performance computing (HPC) systems in media & entertainment vertical to propel market growth

Australia and New Zealand (ANZ) are the early adopters of HPC solutions across the media & entertainment, manufacturing, and healthcare & life sciences verticals. Australian enterprises have readily adopted HPC solutions in the last few years, and the trend is expected to gain more traction as vendors across the globe are expanding their businesses in Australia due to the presence of a large customer base. Rapid technological advancements and a high degree of research and development (R&D) are also expected to create growth opportunities for the HPC market in ANZ.

Australia and New Zealand are known for their hospitality industry. These countries have many 5 star hotels, resorts, and lounges. During the spring season, both countries experience a tremendous influx of tourists. This puts strain on hotel resources to accommodate all these tourists simultaneously. Hence, various computing systems are used in this industry to keep track of all the operations. This generates tremendous data for processing. HPC solutions are highly preferred in this industry to elevate the burden in such situations as they are capable of processing large volumes of data in a short period.

## 12.4.5 JAPAN

### 12.4.5.1 Growing use of high-performance computing (HPC) solutions in metrology to boost market growth

In terms of overall IT spending and market size, Japan is one of the world's largest markets embracing enterprise technology. Japanese companies are taking a keen interest in adopting modern IT technologies. In Japan, Fujitsu has completed the construction of a supercomputer system for the Meteorological Research Institute of Japan, the country's premier weather and climate forecasting agency. Rapid technological advancements, increased R&D expenditure, and government initiatives are supporting the growth of high-speed computation infrastructure across key verticals in the country.

Japan is very susceptible to earthquakes. The country experiences the highest number of earthquakes per year across the globe. Hence, it is essential for the government to know about potential earthquake situations and prepare contingency plans to ensure the least damage and loss of life. Over several years, the Japanese Meteorological Institute has collected large volumes of data regarding patterns of earthquake occurrences. High-speed computation power is required to process large amounts of data and create countermeasures and contingency plans. Here, HPC solutions help in processing this data in a

comparatively short period, giving more time to the agency to develop action plans for deploying countermeasures in case of earthquake occurrences.

## **12.4.6 CHINA**

### **12.4.6.1 Growing adoption of digital technologies such as AI, IoT, and big data to drive the market growth in China**

Huge investments in the private sector, robust government support, and the availability of a large youth population are a few factors driving the demand for new and emerging technologies in China. The country has always been keen on leading the technology market. It has been pioneering the development of HPC products and solutions that can be deployed across several key industry verticals in the country. China has seen tremendous growth in IT infrastructure as well as rapid adoption of IT technologies and related services across the country. With a more than 1.4 billion population, China is a growing economy that is rapidly adopting digital technologies such as AI, big data analytics, and IoT. The country invests large amounts of financial resources in the latest technologies to maintain its leading position in the technology sector. With the growth of various digital technologies and enterprises across the country, China is in a pivotal position to leverage the capabilities of HPC solutions across its enterprises. China is highly efficient and competitive across various industry verticals, such as healthcare and life sciences, and manufacturing, which can fuel the growth of the HPC market in the country.

## **12.4.7 REST OF ASIA PACIFIC**

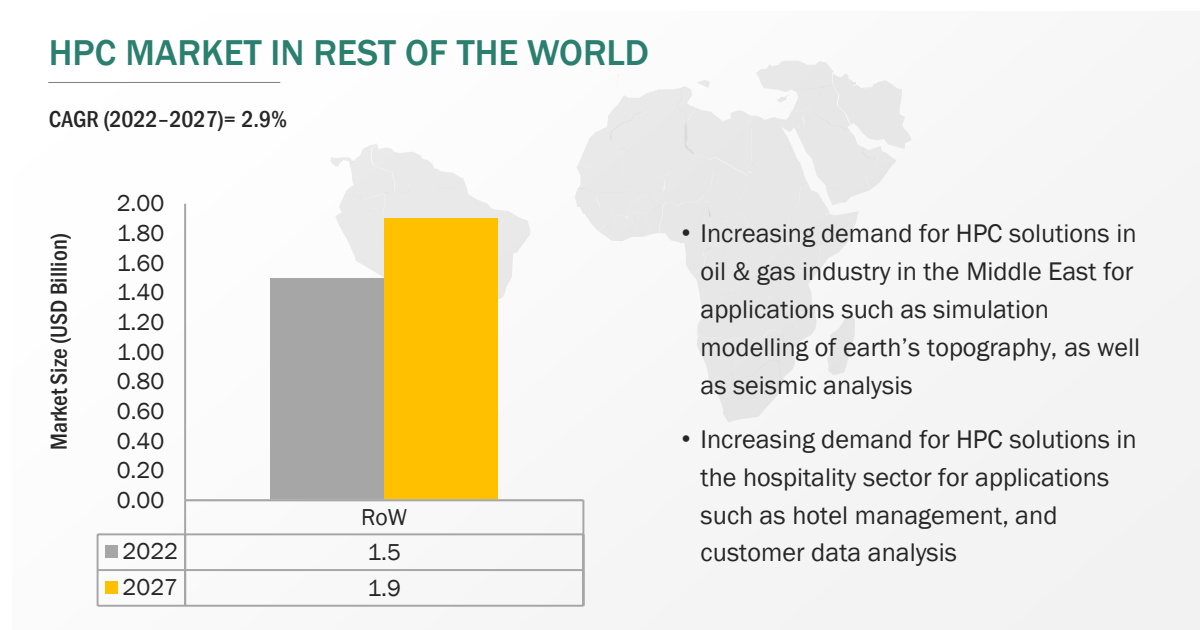
The scope of the Rest of Asia Pacific market includes India, Singapore, South Korea, and other Asian countries. There is an increasing demand for data centers, big data analytics, cloud, IoT, AI, and edge computing technologies in these countries. Enterprises are the main entities promoting R&D investments for technology advancements in various countries across the region. India is one of the rapidly growing economies with substantial scope for the HPC market growth due to its large population and the growth in the adoption of various digital technologies across key industry verticals. Over the last decade, India has tremendously revamped its IT infrastructure and is now a hub of IT innovations. Hence there is an immense growth potential for HPC solutions in the country.

Similarly, Singapore remains a major connectivity hub in Asia Pacific. South Korea is one of the major IT-driven economies in the region that offer the best legal and regulatory environment for developing and deploying cloud solutions. With the rising adoption of cloud solutions in various key industry verticals, the country will experience steady growth in demand for HPC solutions and services. Rapid technology advancements and a high degree of R&D expenditure are expected to create great opportunities for growth of the HPC market in the region.

## 12.5 REST OF WORLD

The market in the Rest of World has been further classified into the Middle East & Africa and South America. Countries in these regions are mostly underdeveloped or developing; therefore, the market for HPC is in the nascent stage across various countries in the region. However, few countries in this region, such as the UAE, Saudi Arabia, Qatar, and Brazil, have substantial growth potential for the HPC market owing to the tremendous revenue generation opportunities from the hospitality industry.

**FIGURE 39** SNAPSHOT HIGH-PERFORMANCE COMPUTING MARKET IN REST OF WORLD



Source: Interviews with Experts, Secondary Research, White Papers, Journals and Magazines, and MarketsandMarkets Analysis

**TABLE 83** ROW: HIGH-PERFORMANCE COMPUTING MARKET, BY COUNTRY, 2018–2021 (USD BILLION)

| Region        | 2018        | 2019        | 2020        | 2021        |
|---------------|-------------|-------------|-------------|-------------|
| Middle East   | 0.55        | 0.59        | 0.64        | 0.70        |
| South America | 0.58        | 0.63        | 0.69        | 0.74        |
| <b>Total</b>  | <b>1.12</b> | <b>1.22</b> | <b>1.33</b> | <b>1.44</b> |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 84** ROW: HIGH-PERFORMANCE COMPUTING MARKET, BY COUNTRY, 2022–2027 (USD BILLION)

| Region        | 2022        | 2023        | 2024        | 2025        | 2026        | 2027        | CAGR<br>(2022–2027) |
|---------------|-------------|-------------|-------------|-------------|-------------|-------------|---------------------|
| Middle East   | 0.73        | 0.76        | 0.79        | 0.82        | 0.85        | 0.89        | 4.0%                |
| South America | 0.78        | 0.82        | 0.86        | 0.90        | 0.94        | 0.98        | 4.6%                |
| <b>Total</b>  | <b>1.51</b> | <b>1.58</b> | <b>1.65</b> | <b>1.72</b> | <b>1.79</b> | <b>1.87</b> | <b>4.3%</b>         |

Source: Secondary Research, Interviews with Experts, and MarketsandMarkets Analysis

### 12.5.1 MIDDLE EAST & AFRICA

The HPC market in the MEA region is projected to grow at a considerable CAGR of 7.9% during the forecast period. The rapid emergence of the media and entertainment industries in this region has led to a significant increase in the demand for advanced high-speed computation solutions propelling the growth of HPC in this region. These solutions provide a competitive edge as they enable the processing of large volumes of complex data. According to the Cisco Global Cloud Index in 2019, the data center traffic in MEA was expected to reach ~400 exabytes in 2021. This rapid increase in data center traffic is expected to pave the way for the growth of HPC solutions in this region. The African data center industry is also expected to grow steadily, owing to the increasing investments and the presence of leading telecom operators and financial institutions in the region. Moreover, the rising demand from content providers owing to the exponential growth in smartphone users with higher content delivery requirements has further caused a surge in demand for HPC infrastructure in this region.

### 12.5.2 DRIVERS: MIDDLE EAST & AFRICA HIGH-PERFORMANCE COMPUTING MARKET

- New cloud offerings from key vendors in MEA are driving government-funded public cloud initiatives and the adoption of emerging technologies.
- Government collaborations with enterprises would boost technological advancements—for instance, in March 2020, Google Cloud Platform signed a strategic collaboration agreement with the Qatar Free Zones Authority (QFZA) to launch its services in the Middle East.
- Initiatives such as Saudi Vision 2030, Dubai Vision 2021, and New Kuwait Vision 2035 are expected to boost the demand for AI, IoT, and big data, which is expected to fuel the growth of the HPC market in the MEA.

### 12.5.3 COVID-19 IMPACT

- Organizations in Saudi Arabia are using cloud computing services to navigate challenges caused by COVID-19.
- In the UAE's technology sector, Emirati Company Group 42 (G42) has planned to leverage the existing AI and cloud computing technologies to prevent the COVID-19 from spreading and ensure medical supplies reach the areas where they are most needed.
- Verticals such as BFSI, government, and energy & utilities are expected to increase their cloud spending in the next 2–3 years to sustain critical business functions in case of a future COVID-19-like scenario.

### 12.5.4 MIDDLE EAST & AFRICA: REGULATIONS

| REGULATION/LAW                                              | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Protection of Personal Information Act (POPI), South Africa | The POPI Act was formed in 2013. It is an equivalent act to the EU GDPR of South Africa. It sets some prior conditions for responsible parties to lawfully process the personal information of data subjects.                                                                                                                                                                                                                                                   |
| Cloud Computing Regulatory Framework, Saudi Arabia          | Communications and Information Technology Commission (CITC) considers that the adoption of the Cloud Computing Regulatory Framework would provide increased regulatory clarity and encourage the adoption of cloud computing services in the country. It is based on best international practices and public consultation analysis, which includes the rights and obligations of service providers, individual customers, government entities, and enterprises. |

*Note: The list is non-exhaustive and derived on a best-effort basis.*

*Source: Secondary Research and MarketsandMarkets Analysis*

### 12.5.5 SOUTH AMERICA

The HPC market in South America is growing substantially and currently experiencing steady growth. Most of the HPC vendors are reaching out to this region with their solutions, owing to the presence of a large number of new-built data centers in this region. Moreover, many large-scale enterprises are expanding their presence in this region, which is expected to boost the regional market's growth. Countries such as Brazil and Argentina in South America are rapidly adopting HPC solutions and are expected to contribute to this market significantly in the coming years, as most of the HPC solution providers are looking to expand their market presence and marketing channels in this region. This region is estimated to account for a small share of 1.5% in 2022 and grow at a CAGR of 7.3% during the forecast period.

### 12.5.6 DRIVERS: SOUTH AMERICA HIGH-PERFORMANCE COMPUTING MARKET

- Initiatives by different governments in South America to improve business operations utilizing digital technologies are expected to propel the market's growth.
- The growing awareness about cloud technology and the increasing shift toward digitization are fueling the adoption of cloud technologies by SMEs in this region.

### 12.5.7 COVID-19 IMPACT

- World Bank assistance and various government COVID-19 control programs to minimize economic losses due to work closures, project delays, and supply chain disruptions would drive the adoption of HPC solutions across public and private sectors in the South American region.
- Owing to the work from home policy, the demand for communication solutions had increased rapidly during the COVID-19 pandemic. Enterprises from media & entertainment and BFSI verticals are adopting IT technologies such as AI, big data analytics, and IoT to improve their business performance in the region leading to the increasing demand for HPC solutions and services.

### 12.5.8 SOUTH AMERICA: REGULATIONS

| REGULATION/LAW                                                            | DESCRIPTION                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lei Geral de Proteção de Dados (LGPD) General Data Protection Law, Brazil | Brazil's LGPD is a data protection law that was formed to regulate the personal data of its citizens. The LGPD attempts to unify over 40 different statutes that currently govern personal data, both online and offline, by replacing certain regulations. |

*Note: The list is non-exhaustive and derived on a best-effort basis.*

*Source: Secondary Research and MarketsandMarkets Analysis*

## 13 COMPETITIVE LANDSCAPE

### 13.1 KEY PLAYER STRATEGIES/RIGHT TO WIN

In this section, we have provided insights regarding the “strategic view” of the competitive scenario in the HPC market.

| KEY PLAYERS                                                                         | PRODUCT TYPE  | PARTNERSHIPS                                                                                                                                                                                                                                 | REGION                                                             | MANUFACTURING                                                                                                              |
|-------------------------------------------------------------------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
|    | HPC Solution  | AMD acquired Xilinx (US), a leading provider of adaptive and edge computing solutions. The acquisition will help AMD boost their portfolio for HPC solutions integrated by Xilinx's edge computing capabilities                              | US and China account for ~53% of the company's revenue             | The company has manufacturing facilities in North America, Europe, and APAC, most of which are located in the US and China |
|    | HPC Solutions | Intel acquired Tower Semiconductors (Israel), a leading manufacturer of semiconductor components and accessories. The move will help Intel bolster the development of their HPC Hardware solutions by providing required hardware components | North America and APAC contribute to ~63% of the company's revenue | The company has manufacturing facilities in APAC, North America, and Europe, most of which are located in the US and China |
|  | HPC Solutions | Hewlett Packard Enterprises acquired Zerto (US), a leading provider of cloud solutions. This will help HPE expand their cloud based HPC offering across several key industry verticals                                                       | North America and EMEA contribute to ~69% of the company's revenue | The company has manufacturing facilities in North America, EMEA, and APAC, most of which are located in the US             |
|  | HPC Solutions | IBM acquired Sentaca (US), a leading provider of cloud migration services. This acquisition will help IBM integrate their HPC solutions in cloud at a faster pace                                                                            | North America and EMEA contribute to ~80% of the company's revenue | The company has manufacturing facilities in North America, EMEA, and APAC, most of which are located in the US             |

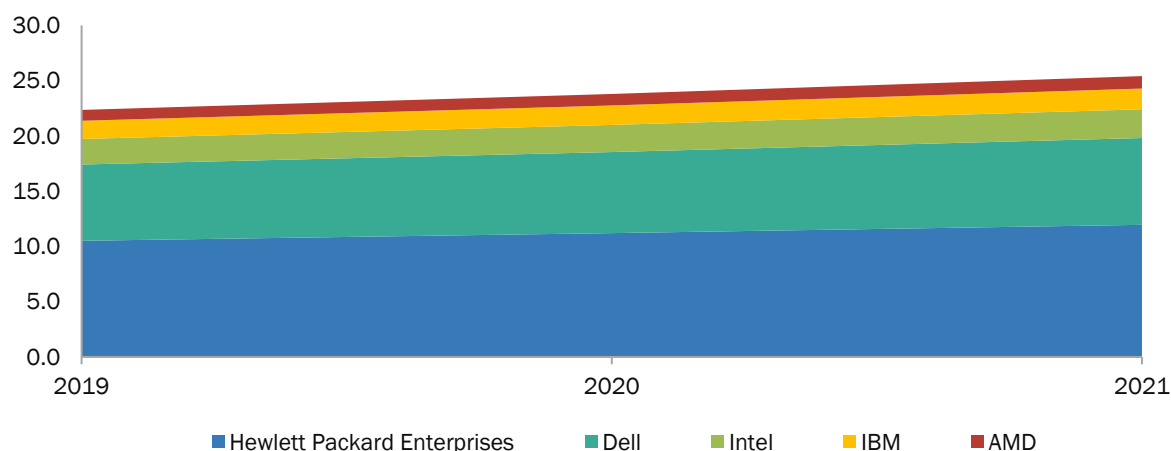
Source: Annual Reports, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

### 13.2 OVERVIEW

This section of the report provides an overview of the competitive scenario in the HPC market. It analyzes growth strategies adopted by companies to enhance their position in the HPC market and ensure long-term growth and success. A market share analysis has been carried out based on the revenues generated by the key players from various business segments in the overall market. The section also features a benchmarking of the strategies adopted by the key players in the HPC market, based on an analysis of developments such as product launches, agreements, and partnerships.

### 13.3 TOP 5 COMPANY ANALYSIS

**FIGURE 40** TOP 5 COMPANY REVENUE ANALYSIS FOR HPC MARKET, 2019–2021



Source: Annual Reports, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis

Hewlett Packard Enterprises, Dell, Intel, IBM, and AMD are the top 5 companies in the HPC market. These players led the market in 2021, owing to their diverse product portfolios. They offer state-of-the-art HPC products, solutions, and services. The top 5 companies mainly focus on product launches, agreements, and partnerships to expand their footprint and business reach.

### 13.4 MARKET SHARE ANALYSIS (2021)

This section analyzes the market share of the key players in the HPC market for 2021. Most prominently active players in this market are public and privately held companies. Thus, the share has been estimated based on several parameters, such as overall revenues, segmental revenues, research and development expenditure, product offerings, geographic presence, and recent activities undertaken by the companies in the market. The companies have been quantified on these parameters and based on the results; the market share has been derived. The HPC market is consolidated as the top five players account for more than 50% of the market share. Also, wide geographical presence, long-term partnerships, and emerging HPC use cases are driving the revenue for these vendors.

**TABLE 85** HPC MARKET: MARKET SHARE ANALYSIS

| DEGREE OF COMPETITION               | COMPETITIVE |
|-------------------------------------|-------------|
| Total Market Share of Top 5 Players | 65–70%      |
| Hewlett Packard Enterprises         | 30–35%      |
| Dell                                | 20–25%      |
| Intel                               | 07–08%      |
| IBM                                 | 05–06%      |
| AMD                                 | 03–04%      |

**DEGREE OF COMPETITION IS DEFINED AS BELOW:**

Fragmented: When the top 5 players have a total market share <25%

Competitive: When the top 5 players have a total of 25–50% market share

Consolidated: When the top 5 players have a total market share >50%

*Source: Annual Reports, Press Releases, Investor Presentations, Interviews with Experts, and MarketsandMarkets Analysis*

HPE provides a range of HPC solutions to cater to the growing customer requirements. The company offers Superdome Flex, SGI 8600, Apollo 6000, Apollo 6500, Apollo 4000, Apollo 2000, and Apollo 10 systems to provide HPC solutions tailored as per customer needs. These solutions are used to solve large scientific, engineering, and data analysis problems. HPE invests heavily in R&D activities. In 2021, the company's R&D expenditure amounted to USD 2.0 billion. The growth in R&D expenditure is primarily due to the increase in networking and technology services in the enterprise group segment. This increase in expenditure assists the company in enhancing its product and services portfolio.

Dell offers PowerEdge servers and Precision workstations, high-performance storage, Genomic Data Analysis Platform, and data center switches and services that cater to HPC needs. The Dell PowerEdge servers and Precision workstations are available in different form factors and can be configured to meet the flexibility and performance needs of users. These servers are designed for HPC environments with broad requirements, including more local Input/Output (I/O) capacity and expandability. On the other hand, Precision workstations run high-demand, industry-specific applications for scientific calculations, 3D industrial designs, remote visualization, engineering simulations, and digital content creation. To optimize infrastructure and minimize costly disruptions, Dell offers HPC deployment services with validated processes and best practices to efficiently plan, design, and implement HPC solutions.

Intel manufactures Xeon Scalable processors that provide workload-optimized performance and throughput for advanced analytics, data compression, and HPC applications. The company's Xeon Phi processor is a bootable host processor delivering massive vectorization and parallelism to support demanding HPC applications.

IBM is an eminent player in the HPC market, as it provides product offerings in the market across the globe. The company is currently focusing on catering to its customers with highly competent technology-enabled products and services to sustain competition. It aims at delivering innovative solutions and improved business continuity while planned and unplanned IT outages, thereby satisfying its customers.

AMD offers the AMD Opteron processor family with up to 16 cores and four memory channels to deliver high-compute capability, performance, and flexibility. These processors find applications in handling dense and multi-threaded workloads with large, complex datasets associated with supercomputing and HPC. The company also offers the AMD Ryzen processor, AMD CDNA 2 Architecture, AMD Infinity Architecture, and AMD ROCm family that caters to the HPC market. Its high-performance x86 Core "Zen" architecture is an x86 processor designed for HPC products. AMD provides AMD GPUs, which cater to compute-intensive applications and assist scientists and engineers in getting quick results. The company also provides EPYC server processors and Radeon Instinct GPU accelerators. These products cater to the demands of heterogeneous computing for machine intelligence and HPC.

## 13.5 COMPANY EVALUATION QUADRANT, 2021

The company evaluation quadrant section provides an overview of the prevailing competitive scenario in the HPC market.

### 13.5.1 STAR

Stars are the leading players in the HPC market. They have robust product portfolios, a wide presence, and adopt effective business growth strategies. Star companies primarily focus on acquiring a leading market position through their high financial capabilities and well-established brand equity. They account for the largest shares of the market.

### 13.5.2 PERVASIVE

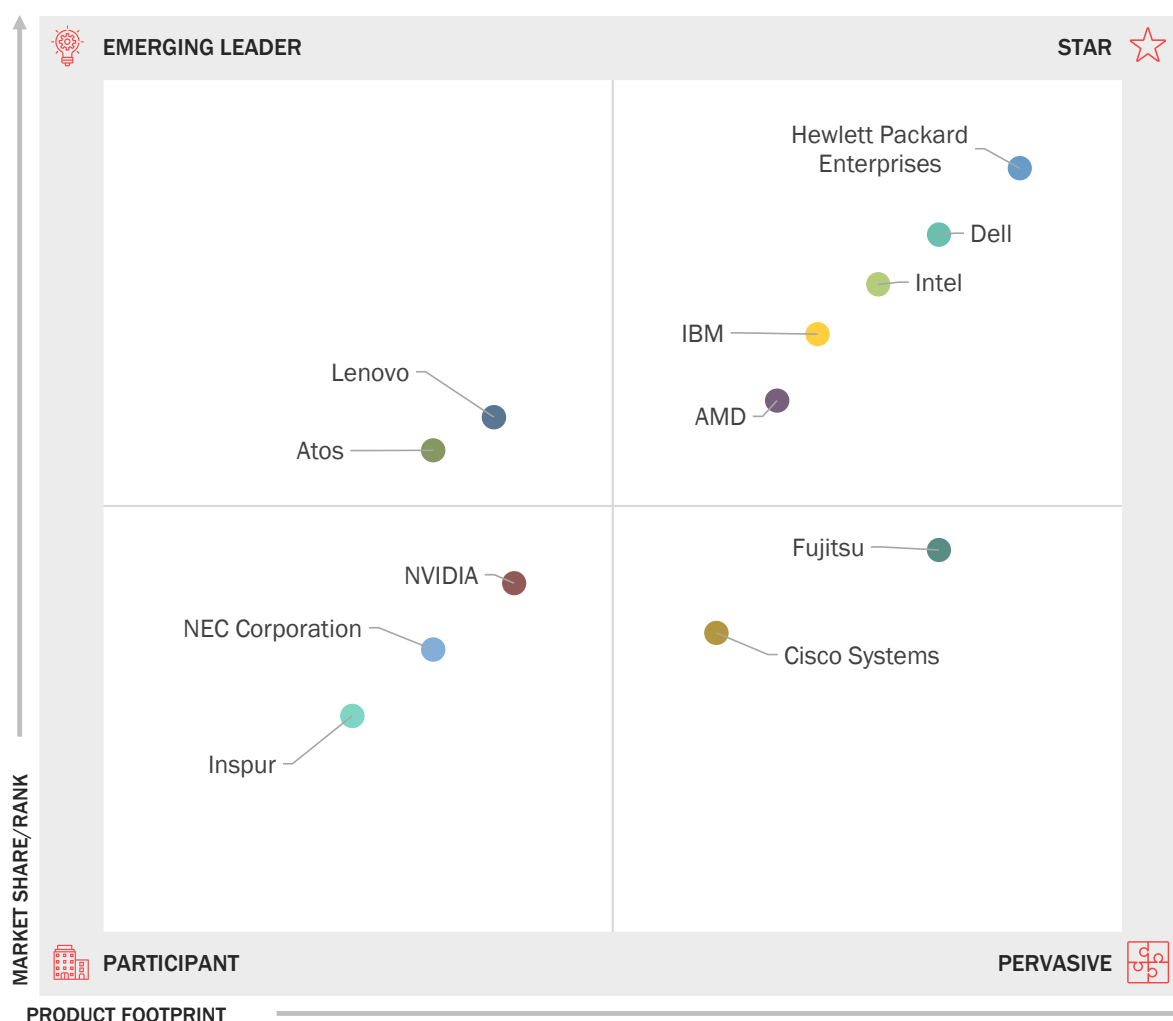
Pervasive companies are well-established players with effective business strategies and high research and development capabilities. However, they have limited product portfolios. Pervasive companies hold significant shares in the market but have a less regional presence.

### 13.5.3 EMERGING LEADER

The HPC market is moderately dependent on emerging leaders who bring innovations in the form of product launches and developments. They demonstrate substantial product innovations but have nominal growth strategies for their overall businesses. Emerging leaders have a substantial regional presence with a small market share.

### 13.5.4 PARTICIPANT

Participant companies offer niche products and have gradually started gaining a position in the HPC market. Their business strategies are not as effective as those of the established players, and they hold a small share of the market.

**FIGURE 41** HPC MARKET COMPANY EVALUATION QUADRANT, 2021

Source: Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

## 13.6 SME EVALUATION QUADRANT, 2021

The SME evaluation quadrant provides information about players that offer HPC-related products and solutions on a smaller scale than key players in the market. Vendor evaluations are based on two broad categories: strength of product portfolio and business strategy excellence.

### 13.6.1 PROGRESSIVE COMPANY

Progressive companies offer strong product portfolios in the HPC market. These vendors have been marking their presence in the market by offering customized solutions according to customers' requirements and adopting growth strategies to deliver advanced products in the market.

### 13.6.2 RESPONSIVE COMPANY

Responsive companies possess innovative solutions to cater to the application-specific requirements of industries. These companies focus on their product portfolios and have the potential to develop strong business strategies to expand their business and stay on par with progressive companies. Responsive companies have been at the forefront of the deployment of their solutions in different industries.

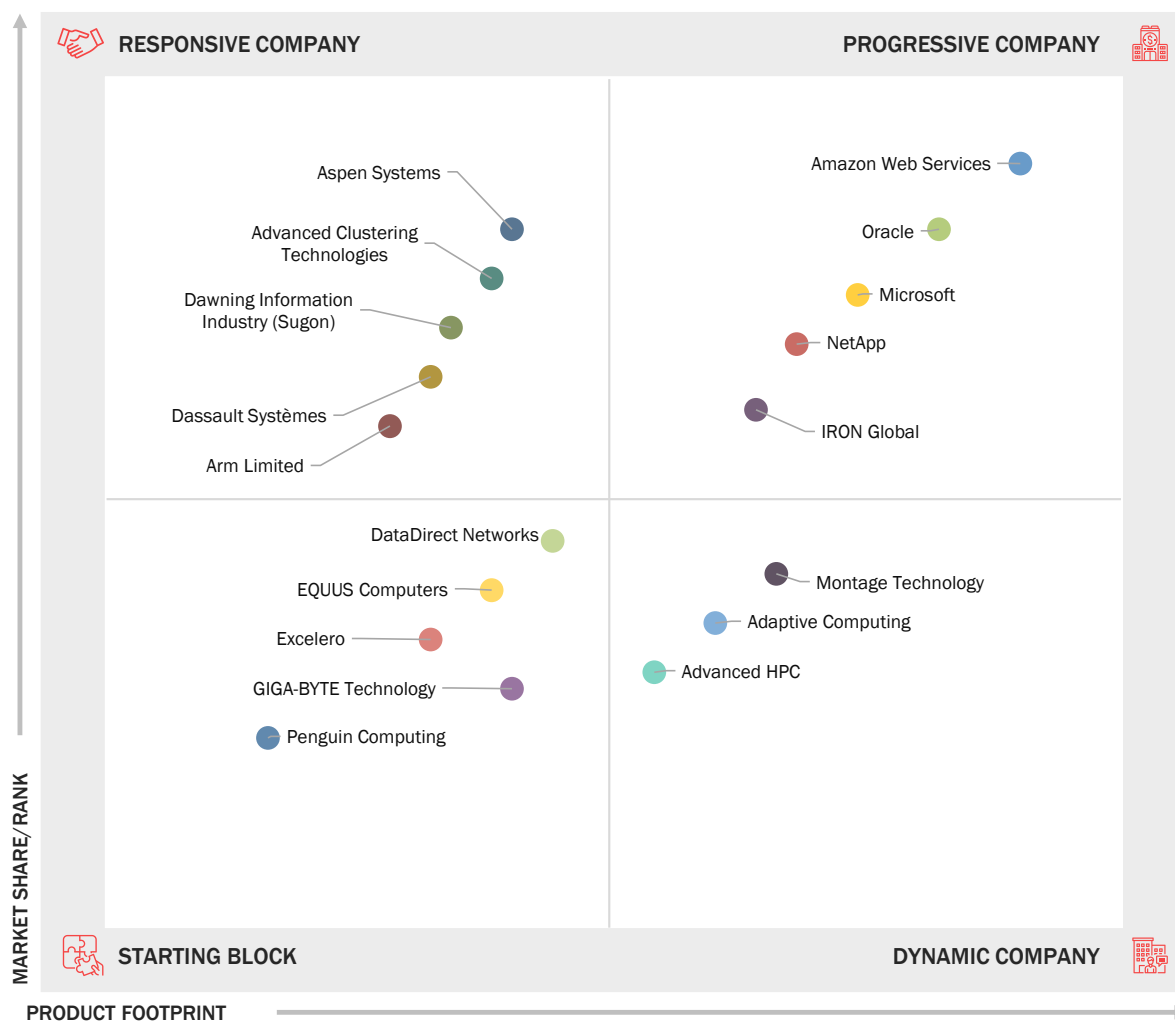
### 13.6.3 DYNAMIC COMPANY

Dynamic companies offer innovative solution portfolios and have an extensive network of channel partners and resellers to increase the deployment of their solutions across a multitude of verticals. They have been consistently witnessing revenue growth in the HPC market and use organic and inorganic strategies to strengthen their market position.

### 13.6.4 STARTING BLOCK

Starting blocks have weaker business strategies than other established vendors. They are new entrants in the market and, hence, require some time to gain significant traction.

**FIGURE 42** HPC MARKET, SME EVALUATION QUADRANT, 2021



Source: Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

**FIGURE 43** HPC MARKET: DETAILED LIST OF KEY STARTUPS/SMES

| COMPANY NAME                     | CATEGORY           | OWNER<br>SHIP<br>TYPE | HQ LOCATION                           | FOUNDING<br>YEAR | EMPLOYEES | FINANCING<br>STATUS | LATEST<br>FUNDING<br>ROUND | TOTAL<br>FUNDING<br>(USD) |
|----------------------------------|--------------------|-----------------------|---------------------------------------|------------------|-----------|---------------------|----------------------------|---------------------------|
| SiMa.ai                          | System<br>Provider | Private               | San Jose,<br>California ,US           | 2018             | 101-250   | Venture<br>Capital  | Series B                   | USD 110<br>Million        |
| Exotanium.                       | System<br>Provider | Private               | Ithaca, New<br>York, US               | 2019             | 11-50     | Venture<br>Capital  | SEED                       | USD 5.5<br>Million        |
| Inspire<br>Semiconductor,<br>Inc | System<br>Provider | Private               | Santa<br>Monica, Califor<br>nia, US   | 2020             | 101-250   | Venture<br>Capital  | Series A                   | USD<br>38.2<br>Million    |
| EdgeMode                         | System<br>Provider | Private               | Chicago,<br>Illinois, US              | 2020             | 11-50     | Venture<br>Capital  | NA                         | NA                        |
| ConstelCom                       | System<br>Provider | Private               | North Crawley,<br>Bedfordshire,<br>UK | 2020             | 1-10      | NA                  | NA                         | NA                        |
| Microsurgeonbot<br>Inc.          | System<br>Provider | Private               | Los Angeles,<br>California, US        | 2017             | 11-50     | NA                  | NA                         | NA                        |
| Rescale                          | System<br>Provider | Private               | San Francisco,<br>California, US      | 2011             | 101-250   | Venture<br>Capital  | Series C                   | USD<br>157.4<br>Million   |
| UberCloud                        | System<br>Provider | Private               | Los Altos, US                         | 2013             | 51-100    | Venture<br>Capital  | NA                         | USD 2<br>Million          |
| RStor                            | System<br>Provider | Private               | San Francisco,<br>US                  | 2017             | 11-50     | Venture<br>Capital  | NA                         | USD 45<br>Million         |
| XTREME-D                         | System<br>Provider | Private               | Tokyo ,Japan                          | 2015             | 11-50     | Venture<br>Capital  | Series A                   | USD 3.7<br>Million        |

Source: Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

**FIGURE 44** HPC MARKET: COMPETITIVE BENCHMARKING OF KEY STARTUPS/SMES

| COMPANY NAME               | VERTICAL             |                                            |                      |               |                       |                            |                    |                |        | GEOGRAPHY     |        |              |     |
|----------------------------|----------------------|--------------------------------------------|----------------------|---------------|-----------------------|----------------------------|--------------------|----------------|--------|---------------|--------|--------------|-----|
|                            | GOVERNMENT & DEFENSE | BANKING, FINANCIAL SERVICES, AND INSURANCE | EDUCATION & RESEARCH | MANUFACTURING | MEDIA & ENTERTAINMENT | HEALTHCARE & LIFE SCIENCES | ENERGY & UTILITIES | EARTH SCIENCES | OTHERS | NORTH AMERICA | EUROPE | ASIA PACIFIC | ROW |
| SiMa.ai                    | ✓                    |                                            | ✓                    | ✓             | ✓                     | ✓                          |                    |                | ✓      | ✓             |        |              |     |
| Exotanium                  |                      | ✓                                          |                      | ✓             | ✓                     |                            |                    |                | ✓      | ✓             |        |              |     |
| Inspire Semiconductor, Inc | ✓                    | ✓                                          | ✓                    | ✓             | ✓                     | ✓                          |                    | ✓              |        | ✓             |        |              |     |
| EdgeMode                   |                      | ✓                                          |                      |               |                       |                            |                    |                | ✓      | ✓             |        |              |     |
| ConstelCom                 |                      |                                            | ✓                    | ✓             | ✓                     | ✓                          | ✓                  | ✓              |        |               | ✓      |              |     |
| Microsurgeonbot Inc.       | ✓                    |                                            | ✓                    |               |                       | ✓                          |                    |                | ✓      | ✓             |        |              |     |
| Rescale                    | ✓                    | ✓                                          | ✓                    | ✓             | ✓                     | ✓                          | ✓                  | ✓              | ✓      | ✓             | ✓      |              |     |
| UberCloud                  |                      |                                            | ✓                    | ✓             | ✓                     |                            |                    |                |        | ✓             |        |              |     |
| RStor                      |                      | ✓                                          | ✓                    |               | ✓                     | ✓                          | ✓                  |                | ✓      | ✓             |        |              |     |
| XTREME-D                   |                      |                                            | ✓                    | ✓             |                       |                            | ✓                  |                | ✓      |               |        |              | ✓   |

Source: Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 86** HPC MARKET: COMPANY FOOTPRINT

| COMPANY                                   | VERTICAL FOOTPRINT | REGIONAL FOOTPRINT | COMPANY FOOTPRINT |
|-------------------------------------------|--------------------|--------------------|-------------------|
| Weightage                                 | 50%                | 50%                | 100%              |
| Advanced Micro Devices, Inc               | 4.00               | 4.00               | 4.00              |
| Intel Corporation                         | 4.00               | 4.00               | 4.00              |
| Hewlett Packard Enterprise Development LP | 3.56               | 4.00               | 3.78              |
| IBM                                       | 4.00               | 4.00               | 4.00              |
| Dell                                      | 4.00               | 4.00               | 4.00              |
| Lenovo                                    | 4.00               | 4.00               | 4.00              |
| Fujitsu                                   | 4.00               | 4.00               | 4.00              |
| Atos SE                                   | 3.56               | 4.00               | 3.78              |
| Cisco Systems, Inc.                       | 4.00               | 4.00               | 4.00              |
| NVIDIA                                    | 3.56               | 4.00               | 3.78              |
| NEC Corporation                           | 2.67               | 4.00               | 3.33              |
| Amazon Web Services, Inc.                 | 3.56               | 4.00               | 3.78              |
| Oracle                                    | 3.56               | 4.00               | 3.78              |
| Microsoft                                 | 4.00               | 4.00               | 4.00              |

|                                        |      |      |      |
|----------------------------------------|------|------|------|
| Inspur                                 | 3.11 | 4.00 | 3.56 |
| NetApp                                 | 3.11 | 4.00 | 3.56 |
| IRON Global                            | 3.56 | 4.00 | 3.78 |
| Aspen Systems Inc.                     | 3.56 | 4.00 | 3.78 |
| Advanced Clustering Technologies, Inc. | 3.11 | 4.00 | 3.56 |

Source: Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 87** HPC MARKET: COMPANY VERTICAL FOOTPRINT

| COMPANY                                   | GOVERNMENT & DEFENSE | BANKING, FINANCIAL SERVICES, AND INSURANCE | EDUCATION & RESEARCH | MANUFACTURING | MEDIA & ENTERTAINMENT | HEALTH CARE & LIFE SCIENCES | ENERGY & UTILITIES | EARTH SCIENCES | OTHERS | OVERALL VERTICAL FOOTPRINT |
|-------------------------------------------|----------------------|--------------------------------------------|----------------------|---------------|-----------------------|-----------------------------|--------------------|----------------|--------|----------------------------|
| Advanced Micro Devices, Inc               | Y                    | Y                                          | Y                    | Y             | Y                     | Y                           | Y                  | Y              | Y      | 4.00                       |
| Intel Corporation                         | Y                    | Y                                          | Y                    | Y             | Y                     | Y                           | Y                  | Y              | Y      | 4.00                       |
| Hewlett Packard Enterprise Development LP | Y                    | Y                                          | Y                    | Y             | Y                     | Y                           | Y                  | Y              | Y      | 4.00                       |
| IBM                                       | Y                    | Y                                          | Y                    | Y             | Y                     | Y                           | Y                  | Y              | Y      | 4.00                       |
| Dell                                      | Y                    | Y                                          | Y                    | Y             | Y                     | Y                           | Y                  | Y              | Y      | 4.00                       |
| Lenovo                                    | Y                    | Y                                          | Y                    | Y             | Y                     | Y                           | Y                  | Y              | Y      | 4.00                       |
| Fujitsu                                   | Y                    | Y                                          | Y                    | Y             | Y                     | Y                           | Y                  | N              | Y      | 3.56                       |
| Atos SE                                   | Y                    | Y                                          | Y                    | Y             | Y                     | Y                           | Y                  | N              | Y      | 3.56                       |
| Cisco Systems, Inc.                       | Y                    | Y                                          | Y                    | Y             | Y                     | Y                           | Y                  | N              | Y      | 3.56                       |
| NVIDIA                                    | Y                    | Y                                          | Y                    | Y             | Y                     | Y                           | Y                  | Y              | Y      | 4.00                       |
| NEC Corporation                           | Y                    | Y                                          | Y                    | N             | Y                     | Y                           | Y                  | Y              | Y      | 3.56                       |
| Amazon Web Services,                      | Y                    | Y                                          | Y                    | Y             | Y                     | Y                           | Y                  | Y              | Y      | 4.00                       |

|                                        |   |   |   |   |   |   |   |   |   |      |
|----------------------------------------|---|---|---|---|---|---|---|---|---|------|
| Inc.                                   |   |   |   |   |   |   |   |   |   |      |
| Oracle                                 | Y | Y | Y | Y | Y | Y | Y | Y | Y | 4.00 |
| Microsoft                              | Y | Y | Y | Y | Y | Y | Y | Y | Y | 4.00 |
| Inspur                                 | Y | Y | Y | Y | N | Y | Y | N | Y | 3.11 |
| NetApp                                 | Y | Y | Y | Y | N | Y | N | N | Y | 2.67 |
| IRON Global                            | Y | Y | Y | Y | Y | Y | Y | N | Y | 3.56 |
| Aspen Systems Inc.                     | Y | Y | Y | Y | Y | Y | N | Y | Y | 3.56 |
| Advanced Clustering Technologies, Inc. | Y | Y | Y | Y | N | Y | Y | N | Y | 3.11 |

Source: Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

**TABLE 88** HPC MARKET: COMPANY REGIONAL FOOTPRINT

| COMPANY                                   | NORTH AMERICA | EUROPE | ASIA PACIFIC | MIDDLE EAST & AFRICA | SOUTH AMERICA | OVERALL REGIONAL FOOTPRINT |
|-------------------------------------------|---------------|--------|--------------|----------------------|---------------|----------------------------|
| Advanced Micro Devices, Inc               | Y             | Y      | Y            | Y                    | Y             | 4.00                       |
| Intel Corporation                         | Y             | Y      | Y            | Y                    | Y             | 4.00                       |
| Hewlett Packard Enterprise Development LP | Y             | Y      | Y            | Y                    | Y             | 4.00                       |
| IBM                                       | Y             | Y      | Y            | Y                    | Y             | 4.00                       |
| Dell                                      | Y             | Y      | Y            | Y                    | Y             | 4.00                       |
| Lenovo                                    | Y             | Y      | Y            | Y                    | Y             | 4.00                       |
| Fujitsu                                   | Y             | Y      | Y            | Y                    | Y             | 4.00                       |
| Atos SE                                   | Y             | Y      | Y            | Y                    | Y             | 4.00                       |
| Cisco Systems, Inc.                       | Y             | Y      | Y            | Y                    | Y             | 4.00                       |
| NVIDIA                                    | Y             | Y      | Y            | Y                    | Y             | 4.00                       |
| NEC Corporation                           | Y             | Y      | Y            | Y                    | Y             | 4.00                       |
| Amazon Web Services, Inc.                 | Y             | Y      | Y            | Y                    | Y             | 4.00                       |
| Oracle                                    | Y             | Y      | Y            | Y                    | Y             | 4.00                       |
| Microsoft                                 | Y             | Y      | Y            | Y                    | Y             | 4.00                       |
| Inspur                                    | Y             | Y      | Y            | Y                    | Y             | 4.00                       |
| NetApp                                    | Y             | Y      | Y            | Y                    | Y             | 4.00                       |

|                                        |   |   |   |   |   |      |
|----------------------------------------|---|---|---|---|---|------|
| IRON Global                            | Y | Y | Y | Y | Y | 4.00 |
| Aspen Systems Inc.                     | Y | Y | Y | Y | Y | 4.00 |
| Advanced Clustering Technologies, Inc. | Y | Y | Y | Y | Y | 4.00 |

Source: Press Releases, Interviews with Experts, and MarketsandMarkets Analysis

## 13.7 COMPETITIVE SCENARIO

The HPC market is dynamic, the reason being the growing investment in HPC technology and the emerging uses cases across the industry segments. This section represents the growth strategies adopted by leading players from March 2019 to March 2022 to expand their global presence and increase their market shares. Key strategies include product launches, agreements, and partnerships.

**TABLE 89** HPC MARKET: PRODUCT LAUNCHES, MARCH 2019–MARCH 2022

| MONTH AND YEAR | COMPANY NAME | PRODUCT TYPE             | PRODUCT NAME                               | PRODUCT DESCRIPTION                                                                                                                                                                                                                                                     |
|----------------|--------------|--------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| March 2022     | AMD          | GPU                      | AMD Radeon PRO W6600X GPU                  | AMD has launched AMD Radeon PRO W6600X GPU. The GPU is built on AMD RDNA 2 architecture and AMD infinity Cache. The GPU provides excellent visual rendering with a decent performance/power ratio                                                                       |
| June 2021      | Intel        | Processor                | Latest 3rd Gen Intel Xeon Scalable         | Intel enhanced its Intel Xeon Scalable Processor lineup by launching 3rd Gen Intel Xeon Scalable processors. They exhibit high performance for HPC workloads with increased memory bandwidth, new core architecture, improved processor cores, and faster input/output. |
| November 2020  | AMD          | Processor                | AMD Ryzen Embedded V2000 series processor. | AMD launched the AMD Ryzen Embedded V2000 series processor developed using 7 nm process technology. The processor is used for embedded applications, such as Thin Client, MiniPC, and Edge systems.                                                                     |
| June 2019      | IBM          | Commercial supercomputer | Pangea III                                 | IBM launched Pangea III, a powerful commercial supercomputer for a global energy company. IBM's POWER9-based supercomputer Pangea III accurately locates resources and can access potential revenue opportunities.                                                      |

Source: Press Releases

**TABLE 90** HPC: DEALS, MARCH 2019–MARCH 2022

| MONTH AND YEAR | DEAL TYPE   | COMPANY 1 NAME | COMPANY 2 NAME         | DESCRIPTION                                                                                                                                                                                                                                                                 | DEAL SIZE      |
|----------------|-------------|----------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| February 2022  | Acquisition | AMD            | Xilinx (US)            | AMD has completed the acquisition of Xilinx. The acquisition has led to the expansion of AMD's product portfolio and customer base across several regions.                                                                                                                  | USD 35 Billion |
| February 2022  | Acquisition | IBM            | Sentaca (US)           | IBM acquired Sentaca, a software consulting firm and cloud migration specialist that supports telcos and communication service providers. Sentaca, in conjunction with IBM, will provide scalable, cloud- and edge-enabled process automation and process safety solutions. | NA             |
| January 2022   | Acquisition | Atos           | Cloudreach (UK)        | Atos acquired Cloudreach, a leading multi-cloud services company specializing in public cloud application development and cloud migration.                                                                                                                                  | NA             |
| November 2021  | Acquisition | IBM            | SXiQ (Australia)       | IBM acquired SXiQ, a multi-cloud MSP and IT consulting firm for AWS and Azure migrations company. This acquisition expands customer reach of IBM in cloud business.                                                                                                         | NA             |
| October 2020   | Partnership | Dell           | Verne Global (Iceland) | Dell partnered with Verne Global, a technology company that offers data centers. The partnership will help both the companies accelerate their client's business operations by leveraging the capabilities of HPC and AI                                                    | NA             |

Note: NA=Not Available

Source: Press Releases

## 14 COMPANY PROFILES

### 14.1 KEY PLAYERS

#### 14.1.1 ADVANCED MICRO DEVICES

##### 14.1.1.1 Business overview

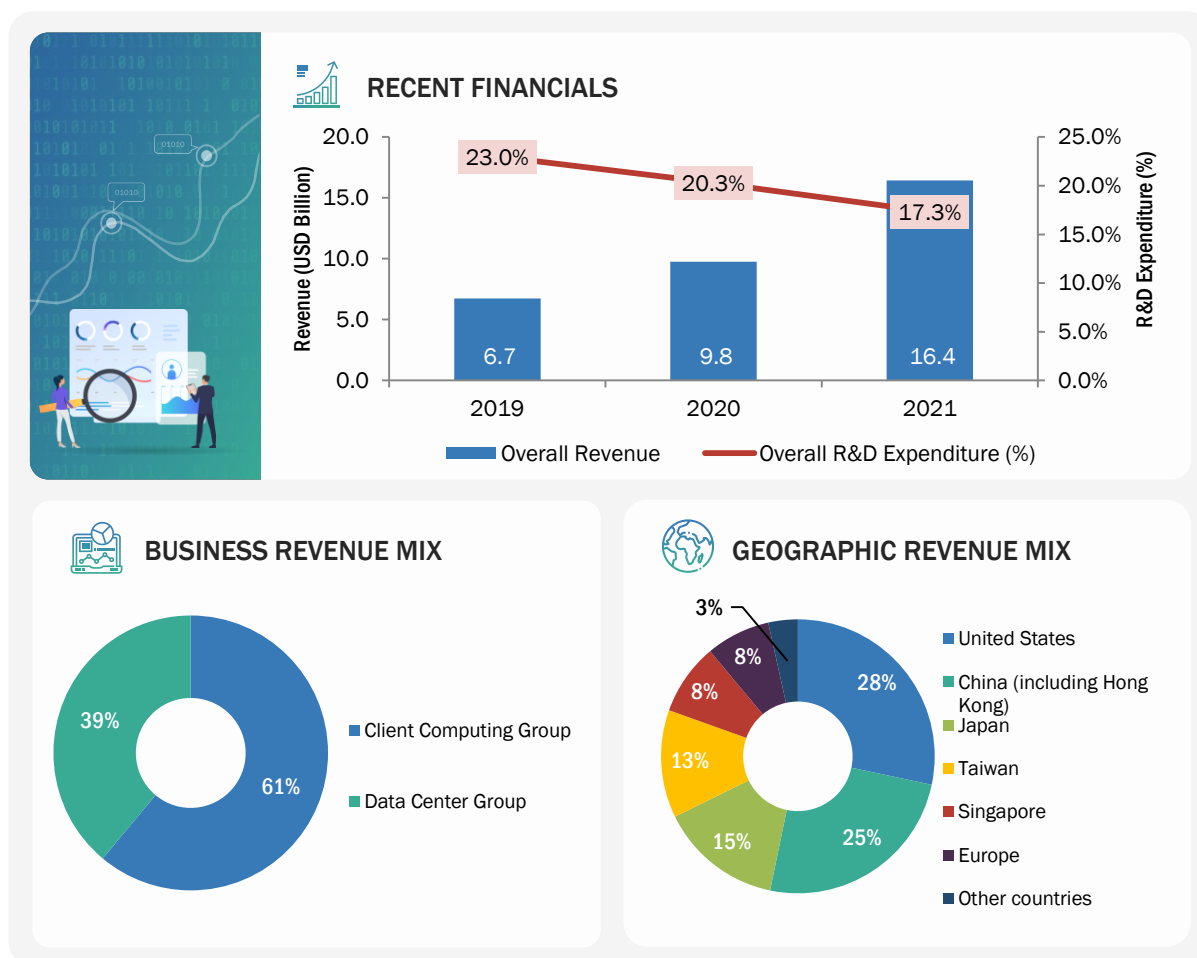
Advanced Micro Devices (AMD) is a global semiconductor manufacturing company. It focuses on providing high-performance computing technologies, software, and products such as supercomputers, AMD CDNA 2 Architecture, AMD Infinity Architecture, and AMD ROCm. AMD primarily offers the x86 microprocessor as a standalone device or incorporated into an Accelerated Processing Unit (APU); discrete Graphics Processing Units (GPUs), chipsets, and professional graphics; and server and embedded processors and semi-custom System-on-Chip (SoC) products and technologies for game consoles. The company also licenses sections of its intellectual property.

AMD sells its products through its direct sales force, sales representatives, and independent distributors. The company primarily caters to original design manufacturers (ODMs), original equipment manufacturers (OEMs), system builders, and independent distributors. It has a strong presence in the US, China, and Japan. The company has a vast supply chain network spread across North America, Europe, and APAC.

**TABLE 91** ADVANCED MICRO DEVICES: BUSINESS OVERVIEW

|                      |                              |
|----------------------|------------------------------|
| Founding Year        | 1969                         |
| Headquarters Country | US                           |
| Headquarters City    | Sunnyvale                    |
| Ownership Type       | Public                       |
| Employees            | 15,500 (As of December 2021) |

Source: Company Website

**FIGURE 45** ADVANCED MICRO DEVICES: COMPANY SNAPSHOT

Note: The "recent financials" bar chart represents the total revenue. In the recent financials block, the R&D expenditure has been considered a percentage of the company's total revenue. The fiscal year of the company ends on December 31. Due to rounding off, percentages presented in the pie charts may not reflect precise segmental revenues; therefore, a slight variation in segmental revenues may be observed.

## 14.1.1.2 Products offered

**TABLE 92** ADVANCED MICRO DEVICES: PRODUCTS OFFERED

| PRODUCT                   | DESCRIPTION                                                                                                                                                                                                                                            | PRODUCT TYPE                   | APPLICATION                                                                                                      |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------|
| AMD INSTINCT              | AMD Instinct is engineered for data center computing, supercharging high-performance computing (HPC), and AI workloads to boost discoveries. The AMD Instinct can provide services from single server solutions to the world's largest supercomputers. | Accelerator                    | <ul style="list-style-type: none"> <li>Data Center</li> <li>HPC</li> <li>Artificial intelligence (AI)</li> </ul> |
| AMD CDNA Architecture     | AMD CDNA system architecture is designed for extreme scalability and exhibits high performance over the previous generation accelerators.                                                                                                              | HPC Offering                   | <ul style="list-style-type: none"> <li>Data Center</li> <li>HPC</li> <li>AI</li> </ul>                           |
| AMD Infinity Architecture | AMD Infinity Architecture offers an advanced platform for GPU's to work in tandem with each other helping them to share data efficiently.                                                                                                              | HPC Offerings                  | <ul style="list-style-type: none"> <li>Data Center</li> <li>HPC</li> <li>AI</li> </ul>                           |
| AMD ROCm                  | AMD ROCm is an open-source software development platform for HPC/Hyperscale-class GPU computing.                                                                                                                                                       | HPC Offerings                  | <ul style="list-style-type: none"> <li>Data Center</li> <li>HPC</li> <li>AI</li> </ul>                           |
| AMD EPYC processor        | AMD EPYC is the world's highest-performing x86 server CPU.                                                                                                                                                                                             | HPC Offerings                  | <ul style="list-style-type: none"> <li>Data Center</li> <li>HPC</li> <li>AI</li> </ul>                           |
| AMD CPUs and GPUs         | The central processing unit (CPU) generally comprises the main memory, control unit, and arithmetic-logic unit.                                                                                                                                        | Central Processing Unit (CPU)  | <ul style="list-style-type: none"> <li>Data Center</li> <li>HPC</li> </ul>                                       |
| AMD GPUs                  | A graphics processing unit (GPU) is a specialized electronic circuit designed to rapidly manipulate and alter memory to accelerate the creation of images.                                                                                             | Graphics Processing Unit (GPU) | <ul style="list-style-type: none"> <li>Data Center</li> <li>HPC</li> </ul>                                       |

Source: Company Website

### 14.1.1.3 Recent developments

#### 14.1.1.3.1 Advanced Micro Devices: product launches

| MONTH & YEAR  | COMPANY | PRODUCT TYPE | PRODUCT NAME                              | PRODUCT DESCRIPTION                                                                                                                                                                                                          |
|---------------|---------|--------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| March 2022    | AMD     | GPU          | AMD Radeon PRO W6600X GPU                 | AMD has launched AMD Radeon PRO W6600X GPU. The GPU is built on AMD RDNA 2 architecture and AMD infinity Cache. The GPU provides excellent visual rendering with a decent performance/power ratio                            |
| January 2022  | AMD     | Processor    | AMD Ryzen 6000                            | AMD launched its AMD Ryzen 6000 processor powered by its Zen 3+ core. Zen 3+ core is built on AMD RDNA 2 architecture with integrated graphics. The processor delivers optimal gaming performance with a decent battery life |
| November 2020 | AMD     | Accelerator  | AMD Instinct MI100 with ROCm 4.0.         | AMD launched the AMD Instinct MI100 accelerator with the ROCm 4.0 open ecosystem to support scientific workloads.                                                                                                            |
| November 2020 | AMD     | Processor    | AMD Ryzen Embedded V2000 Series Processor | AMD launched the AMD Ryzen Embedded V2000 series processor developed using 7 nm process technology. The processor is used for embedded applications, such as Thin Client, MiniPC, and Edge systems.                          |

Source: Company Website and Press Releases

#### 14.1.1.3.2 Advanced Micro Devices: Deals

| MONTH & YEAR  | DEAL TYPE     | COMPANY 1 | COMPANY 2 | DESCRIPTION                                                                                                                                                                                                                                                               | DEAL SIZE      |
|---------------|---------------|-----------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| February 2022 | Acquisition   | AMD       | Xilinx    | AMD has completed the acquisition of Xilinx. The acquisition has led to the expansion of AMD's product portfolio and customer base across several regions.                                                                                                                | USD 35 Billion |
| November 2021 | Collaboration | AMD       | MediaTek  | AMD signed a collaboration agreement with MediaTek, an industry-leading Wi-Fi solutions provider. Under this collaboration, AMD's RZ600 Series Wi-Fi 6E modules are integrated with MediaTek's new Filogic 330P chipset.                                                  | NA             |
| November 2020 | Agreement     | AMD       | IBM       | AMD and IBM signed an agreement to enhance and extend the security and AI offerings of both companies. The joint development enhanced and supported a broad range of accelerators across HPC and enterprise-critical capabilities, such as virtualization and encryption. | NA             |

Source: Company Website and Press Releases

#### 14.1.1.4 MnM view

##### 14.1.1.4.1 Key strengths/Right to win

AMD offers the AMD Opteron processor family with up to 16 cores and four memory channels to deliver high-compute capability, performance, and flexibility. These processors find applications in handling dense and multi-threaded workloads with large, complex datasets associated with supercomputing and HPC. The company also offers the AMD Ryzen processor, AMD CDNA 2 Architecture, AMD Infinity Architecture, and AMD ROCm family that caters to the HPC market. Its high-performance x86 Core “Zen” architecture is an x86 processor designed for HPC products. AMD provides AMD GPUs, which cater to compute-intensive applications and assist scientists and engineers in getting quick results. The company also provides EPYC server processors and Radeon Instinct GPU accelerators. These products cater to the demands of heterogeneous computing for machine intelligence and HPC.

##### 14.1.1.4.2 Strategic choices made

AMD is a major contributor to the HPC market and intends to become a leader in the HPC and graphics markets in the coming years. The company strives to re-enter the high-performance and server computing market with the x86 processor Core “Zen.” To increase its focus on designing high-performance products, the company adopts inorganic strategies such as partnerships, agreements, and acquisitions. For instance, in February 2022, AMD completed the acquisition of Xilinx to expand its product portfolio and customer base across the globe. In AMD’s product offerings are used to power many supercomputers, enabling critical research and operations for a variety of scientific research across several industries. The company has won many awards for its HPC portfolio; this highlights its focus on innovation in the HPC market.

##### 14.1.1.4.3 Weaknesses and competitive threats

AMD relies on third-party manufacturers and suppliers to ensure adequate product supply and cater to customer demand. The company is highly dependent on non-US operations to generate a major portion of its overall revenue. For instance, it generated around 71% of its overall revenue from non-US operations in 2021.

## 14.1.2 INTEL

### 14.1.2.1 Business overview

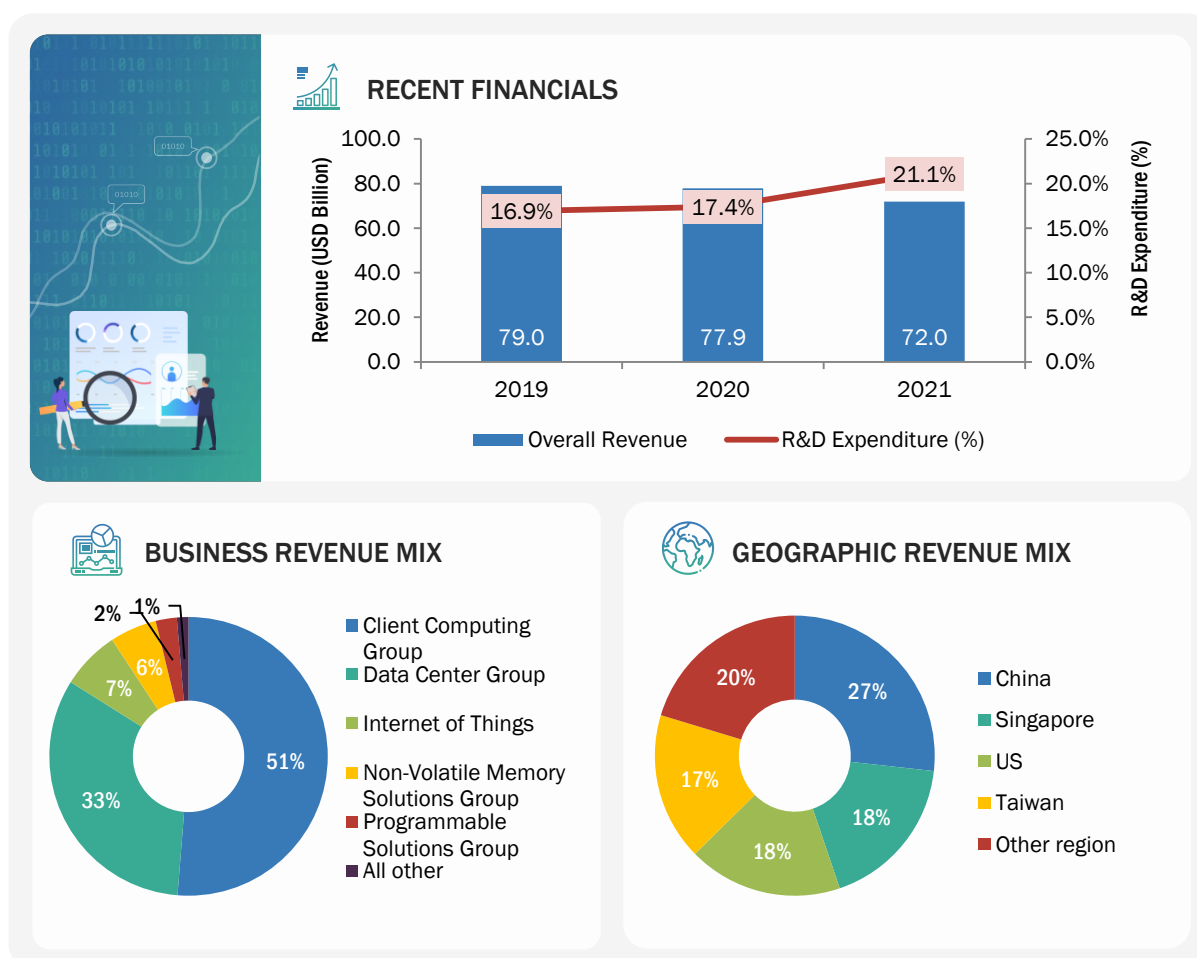
Intel is a leading player in the field of high-performance computing. It was founded in 1968 and is headquartered in Santa Clara, California, US. The company offers its networking, computer, and communication platforms to many customers, including original design manufacturers (ODMs), original equipment manufacturers (OEMs), cloud service providers, and communication, industrial, and automobile equipment manufacturers. Intel designs and develops key products and technologies that power cloud solutions across key industry verticals. The company is a pioneer in high-performance computing. It executes projects across industries such as oil & gas, chemical & petrochemical, power, life sciences, and metals.

Intel has a number of established sales and business development offices across 50+ countries across North America, Europe, APAC, and the Middle East. The company's network of resellers and distributors is spread across 80+ countries, offering various innovative products across matured as well as growing markets.

**TABLE 93** INTEL: BUSINESS OVERVIEW

|                      |                               |
|----------------------|-------------------------------|
| Founding Year        | 1968                          |
| Headquarters Country | US                            |
| Headquarters City    | Santa Clara                   |
| Ownership Type       | Public                        |
| Employees            | 121,100 (As of December 2021) |

Source: Company Website

**FIGURE 46** INTEL: COMPANY SNAPSHOT

Note: The "recent financials" bar chart represents the total revenue. In the recent financials block, the R&D expenditure has been considered a percentage of the company's total revenue. The fiscal year of the company ends on December 31.

Due to rounding off, percentages presented in the pie charts may not reflect precise segmental revenues; therefore, a slight variation in segmental revenues may be observed.

Source: Company Website, Press Releases, and MarketsandMarkets Analysis

### 14.1.2.2 Products offered

**TABLE 94** INTEL: PRODUCT OFFERINGS

| PRODUCT                        | DESCRIPTION                                                                                                                     | PRODUCT TYPE | APPLICATION                                                                                                                   |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------|
| Intel Xeon Scalable Processors | Intel Xeon Scalable processor provides scalability and optimum performance for compute, storage, memory, network, and security. | HPC Offering | <ul style="list-style-type: none"> <li>High-end Military Servers</li> <li>Industrial Servers</li> <li>Workstations</li> </ul> |
| Intel Deep Learning Boost      | Intel Xeon Scalable processors are built specifically to run complex AI-powered solutions on existing hardware.                 | HPC Offering | <ul style="list-style-type: none"> <li>Network Infrastructure</li> <li>Data Center</li> <li>Telecommunication</li> </ul>      |
| Intel Optane Technology        | Intel Optane memory recognizes, remembers, and provides speedy access to frequently used files, applications, and games.        | HPC Offering | <ul style="list-style-type: none"> <li>Network Infrastructure</li> <li>Data Center</li> <li>Telecommunication</li> </ul>      |

|                              |                                                                                                            |              |                                                                                                                          |
|------------------------------|------------------------------------------------------------------------------------------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------|
| Intel Omni-Path Architecture | Intel Omni-Path Architecture has been designed to meet the requirements of small- to large-scale clusters. | HPC Offering | <ul style="list-style-type: none"> <li>Network Infrastructure</li> <li>Data Center</li> <li>Telecommunication</li> </ul> |
| Intel FPGAs                  | Intel FPGAs move and store more data with greater efficiency.                                              | HPC Offering | <ul style="list-style-type: none"> <li>Network Infrastructure</li> <li>Data Center</li> <li>Telecommunication</li> </ul> |
| HPC Software and Tools       | HPC software and tools are developed to verify cluster components working together seamlessly.             | HPC Offering | <ul style="list-style-type: none"> <li>Network Infrastructure</li> <li>Data Center</li> <li>Telecommunication</li> </ul> |
| Intel Select Solutions       | Intel Select Solutions optimize workload configurations and integrate the latest hardware and software.    | HPC Offering | <ul style="list-style-type: none"> <li>Network Infrastructure</li> <li>Data Center</li> <li>Telecommunication</li> </ul> |

Source: Company Website

### 14.1.2.3 Recent developments

#### 14.1.2.3.1 Intel: Product launches

| MONTH & YEAR  | COMPANY | PRODUCT TYPE | PRODUCT NAME                       | PRODUCT DESCRIPTION                                                                                                                                                                                                                                                                             |
|---------------|---------|--------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| June 2021     | Intel   | Processor    | Latest 3rd Gen Intel Xeon Scalable | Intel enhanced its Intel Xeon Scalable Processor lineup by launching 3rd Gen Intel Xeon Scalable processors. They exhibit high performance for HPC workloads with increased memory bandwidth, new core architecture, improved processor cores, and faster input/output.                         |
| November 2020 | Intel   | GPU          | Intel oneAPI Toolkits              | Intel released Intel oneAPI toolkits with new capabilities. The capabilities include discrete Graphics Processing Unit (GPU) for data centers and the Intel Server GPU microarchitecture. They are explicitly designed for high-density, low-latency Android cloud gaming, and media streaming. |

#### 14.1.2.3.2 Intel: Deals

| MONTH & YEAR  | DEAL TYPE   | COMPANY 1         | COMPANY 2                    | DESCRIPTION                                                                                                                                                                                                                                 | DEAL SIZE |
|---------------|-------------|-------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| February 2022 | Acquisition | Intel Corporation | Tower Semiconductor (Israel) | Intel has acquired Tower Semiconductor. This acquisition has helped Intel integrate its products with Tower Semiconductors' complementary products. This collaboration also helped increase Intel's geographic presence in the Middle East. | NA        |

|               |               |                   |                      |                                                                                                                                                                                                                                                                                                                                                                                          |    |
|---------------|---------------|-------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| November 2020 | Collaboration | Intel Corporation | HPE and Argonne (US) | Intel collaborated with HPE, an American multinational enterprise information technology company, and Argonne, a science and engineering research national laboratory, which is an engineering research wing of the US Department of Energy. These companies collaborated to develop Aurora, an Exascale computer system to make scientific breakthroughs, innovations, and discoveries. | NA |
|---------------|---------------|-------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|

Source: Company Website and Press Releases

#### 14.1.2.4 MnM view

##### 14.1.2.4.1 Key strength/right to win

Intel Xeon Scalable processors provide workload-optimized performance and throughput for advanced analytics, data compression, and HPC applications. The company's Xeon Phi processor is a bootable host processor delivering massive vectorization and parallelism to support demanding HPC applications.

##### 14.1.2.4.2 Strategic choices made

Intel's core strategic focus is directed toward harnessing the value of Moore's law by creating devices with additional functionalities while controlling cost, size, and power. It is consistently focused on growth and has invested heavily in R&D to fuel its drive for developing new and innovative products. The company continues to invest over 18–20% of its revenue in R&D activities. Intel's focus on the development of new products is evident from the fact that it launched the Intel Xeon Scalable processors for catering to the HPC environment.

##### 14.1.2.4.3 Weaknesses and competitive threats

The fluctuations in the market values of domestic as well as international investments may affect negatively by liquidity, credit deterioration or losses, interest rate changes, financial results, and political risks. Defective products or errata may damage the company's reputation and business in the market. The risks associated with organic and inorganic growth strategies could fail to achieve the company's financial objectives and adversely impact business operations.

### 14.1.3 HEWLETT PACKARD ENTERPRISE (HPE)

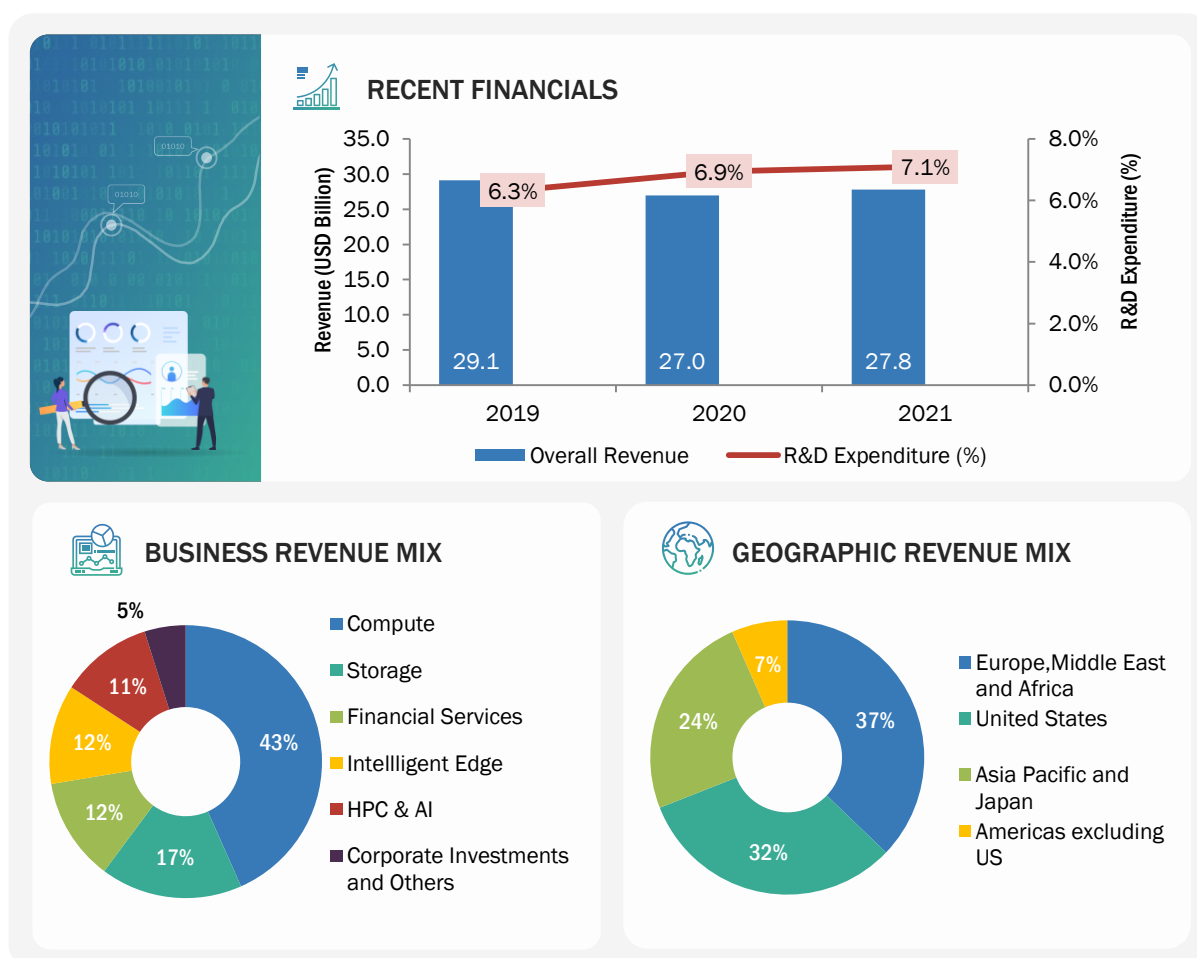
#### 14.1.3.1 Business overview

Hewlett Packard Enterprise (HPE) was founded in 2015 and is headquartered in California, US. It is a public company listed on NYSE: HPE. In November 2015, HPE split off as an independent company from Hewlett Packard (HP) Company. HPE provides IT solutions to businesses and public enterprises. It operates through 6 segments: Compute, Storage, Financial Services, Intelligent Edge, HPC & AI, and Corporate Investments and Others. It offers a diversified portfolio of products and services. The Enterprise Group segment provides the infrastructure needed to optimize IT environments, as well as several products and services, such as servers, networking, storage, and technology services. The company has a major presence in the US, and offices in several regions, including Africa, APAC, Europe, and the Middle East.

**TABLE 95** HEWLETT PACKARD ENTERPRISE: BUSINESS OVERVIEW

|                      |                             |
|----------------------|-----------------------------|
| Founding Year        | 2015                        |
| Headquarters Country | US                          |
| Headquarters City    | Spring City                 |
| Ownership Type       | Public                      |
| Employees            | 60,400 (As of October 2021) |

Source: Company Website

**FIGURE 47** HEWLETT PACKARD ENTERPRISE: COMPANY SNAPSHOT

Note: The "recent financials" bar chart represents the total revenue. In the recent financials block, the R&D expenditure has been considered a percentage of the company's total revenue. The fiscal year of the company ends on October 31.

Due to rounding off, percentages presented in the pie charts may not reflect precise segmental revenues; therefore, a slight variation in segmental revenues may be observed.

Source: Company Website, Press Releases, and MarketsandMarkets Analysis

#### 14.1.3.2 Products offered

**TABLE 96** HEWLETT PACKARD ENTERPRISE: PRODUCT OFFERINGS

| PRODUCT                            | DESCRIPTION                                                                                                                  | PRODUCT TYPE | APPLICATION                                                                                                              |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------|
| HPE Apollo                         | The HPE Apollo high-density server family is developed to deliver the highest levels of performance and efficiency.          | Servers      | <ul style="list-style-type: none"> <li>Network Infrastructure</li> <li>Data Center</li> <li>Telecommunication</li> </ul> |
| HPE Performance Cluster Management | HPE Performance Cluster Manager enables complete provisioning, management, and monitoring clusters scaling to 100,000 nodes. | Software     | <ul style="list-style-type: none"> <li>Network Infrastructure</li> <li>Data Center</li> <li>Telecommunication</li> </ul> |
| InfiniBand                         | InfiniBand are switches that deliver complete chassis and fabric management in networking.                                   | Networking   | <ul style="list-style-type: none"> <li>Network Infrastructure</li> <li>Data Center</li> <li>Telecommunication</li> </ul> |

Source: Company Website

### 14.1.3.3 Recent developments

#### 14.1.3.3.1 Hewlett Packard Enterprise: Product launches

| MONTH & YEAR   | COMPANY | PRODUCT TYPE   | PRODUCT NAME                         | PRODUCT DESCRIPTION                                                                                                               |
|----------------|---------|----------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| September 2021 | HPE     | Cloud Platform | HPE GreenLake Edge-to-Cloud Platform | HPE launched HPE GreenLake Edge-to-Cloud Platform. It provides data modernization, analytics, and data protection cloud services. |

Source: Company Website and Press Releases

#### 14.1.3.3.2 Hewlett Packard Enterprise: Deals

| MONTH & YEAR   | DEAL TYPE   | COMPANY 1 | COMPANY 2                    | DESCRIPTION                                                                                                                                                                                       | DEAL SIZE |
|----------------|-------------|-----------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| September 2021 | Acquisition | HPE       | Zerto (US)                   | HPE acquired Zerto. This acquisition has helped the company to expand its cloud data management and protection product portfolio.                                                                 | NA        |
| July 2020      | Partnership | HPE       | University of Edinburgh (UK) | HPE partnered with the University of Edinburgh, EPCC, Scotland. HPE offers end-to-end HPC infrastructure, as well as AI solutions , powered by HPE Apollo Systems and HPE Superdome Flex Servers. | NA        |

Source: Company Website and Press Releases

### 14.1.3.4 MnM view

#### 14.1.3.4.1 Key strengths/right to win

HPE provides a range of HPC solutions to cater to the growing customer requirements. The company offers Superdome Flex, SGI 8600, Apollo 6000, Apollo 6500, Apollo 4000, Apollo 2000, and Apollo 10 systems to provide HPC solutions tailored as per customer needs. These solutions are used to solve large scientific, engineering, and data analysis problems. HPE invests heavily in R&D activities. In 2019, the company's R&D expenditure amounted to USD 1.8 billion. The growth in R&D expenditure is primarily due to the increase in networking and technology services in the enterprise group segment. This increase in expenditure assists the company in enhancing its product and services portfolio.

#### 14.1.3.4.2 Strategic choices made

HPE made a major advancement in the HPC market by acquiring Cray. This extended and strengthened HPE's current position in the HPC market. HPE adopts effective inorganic growth strategies, such as partnerships, collaborations, and acquisitions, to maintain a leadership position in the market. For instance, in July 2020, HPE partnered with the University of Edinburgh EPCC in Scotland. HPE offers end-to-end infrastructure, such as HPE and AI solutions, powered by HPE Apollo Systems and HPE Superdome Flex Servers. These organic growth strategies help the company consolidate its presence in the Americas and Europe.

#### 14.1.3.4.3 Weaknesses and competitive threats

As a result of the company's international business operations, political and economic changes may harm its revenue, costs, expenses, and financial position. The company is dependent on third-party suppliers, and its financial condition could suffer if it fails to manage suppliers.

## 14.1.4 IBM

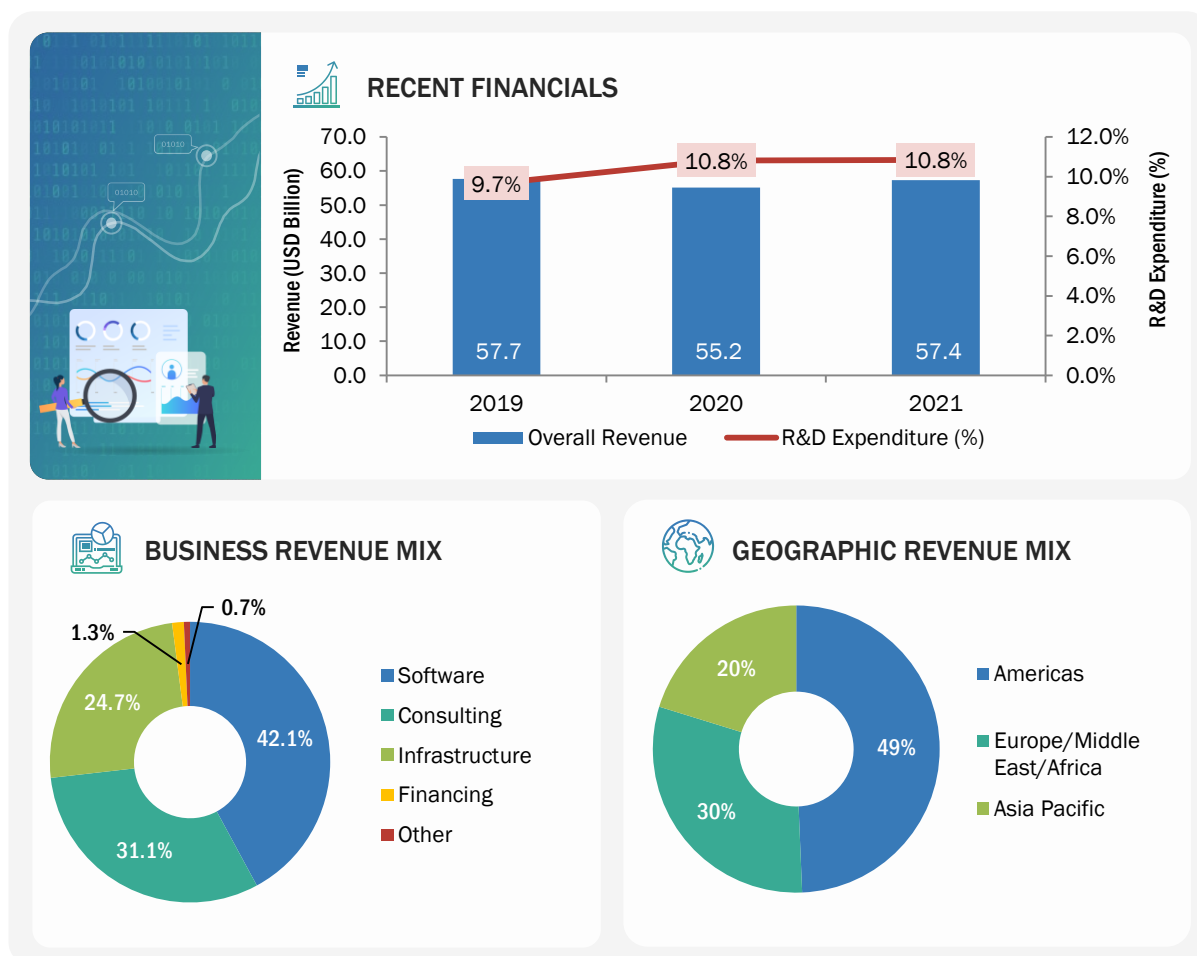
### 14.1.4.1 Business overview

Founded in 1911 and headquartered in New York, US, IBM offers a wide range of IT services, software, systems, and fundamental research and financial services to its commercial clients. The company operates through the Software, Consulting, Infrastructure, Financing, and Other segments. It offers a comprehensive set of unified solutions and services to various verticals, such as aerospace & defense, automotive, banking, chemical & petroleum, communications, consumer goods, and education. Moreover, the company also caters to the electronics, energy & utilities, finance, government, healthcare, insurance, life sciences, media & entertainment, metals & mining, retail, and travel & transportation verticals. IBM majorly operates in APAC, Europe, and North America, with a presence in more than 170 countries. It has an employee base of 364,800 people as of 2021.

**TABLE 97** IBM: BUSINESS OVERVIEW

|                      |                               |
|----------------------|-------------------------------|
| Founding Year        | 1911                          |
| Headquarters Country | US                            |
| Headquarters City    | Armonk, New York              |
| Ownership Type       | Public                        |
| Employees            | 364,800 (As of December 2021) |

Source: Company Website

**FIGURE 48** IBM: COMPANY SNAPSHOT

Note: The "recent financials" bar chart represents the total revenue, and the R&D expenditure has been considered a percentage of the company's total revenue. The fiscal year of the company ends on December 31.

Due to rounding off, percentages presented in the pie charts may not reflect precise segmental revenues; therefore, a slight variation in segmental revenues may be observed.

Source: Company Website, Press Releases, and MarketsandMarkets Analysis

#### 14.1.4.2 Products offered

**TABLE 98** IBM: PRODUCT OFFERINGS

| PRODUCT             | DESCRIPTION                                                                                  | PRODUCT TYPE    | APPLICATION                                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------|
| IBM POWER9          | POWER9 is a family of superscalar, multithreading, and multi-core microprocessors.           | Microprocessors | <ul style="list-style-type: none"> <li>Network infrastructure</li> <li>Data center</li> <li>Telecommunication</li> </ul> |
| IBM Elastic Storage | IBM Elastic Storage System (ESS) comprises of capacity nodes and low latency analysis nodes. | Storage System  | <ul style="list-style-type: none"> <li>Network infrastructure</li> <li>Data center</li> <li>Telecommunication</li> </ul> |
| IBM Aspera          | IBM Aspera is an interface to share and access files.                                        | Platform        | <ul style="list-style-type: none"> <li>Network infrastructure</li> <li>Data center</li> <li>Telecommunication</li> </ul> |

|                          |                                                                                                                                             |                              |                                                                                                                          |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| IBM Cloud Object Storage | IBM Cloud Object Storage is a service offered by IBM for storing and accessing unstructured data.                                           | Cloud Storage                | <ul style="list-style-type: none"> <li>Network infrastructure</li> <li>Data center</li> <li>Telecommunication</li> </ul> |
| IBM Spectrum Symphony    | IBM Spectrum Symphony is an enterprise-class management system for operating compute-intensive and data-intensive distributed applications. | Application                  | <ul style="list-style-type: none"> <li>Network infrastructure</li> <li>Data center</li> <li>Telecommunication</li> </ul> |
| IBM Spectrum LSF         | IBM Spectrum LSF is a workload management platform.                                                                                         | workload management platform | <ul style="list-style-type: none"> <li>Network infrastructure</li> <li>Data center</li> <li>Telecommunication</li> </ul> |

Source: Company Website

### 14.1.4.3 Recent developments

#### 14.1.4.3.1 IBM: Product launches

| MONTH & YEAR  | COMPANY | PRODUCT TYPE                         | PRODUCT NAME | PRODUCT DESCRIPTION                                                                                                                                                                                                |
|---------------|---------|--------------------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| December 2021 | IBM     | IBM Z and Cloud Modernization Centre | IBM Z        | IBM has introduced the IBM Z and Cloud Modernization Centre software integrated with a vast array of tools, training modules, and other resources.                                                                 |
| June 2019     | IBM     | Commercial Supercomputer             | Pangea III   | IBM launched Pangea III, a powerful commercial supercomputer for a global energy company. IBM's POWER9-based supercomputer Pangea III accurately locates resources and can access potential revenue opportunities. |

Source: Company Website and Press Releases

#### 14.1.4.3.2 IBM: Deals

| MONTH & YEAR   | DEAL TYPE   | COMPANY 1 | COMPANY 2         | DESCRIPTION                                                                                                                                                                                                                                                                 | DEAL SIZE |
|----------------|-------------|-----------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| February 2022  | Acquisition | IBM       | Sentaca (US)      | IBM acquired Sentaca, a software consulting firm and cloud migration specialist that supports telcos and communication service providers. Sentaca, in conjunction with IBM, will provide scalable, cloud- and edge-enabled process automation and process safety solutions. | NA        |
| November, 2021 | Acquisition | IBM       | SXiQ, (Australia) | IBM acquired SXiQ, a leading cloud and digital services specialist. This acquisition has expanded IBM's customer base in the cloud business.                                                                                                                                | NA        |

Source: Company Website and Press Releases

#### 14.1.4.4 MNM view

##### 14.1.4.4.1 Key strengths/right to win

IBM is an eminent player in the HPC market, as it provides product offerings in the market across the globe. The company is currently focusing on catering to its customers with highly competent technology-enabled products and services to sustain competition. It aims at delivering innovative solutions and improved business continuity while planned and unplanned IT outages, thereby satisfying its customers.

##### 14.1.4.4.2 Strategic choices made

The company focuses on strengthening its organic growth strategies, such as constant product enhancements and new product launches. These growth strategies have helped IBM hold its position in the HPC market, while they have also helped the company contentiously offer improved HPC products to customers. The company's growth strategy is to deliver high-value solutions to enterprise clients across all industry verticals. IBM focuses on delivering innovations in technology with superior solutions that enhance client and industrial outcomes.

##### 14.1.4.4.3 Weaknesses and competitive threats

The company was overdependent on the Americas for revenue generation in 2021. Its revenue generated from the Americas is almost 49% of its overall revenue. This may affect the company's growth in other regions.

## 14.1.5 DELL

### 14.1.5.1 Business overview

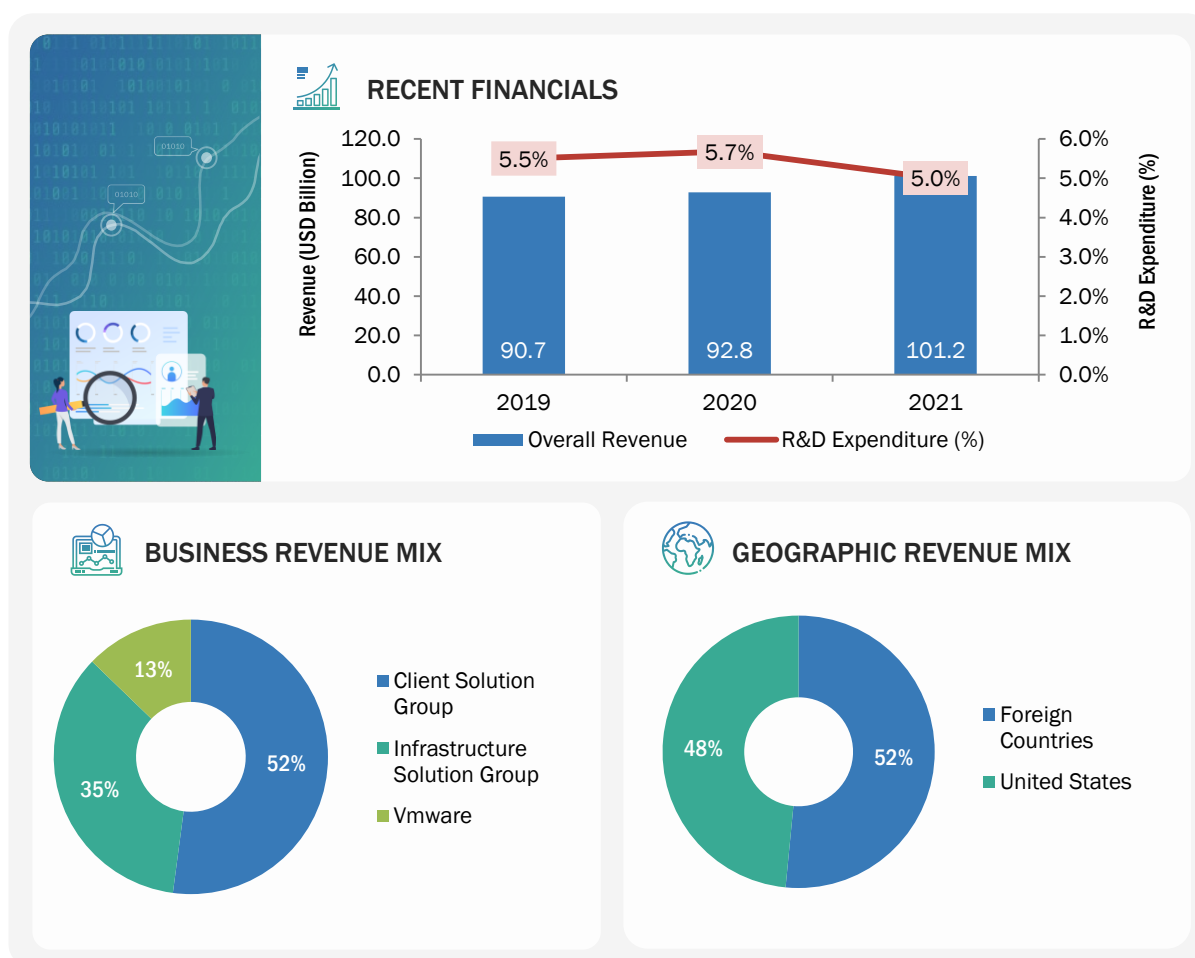
Dell, headquartered in Round Rock, Texas, US, was founded by the merger of EMC Corporation and Dell Computers. It is one of the major manufacturers and providers of computer systems and services. The company provides data center solutions that help enterprises transform their business. Dell's solutions for data centers include servers, networking, storage, HPC, converged solutions, enterprise management systems, data virtualization systems, data center managed services, performance monitoring systems, and Windows servers.

Dell has a presence across all major regions, including North America, Europe, APAC, and the Middle East & Africa. The company has operations in more than 100+ countries and a vast network of distributors and resellers across matured and developing markets.

**TABLE 99**     **DELL: BUSINESS OVERVIEW**

|                      |                              |
|----------------------|------------------------------|
| Founding Year        | 2016                         |
| Headquarters Country | US                           |
| Headquarters City    | Texas                        |
| Ownership Type       | Public                       |
| Employees            | 158,000 (As of January 2022) |

Source: Company Website

**FIGURE 49** DELL: COMPANY SNAPSHOT

Note: The "recent financials" bar chart represents the total revenue, and the R&D expenditure has been considered a percentage of the company's total revenue. The fiscal year of the company ends on January 31. Due to rounding off, percentages presented in the pie charts may not reflect precise segmental revenues; therefore, a slight variation in segmental revenues may be observed.

Source: Company Website, Press Releases, and MarketsandMarkets Analysis

#### 14.1.5.2 Products offered

**TABLE 100** DELL: PRODUCT OFFERINGS

| PRODUCT                          | DESCRIPTION                                                                                                                                                                                        | PRODUCT TYPE | APPLICATION                                                                                                                                                      |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dell EMC Ready Solutions for HPC | Dell EMC Ready Solutions for high-performance computing enable customers to achieve faster time to profit or discovery with the comprehensive, modular, platform- and data-agnostic HPC solutions. | Application  | <ul style="list-style-type: none"> <li>Data storage</li> <li>Information Security</li> <li>Virtualization</li> <li>Analytics</li> <li>Cloud Computing</li> </ul> |
| Dell EMC PowerEdge server        | Dell EMC PowerEdge server is scalable business architecture, intelligent automation, and integrated security for workloads.                                                                        | Server       | <ul style="list-style-type: none"> <li>Network infrastructure</li> <li>Data center</li> <li>Telecommunication</li> </ul>                                         |
| Bright Cluster Manager           | Bright Cluster Manager provides infrastructure management technology for Dell's HPC Systems.                                                                                                       | HPC Offering | <ul style="list-style-type: none"> <li>Network infrastructure</li> </ul>                                                                                         |

- Data center
- Telecommunication

Source: Company Website

### 14.1.5.3 Recent developments

#### 14.1.5.3.1 DELL: Product launches

| MONTH & YEAR  | COMPANY | PRODUCT TYPE | PRODUCT NAME | PRODUCT DESCRIPTION                                                                                                                                                                           |
|---------------|---------|--------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| June 2021     | Dell    | HPC Software | Omnia        | Dell Technologies launched Omnia, a new software underpinning workload deployment and management and composable infrastructure capabilities.                                                  |
| November 2019 | Dell    | HPC          | Dell EMC     | Dell enhanced its HPS offerings by launching improved Dell EMC storage solutions, simplified AI deployments, and the Dell EMC PowerEdge server portfolio to speed up application performance. |

Source: Company Website

#### 14.1.5.3.2 Dell: Deals

| MONTH & YEAR | DEAL TYPE   | COMPANY 1 | COMPANY 2    | DESCRIPTION                                                                                                                                                                                                                              | DEAL SIZE |
|--------------|-------------|-----------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| October 2020 | Partnership | Dell      | Verne Global | Dell partnered with Verne Global, a technology company that offers data centers. The partnership will help both the companies accelerate their clients business operations by leveraging the capabilities of HPC and AI                  | NA        |
| August 2020  | Partnership | Dell      | R Systems    | Dell partnered with R Systems, a global digital transformation company. With the technical expertise of both the companies and leveraging HPC capabilities, they can address demands from commercial and institutional research clients. | NA        |

Source: Company Website and Press Releases

#### 14.1.5.4 MNM view

##### 14.1.5.4.1 Key strengths/right to win

Dell offers PowerEdge servers and Precision workstations, high-performance storage, Genomic Data Analysis Platform, and data center switches and services that cater to HPC needs. The Dell PowerEdge servers and Precision workstations are available in different form factors and can be configured to meet the flexibility and performance needs of users. These servers are designed for HPC environments with broad requirements, including more local Input/Output (I/O) capacity and expandability. On the other hand, Precision workstations run high-demand, industry-specific applications for scientific calculations, 3D industrial designs, remote visualization, engineering simulations, and digital content creation. To optimize infrastructure and minimize costly disruptions, Dell offers HPC deployment services with validated processes and best practices to efficiently plan, design, and implement HPC solutions.

##### 14.1.5.4.2 Strategic choices made

Dell caters to various industry verticals, including education, financial services, energy, government, web development, manufacturing, telecom and media, retail, and healthcare. It identifies the varied customer needs for HPC and focuses on customized approaches to provide HPC solutions. The company's focus on adopting organic growth strategies helps it maintain a leadership position and increase its footprints in the HPC market. As a part of its strategies, the company has also partnered with leading HPC technology and application vendors to provide solutions that efficiently cater to customers' required workloads and needs.

##### 14.1.5.4.3 Weaknesses and competitive threats

The company faces product and price competition in all business areas from both branded and generic competitors. This competitive pressure may affect the company's industry unit share position, revenue, and profitability. The changing global economic conditions may harm the company's business and reduce its net revenue and profitability.

## 14.1.6 LENOVO

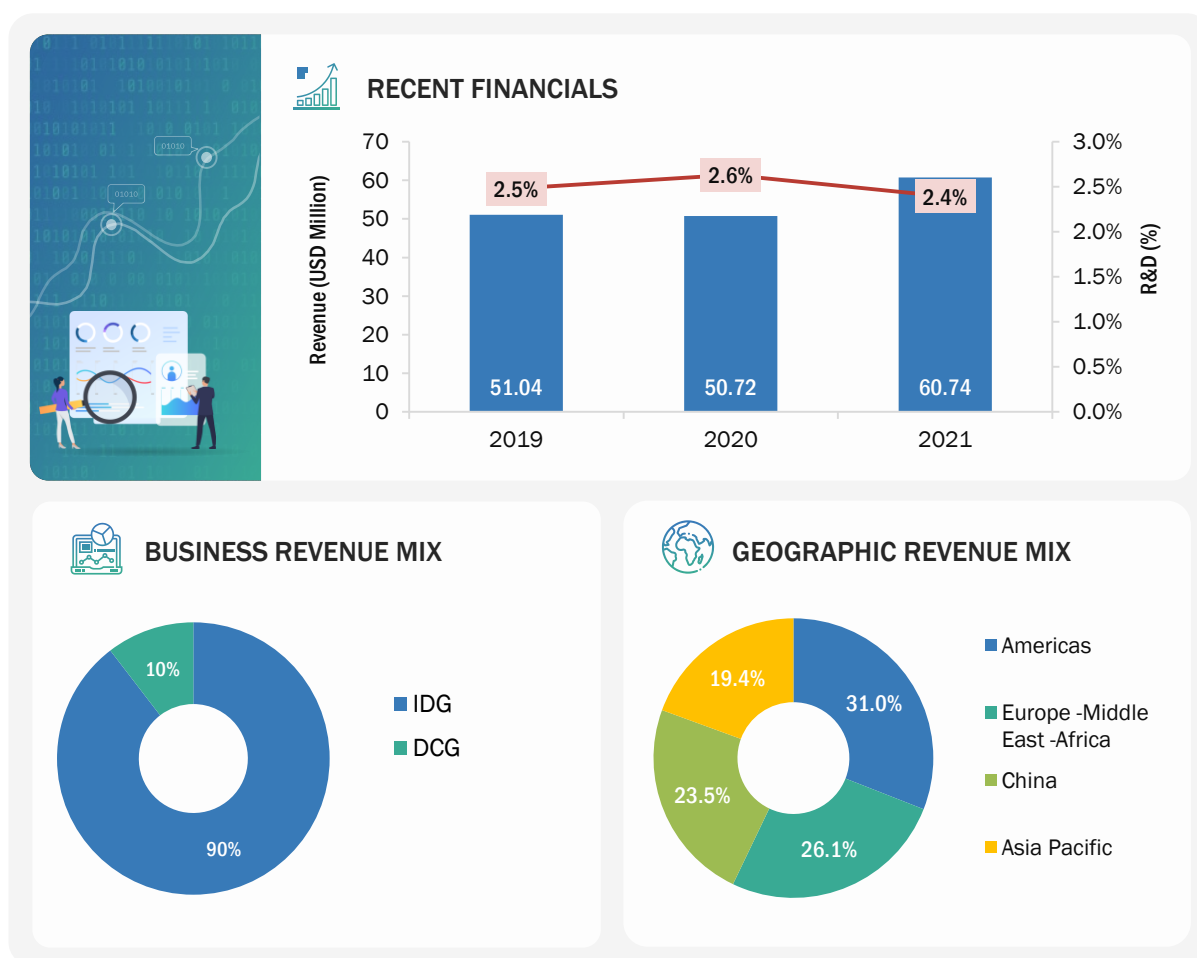
### 14.1.6.1 Business overview

Lenovo, founded in 1984, is headquartered in Beijing, China. It is a Fortune 500 company and among the leading providers of innovative, commercial consumer, and enterprise technologies. The company offers a wide range of products and services, including PCs, servers, workstations, smart TVs, storage devices, smartphones, tablets, and apps. The company operates through various business groups, such as Intelligent Devices Group (IDG) and Data Center Group (DCG). As of March 2021, Lenovo had an employee base of 158,000 people across the globe. The company marks a major presence in the US, with offices located worldwide.

**TABLE 101 LENOVO: BUSINESS OVERVIEW**

|                      |                            |
|----------------------|----------------------------|
| Founding Year        | 1984                       |
| Headquarters Country | China                      |
| Headquarters City    | Quarry Bay                 |
| Ownership Type       | Public                     |
| Employees            | 158,000 (As of March 2021) |

Source: Company Website

**FIGURE 50** LENOVO: COMPANY SNAPSHOT

Note: The "recent financials" bar chart represents the total revenue. In the recent financials block, the R&D expenditure has been considered a percentage of the company's total revenue. The fiscal year of the company ends on March 31. Due to rounding off, percentages presented in the pie charts may not reflect precise segmental revenues; therefore, a slight variation in segmental revenues may be observed.

Source: Company Website, Press Releases, and MarketsandMarkets Analysis

#### 14.1.6.2 Products offered

**TABLE 102** LENOVO: PRODUCT OFFERINGS

| PRODUCT                          | DESCRIPTION                                                                                                           | PRODUCT TYPE | APPLICATION                                                                                                              |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------|
| Think System Servers             | This single-processor rack server, in a compact 1U form factor, is built for expanding businesses or edge deployment. | Server       | <ul style="list-style-type: none"> <li>Network Infrastructure</li> <li>Data Center</li> <li>Telecommunication</li> </ul> |
| TruScale Infrastructure Services | TruScale Infrastructure Services is the latest data center hardware through a pay-as-you-go business model.           | Server       | <ul style="list-style-type: none"> <li>Network Infrastructure</li> <li>Data Center</li> <li>Telecommunication</li> </ul> |

Source: Company Website

### 14.1.6.3 Recent developments

#### 14.1.6.3.1 Lenovo: Product launches

| MONTH & YEAR | COMPANY | PRODUCT TYPE | PRODUCT NAME    | PRODUCT DESCRIPTION                                                                                                                                                                         |
|--------------|---------|--------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| January 2022 | Lenovo  | HPC Software | Lenovo TruScale | Lenovo has launched Lenovo TruScale High-Performance Computing-as-a-Service (HPCaaS), delivering the power of supercomputing to organizations of all sizes through a cloud-like experience. |
| June 2020    | Lenovo  | HPC          | Lenovo Yoga 5G  | Lenovo launched Lenovo Yoga 5G, a 5G computing product. Lenovo Yoga 5G is available across the globe through 5G network service providers, such as Verizon, EE, Sunrise, and CMCC.          |

Source: Company Website

#### 14.1.6.3.2 Lenovo: Deals

| MONTH & YEAR | DEAL TYPE   | COMPANY 1 | COMPANY 2          | DESCRIPTION                                                                                                                                                                                                                         | DEAL SIZE |
|--------------|-------------|-----------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| October 2020 | Partnership | Lenovo    | DreamWorks (US)    | Lenovo partnered with DreamWorks, an American animation studio and a subsidiary of Universal Pictures. Lenovo's HPC and data center offerings play a vital role in supporting the content creation process.                         | NA        |
| May 2019     | Partnership | Lenovo    | Team SoloMid (TSM) | Lenovo partnered with TSM, a professional eSports organization based in the US. TSM is Lenovo's first official eSports sponsorship in North America. Lenovo is increasing its investments with a high focus on the gaming industry. | NA        |

Source: Company Website and Press Releases

## 14.1.7 FUJITSU

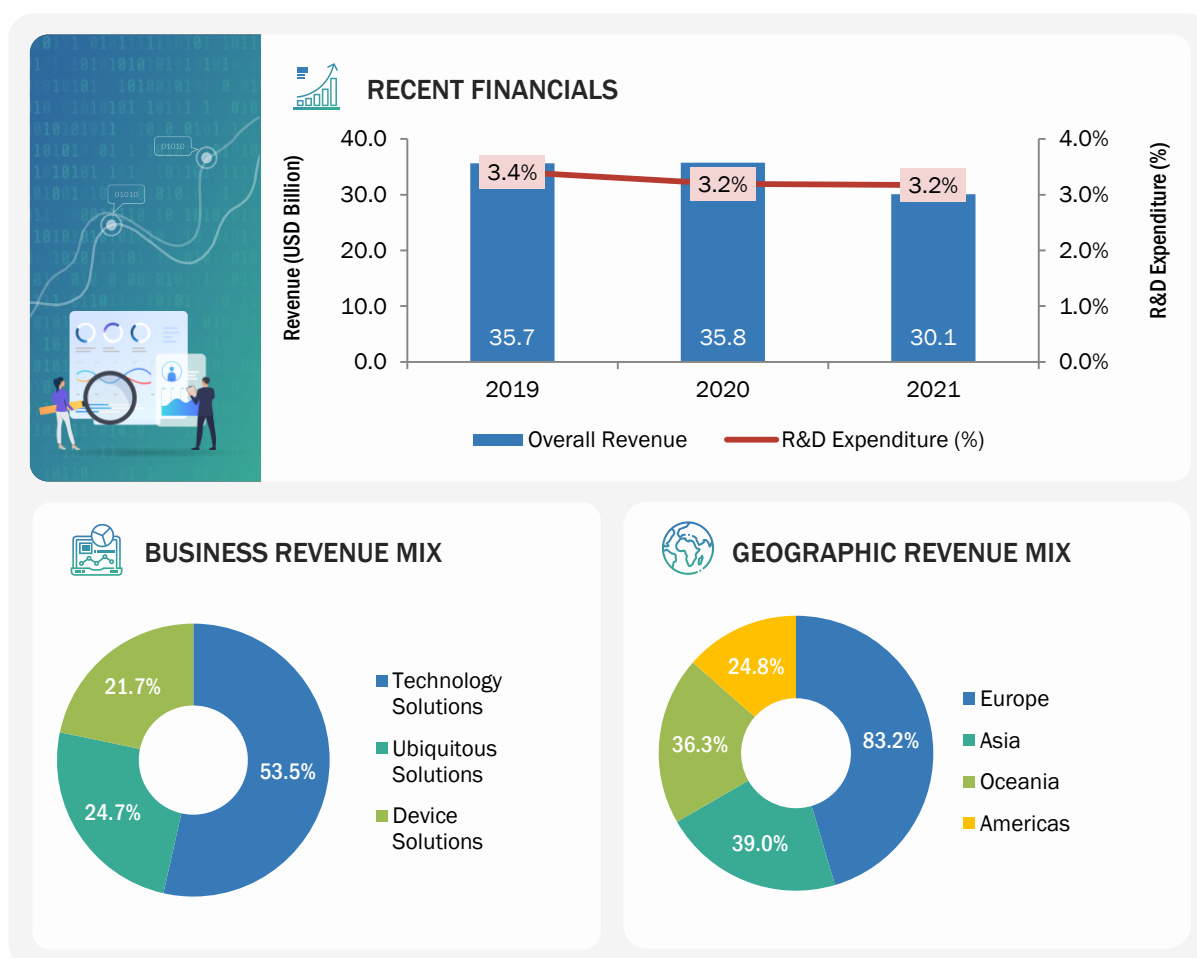
### 14.1.7.1 Business overview

Fujitsu was founded in 1935 and is headquartered in Tokyo, Japan. It is an ICT company that offers a wide range of technologies, solutions, and services worldwide. The company operates through 3 segments, namely, Technology Solutions, Ubiquitous Solutions, and Device Solutions. The Technology Solutions segment provides IT-driven solutions to large corporates. The Ubiquitous Solutions segment offers various products, such as PCs, mobile phones, car audio and navigation systems, mobile communication equipment, and automobile electronics. The Device Solutions segment provides optical transceiver modules and printed circuit boards (PCBs). Fujitsu primarily caters to the automotive, finance, healthcare, retail, life sciences, telecommunications, and automated identification technology verticals. The company has operations in APAC, Europe, and North America.

**TABLE 103** FUJITSU: BUSINESS OVERVIEW

|                      |                            |
|----------------------|----------------------------|
| Founding Year        | 1915                       |
| Headquarters Country | Japan                      |
| Headquarters City    | Tokyo                      |
| Ownership Type       | Public                     |
| Employees            | 123,671 (As of March 2021) |

Source: Company Website

**FIGURE 51** FUJITSU: COMPANY SNAPSHOT

Note: The "recent financials" bar chart represents the total revenue. In the recent financials block, the R&D expenditure has been considered a percentage of the company's total revenue. The fiscal year of the company ends on March 31. The currency conversion rates used for 2019, 2020, and 2021 were YEN 1 = USD 0.00902, YEN 1 = USD 0.00927, and YEN 1 = USD 0.00907, respectively. Due to rounding off, percentages presented in the pie charts may not reflect precise segmental revenues; therefore, a slight variation in segmental revenues may be observed.

Source: Company Website, Press Releases, and MarketsandMarkets Analysis

#### 14.1.7.2 Products offered

**TABLE 104** FUJITSU: PRODUCT OFFERINGS

| PRODUCT                               | DESCRIPTION                                                                                                                                       | PRODUCT TYPE  | APPLICATION                                                                                                                                                      |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FUJITSU Supercomputer PRIMEHPC FX1000 | PRIMEHPC FX1000 supercomputer helps in the fields of research and development using supercomputers across various key industry verticals          | Supercomputer | Research and Development                                                                                                                                         |
| PRIMEFLEX for HPC Solutions           | PRIMEFLEX for HPC provides validated integrated solutions based on predefined and optimized HPC cluster configurations for specific applications. | Application   | <ul style="list-style-type: none"> <li>Data Storage</li> <li>Information Security</li> <li>Virtualization</li> <li>Analytics</li> <li>Cloud Computing</li> </ul> |

Source: Company Website

### 14.1.7.3 Recent development

#### 14.1.7.3.1 Fujitsu: Product launches

| MONTH & YEAR | COMPANY | PRODUCT TYPE           | PRODUCT NAME                         | DESCRIPTION                                                                                                                                                                          |
|--------------|---------|------------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| July 2020    | FUJITSU | System                 | Private 5G System                    | Fujitsu launched a high-speed computing technology, private 5G system, to optimize the deployment of the millimeter wave systems                                                     |
| May 2020     | FUJITSU | Data Services Platform | ETERNUS Data Services Platform (DSP) | Fujitsu launched Fujitsu Storage ETERNUS DSP, a solution that can be used in the deployment and maintenance of storage infrastructure with reduced burden of storage administration. |

Source: Company Website and Press Releases

#### 14.1.7.3.2 Fujitsu: Deals

| MONTH & YEAR  | DEAL TYPE   | COMPANY 1 | COMPANY 2                      | DESCRIPTION                                                                                                                                                                                                                                                                   | DEAL SIZE |
|---------------|-------------|-----------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| February 2021 | Partnership | FUJITSU   | Xilinx (US)                    | FUJITSU and Xilinx announced the plan to collaborate on 5G development in the US.                                                                                                                                                                                             | NA        |
| November 2020 | Partnership | FUJITSU   | University of Toronto (Canada) | Fujitsu, in collaboration with the University of Toronto, developed the search technology. The search technology helps achieve megabit class performance for the large-scale problems of Digital Annealer, the world's first quantum-inspired digital technology architecture | NA        |

Source: Company Website and Press Releases

## 14.1.8 ATOS

### 14.1.8.1 Business overview

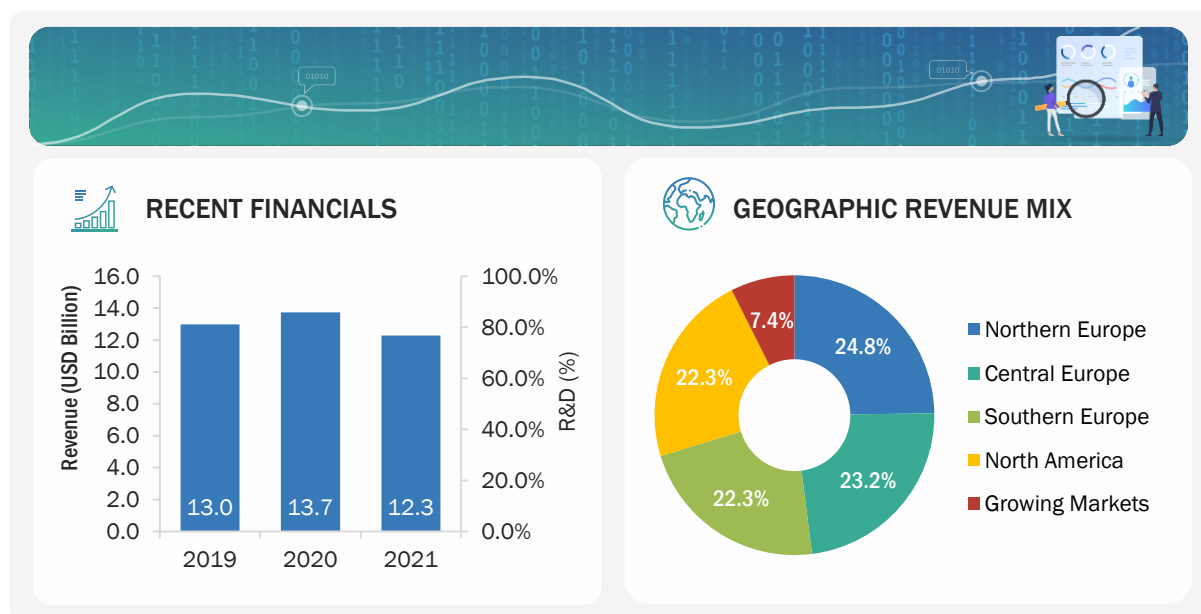
Atos was founded in 1997 and is headquartered in Bezons, France. The company offers infrastructure and data management systems, business cloud platform solutions, and big data and cybersecurity solutions to customers across the globe. Atos provides its solutions to customers from various verticals, including aerospace, defense, banking, insurance, telecommunication, manufacturing, life sciences, transport, media, and energy. As of 2021, the company had an employee base of 86,700 people working for its global clientele spread in over 73 countries.

**TABLE 105 ATOS: BUSINESS OVERVIEW**

|                      |                               |
|----------------------|-------------------------------|
| Founding Year        | 1997                          |
| Headquarters Country | France                        |
| Headquarters City    | Bezons                        |
| Ownership Type       | Public                        |
| Employees            | 86,700 (As of September 2021) |

Source: Company Website

**FIGURE 52 ATOS: COMPANY SNAPSHOT**



Note: The "recent financials" bar chart represents the total revenue. In the recent financials block, the R&D expenditure has been considered a percentage of the company's total revenue. The fiscal year of the company ends on December 31.

The currency conversion rates used for 2019, 2020, and 2021 were EUR 1 = USD 1.11986, EUR 1 = USD 1.22824, and EUR = USD 1.1324, respectively.

Due to rounding off, percentages presented in the pie charts may not reflect precise segmental revenues; therefore, a slight variation in segmental revenues may be observed.

Source: Company Website, Press Releases, and MarketsandMarkets Analysis

### 14.1.8.2 Products offered

**TABLE 106** ATOS: PRODUCT OFFERINGS

| PRODUCT                      | DESCRIPTION                                                                                                                                     | PRODUCT TYPE              | APPLICATION                                                                                                          |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------|
| Atos XCS                     | Atos XCS is a web portal designed for HPC cloud front-ends.                                                                                     | Web Portal for HPC and AI | <ul style="list-style-type: none"> <li>Virtualization</li> <li>Analytics</li> <li>Cloud Computing</li> </ul>         |
| BullSequana X supercomputers | Atos next-gen hybrid supercomputing platform BullSequana X delivers unprecedented computing power at any scale, up to an Exascale-class system. | Supercomputing Platform   | <ul style="list-style-type: none"> <li>Data storage</li> <li>Information Security</li> <li>Virtualization</li> </ul> |
| HPC-as-a-service             | The company offers HPC-as-a-service alongside AI and ML algorithms across key verticals                                                         | Service                   | <ul style="list-style-type: none"> <li>Virtualization</li> <li>Analytics</li> </ul>                                  |
| Mobull                       | Mobull is an ingenious containerized data center solution that provides companies a high computing power.                                       | Data Center               | <ul style="list-style-type: none"> <li>Data storage</li> <li>Information Security</li> </ul>                         |

Source: Company Website

### 14.1.8.3 Recent development

#### 14.1.8.3.1 Atos: Product launches

| MONTH & YEAR  | COMPANY | PRODUCT TYPE  | PRODUCT NAME        | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                  |
|---------------|---------|---------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| February 2022 | Atos    | Supercomputer | BullSequana XH3000  | Atos has revealed its new Exascale-class supercomputer, the BullSequana XH3000, a hybrid computing platform for advanced research in fields such as weather forecasting and climate change, drug discovery, and genomics.                                                                                                                                    |
| November 2020 | Atos    | HPC Software  | HPC Software Suites | Atos launched HPC Software Suites to enable customers and users to effectively manage their supercomputing environments, optimize performance, and reduce energy consumption. HPC Software Suites include the following: Smart Data Management Suite, Smart Energy Management Suite, Smart Performance Management Suite, and Smart Management Center xScale. |

Source: Company Website and Press Releases

## 14.1.8.3.2 Atos: Deals

| MONTH & YEAR  | DEAL TYPE    | COMPANY 1 | COMPANY 2       | DESCRIPTION                                                                                                                                                                                                                                                                         | DEAL SIZE |
|---------------|--------------|-----------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| January 2022  | Acquisitions | Atos      | Cloudreach (UK) | Atos acquired Cloudreach, a leading multi-cloud services company specializing in public cloud application development and cloud migration.                                                                                                                                          | NA        |
| November 2020 | Partnership  | Atos      | Padqal (France) | Atos collaborated with Padqal, a French startup. Both companies together developed a quantum accelerator based on the neutral atom technology for HPC systems. This new quantum accelerator can achieve megabit class performance for the large-scale problems of Digital Annealer. | NA        |

Source: Company Website and Press Releases

## 14.1.9 CISCO

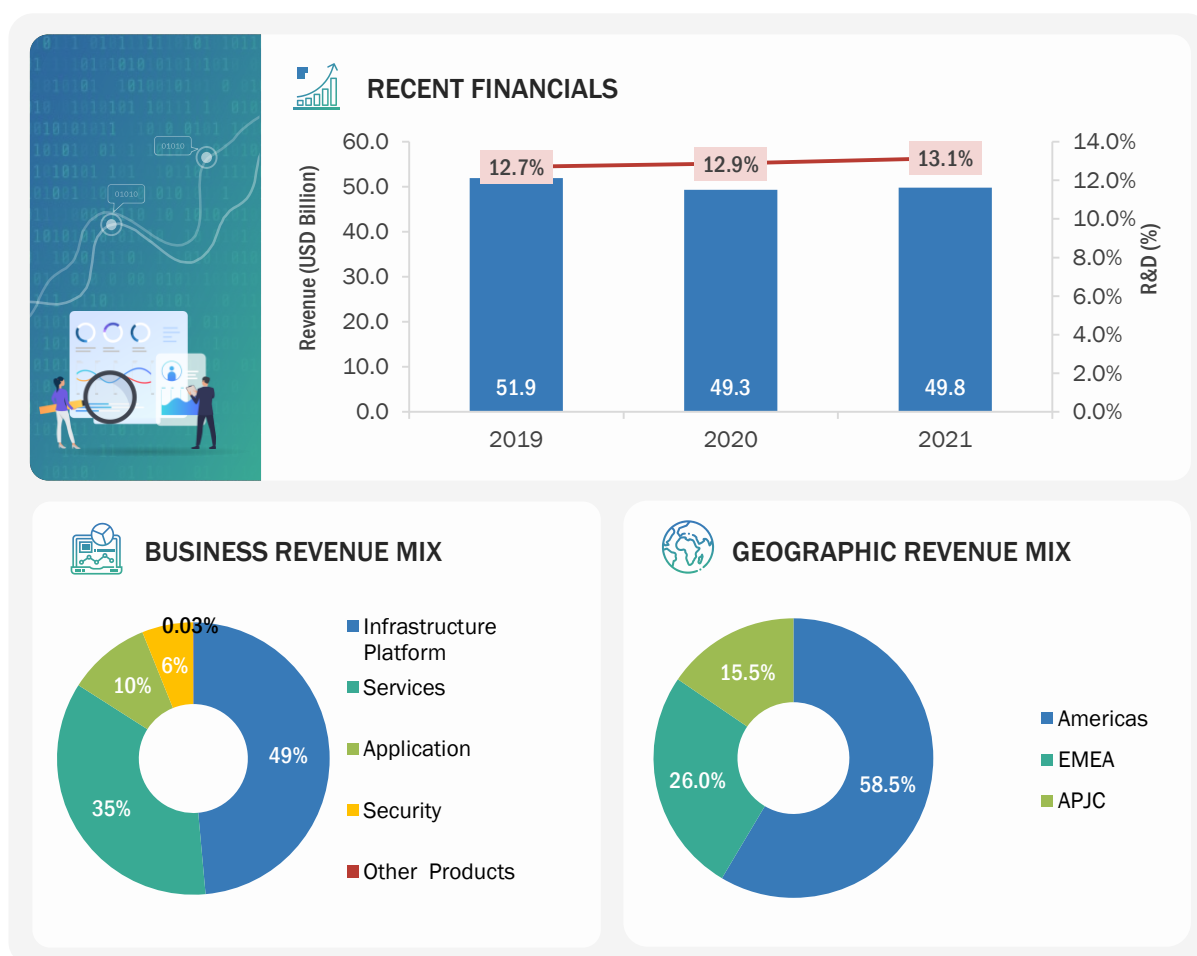
### 14.1.9.1 Business overview

Cisco Systems was established in 1984 and is headquartered in San Jose, California, US. It offers a wide range of products and services, as well as delivers integrated solutions that help in connecting commercial networks across the globe. The company operates in various business segments, such as Infrastructure Platform, Services, Application, Security, and Other Products. Cisco Systems provides cloud security solutions that help protect the users' data and apps in the cloud against malware and data breaches. Its solutions are used in different verticals, including retail, government, defense, financial services, healthcare, hospitality, energy, and education. As of 2021, the company had an employee base of 54,000 people in more than 400 offices.

**TABLE 107** CISCO: BUSINESS OVERVIEW

|                      |                              |
|----------------------|------------------------------|
| Founding Year        | 1987                         |
| Headquarters Country | US                           |
| Headquarters City    | Houston                      |
| Ownership Type       | Public                       |
| Employees            | 54,000 (As of December 2021) |

Source: Company Website

**FIGURE 53 CISCO: COMPANY SNAPSHOT**

Note: The "recent financials" bar chart represents the total revenue, and the R&D expenditure has been considered a percentage of the company's total revenue. The fiscal year of the company ends on December 31. Due to rounding off, percentages presented in the pie charts may not reflect precise segmental revenues; therefore, a slight variation in segmental revenues may be observed

#### 14.1.9.2 Products offered

**TABLE 108 CISCO: PRODUCT OFFERINGS**

| PRODUCT                                    | DESCRIPTION                                                                                                                       | PRODUCT TYPE      | APPLICATION                                                                                                           |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------|
| Nexus Ultra-low Latency Ethernet Solutions | It is an ultra-low latency network platform integrated with Cisco Ethernet switches—including Cisco Nexus 9000 and 3000 families. | Network Platform  | <ul style="list-style-type: none"> <li>Data storage</li> <li>Information Security</li> <li>Cloud Computing</li> </ul> |
| Userspace NIC (usNIC)                      | Cisco Userspace NIC (Cisco usNIC) feature improves the performance of software applications that run on the Cisco UCS servers.    | Software Solution | <ul style="list-style-type: none"> <li>Data Storage</li> <li>Analytics</li> <li>Cloud Computing</li> </ul>            |

Source: Company Website

### 14.1.9.3 Recent development

#### 14.1.9.3.1 Cisco: Deals

| MONTH & YEAR | DEAL TYPE   | COMPANY 1 | COMPANY 2    | DESCRIPTION                                                                                                                                                                                                      | DEAL SIZE |
|--------------|-------------|-----------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| May 2020     | Acquisition | Cisco     | ThousandEyes | Cisco acquired ThousandEyes, an internet and cloud intelligence platform provider. With the acquisition, Cisco enhanced its network performance and monitoring capabilities across enterprises and in the cloud. | NA        |

Source: Company Website and Press Releases

## 14.1.10 NVIDIA

### 14.1.10.1 Business overview

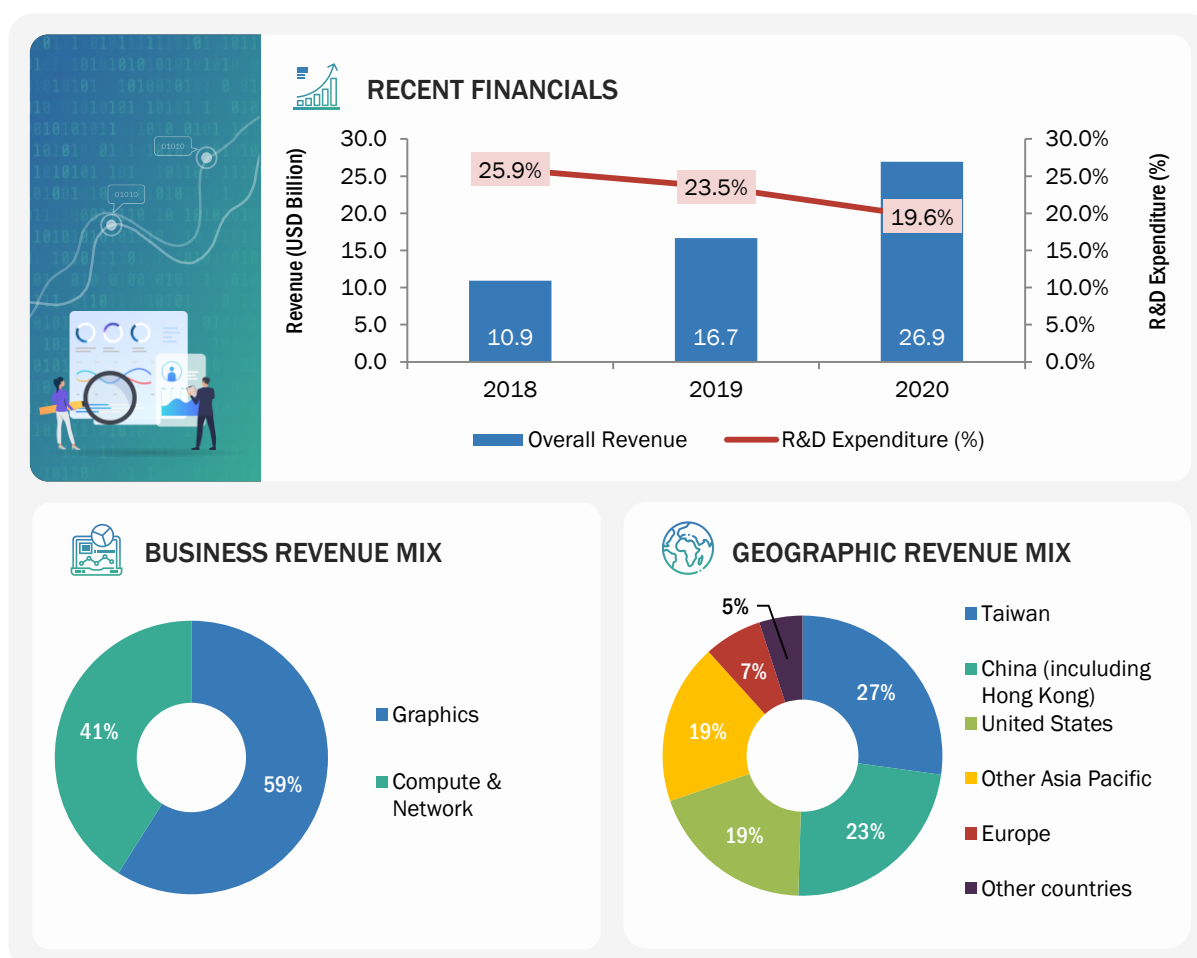
Founded in 1993 and headquartered in California, US, NVIDIA operates globally as a visual computing company. The company operates through 2 segments, namely, GPU and Tegra Processor. The GPU business unit offers Tesla for accelerated computing and deep learning, which leverages the parallel computing capabilities of GPUs for general-purpose computing. The company caters to verticals such as transportation, healthcare, manufacturing, logistics, and public safety.

NVIDIA's global operations are spread across three regions: Asia Pacific, the Americas, and Europe. It generates the highest sales revenue from Japan, China, South Korea, and Brazil.

**TABLE 109** NVIDIA: BUSINESS OVERVIEW

|                      |                           |
|----------------------|---------------------------|
| Founding Year        | 1933                      |
| Headquarters Country | Japan                     |
| Headquarters City    | Kyoto                     |
| Ownership Type       | Public                    |
| Employees            | 28,254 (As of March 2021) |

Source: Company Website

**FIGURE 54** NVIDIA: COMPANY SNAPSHOT

Note: The "recent financials" bar chart represents the total revenue, and the R&D expenditure has been considered a percentage of the company's total revenue. The fiscal year of the company ends on December 31.

Due to rounding off, percentages presented in the pie charts may not reflect precise segmental revenues; therefore, a slight variation in segmental revenues may be observed.

#### 14.1.10.2 Products offered

**TABLE 110** NVIDIA: PRODUCT OFFERINGS

| PRODUCT                     | DESCRIPTION                                                                                                                                                                | PRODUCT TYPE                | APPLICATION                                                                                                                                   |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| NVIDIA CUDA 11              | CUDA is a parallel computing platform and application programming interface that allows the software to use certain types of the graphics processing units for processing, | Parallel Computing Platform | <ul style="list-style-type: none"> <li>Data Storage</li> <li>Information Security</li> <li>Virtualization</li> <li>Cloud Computing</li> </ul> |
| NVIDIA A100 Tensor Core GPU | NVIDIA A100 Tensor Core GPU delivers GPU acceleration across various CPU cores to power the world's highest-performing elastic data centers.                               | GPU                         | <ul style="list-style-type: none"> <li>Data Center</li> <li>Graphics and Video Rendering</li> </ul>                                           |
| NVIDIA DGX A100             | NVIDIA DGX A100 is an AI system built on the NVIDIA A100 Tensor Core GPU.                                                                                                  | HPC Application             | <ul style="list-style-type: none"> <li>Analytics</li> <li>Data storage</li> <li>Information Security</li> </ul>                               |

|                         |                                                                                                                                                                        |                 |                                                                                                            |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------|
|                         |                                                                                                                                                                        |                 | <ul style="list-style-type: none"> <li>Virtualization</li> </ul>                                           |
| NVIDIA Data Center GPUs | The NVIDIA Data Center GPUs give data centers the power to accelerate deep learning, machine learning, and high-performance computing (HPC) workloads.                 | GPU             | <ul style="list-style-type: none"> <li>Data Centre</li> <li>Graphics and Video Rendering</li> </ul>        |
| NVIDIA HPC SDK          | The NVIDIA HPC SDK includes instructions for developing, profiling, and deploying software using the HPC Container Maker to simplify the creation of container images. | HPC Application | <ul style="list-style-type: none"> <li>Data storage</li> <li>Analytics</li> <li>Cloud Computing</li> </ul> |

Source: Company Website

### 14.1.10.3 Recent development

#### 14.1.10.3.1 NVIDIA: Product launches

| MONTH & YEAR  | COMPANY | PRODUCT TYPE            | PRODUCT NAME          | DESCRIPTION                                                                                                                                                                                        |
|---------------|---------|-------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| June 2021     | NVIDIA  | Supercomputing Platform | HGX A100 Systems      | NVIDIA launched the NVIDIA HGX AI supercomputing platform with new technologies that combine AI with high-performance computing, making supercomputing more useful to many industries.             |
| April 2021    | NVIDIA  | CPU                     | Data Center CPU       | NVIDIA launched its first data center CPU, an Arm-based processor that will deliver 10x the performance of today's fastest servers on the most complex AI and high-performance computing workloads |
| December 2020 | NVIDIA  | Analysis Tool           | Nsight Systems2020.5. | NVIDIA launched Nsight Systems 2020.5, a performance analysis tool. The tool helps developers tune and scale software across CPUs and GPUs.                                                        |

Source: Company Website and Press Releases

## 14.1.10.3.2 NVIDIA: Deals

| MONTH & YEAR | DEAL TYPE   | COMPANY 1 | COMPANY 2             | DESCRIPTION                                                                                                                                                                                                                                                      | DEAL SIZE |
|--------------|-------------|-----------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| January 2022 | Acquisition | NVIDIA    | Bright Computing (US) | Nvidia has acquired Bright Computing, a leader in software for managing high-performance computing systems                                                                                                                                                       | NA        |
| October 2020 | Partnership | NVIDIA    | CINECA (Italy)        | NVIDIA collaborated with CINECA, an Italian inter-university consortium. NVIDIA deployed an accelerated computing platform to develop an AI supercomputer.                                                                                                       | NA        |
| March 2019   | Acquisition | NVIDIA    | Mellanox (Israel)     | NVIDIA acquired Mellanox, an Israeli American multinational supplier of computer networking products. With the acquisition, NVIDIA optimized data center scale workloads across the entire computing, networking, and storage stack to achieve high performance. | NA        |

Source: Company Website and Press Releases

## 14.2 OTHER PLAYERS

### 14.2.1 NEC CORPORATION

**TABLE 111** NEC CORPORATION: BUSINESS OVERVIEW

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founding Year        | 1899                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Headquarters Country | Japan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Headquarters City    | Tokyo                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Ownership Type       | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Revenue              | USD 27.166 Billion (2021)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Employees            | 115,000 (As of March 2021)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Business Overview    | NEC Corporation is one of the leading manufacturers of telecommunications equipment and related software and services. NEC Corporation was known as Nippon Electric Company, Limited before rebranding in 1983 as NEC. The company provides IT and network solutions, including artificial intelligence (AI), cloud computing, Internet of things (IoT) platforms, and telecommunication equipment and software. The company has launched its first HPC offering with the development of a product line of vector computers, and a major milestone was the development of the SX-Aurora TSUBASA. |
| Products Offered     | <ul style="list-style-type: none"> <li>▪ SX-Aurora TSUBASA</li> <li>▪ Vector Computer</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Geographic Presence  | North America, Europe, APAC, MEA, and Latin America                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

Source: Company Website, Annual Reports, Company Publications, and Press Releases

## 14.2.2 AMAZON WEB SERVICES

**TABLE 112** AMAZON WEB SERVICES: BUSINESS OVERVIEW

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founding Year        | 2006                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Headquarters Country | US                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Headquarters City    | Seattle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Ownership Type       | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Revenue              | USD 469,822 Million (2021)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Employees            | 1,608,000 (As of December 2021)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Business Overview    | <p>Amazon Web Services (AWS) is the world's most comprehensive and broadly adopted cloud platform, offering over 200 fully featured services from data centers. The company offers a broad set of global cloud-based products, including compute, management tools, storage, databases, analytics, networking, mobile, developer tools, IoT, security, and enterprise applications. Furthermore, the company also provides complex simulations and deep learning workloads in the cloud with a complete suite of high-performance computing (HPC) products and services on AWS.</p> |
| Products Offered     | <ul style="list-style-type: none"> <li>▪ Amazon EC2</li> <li>▪ Elastic Fabric Adapter</li> <li>▪ AWS Parallel Cluster</li> <li>▪ Amazon FSx for Lustre</li> <li>▪ AWS Batch</li> <li>▪ NICE DCV</li> </ul>                                                                                                                                                                                                                                                                                                                                                                          |
| Geographic Presence  | North America, Europe, APAC, MEA, and Latin America                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

Source: Company Website, Annual Reports, Company Publications, and Press Releases

### 14.2.3 ORACLE

**TABLE 113 ORACLE: BUSINESS OVERVIEW**

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founding Year        | 1977                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Headquarters Country | US                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Headquarters City    | Austin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Ownership Type       | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Revenue              | USD 34.10 Billion (2021)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Employees            | 132,000 (As of May 2021)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Business Overview    | Oracle engages in the provision of products and services that address all aspects of corporate information technology environments. It operates through the following business segments: Cloud and License, Hardware, and Services. The Hardware segment provides hardware products and hardware-related software products, including Oracle engineered systems, servers, storage, industry-specific hardware, operating systems, virtualization, management and other hardware-related software, and related hardware support. The Services segment offers consulting, advanced support, and education services. |
| Products Offered     | Oracle Cloud Infrastructure (OCI)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Geographic Presence  | North America, Europe, APAC, MEA, and Latin America                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

Source: Company Website, Annual Reports, Company Publications, and Press Releases

### 14.2.4 MICROSOFT

**TABLE 114 MICROSOFT: BUSINESS OVERVIEW**

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founding Year        | 1975                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Headquarters Country | US                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Headquarters City    | Washington                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Ownership Type       | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Revenue              | USD 168.09 Billion (2021)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Employees            | 181,000 (As of June 2021)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Business Overview    | Microsoft Corporation, a technology company, develops and supports a range of software products, devices, services, and solutions. The company offers operating systems for servers, computing devices, phones, and other intelligent devices. The company also provides server applications for distributed computing environments; productivity applications; business solution applications; desktop and server management tools; software development tools; video games; and online advertising services. |
| Products Offered     | Azure High-performance Computing (HPC)                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Geographic Presence  | North America, Europe, APAC, MEA, and Latin America                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

Source: Company Website, Annual Reports, Company Publications, and Press Releases

### 14.2.5 INSPUR

**TABLE 115** INSPUR: BUSINESS OVERVIEW

|                      |                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founding Year        | 1945                                                                                                                                                                                                                                                                                                                                                                      |
| Headquarters Country | China                                                                                                                                                                                                                                                                                                                                                                     |
| Headquarters City    | Jinan                                                                                                                                                                                                                                                                                                                                                                     |
| Ownership            | Public                                                                                                                                                                                                                                                                                                                                                                    |
| Revenue              | USD 23.6 Billion (2021)                                                                                                                                                                                                                                                                                                                                                   |
| Business Overview    | Inspur is headquartered in Jinan (Shandong), China. The company develops enterprise management software and other products. It offers cloud computing and big data solutions alongside servers and storage solutions for AI-based applications. The company's server offerings include GPU servers, rack-mounted servers, open-computing servers, and multi-node servers. |
| Products Offered     | <ul style="list-style-type: none"> <li>▪ System Platform</li> <li>▪ Management Suite</li> <li>▪ Application Optimization</li> <li>▪ Industrial Appliances</li> </ul>                                                                                                                                                                                                      |
| Geographic Presence  | North America, Europe, APAC, MEA, and Latin America                                                                                                                                                                                                                                                                                                                       |

Source: Company Website, Annual Reports, Company Publications, and Press Releases

### 14.2.6 NETAPP

**TABLE 116** NETAPP: BUSINESS OVERVIEW

|                        |                                                                                                                                                                                                                                                                                                                              |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founding Year          | 1992                                                                                                                                                                                                                                                                                                                         |
| Headquarters Country   | US                                                                                                                                                                                                                                                                                                                           |
| Headquarters City      | California                                                                                                                                                                                                                                                                                                                   |
| Ownership Type         | Public                                                                                                                                                                                                                                                                                                                       |
| Overall Annual Revenue | USD 5.74 Billion (2021)                                                                                                                                                                                                                                                                                                      |
| Business Overview      | NetApp is headquartered in Sunnyvale (California), US. The company specializes in network storage, cloud computing, storage efficiency, information management, data storage, and hybrid cloud. It offers customers the most appropriate data storage solutions, such as cloud data services, storage systems, and software. |
| Products Offered       | NetApp HPC                                                                                                                                                                                                                                                                                                                   |
| Geographic Presence    | North America, Europe, APAC, MEA, and Latin America                                                                                                                                                                                                                                                                          |

Source: Company Website, Annual Reports, Company Publications, and Press Releases

### 14.2.7 IRON GLOBAL

**TABLE 117** IRON: BUSINESS OVERVIEW

|                      |                                                                                                                                                                                                                                                                                                                                                              |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founding Year        | 2002                                                                                                                                                                                                                                                                                                                                                         |
| Headquarters Country | US                                                                                                                                                                                                                                                                                                                                                           |
| Headquarters City    | California                                                                                                                                                                                                                                                                                                                                                   |
| Ownership Type       | Private                                                                                                                                                                                                                                                                                                                                                      |
| Business Overview    | IRON Global is headquartered in Fremont (California), US. The company specializes in the provision of application-specific embedded system platforms. It offers product life cycle management services to the networking, storage, security, medical, and communications verticals worldwide. It also offers support to its customers across 180+ countries. |
| Products Offered     | <ul style="list-style-type: none"> <li>▪ HPC Server Platforms</li> <li>▪ iCluster Infrastructure Blocks</li> <li>▪ Software Building Blocks</li> <li>▪ HPC System Integration Services</li> <li>▪ iCluster Configuration Lab</li> </ul>                                                                                                                      |
| Geographic Presence  | North America, Europe, APAC, MEA, and Latin America                                                                                                                                                                                                                                                                                                          |

Source: Company Website, Annual Reports, Company Publications, and Press Releases

### 14.2.8 ASPEN SYSTEM

**TABLE 118** ASPEN SYSTEM: BUSINESS OVERVIEW

|                      |                                                                                                                                                                                                                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founding Year        | 1982                                                                                                                                                                                                                                                                              |
| Headquarters Country | US                                                                                                                                                                                                                                                                                |
| Headquarters City    | Colorado                                                                                                                                                                                                                                                                          |
| Ownership Type       | Private                                                                                                                                                                                                                                                                           |
| Business Overview    | Aspen Systems is headquartered in Wheat Ridge (Colorado), US. The company is specialized in HPC, supercomputing, clusters, and facility design services. It sells, manufactures, designs, installs, and supports computing, storage, networking, and software and infrastructure. |
| Products Offered     | <ul style="list-style-type: none"> <li>▪ Linux Cluster Solutions</li> <li>▪ Networking</li> <li>▪ HPC Processors</li> <li>▪ Software Solutions</li> <li>▪ HPC Applications</li> </ul>                                                                                             |
| Geographic Presence  | North America                                                                                                                                                                                                                                                                     |

Source: Company Website, Annual Reports, Company Publications, and Press Releases

### 14.2.9 ADVANCED CLUSTERING TECHNOLOGIES

**TABLE 119** ADVANCED CLUSTERING TECHNOLOGIES: BUSINESS OVERVIEW

|                      |                                                                                                                                                                                                                                                                                                                                          |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founding Year        | 2001                                                                                                                                                                                                                                                                                                                                     |
| Headquarters Country | US                                                                                                                                                                                                                                                                                                                                       |
| Headquarters City    | Missouri                                                                                                                                                                                                                                                                                                                                 |
| Ownership Type       | Private                                                                                                                                                                                                                                                                                                                                  |
| Business Overview    | Advanced Clustering Technologies is headquartered in Kansas City (Missouri), US. The company specializes in Linux clusters, HPC software tools, HPC hardware, cloud HPC, GPU computing, and HPC cluster management software. It develops systems for some of the most prestigious national laboratories, universities, and corporations. |
| Products Offered     | HPC Clusters                                                                                                                                                                                                                                                                                                                             |
| Geographic Presence  | North America                                                                                                                                                                                                                                                                                                                            |

Source: Company Website, Annual Reports, Company Publications, and Press Releases

### 14.2.10 DAWNING INFORMATION INDUSTRY CO. LTD. (SUGON)

**TABLE 120** SUGON: BUSINESS OVERVIEW

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founding Year        | 1990                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Headquarters Country | China                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Headquarters City    | Beijing                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Ownership            | Private                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Business Overview    | Sugon is also known as Dawning Information Industry. The company offers HPC, server, storage, cloud computing, and big data in China. The company is closely working on National programs and formed foundations with strong support from the Chinese Academy of Sciences. In HPC, Sugon is one of the manufacturers in the APAC region. Sugon focused on the enhancement to create outstanding values for global customers. |
| Products Offered     | <ul style="list-style-type: none"> <li>▪ Rack server (I840-G30, A620-G30, A320-G30, I620-G30)</li> <li>▪ Blade Server (TC6600, TC4600E G3)</li> <li>▪ High Density (TC 4600 T G3, TC 5600-I)</li> <li>▪ Sugon S640-G30 Storage server</li> <li>▪ Sugon Advance Computing solution (Dawning 6000 series HPC computer, Silicon Cube series advance computing computer)</li> </ul>                                              |
| Geographic Presence  | North America, Europe, APAC, MEA, and Latin America                                                                                                                                                                                                                                                                                                                                                                          |

Source: Company Website, Annual Reports, Company Publications, and Press Releases

### 14.2.11 DASSAULT SYSTEMS

**TABLE 121 DASSAULT SYSTEMS: BUSINESS OVERVIEW**

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founding Year        | 1881                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Headquarters Country | France                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Headquarters City    | Vélizy-Villacoublay                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Ownership Type       | Public                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Revenue              | USD 4.86 Billion (2021)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Employees            | 20,000 (As of December 2021)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Business Overview    | <p>Dassault Systems provides software applications and services designed to support its innovation processes. The company is a specialist in 3D design, PLM, simulation, architecture, engineering and construction, and industrial equipment. The company's offerings include software applications and services, ranging from digital, conceptual design drawings to full digital mock-up; virtual testing of products; and end-to end global industrial operations, including manufacturing management and operations planning and optimization; and marketing and sales, from digital marketing and advertising to the end-consumer shopping experience. The company caters to industry verticals, such as life sciences, industrial equipment, transportation, retail, marine and offshore, aerospace &amp; defense, architecture, and engineering and construction.</p> |
| Products Offered     | <ul style="list-style-type: none"> <li>▪ 3DEXPERIENCE</li> <li>▪ Catia</li> <li>▪ Solidworks</li> <li>▪ Simulia</li> <li>▪ Abaqus</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Geographic Presence  | North America, Europe, APAC, MEA, and Latin America                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

Source: Company Website, Annual Reports, Company Publications, and Press Releases

### 14.2.12 ARM LIMITED

**TABLE 122** ARM LIMITED: BUSINESS OVERVIEW

|                      |                                                                                                                                                                                                                                                                                                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founding Year        | 1990                                                                                                                                                                                                                                                                                                                                                      |
| Headquarters Country | England                                                                                                                                                                                                                                                                                                                                                   |
| Headquarters City    | Cambridge                                                                                                                                                                                                                                                                                                                                                 |
| Ownership            | Private                                                                                                                                                                                                                                                                                                                                                   |
| Business Overview    | Arm is a specialist in Intellectual Property (IP), CPU, electronics design, software, and IoT. The company is a semiconductor enterprise that designs and develops microprocessors, cell libraries, embedded memories, and related tools. The company offers support and training services for its customers. The company has offices across all regions. |
| Products Offered     | Arm HPC solutions                                                                                                                                                                                                                                                                                                                                         |
| Geographic Presence  | North America, Europe, APAC, MEA, and Latin America                                                                                                                                                                                                                                                                                                       |

Source: Company Website, Annual Reports, Company Publications, and Press Releases

### 14.2.13 MONTAGE TECHNOLOGY

**TABLE 123** MONTAGE TECHNOLOGY: BUSINESS OVERVIEW

|                      |                                                                                                                                                                                                                           |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founding Year        | 1948                                                                                                                                                                                                                      |
| Headquarters Country | China                                                                                                                                                                                                                     |
| Headquarters City    | Shanghai                                                                                                                                                                                                                  |
| Ownership Type       | Public                                                                                                                                                                                                                    |
| Revenue              | USD 868.2 Million (As of September 2021)                                                                                                                                                                                  |
| Business Overview    | Montage Technology is an Integrated Circuit (IC) design company dedicated to providing high-performance, low-power IC solutions for cloud computing and data center markets. The company has an office in Sant Jose (US). |
| Products Offered     | <ul style="list-style-type: none"> <li>▪ Jintide CPU and HSDIMM</li> <li>▪ PCIe Retimer</li> </ul>                                                                                                                        |
| Geographic Presence  | APAC and North America                                                                                                                                                                                                    |

Source: Company Website, Annual Reports, Company Publications, and Press Releases

### 14.2.14 ADAPTIVE COMPUTING

**TABLE 124** ADAPTIVE COMPUTING: BUSINESS OVERVIEW

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founding Years       | 2001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Headquarters Country | US                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Headquarters City    | Florida                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Ownership Type       | Private                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Business Overview    | <p>Adaptive Computing offers advanced HPC applications and tools to clients. The company is specialized in HPC, cloud, data center, large scale scheduling, batch systems, workload management, big data analysis, technical computing, big workflow, remote visualization, data migration, cloud migration, elastic computing, and grid management. The company offers products and solutions to customers around the globe, with sales, consulting, and support services that ensure success. The company caters to industry verticals such as BFSI, oil and gas, manufacturing, government, and education.</p> |
| Products Offered     | <ul style="list-style-type: none"> <li>▪ Moab HPC Suite</li> <li>▪ NODUS cloud solutions</li> <li>▪ VIEWPOINT</li> <li>▪ ACCOUNTING MANAGER</li> <li>▪ WIDE AREA GRID</li> <li>▪ POWER MANAGEMENT</li> <li>▪ REMOTE VISUALIZATION</li> <li>▪ REPORTING AND ANALYTICS</li> <li>▪ NITRO HIGH THROUGHPUT</li> <li>▪ TORQUE RESOURCE MANAGER</li> </ul>                                                                                                                                                                                                                                                               |
| Geographic Presence  | North America, Europe, APAC, MEA, and Latin America                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

Source: Company Website, Annual Reports, Company Publications, and Press Releases

### 14.2.15 ADVANCED HPC

**TABLE 125** ADVANCED HPC: BUSINESS OVERVIEW

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founding Year        | 2009                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Headquarters Country | US                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Headquarters City    | California                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Ownership Type       | Private                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Business Overview    | Advanced HPC is a provider of customized HPC solutions. The company is specialized in Linux clusters, servers and workstations, Directly Attached Storage (DAS), Network Attached Storage (NAS), and Storage Area Networks (SAN), tape backup, backup software, virtualization, parallel file system, data storage, data archive, data center infrastructure, and networking. The company offers performance computing, storage systems, and data management solutions to industry verticals such as healthcare, government, defense, and educational institutions. |
| Products Offered     | <ul style="list-style-type: none"> <li>▪ High-Performance Servers</li> <li>▪ Data Storage &amp; Archive</li> <li>▪ Switches and Cables</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Geographic Presence  | North America                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

Source: Company Website, Annual Reports, Company Publications, and Press Releases

### 14.2.16 DATADIRECT NETWORKS

**TABLE 126** DATADIRECT NETWORKS: BUSINESS OVERVIEW

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founding Year        | 1998                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Headquarters Country | US                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Headquarters City    | California                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Ownership Type       | Private                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Business Overview    | DataDirect Networks (DDN) offers big data storage to data-intensive, global organizations. The company specializes in Genetic Research, Postproduction, Real-Time Broadcast, High Definition (HD), 2K, 4K & 3D Workflows, Full Motion Video, Physical Security, Geospatial Intelligence, Financial Analytics, Geophysics, Biometrics, Supercomputing, Animation, Social Networking and Web Content, and Online Gaming. The company serves IT, education, healthcare, BFSI, government, and manufacturing. |
| Products Offered     | DDN's High-Performance Data Management                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Geographic Presence  | North America, Europe, APAC, MEA, and Latin America                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

Source: Company Website, Annual Reports, Company Publications, and Press Releases

### 14.2.17 EQUUS COMPUTERS

**TABLE 127** EQUUS COMPUTERS: BUSINESS OVERVIEW

|                      |                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founding Year        | 1989                                                                                                                                                                                                                                                                                                                                                                |
| Headquarters Country | US                                                                                                                                                                                                                                                                                                                                                                  |
| Headquarters City    | Minneapolis                                                                                                                                                                                                                                                                                                                                                         |
| Ownership Type       | Private                                                                                                                                                                                                                                                                                                                                                             |
| Business Overview    | The company is specialized in servers and storage, software-defined infrastructure, data centers, and client computer systems. It caters to industries such as healthcare, IT, government, and education. The company has sales offices in Illinois, Kansas, Minnesota, Missouri, Texas, and Washington, while manufacturing facilities in California and Illinois. |
| Products Offered     | <ul style="list-style-type: none"> <li>▪ WHITEBOX OPEN PLATFORMS</li> <li>▪ HIGH-DENSITY HDSS</li> </ul>                                                                                                                                                                                                                                                            |
| Geographic Presence  | North America and Europe                                                                                                                                                                                                                                                                                                                                            |

Source: Company Website, Annual Reports, Company Publications, and Press Releases

### 14.2.18 EXCELERO

**TABLE 128** EXCELERO: BUSINESS OVERVIEW

|                      |                                                                                                                                                                                                                                                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founding Years       | 2014                                                                                                                                                                                                                                                                                                                                  |
| Headquarters Country | US                                                                                                                                                                                                                                                                                                                                    |
| Headquarters City    | California                                                                                                                                                                                                                                                                                                                            |
| Ownership Type       | Private                                                                                                                                                                                                                                                                                                                               |
| Business Overview    | Excelero is specialized in storage, flash, low latency, remote flash, Server SAN, block storage, Database-as-a-Service (DBaaS), and Cloud. The company offers distributed block storage solutions for web-scale applications, data analytics, and ML applications. The company mainly caters to the media and entertainment vertical. |
| Products Offered     | NVMesh                                                                                                                                                                                                                                                                                                                                |
| Geographic Presence  | North America                                                                                                                                                                                                                                                                                                                         |

Source: Company Website, Annual Reports, Company Publications, and Press Releases

### 14.2.19 GIGA-BYTE

**TABLE 129** GIGA-BYTE: BUSINESS OVERVIEW

|                      |                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founding Year        | 1986                                                                                                                                                                                                                                                                                                                                                                                                      |
| Headquarters Country | Taiwan                                                                                                                                                                                                                                                                                                                                                                                                    |
| Headquarters City    | New Taipei                                                                                                                                                                                                                                                                                                                                                                                                |
| Ownership Type       | Public                                                                                                                                                                                                                                                                                                                                                                                                    |
| Revenue              | USD 84 Billion (2020)                                                                                                                                                                                                                                                                                                                                                                                     |
| Business Overview    | The company offers motherboards, graphics cards, desktops, laptops, PC peripherals and components, servers, embedded-system, gaming computers, gaming laptops, and server solutions. The company extended its scope to business servers and cloud systems with hardware and software solutions that can integrate AI and IoT applications to develop the ideal infrastructure for future prospects in 5G. |
| Products Offered     | <ul style="list-style-type: none"> <li>▪ H-Series High-Density Servers</li> <li>▪ G-Series GPU Servers</li> <li>▪ H262_Z61</li> <li>▪ G291_280</li> </ul>                                                                                                                                                                                                                                                 |
| Geographic Presence  | North America, Europe, APAC, MEA, and Latin America                                                                                                                                                                                                                                                                                                                                                       |

Source: Company Website, Annual Reports, Company Publications, and Press Releases

### 14.2.20 PENGUIN COMPUTING

**TABLE 130** PENGUIN COMPUTING: BUSINESS OVERVIEW

|                      |                                                                                                                                                                                                                                                                                                                                                       |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Founding Year        | 1998                                                                                                                                                                                                                                                                                                                                                  |
| Headquarters Country | US                                                                                                                                                                                                                                                                                                                                                    |
| Headquarters City    | California                                                                                                                                                                                                                                                                                                                                            |
| Ownership Type       | Public                                                                                                                                                                                                                                                                                                                                                |
| Revenue              | USD 4.86 Billion (2021)                                                                                                                                                                                                                                                                                                                               |
| Employees            | 20,000 (As of December 2021)                                                                                                                                                                                                                                                                                                                          |
| Business Overview    | Penguin Computing is an American private supplier of enterprise, artificial intelligence (AI), high-performance computing (HPC), software-defined storage, and cloud computing solutions with a comprehensive portfolio of products, ranging from Linux servers and workstations to integrated, turnkey HPC clusters and cluster management software. |
| Products Offered     | <ul style="list-style-type: none"> <li>▪ Tundra AP</li> <li>▪ Tundra ES for HPC</li> <li>▪ OCP-based Servers for Tundra</li> </ul>                                                                                                                                                                                                                    |
| Geographic Presence  | North America                                                                                                                                                                                                                                                                                                                                         |

Source: Company Website, Annual Reports, Company Publications, and Press Releases

## 15 APPENDIX

### 15.1 INSIGHTS OF INDUSTRY EXPERTS



"The demand for HPC is growing at a significant rate in healthcare sector due to the growing need for high-speed processing of patient datasets to reduce processing wait time."



"An unprecedented disturbance in the market for high-speed computing will be experienced due to Russia-Ukraine conflict owing to the decline in supply of palladium required for semiconductor manufacturing."



"North America is expected to hold the largest share of the global high-performance computing market in the post-COVID-19 period."



"North America is expected to hold the largest share of the global high-performance computing market in the post-COVID-19 period."

Source: Interviews with Experts

## 15.2 DISCUSSION GUIDE

**Q. 1.** Briefly introduce your company and its role in the high-performance computing market.

Response: \_\_\_\_\_

**Q. 2.** How does your company differentiate itself from competitors?

Primary's Viewpoint \_\_\_\_\_

**Q. 3.** How is your company positioning itself to take advantage of the growing demand for high-performance computing around the globe?

Primary's Viewpoint \_\_\_\_\_

### MARKET DYNAMICS

**Q. 4.** According to you, what are the driving factors, restraints, opportunities, and challenges related to the high-performance computing market?

| PARTICULAR    | DESCRIPTION |
|---------------|-------------|
| Drivers       |             |
| Restraints    |             |
| Opportunities |             |
| Challenges    |             |

### MARKET SEGMENTATION, SIZE, AND GROWTH

**Q. 5.** We have segregated the high-performance computing market by component, computation type, deployment, organization size, and vertical. Please provide a percentage split for the following segments for 2021 and 2022. What are the figures likely to be by 2027? Any further details would be helpful.

| HIGH-PERFORMANCE COMPUTING MARKET,<br>BY COMPONENT  | 2021 | 2022 | 2027 |
|-----------------------------------------------------|------|------|------|
| Solutions                                           |      |      |      |
| Software                                            |      |      |      |
| Service                                             |      |      |      |
| HIGH-PERFORMANCE COMPUTING MARKET,<br>BY DEPLOYMENT | 2021 | 2022 | 2027 |
| Cloud                                               |      |      |      |
| On-Premises                                         |      |      |      |

| HIGH-PERFORMANCE COMPUTING MARKET,<br>BY ORGANIZATION SIZE | 2021 | 2022 | 2027 |
|------------------------------------------------------------|------|------|------|
| Small- and Medium-sized Enterprise                         |      |      |      |
| Large Enterprise                                           |      |      |      |

| HIGH-PERFORMANCE COMPUTING MARKET,<br>BY VERTICAL | 2021 | 2022 | 2027 |
|---------------------------------------------------|------|------|------|
| Banking, Finance, Services, and Insurance         |      |      |      |
| Earth Science                                     |      |      |      |
| Education & Research                              |      |      |      |
| Energy & Utilities                                |      |      |      |
| Government & Defense                              |      |      |      |
| Healthcare & Life sciences                        |      |      |      |
| Manufacturing                                     |      |      |      |
| Media & Entertainment                             |      |      |      |
| Others                                            |      |      |      |

Note: Others include retail, transportation, travel & hospitality, and real estate

Further details: \_\_\_\_\_

**Q. 6.** We have segregated the high-performance computing market into the following geographies. Please provide a percentage split for the regions for 2021 and 2022. What are the figures likely to be in 2027? Any further details would be helpful.

| REGION              | 2021 | 2022 | 2027 |
|---------------------|------|------|------|
| North America       |      |      |      |
| Europe              |      |      |      |
| Asia Pacific (APAC) |      |      |      |
| Rest of World (RoW) |      |      |      |

Note: Rest of the World includes South America, Middle East, and Africa

Further details: \_\_\_\_\_

**Q. 7.** We have segmented the abovementioned regional markets into the following countries. Please provide the percentage split of these countries for 2021 and 2022. What are the figures likely to be in 2027? Any further details would be helpful.

#### North America

| COUNTRY | 2021 | 2022 | 2027 |
|---------|------|------|------|
| US      |      |      |      |
| Canada  |      |      |      |
| Mexico  |      |      |      |

Further details: \_\_\_\_\_

#### EUROPE:

| COUNTRY        | 2021 | 2022 | 2027 |
|----------------|------|------|------|
| Germany        |      |      |      |
| UK             |      |      |      |
| France         |      |      |      |
| Italy          |      |      |      |
| Rest of Europe |      |      |      |

*Note: Rest of Europe include all the countries not mentioned above*

Further details: \_\_\_\_\_

#### ASIA PACIFIC (APAC):

| COUNTRY              | 2021 | 2022 | 2027 |
|----------------------|------|------|------|
| China                |      |      |      |
| Japan                |      |      |      |
| South Korea          |      |      |      |
| Rest of Asia Pacific |      |      |      |

*Note: Rest of Asia Pacific include all the countries not mentioned above*

Further details: \_\_\_\_\_

#### REST OF THE WORLD (ROW):

| COUNTRY              | 2021 | 2022 | 2027 |
|----------------------|------|------|------|
| South America        |      |      |      |
| Middle East & Africa |      |      |      |

Further details: \_\_\_\_\_

- Q. 8.** What, according to you, will be the growth rate/CAGR of the high-performance computing market in the next 5 years? Will demand for high-performance computing solutions, software, and services rise rapidly?

Primary's Viewpoint \_\_\_\_\_

- Q. 9.** Please list the top five players in the high-performance computing market.

| COMPANIES | MARKET SHARE % | RANK |
|-----------|----------------|------|
|           |                |      |
|           |                |      |
|           |                |      |
|           |                |      |

- Q. 10.** What are your key strategies to enhance market share in terms of reach, presence, and/or revenue?

Primary's Viewpoint \_\_\_\_\_

- Q. 11.** Please indicate any prominent developments related to international unions/associations, regulations, government policies, and/or private organizations.

Primary's Viewpoint \_\_\_\_\_

- Q. 12.** What, according to you, are the key pain areas for stakeholders in this market?

Additional Information about the Market: \_\_\_\_\_

- Q. 13.** What, according to you, is the impact of the COVID-19 pandemic on the high-performance computing market?

Additional information about the market: \_\_\_\_\_

## 15.3 KNOWLEDGE STORE: MARKETSandMARKETS' SUBSCRIPTION PORTAL

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- Energy & Power
- Food & Beverage
- Healthcare
- Information & Communications Technology
- Packaging, Construction, Mining & Gases
- Retail & eCommerce
- Semiconductor & Electronics**

 Augmented/Virtual Reality
  Battery & Wireless Charging
  Communication & Connectivity Technology
  Display Technology
  Drones & Robotics
  Electrical System & Components

 Electronics System & Components
  Industrial Automation
  Information System & Analytics
  Security & Surveillance
  Semiconductor Materials & Components
  Sensors and Control

 Smart Home
  Smart Lighting
  Testing, Inspection & Certification

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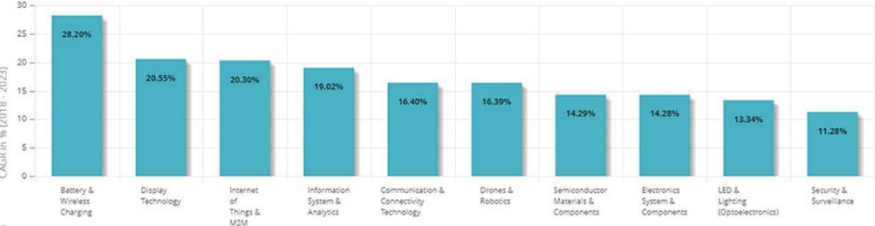
### HIGH GROWTH MARKETS

**Industries** 

- Aerospace & Defence (285)
- Agriculture (210)
- Automotive & Transportation (270)
- Chemicals & Material (1119)
- Energy & Power (363)
- Food & Beverage (309)
- Healthcare (928)
- Information & Communications Technology (852)
- Packaging, Construction, Mining & Gases (158)
- Semiconductor & Electronics (915)**
  - ☒ Battery & Wireless Charging (13)
  - ☒ Communication & Connectivity Technology (81)
  - ☒ Display Technology (72)
  - ☒ Drones & Robotics (22)
  - ☒ Electrical System & Components (56)
  - ☒ Electronics System & Components (161)
  - ☒ Industrial Automation (104)
  - ☒ Information System & Analytics (40)
  - ☒ Internet of Things & M2M (34)
  - ☒ LED & Lighting (Optoelectronics) (50)
  - ☒ Security & Surveillance (71)
  - ☒ Semiconductor Materials & Components (81)
  - ☒ Sensors and Control (89)
  - ☒ Testing, Inspection & Certification (41)

**Subscription Status**

- ☐ Subscribed
- ☐ Not Subscribed



| Market                                  | CAGR % (2018 - 2023) |
|-----------------------------------------|----------------------|
| Battery & Wireless Charging             | 28.20%               |
| Display Technology                      | 20.55%               |
| Internet of Things & M2M                | 20.30%               |
| Information System & Analytics          | 19.02%               |
| Communication & Connectivity Technology | 16.40%               |
| Drones & Robotics                       | 16.39%               |
| Semiconductor Materials & Components    | 14.29%               |
| Electronics System & Components         | 14.28%               |
| LED & Lighting (Optoelectronics)        | 13.34%               |
| Security & Surveillance                 | 11.28%               |

Search Reports

| REPORT TITLE                                                                                                                                                                                                                                                                                                                                                    | DOMAIN                      | MARKET SIZE (USD Bn - 2018) | CAGR % | PUBLISH DATE |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------------|--------|--------------|
|  <b>LTE Advanced Pro Market by Communication Infrastructure (Small Cell, Macro Cell, Ran Equipment, DAS), Core Network Technology (SDN, NFV), Deployment Location (Urban Areas, Public Spaces, Rural Areas, Residential) and Geography - Global Forecast to 2023</b>         | Semiconductor & Electronics | 0.15                        | 121.6  | Oct 2018     |
|  <b>G-Fast Chipset Market by Deployment Type (DPU and CPE), End User (Residential, and Enterprise/Commercial), Copper-Line Length (Lines Shorter Than 100 Meters, Lines Of 100 Meters-150 Meters, Lines Longer Than 250 Meters) - Global Forecast to 2022</b>                | Semiconductor & Electronics | 0.19                        | 116.5  | Nov 2016     |
| <b>5G Fixed Wireless Access (FWA) Market by Offering (Hardware: Access Units &amp; CPE, Services), Demography (Urban, Semi-Urban, Rural), Application (Residential, Commercial, Industrial, Government), and Region - Global Forecast to 2026</b>                                                                                                               | Semiconductor & Electronics | 0.2                         | 97.47  | Feb 2019     |
|  <b>Free Space Optics (FSO) and Visible Light Communication (VLC)/Light Fidelity (Li-Fi) Market by Component (LED, Photodetector, Microcontroller, and Software), Transmission Type, Application, and Geography - Global Forecast to 2023</b>                                | Semiconductor & Electronics | 2.83                        | 93.34  | Sep 2018     |
|  <b>Neuromorphic Computing Market by Offering (Hardware, Software), Application (Image Recognition, Signal Recognition, Data Mining), Industry (Aerospace &amp; Defense, IT &amp; Telecom, Automotive, Medical &amp; Industrial) and Geography - Global Forecast to 2022</b> | Semiconductor & Electronics | 0.03                        | 86     | Sep 2016     |
|  <b>Micro-LED Market by Application (Display (Smartwatch, NTE Devices (AR, VR), Smartphone, Television, HUD, Digital Signage, Laptop &amp; Monitor) and Lighting (General, Automotive)), Display Panel Size, Vertical, and Geography - Global Forecast to 2025</b>           | Semiconductor & Electronics | 0.33                        | 80.1   | May 2018     |
|  <b>Blockchain in Manufacturing Market by Application (Business Process Optimization, Logistics and Supply Chain Management, Counterfeit Management), End Use (Automotive, Energy &amp; Power, Industrial, Pharmaceuticals), and Region - Global Forecast to 2025</b>        | Semiconductor & Electronics | 0.01                        | 80     | Sep 2018     |

## 15.4 AVAILABLE CUSTOMIZATIONS

With the given market data, MarketsandMarkets offers customizations based on the company's specific needs. The following customization options are available for the report.

### COMPANY INFORMATION

- Detailed analysis and profiling of additional market players (up to five)

## 15.5 RELATED REPORTS

| NO. | REPORT TITLE                                                                                                                                                                                                                                                                                                                                                                                                                                 | PUBLISHED DATE |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 1   | <b>NEUROMORPHIC COMPUTING MARKET - GLOBAL FORECAST TO 2026</b><br>By Offering, Deployment, Application (Image Recognition, Signal Recognition, Data Mining), Vertical (Aerospace, Military, & Defense, Automotive, Medical) and Geography<br><a href="https://www.marketsandmarkets.com/Market-Reports/neuromorphic-chip-market-227703024.html">https://www.marketsandmarkets.com/Market-Reports/neuromorphic-chip-market-227703024.html</a> | May 2021       |
| 2   | <b>QUANTUM COMPUTING MARKET- GLOBAL FORECAST TO 2026</b><br>By Offering (Systems and Services), Deployment (On Premises and Cloud Based), Application, Technology, End-use Industry and Region<br><a href="https://www.marketsandmarkets.com/Market-Reports/quantum-computing-market-144888301.html">https://www.marketsandmarkets.com/Market-Reports/quantum-computing-market-144888301.html</a>                                            | February 2021  |

## 15.6 AUTHOR DETAILS

### **Apurva Agarwal**

*Associate Director,  
Semiconductor and  
Electronics*

Apurva is a Management and Engineering professional with more than 14 years of experience in TMT with a focus on core tech areas such as Comms, Compute, Networking and Storage. He is currently focusing on the impact of Emerging Technologies such as IoT, AI & 5G on the global technology value chain. He has previously worked with STMicroelectronics and Samsung, and as a TMT equity analyst for investment banks.

Apurva has done his Engineering from Birla Institute of Technology & Science (BITS), Pilani and has completed his post-graduation in management from Indian Institute of Management, Bangalore.

### **Contributors**

Raghuraj Singh,  
*Associate Manager*

Harshal Gadgil,  
*Research Analyst*

Anant Kajale,  
*Research Associate*

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